Elementary Analysis

— ***EYNTKA*** **Series** —

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Here (functions, numbers, sets, etc.)

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The Real Numbers

Arithmetic in $\mathbb{Q}$ settles every question it raises about itself. Sums, products and quotients of rationals are rational, the order behaves under both operations, and a rational is determined by finitely much information. The failure appears at the first question that is not about arithmetic. A square of side 1 has a diagonal whose square is 2, and proposition 1.1.3 shows that no rational has square 2. What goes wrong is sharper than a missing element. Consider the set of rationals whose square is less than 2. It is bounded above, by 2 for instance. It has no smallest upper bound in $\mathbb{Q}$: given any rational upper bound, an explicit smaller one can be produced. So $\mathbb{Q}$ carries a set with upper bounds and no least upper bound, and that is a defect visible entirely from inside $\mathbb{Q}$, with no appeal to a larger system.

Repairing that defect is what this chapter does, and the repair is what every later chapter spends. The rationals are taken here as a given ordered field, as are the set-theoretic notions of function, bijection and countability. The real numbers are built from them, as equivalence classes of Cauchy sequences of rationals, so that completeness becomes a theorem about the construction, where the axiomatic route imposes it. The pattern of the chapter, and of the book, is to state first what fails in $\mathbb{Q}$ and then what completeness gives in its place.

The construction chosen here names a real number by a sequence of rationals closing in on it. The decimal expansions $1, 1.4, 1.41, 1.414, \ldots$ are such a sequence, and the construction is the observation that the sequence is already a serviceable name for the number it fails to reach. An older construction instead cuts $\mathbb{Q}$ in two at the missing number. Appendix A carries it, and theorem 1.5.2 shows the two produce the same $\mathbb{R}$.

Section 1.1 builds $\mathbb{R}$ from Cauchy sequences of rationals, gives it its field operations and its order, and places $\mathbb{Q}$ inside it. Section 1.2 proves the least upper bound property, which under this construction takes a bisection argument, where the cut route would allow a one-line union. Section 1.3 develops the arithmetic of suprema and the absolute value, the estimates later chapters use more than any other result here, and extracts n-th roots. Section 1.4 draws out the Archimedean property in its sharpened form, the density of $\mathbb{Q}$ and of the irrationals, the nested interval theorem and the uncountability of

$\mathbb{R}$. Section 1.5⋆ closes by showing that the construction was forced: any two complete ordered fields are isomorphic, so $\mathbb{R}$ is not one choice among many.

1.1 Construction of the Real Numbers

Five things are taken as given throughout, and every argument below uses only these. First, $\mathbb{Q}$ is an ordered field in the sense of definition 1.1.1. Second, $\mathbb{Q}$ is Archimedean, also in the sense of that definition. Third, every rational number can be written as m/n with $m \in \mathbb{Z}$ and n a positive integer. Fourth, a statement about $\mathbb{Z}$: every nonempty set of positive integers has a least element. Fifth, also about $\mathbb{Z}$, every integer is either even or odd, and a rational may be written in lowest terms in exactly one way. The parity clause is used only in proposition 1.1.3. The uniqueness half of the fifth is what makes Thomae's function well defined in section 3.2, which is also where the least element of the fourth is used a second time.

Definition 1.1.1: Ordered Field and the Archimedean Property

An *ordered field* is a field F together with a relation $<$ such that for all $x, y, z \in F$:

1. exactly one of $x < y$, $x = y$, $y < x$ holds;
2. if $x < y$ and $y < z$ then $x < z$;
3. if $x < y$ then $x + z < y + z$;
4. if $x < y$ and $0 < z$ then $xz < yz$.

Here $x > y$ means $y < x$, and $x \leq y$ means that $x < y$ or $x = y$. The element $x - y$ is $x + (-y)$, and $|x|$ denotes x if $x \geq 0$ and $-x$ otherwise. An ordered field F is *Archimedean* if for every $x \in F$ there is a positive integer n with $x < n \cdot 1$.

The consequences recorded next hold in every ordered field, so they are available for $\mathbb{Q}$ now and for $\mathbb{R}$ once theorem 1.1.10 is proved. Every later inequality manipulation in this book cites this proposition, since what a step uses is one of these numbered facts.

Proposition 1.1.2: Consequences of the Ordered Field Axioms

Let F be an ordered field and let $x, y, z, w \in F$.

1. Cancellation holds: $x + z = y + z$ implies $x = y$, and if $z \neq 0$ then $xz = yz$ implies $x = y$.
2. The elements 0, 1, $-x$ and, for $x \neq 0$, x^{-1} are unique, and $-(-x) = x$.
3. $x \cdot 0 = 0$, and $xy = 0$ forces $x = 0$ or $y = 0$.
4. $(-x)y = -(xy)$ and $(-x)(-y) = xy$.
5. If $z > 0$ and $x < y$ then $xz < yz$; if $z < 0$ and $x < y$ then $yz < xz$.
6. $x < y$ if and only if $0 < y - x$, and $x < y$ if and only if $-y < -x$. Moreover $x \neq 0$ implies $x^2 > 0$, so $1 > 0$.
7. If $0 < x < y$ then $0 < y^{-1} < x^{-1}$.
8. If $x < y$ then $x < (x + y)(1 + 1)^{-1} < y$.
9. $|x + y| \leq |x| + |y|$, and $|xy| = |x||y|$.

***Proof*:**
(1) and (2). Adding $-z$ to $x + z = y + z$ gives $x = y$. Multiplying $xz = yz$ by z^{-1} gives $x = y$. If $0'$ is a second additive identity then $0 = 0 + 0' = 0'$, and the same argument gives uniqueness of 1. If y and y' both satisfy $x + y = 0 = x + y'$ then $y = y'$ by (1), and $-(-x) = x$ because x satisfies the defining property of $-(-x)$. For $x \neq 0$ the element x^{-1} is the only y with $xy = 1$, since $y = y \cdot 1 = y(xx^{-1}) = (yx)x^{-1} = x^{-1}$.

(3). From $x \cdot 0 = x(0 + 0) = x \cdot 0 + x \cdot 0$ and cancellation, $x \cdot 0 = 0$. If $xy = 0$ and $x \neq 0$ then $y = x^{-1}(xy) = x^{-1} \cdot 0 = 0$.

(4). $xy + (-x)y = (x + (-x))y = 0 \cdot y = 0$, so $(-x)y = -(xy)$. Applying this twice gives $(-x)(-y) = -(x(-y)) = -(-(xy)) = xy$.

(5). If $x < y$ and $z > 0$ then $0 < y - x$ by axiom (3), so $0 < (y - x)z$ by axiom (4), and $(y - x)z = yz - xz$ by distributivity and (4), whence $xz < yz$. If $z < 0$ then $-z > 0$, so $(y - x)(-z) > 0$. By (4) this element is $-(yz - xz)$, so $yz - xz < 0$, that is $yz < xz$.

(6). Adding $-x$ to $x < y$ gives $0 < y - x$, and adding x to $0 < y - x$ gives $x < y$. Adding $-x - y$ to $x < y$ gives $-y < -x$. If $x > 0$ then $x^2 > 0$ by axiom (4). If $x < 0$ then $-x > 0$ and $x^2 = (-x)(-x) > 0$ by (4) again. Since $1 \neq 0$ and $1 = 1 \cdot 1$, this gives $1 > 0$.

(7). If $0 < x$ then $x^{-1} \neq 0$, and $x^{-1} < 0$ would give $1 = x\,x^{-1} < 0$ by (5), contradicting (6). So $x^{-1} > 0$. Now $0 < x < y$ gives $x\,(xy)^{-1} < y\,(xy)^{-1}$ by (5), which is $y^{-1} < x^{-1}$.

(8). From $x < y$, adding x gives $x + x < x + y$ and adding y gives $x + y < y + y$. Since $1 + 1 > 0$ by (6), multiplying by $(1 + 1)^{-1}$, which is positive by (7), preserves both inequalities, and $(x + x)(1 + 1)^{-1} = x$.

(9). Both sides of the triangle inequality are unchanged by replacing (x, y) with $(-x, -y)$, so it suffices to treat $x + y \geq 0$, where $|x + y| = x + y \leq |x| + |y|$ because $x \leq |x|$ and $y \leq |y|$. The

multiplicative identity follows by checking the four sign cases with (4), completing the proof.

The Archimedean property is a hypothesis on $\mathbb{Q}$ separate from the field and order axioms, and it is worth saying exactly where the construction spends it, since most of what follows does not. Neither lemma 1.1.6 nor the trichotomy argument of theorem 1.1.10 needs it: both hold in any ordered field. Density of $\mathbb{Q}$ in itself does not need it either, being proposition 1.1.2(8). It is used in exactly three places. Example 1.1.5(2) and (3) need an index beyond which a prescribed rational bound is beaten. Proposition 1.1.12 transports the property from $\mathbb{Q}$ to $\mathbb{R}$. And theorem 1.2.4 needs an index N with $(1+1)^{-(N-1)}$ smaller than a given rational, which is what makes its bisection terminate.

Arithmetic in $\mathbb{Q}$ answers every question about $\mathbb{Q}$ that arithmetic can pose, and fails at the first question that geometry poses.

Proposition 1.1.3: $\sqrt{2}$ Is Irrational

There is no $x \in \mathbb{Q}$ with $x^2 = 2$.

***Proof*:**
Suppose $x \in \mathbb{Q}$ satisfies $x^2 = 2$. Write $x = m/n$ with $m \in \mathbb{Z}$ and n a positive integer, in lowest terms, so that m and n are not both even. Then $m^2 = 2n^2$.

An odd integer $m = 2j+1$ has $m^2 = 4j^2+4j+1 = 2(2j^2+2j)+1$, which is odd. Since $m^2 = 2n^2$ is even, m is therefore not odd, so m is even, say $m = 2k$. Then $4k^2 = 2n^2$, so $n^2 = 2k^2$, and the same argument applied to n shows n is even. So m and n are both even, contradicting the choice of lowest terms, as we sought to show.

What proposition 1.1.3 exhibits is not merely a missing element. The rationals 1, 1.4, 1.41, 1.414, ... obtained by taking successively more decimal digits of a square root of 2 crowd together: any two terms beyond the kth differ by less than 10^{-k}. Inside $\mathbb{Q}$ that sequence approaches nothing at all, and the construction below is the observation that such a sequence is itself a perfectly good name for the number it fails to reach.

Definition 1.1.4: Cauchy Sequence of Rationals

A sequence $(a_n)_{n\geq 1}$ of rational numbers is a *Cauchy sequence* if for every rational $\epsilon > 0$ there is a positive integer N such that $|a_n - a_m| < \epsilon$ whenever $n, m \geq N$. The sequence is *null* if for every rational $\epsilon > 0$ there is a positive integer N with $|a_n| < \epsilon$ for all $n \geq N$.

Two features of the definition deserve stating before it is used. It quantifies over *rational* ϵ only, which is all that is available at this stage, and it never mentions a limit: the condition is entirely about the terms of the sequence and their distances from one another. That is what allows a sequence with no rational limit to satisfy it.

Example 1.1.5: Three Sequences of Rationals

1. The constant sequence $a_n = q$ is Cauchy for every $q \in \mathbb{Q}$, since $|a_n - a_m| = 0$.
2. Let a_n be the largest rational of the form $k/10^n$ with $k \in \mathbb{Z}$ and $(k/10^n)^2 < 2$. For $m \geq n$ the terms a_n and a_m agree to n decimal places, so $|a_n - a_m| \leq 10^{-n}$, and given a rational $\epsilon > 0$ the Archimedean property gives an N with $10^{-N} < \epsilon$. So (a_n) is Cauchy. By proposition 1.1.3 no rational is its limit.
3. The sequence $a_n = 1/n$ is null, again by the Archimedean property: given $\epsilon > 0$ choose N with $N > 1/\epsilon$.

Lemma 1.1.6: Cauchy Sequences Are Bounded

If (a_n) is a Cauchy sequence of rationals then there is a rational M with $|a_n| \leq M$ for every n.

***Proof*:**
Take $\epsilon = 1$ in definition 1.1.4 and let N be the corresponding index. For $n \geq N$ the triangle inequality gives $|a_n| \leq |a_n - a_N| + |a_N| < 1 + |a_N|$. The finitely many terms $a_1, \ldots, a_{N-1}$ have a largest absolute value, since a finite nonempty set of rationals has a largest element. Taking M to be the larger of that value and $1 + |a_N|$ bounds every term, completing the proof.

Two Cauchy sequences should name the same number exactly when they eventually become indistinguishable, and the following makes that precise and shows it behaves as an identification must.

Definition 1.1.7: Equivalence of Cauchy Sequences

Cauchy sequences (a_n) and (b_n) of rationals are *equivalent*, written $(a_n) \sim (b_n)$, if the sequence $(a_n - b_n)$ is null.

Lemma 1.1.8: Equivalence of Cauchy Sequences Is an Equivalence Relation

The relation $\sim$ of definition 1.1.7 is reflexive, symmetric and transitive on the set of Cauchy sequences of rationals.

***Proof*:**
Reflexivity and symmetry are immediate, since $a_n - a_n = 0$ and $|a_n - b_n| = |b_n - a_n|$. For transitivity suppose $(a_n - b_n)$ and $(b_n - c_n)$ are null and let $\epsilon > 0$ be rational. Choose N_1 with $|a_n - b_n| < \epsilon(1+1)^{-1}$ for $n \geq N_1$ and N_2 with $|b_n - c_n| < \epsilon(1+1)^{-1}$ for $n \geq N_2$. For n at least the larger of N_1 and N_2,

$$|a_n - c_n| \leq |a_n - b_n| + |b_n - c_n| < \epsilon$$

by proposition 1.1.2(9), completing the proof.

Definition 1.1.9: Real Numbers

The set of *real numbers* $\mathbb{R}$ is the set of equivalence classes of Cauchy sequences of rationals under $\sim$. The class of (a_n) is written $[a_n]$, and a sequence belonging to a class is called a *representative* of it.

A real number is therefore not a single object of $\mathbb{Q}$ but a whole family of sequences agreeing in the limit, and every definition made on $\mathbb{R}$ below is first written in terms of a representative and then shown not to depend on which representative was chosen. That verification is not a formality: a formula on representatives that failed it would define nothing.

Theorem 1.1.10: $\mathbb{R}$ Is an Ordered Field

Define, for Cauchy sequences (a_n) and (b_n) of rationals,

$$[a_n] + [b_n] = [a_n + b_n], \qquad [a_n] \cdot [b_n] = [a_n b_n]$$

and declare $[a_n] < [b_n]$ when there are a rational $\delta > 0$ and an index N with $b_n - a_n > \delta$ for all $n \geq N$. These are well defined, and with them $\mathbb{R}$ is an ordered field whose zero element is the class of the constant sequence 0 and whose unit element is the class of the constant sequence 1.

***Proof*:**
Step 1: the operations produce Cauchy sequences. If (a_n) and (b_n) are Cauchy then so is $(a_n + b_n)$, since $|(a_n + b_n) - (a_m + b_m)| \leq |a_n - a_m| + |b_n - b_m|$ and each term can be made smaller than $\epsilon(1+1)^{-1}$. For the product, take bounds M_a and M_b from lemma 1.1.6 and write

$$|a_n b_n - a_m b_m| = |a_n(b_n - b_m) + b_m(a_n - a_m)| \leq M_a|b_n - b_m| + M_b|a_n - a_m|$$

so choosing N to make each difference smaller than $\epsilon\,(M_a + M_b + 1)^{-1}$ suffices.

Step 2: the operations are well defined. Suppose $(a_n) \sim (a'_n)$ and $(b_n) \sim (b'_n)$. Then $(a_n + b_n) - (a'_n + b'_n) = (a_n - a'_n) + (b_n - b'_n)$ is a sum of two null sequences and so is null. For the product, with bounds M for (a_n) and M' for (b'_n),

$$|a_n b_n - a'_n b'_n| = |a_n(b_n - b'_n) + b'_n(a_n - a'_n)| \leq M|b_n - b'_n| + M'|a_n - a'_n|$$

and the right-hand side is null. So both operations depend only on the classes.

Step 3: the field axioms. Commutativity, associativity and distributivity hold termwise in $\mathbb{Q}$ and therefore hold for classes. The class of the constant sequence 0 is an additive identity and $[-a_n]$ is an additive inverse of $[a_n]$. Only the multiplicative inverse needs an argument. Let $[a_n] \neq 0$, so (a_n) is not null: there is a rational $\delta > 0$ such that $|a_n| \geq \delta$ for infinitely many n. Choose N with $|a_n - a_m| < \delta(1+1)^{-1}$ for $n, m \geq N$ and pick $m \geq N$ with $|a_m| \geq \delta$. Then for every $n \geq N$,

$$|a_n| \geq |a_m| - |a_n - a_m| > \delta - \delta(1+1)^{-1} = \delta(1+1)^{-1}$$

so $a_n \neq 0$ for $n \geq N$. Define $b_n = a_n^{-1}$ for $n \geq N$ and $b_n = 0$ otherwise. For $n, m \geq N$,

$$|b_n - b_m| = \frac{|a_m - a_n|}{|a_n||a_m|} \leq |a_m - a_n| \cdot \frac{(1+1)^2}{\delta^2}$$

so (b_n) is Cauchy, and $(a_n b_n)$ differs from the constant sequence 1 in at most the first $N-1$ terms, so $[a_n][b_n] = 1$.

Step 4: the order is well defined and total. Suppose $[a_n] < [b_n]$ with witnesses δ and N, and let $(a'_n) \sim (a_n)$ and $(b'_n) \sim (b_n)$. Choose $N' \geq N$ beyond which both $|a_n - a'_n|$ and $|b_n - b'_n|$ are less than $\delta(1+1+1+1)^{-1}$. For $n \geq N'$,

$$b'_n - a'_n = (b_n - a_n) + (b'_n - b_n) + (a_n - a'_n) > \delta - \delta(1+1)^{-1} = \delta(1+1)^{-1}$$

so the relation holds for the primed representatives with witness $\delta(1+1)^{-1}$. For trichotomy, let $[a_n]$ and $[b_n]$ be given and put $c_n = b_n - a_n$. If (c_n) is null the classes are equal. Otherwise, by the argument of Step 3 there are $\delta > 0$ and N with $|c_n| > \delta$ for all $n \geq N$. Since (c_n) is Cauchy it cannot change sign beyond the index where its terms differ by less than δ, so either $c_n > \delta$ for all large n, giving $[a_n] < [b_n]$, or $c_n < -\delta$ for all large n, giving $[b_n] < [a_n]$. Exactly one of the three holds, since a null sequence cannot satisfy either strict condition.

Step 5: the order axioms. Transitivity follows by adding the two witnesses. If $[a_n] < [b_n]$ with witness δ then for any $[c_n]$ the sequence $(b_n + c_n) - (a_n + c_n) = b_n - a_n$ exceeds δ beyond N, giving compatibility with addition. If moreover $0 < [c_n]$ with witness γ then $b_n c_n - a_n c_n = (b_n - a_n)c_n > \delta\gamma$ beyond the larger index, giving compatibility with multiplication, as we sought to show.

The construction has produced a field, and the next proposition places $\mathbb{Q}$ inside it so that the two arithmetics agree.

Proposition 1.1.11: $\mathbb{Q}$ Embeds in $\mathbb{R}$ as an Ordered Subfield

For $q \in \mathbb{Q}$ let q^* denote the class of the constant sequence with every term q. The map $q \mapsto q^*$ is injective and satisfies

$$q^* + r^* = (q+r)^*, \qquad q^* r^* = (qr)^*, \qquad -(q^*) = (-q)^*$$

for all $q, r \in \mathbb{Q}$, together with $q < r$ if and only if $q^* < r^*$. It sends 0 to the zero of $\mathbb{R}$ and 1 to its unit.

***Proof*:**
The three displayed identities hold termwise. If $q < r$ then the constant sequence $r - q$ exceeds the rational $\delta = (r-q)(1+1)^{-1} > 0$ at every index, so $q^* < r^*$. Conversely $q^* < r^*$ excludes $q = r$ and, by trichotomy in $\mathbb{Q}$, excludes $r < q$, which would give $r^* < q^*$. Injectivity follows, since $q \neq r$ forces $q^* \neq r^*$, completing the proof.

From here on a rational q and its image q^* are written the same way, and $\mathbb{Q}$ is treated as a subset of $\mathbb{R}$. Theorem 1.1.10 then transfers every consequence recorded in proposition 1.1.2 to real numbers, including the triangle inequality, which section 1.3 develops in the form later chapters use.

The construction also settles, at once, a property that the cut route obtains only after the least upper bound property is available. It is recorded here because section 1.2 uses it to prove that property, and a proof there that leaned on theorem 1.4.1 would be circular.

Proposition 1.1.12: $\mathbb{R}$ Is Archimedean

For every $x \in \mathbb{R}$ there is a positive integer n with $x < n$.

***Proof*:**
Let (a_n) be a representative of x and take a rational bound M with $|a_k| \le M$ for all k, which exists by lemma 1.1.6. Write $M = p/q$ with $p \in \mathbb{Z}$ and q a positive integer, so $M \le p$. Since $\mathbb{Q}$ is Archimedean there is a positive integer n with $p + 1 < n$. Then $n - a_k \ge n - M \ge n - p > 1$ for every k, so the constant sequence n exceeds (a_k) by the fixed rational 1 at every index, which is exactly $x < n$ in the sense of theorem 1.1.10, completing the proof.

1.1.13 Note: The Construction by Dedekind Cuts A second construction of $\mathbb{R}$ partitions $\mathbb{Q}$ where the one above sequences it: a *cut* is a pair $A|B$ of nonempty sets with $A \cup B = \mathbb{Q}$, every element of A below every element of B, and A having no largest element, and $\mathbb{R}$ is the set of cuts. Appendix A carries it in full. The two constructions produce ordered fields with the least upper bound property, and theorem 1.5.2 shows that any two such fields are isomorphic, so they produce the same $\mathbb{R}$. The cut route reaches the least upper bound property faster, by taking a union of lower sets. The sequence route reaches the Cauchy criterion faster, and generalizes to any metric space, which is the form the rest of this series uses.

Exercise 1.1.1

1. Show that a null sequence of rationals is Cauchy, and that a Cauchy sequence with a subsequence that is null is itself null.
2. Let (a_n) be Cauchy and let (b_n) agree with (a_n) for all but finitely many indices. Show that (b_n) is Cauchy and $[a_n] = [b_n]$. Deduce that altering finitely many terms of a representative changes nothing.
3. Show that the sequence $a_1 = 1$, $a_{n+1} = (a_n + 2a_n^{-1})(1 + 1)^{-1}$ of rationals is Cauchy, and that its class x satisfies $x^2 = 2$. Conclude that $\mathbb{R}$ contains an element $\mathbb{Q}$ does not.
4. Where does the proof of theorem 1.1.10 Step 3 use that (a_n) is not null? Termwise nonvanishing, that $a_n \ne 0$ for every n, is weaker. Give a Cauchy sequence with every term nonzero whose class has no multiplicative inverse.
5. Show that lemma 1.1.6 holds in *any* ordered field, using no Archimedean hypothesis. Then show that example 1.1.5(2) does need it, by exhibiting an ordered field in which no index N

satisfies $10^{-N} < \epsilon$ for a suitable ϵ.

6. Deduce from proposition 1.1.12 that between any two reals $x < y$ with $y - x > 1$ there is an integer, and that for every real $x > 0$ there is a positive integer n with $n^{-1} < x$.

1.2 Completeness and the Least Upper Bound Property

Section 1.1 produced an ordered field containing $\mathbb{Q}$. Every ordered field containing $\mathbb{Q}$ has that much, including $\mathbb{Q}$ itself, so nothing so far distinguishes $\mathbb{R}$. The property that does is the one the chapter opened on, and stating it needs two definitions first.

Definition 1.2.1: Upper and Lower Bound

Let $E \subseteq \mathbb{R}$. An element $u \in \mathbb{R}$ is an *upper bound* of E if $x \leq u$ for every $x \in E$, and E is *bounded above* if it has one. Symmetrically ℓ is a *lower bound* if $\ell \leq x$ for every $x \in E$, and E is *bounded below* if it has one.

An upper bound of E need not belong to E, and a set with one has many: if u is an upper bound and $u \leq v$, then $x \leq u \leq v$ for every $x \in E$, so v is an upper bound as well. The upper bounds of a set therefore form a collection closed upward in the order, and asking whether a set has an upper bound is a weak question. The same two definitions apply within $\mathbb{Q}$, with "rational number" in place of "real number" throughout, and a rational upper bound of a set of rationals is exactly an upper bound in the sense above that happens to lie in $\mathbb{Q}$ (proposition 1.1.11).

Since the upper bounds are closed upward, the informative question is whether there is a smallest one.

Definition 1.2.2: Supremum and Infimum

Let $E \subseteq \mathbb{R}$ be nonempty and bounded above. An element $s \in \mathbb{R}$ is the *supremum* of E, written $\sup E$, if s is an upper bound of E and $s \leq u$ for every upper bound u of E. Symmetrically, for E bounded below, m is the *infimum* $\inf E$ if m is a lower bound and $\ell \leq m$ for every lower bound ℓ.

A supremum is unique when it exists: if s and s' both qualify then $s \leq s'$ and $s' \leq s$, so $s = s'$ by trichotomy. Nothing yet says one exists.

Example 1.2.3: Suprema Present, Absent, and Outside the Field

1. Let $E_1 = \{x \in \mathbb{R} | 0 \leq x \leq 1\}$. Then $\sup E_1 = 1$, and the supremum belongs to E_1.

2. Let $E_2 = \{x \in \mathbb{R} | 0 \leq x < 1\}$. Then $\sup E_2 = 1$ as well, and the supremum does not belong to E_2. To see it, 1 is an upper bound. If $u < 1$ were an upper bound then proposition 1.1.2(8) gives v with $u < v < 1$, so $v \in E_2$ exceeds u, and u is not an upper bound after all.

3. Let $F = \{q \in \mathbb{Q} | q > 0 \text{ and } q^2 < 2\}$, regarded as a set of rationals. Then F has rational upper bounds, for instance 2, and *no* rational supremum. Suppose $u \in \mathbb{Q}$ were one. By proposition 1.1.3, $u^2 \neq 2$. If $u^2 < 2$, put $u' = u + (2 - u^2)(u + 2)^{-1} \cdot (1 + 1)^{-1}$. A computation gives $u' > u$ and $(u')^2 < 2$, so $u' \in F$ exceeds the upper bound u. If $u^2 > 2$, the same

expression gives $u' < u$ with $(u')^2 > 2$, so u' is a smaller upper bound. Either way u fails, so no rational supremum exists.

Part (3) is the failure the chapter opened on, now stated precisely: a set of rationals, bounded above in $\mathbb{Q}$, whose upper bounds have no least element in $\mathbb{Q}$. The theorem below says that $\mathbb{R}$ admits no such set. Under the construction of section 1.1 this takes genuine work, and the proof is the only place in the chapter where a real number is produced by naming a rational sequence and then reading off its class.

Theorem 1.2.4: The Least Upper Bound Property

Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

***Proof*:**
Let $E \subseteq \mathbb{R}$ be nonempty and bounded above, say by u_0, and fix $x_0 \in E$.

Step 1: rational brackets. By proposition 1.1.12 there is a positive integer q with $u_0 < q$, and applying the same proposition to $-x_0$ gives a positive integer p with $-x_0 < p$, that is $-p < x_0$. Put $a_1 = -p$ and $b_1 = q$, both rational. Then b_1 is an upper bound of E, since $u_0 < b_1$, and a_1 is not, since $a_1 < x_0$ and $x_0 \in E$.

Step 2: bisection. Define a_{n+1} and b_{n+1} from a_n and b_n by letting $c_n = (a_n + b_n)(1+1)^{-1}$ and setting

$$(a_{n+1}, b_{n+1}) = \begin{cases} (a_n, c_n) & \text{if } c_n \text{ is an upper bound of } E \\ (c_n, b_n) & \text{otherwise} \end{cases}$$

Each c_n is rational whenever a_n and b_n are, so both sequences are rational throughout. Two properties hold at every stage, by induction: b_n is an upper bound of E and a_n is not, and $b_n - a_n = (b_1 - a_1)(1+1)^{-(n-1)}$. The first is immediate from the case split, and the second because bisection halves the gap.

Step 3: the brackets are Cauchy. The sequence (b_n) is decreasing and (a_n) increasing, and every a_n is below every b_m: for $n \le m$ this is $a_n \le a_m < b_m$, and for $n > m$ it is $a_n < b_n \le b_m$. Hence for $n, m \ge N$ both b_n and b_m lie in the interval from a_N to b_N, so

$$|b_n - b_m| \le b_N - a_N = (b_1 - a_1)(1+1)^{-(N-1)}$$

Given a rational $\epsilon > 0$, the Archimedean property of $\mathbb{Q}$ gives an N with $(b_1 - a_1)(1+1)^{-(N-1)} < \epsilon$, since $(1+1)^{N-1} \ge N$ for every positive integer N. So (b_n) is a Cauchy sequence of rationals, and (a_n) is equivalent to it because $b_n - a_n$ is null by the same estimate. Put $s = [b_n] \in \mathbb{R}$, which by definition 1.1.9 is a real number, and note $s = [a_n]$ as well.

Step 4: s is an upper bound. Let $x \in E$ and suppose $s < x$. Then $x - s > 0$, so by proposition 1.1.12 there is a positive integer k with $(x-s)^{-1} < k$, hence $k^{-1} < x - s$. Choose N with $b_N - a_N < k^{-1}$, which Step 3 permits. Since $s = [a_n]$ and (a_n) is increasing, $a_N \le s$, so

$$b_N = a_N + (b_N - a_N) < s + k^{-1} < s + (x - s) = x$$

contradicting that b_N is an upper bound of E. So $x \le s$ for every $x \in E$.

Step 5: s is the least upper bound. Let u be an upper bound of E and suppose $u < s$. Then $s - u > 0$, and as in Step 4 choose a positive integer k with $k^{-1} < s - u$ and then N with

$b_N - a_N < k^{-1}$. Since $s = [b_n]$ and (b_n) is decreasing, $s \leq b_N$, so

$$a_N = b_N - (b_N - a_N) > s - k^{-1} > s - (s - u) = u$$

So $a_N > u$, and since a_N is not an upper bound of E there is $x \in E$ with $a_N < x$. Then $u < a_N < x$ contradicts that u is an upper bound. Hence $s \leq u$ for every upper bound u, and $s = \sup E$, completing the proof.

The proof used the construction twice and used it in an essential way: Step 3 produced a real number by exhibiting a rational Cauchy sequence, and Steps 4 and 5 compared that number with others through proposition 1.1.12, which section 1.1 obtained from the boundedness of representatives, with no appeal to completeness. Reversing that order would be circular, and theorem 1.4.1 records the point where the Archimedean property is instead read off the least upper bound property, for readers who meet $\mathbb{R}$ axiomatically.

The corresponding statement for lower bounds needs no second argument.

Corollary 1.2.5: The Greatest Lower Bound Property

Every nonempty subset of $\mathbb{R}$ that is bounded below has an infimum in $\mathbb{R}$.

***Proof*:**
Let E be nonempty and bounded below by ℓ, and put $-E = \{-x | x \in E\}$. If $\ell \leq x$ for every $x \in E$ then $-x \leq -\ell$ by proposition 1.1.2(6), so $-E$ is nonempty and bounded above by $-\ell$. Let $s = \sup(-E)$, which exists by theorem 1.2.4. Then $-s$ is a lower bound of E, since $-x \leq s$ gives $-s \leq x$. If ℓ' is any lower bound of E then $-\ell'$ is an upper bound of $-E$, so $s \leq -\ell'$ and hence $\ell' \leq -s$. So $-s = \inf E$, completing the proof.

The form in which the least upper bound property is applied is almost never the definition. What later arguments use is that a supremum is approached from below by elements of the set, and that this property characterizes it.

Proposition 1.2.6: The ϵ-Characterization of the Supremum

Let $E \subseteq \mathbb{R}$ be nonempty and let s be an upper bound of E. Then $s = \sup E$ if and only if for every real $\epsilon > 0$ there is an $x \in E$ with $s - \epsilon < x$.

***Proof*:**
Suppose $s = \sup E$ and let $\epsilon > 0$. Then $s - \epsilon < s$, so $s - \epsilon$ is not an upper bound of E, since s is the least one. So some $x \in E$ satisfies $s - \epsilon < x$.

Conversely suppose the condition holds and let u be an upper bound of E with $u < s$. Taking $\epsilon = s - u > 0$ produces $x \in E$ with $s - (s - u) < x$, that is $u < x$, contradicting that u is an upper bound. So $s \leq u$ for every upper bound u, and $s = \sup E$, as we sought to show.

Exercise 1.2.1

1. Show that a set with a largest element has that element as its supremum, and give a set whose supremum is not a largest element.
2. Let $A \subseteq B \subseteq \mathbb{R}$ with A nonempty and B bounded above. Show that $\sup A \leq \sup B$.
3. Show that $\sup E$ is the unique real number s such that $x \leq s$ for all $x \in E$ and, for every $\epsilon > 0$, some $x \in E$ exceeds $s - \epsilon$.
4. Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above with $\sup A \leq \sup B$. Show that $A \cup B$ is bounded above and that $\sup(A \cup B) = \sup B$.
5. In the bisection of theorem 1.2.4, suppose E has a largest element M. Show that M is an upper bound at every stage, so that the case split sends b_{n+1} to the midpoint whenever the midpoint is at least M, and that the conclusion $s = M$ follows. (Note b_1 is chosen as an integer above u_0 and need not equal M.)
6. Give a nonempty set of rationals, bounded above in $\mathbb{Q}$, whose supremum in $\mathbb{R}$ is irrational, and explain why example 1.2.3(3) does not contradict theorem 1.2.4.
7. Show that $\mathbb{Q}$, with the order it inherits from $\mathbb{R}$, is an ordered field that does not satisfy the least upper bound property, so that theorem 1.2.4 is not a consequence of the ordered field axioms alone.
8. Let E be nonempty and bounded above and let $c > 0$. Show that $\sup(cE) = c \sup E$, where $cE = \{cx \big| x \in E\}$, and find what replaces this when $c < 0$.

1.3 The Calculus of Suprema and Absolute Value

Section 1.2 produces suprema and infima and says nothing about how they interact with the arithmetic of $\mathbb{R}$. The interactions needed later are few: a supremum over a larger set is at least as large, a nonnegative constant passes through a supremum, and the supremum of a set of sums is the sum of the suprema. The third of these is an equality for sets and only an inequality for functions, and the difference between the two statements is what the later chapters have to keep track of when they estimate a sum of two quantities that are large at different points.

The second half of the section introduces the absolute value, proves the triangle inequality in the forms later chapters use, and combines the two themes: the number $\sup_D f - \inf_D f$ is exactly the smallest constant bounding every difference $|f(x) - f(y)|$. The section closes with the first existence theorem proved from the least upper bound property, namely that every nonnegative real has a unique nonnegative n-th root. In $\mathbb{Q}$ the corresponding statement already fails for $n = 2$ and $x = 2$ by proposition 1.1.3, so this is the point at which $\sqrt{2}$ becomes a real number, where in $\mathbb{Q}$ it was a gap.

Every argument below begins the same way: a supremum is an upper bound, so it gives an inequality valid at every point of the set, and it is the *least* upper bound, so no smaller number has that property. The second half of that sentence is applied constantly, and always in the single form recorded in proposition 1.2.6: a supremum is approached from below by elements of the set. Only the reflected statement for the infimum is still owed.

Lemma 1.3.1: Approximation Property of the Infimum

Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $m = \inf A$. Then for every $\epsilon > 0$ there is an $a \in A$ with $m \leq a < m + \epsilon$.

***Proof*:**
Let $\epsilon > 0$. Then $m + \epsilon > m$, so $m + \epsilon$ is not a lower bound of A, since m is the greatest lower bound. So some $a \in A$ satisfies $a < m + \epsilon$, and $m \leq a$ because m is a lower bound, completing the proof.

Read from right to left, proposition 1.2.6 is how a supremum is identified in practice: guess the value, check that it is an upper bound, and then check that some element of the set exceeds any smaller candidate. Read from left to right, it and the lemma above are how a supremum and an infimum are used once they are known. Neither asserts that the bound belongs to the set, and in general it does not. What they give instead is an element of the set within any prescribed tolerance of it.

The sum rule among the three manipulations promised above has a shape worth noting: the elements of $A + B$ are the sums $a + b$ where a and b range independently, so a may be pushed close to $\sup A$ while b is pushed close to $\sup B$, and nothing prevents doing both at once.

Proposition 1.3.2: The Algebra of Suprema for Sets

Let A and B be nonempty subsets of $\mathbb{R}$.

1. If $A \subseteq B$ and B is bounded above, then A is bounded above and $\sup A \leq \sup B$.
2. If A is bounded above and $c \geq 0$, then $cA = \{ca \big| a \in A\}$ is bounded above and $\sup(cA) = c \sup A$.
3. If A is bounded below, then $-A = \{-a \big| a \in A\}$ is bounded above and $\sup(-A) = -\inf A$.
4. If A and B are bounded above, then $A + B = \{a + b \big| a \in A,\ b \in B\}$ is bounded above and $\sup(A + B) = \sup A + \sup B$.

***Proof*:**

1. Every $a \in A$ lies in B and therefore satisfies $a \leq \sup B$, so $\sup B$ is an upper bound of A and A is bounded above. Both sets are nonempty and bounded above, so both suprema exist by theorem 1.2.4, and $\sup A \leq \sup B$ because $\sup A$ is the least upper bound of A while $\sup B$ is one of its upper bounds.
2. Write $s = \sup A$. If $c = 0$ then $cA = \{0\}$, since A is nonempty, and $\sup\{0\} = 0 = 0 \cdot s$. Suppose $c > 0$. For $a \in A$ the inequality $a \leq s$ multiplies by the positive number c to give $ca \leq cs$ (proposition 1.1.2(5)), so cs is an upper bound of cA and cA is bounded above. Let u be any upper bound of cA. Since $c > 0$, the inverse $1/c$ is positive (proposition 1.1.2(7)), so multiplying $ca \leq u$ by $1/c$ gives $a \leq u/c$ for every $a \in A$. Thus u/c is an upper bound of A, whence $s \leq u/c$ and therefore $cs \leq u$. So cs is the least upper bound of cA.
3. Write $m = \inf A$, which exists by corollary 1.2.5 since A is nonempty and bounded below. For $a \in A$ the inequality $m \leq a$ gives $-a \leq -m$, so $-m$ is an upper bound of $-A$. Let u

be an upper bound of $-A$. Then $-a \le u$ for every $a \in A$, hence $-u \le a$ for every $a \in A$, so $-u$ is a lower bound of A and therefore $-u \le m$, that is $-m \le u$. So $-m$ is the least upper bound of $-A$.

4. Write $s = \sup A$ and $t = \sup B$. For $a \in A$ and $b \in B$, adding $a \le s$ and $b \le t$ gives $a + b \le s + t$, so $s + t$ is an upper bound of $A + B$. The set $A + B$ is nonempty, so $\sigma = \sup(A + B)$ exists by theorem 1.2.4 and $\sigma \le s + t$. Suppose $\sigma < s + t$ and put

$$\epsilon = \frac{s + t - \sigma}{2} > 0$$

By proposition 1.2.6 applied to A and then to B, there are $a \in A$ with $a > s - \epsilon$ and $b \in B$ with $b > t - \epsilon$. Adding,

$$a + b > s + t - 2\epsilon = \sigma$$

which contradicts the fact that σ is an upper bound of $A + B$. Hence $\sigma = s + t$, completing the proof.

Part (3) is the reason the theory is usually stated for suprema alone: an infimum is a supremum of the reflected set, so every statement above has a companion obtained by replacing A with $-A$. Part (2) fails as stated for $c < 0$, and part (3) says what replaces it: if $c < 0$ and A is bounded below, then $cA = (-c)(-A)$ with $-c > 0$, so parts (2) and (3) give

$$\sup(cA) = (-c)\sup(-A) = (-c)(-\inf A) = c \inf A$$

Multiplying by a negative number reverses the order, and the effect on the two bounds is to exchange them.

Example 1.3.3: Suprema of Two Intervals and Their Sum

Let $A = [0, 1)$ be the set of $x \in \mathbb{R}$ with $0 \le x < 1$, and let $B = (-1, 2]$ be the set of $x \in \mathbb{R}$ with $-1 < x \le 2$.

The number 1 is an upper bound of A. Given $\epsilon > 0$, put $a = \max\{0, 1 - \epsilon/2\}$. Then $a \ge 0$ and $a < 1$, so $a \in A$. Moreover $a > 1 - \epsilon$, because if $a = 1 - \epsilon/2$ this is the inequality $-\epsilon/2 > -\epsilon$, while if $a = 0$ then $1 - \epsilon/2 \le 0$ forces $\epsilon \ge 2$ and hence $1 - \epsilon \le -1 < 0 = a$. By proposition 1.2.6, $\sup A = 1$, and $1 \notin A$. The number 0 lies in A and is a lower bound of A, so $\inf A = 0$. For B no approximation argument is needed: the number 2 is an upper bound of B and belongs to B, so every upper bound of B is at least 2 and $\sup B = 2$.

Scaling by $c = 3$ gives $3A = [0, 3)$, since $x \mapsto 3x$ carries the condition $0 \le a < 1$ to $0 \le 3a < 3$, and proposition 1.3.2(2) gives $\sup(3A) = 3 \sup A = 3$. For the sum, every $a + b$ with $a \in A$ and $b \in B$ satisfies $-1 < a + b < 3$. Conversely, if $-1 < z \le 2$ then $z = 0 + z$ with $0 \in A$ and $z \in B$, while if $2 < z < 3$ then $z = (z - 2) + 2$ with $z - 2 \in (0, 1) \subseteq A$ and $2 \in B$. Hence $A + B = (-1, 3)$ and $\sup(A + B) = 3 = 1 + 2 = \sup A + \sup B$. The supremum of $A + B$ is not attained even though $\sup B$ is: the equality in proposition 1.3.2(4) is an identity between numbers and says nothing about which of them belong to their sets. Finally $-A = (-1, 0]$, whose supremum is $0 = -\inf A$.

Sets are replaced by functions throughout the rest of the book, and the translation is purely notational: for a nonempty set D and a function $f : D \to \mathbb{R}$, write $f(D) = \{f(x) \big| x \in D\}$ for the range and

$$\sup_D f = \sup f(D), \qquad \inf_D f = \inf f(D)$$

calling f *bounded on* D when $f(D)$ is bounded above and below. What does not translate is the sum rule. The set $(f+g)(D)$ consists of the sums $f(x)+g(x)$ taken at a *common* point x, so it is only part of $f(D)+g(D)$, and the supremum can drop.

Proposition 1.3.4: Suprema and Infima of a Sum of Functions

Let D be a nonempty set and let $f, g\colon D \to \mathbb{R}$ be bounded. Then $f+g$ is bounded and

$$\sup_D(f+g) \leq \sup_D f + \sup_D g, \qquad \inf_D(f+g) \geq \inf_D f + \inf_D g$$

Both inequalities can be strict.

***Proof*:**
Every element of $(f+g)(D)$ has the form $f(x)+g(x)$ for some $x \in D$, and $f(x) \in f(D)$ and $g(x) \in g(D)$, so

$$(f+g)(D) \subseteq f(D) + g(D)$$

The sets $f(D)$ and $g(D)$ are nonempty and bounded above, so proposition 1.3.2(4) applies and gives that $f(D)+g(D)$ is bounded above with supremum $\sup_D f + \sup_D g$. In particular $(f+g)(D)$ is bounded above, being a subset of a set bounded above, and proposition 1.3.2(1) gives

$$\sup_D(f+g) \leq \sup\big(f(D)+g(D)\big) = \sup_D f + \sup_D g$$

For the second inequality, note that $-f$ and $-g$ are bounded because f and g are, and that $(-f)+(-g) = -(f+g)$. The inequality just proved, applied to $-f$ and $-g$, therefore shows that $-(f+g)$ is bounded above, which is to say that $f+g$ is bounded below, and gives

$$\sup_D\big(-(f+g)\big) \leq \sup_D(-f) + \sup_D(-g) \tag{1.1}$$

By proposition 1.3.2(3) applied to ranges, $\sup_D(-h) = -\inf_D h$ for every bounded $h\colon D \to \mathbb{R}$, since $(-h)(D) = -h(D)$. Rewriting all three suprema in (1.1) by this identity turns it into

$$-\inf_D(f+g) \leq -\inf_D f - \inf_D g$$

and multiplying by -1 reverses the inequality to give $\inf_D(f+g) \geq \inf_D f + \inf_D g$. Both can be strict: on $D = [0,1]$ with $f(x) = x$ and $g(x) = 1-x$ the sum is the constant function 1, so $\sup_D(f+g) = 1$ while $\sup_D f + \sup_D g = 2$, and $\inf_D(f+g) = 1$ while $\inf_D f + \inf_D g = 0$. Example 1.3.5 verifies these four values, completing the proof.

Example 1.3.5: A Strict Inequality for $\sup(f+g)$
Let $D = [0,1]$ and define $f, g\colon D \to \mathbb{R}$ by $f(x) = x$ and $g(x) = 1-x$. Then $f(D) = [0,1]$ and $g(D) = [0,1]$, so $\sup_D f = \sup_D g = 1$ and $\inf_D f = \inf_D g = 0$, each attained at an endpoint. The sum is the constant function $(f+g)(x) = 1$, so

$$\sup_D(f+g) = 1 < 2 = \sup_D f + \sup_D g$$

and likewise $\inf_D(f+g) = 1 > 0 = \inf_D f + \inf_D g$. The gap has a single cause: f is large only near $x = 1$ and g only near $x = 0$, and no point of D makes both large at once. The set version has no such obstruction, and indeed $f(D) + g(D) = [0, 2]$, whose supremum is 2.

Distances on the real line are measured by the absolute value, and every ϵ-δ and ϵ-N statement in the chapters that follow is an inequality between absolute values. The definition splits into two cases because the order on $\mathbb{R}$ does, and the point of proposition 1.3.8 is that the case split need almost never be performed again.

Definition 1.3.6: Absolute Value

For $x \in \mathbb{R}$ the *absolute value* of x is

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

For $x, y \in \mathbb{R}$ the number $|x - y|$ is called the *distance* from x to y.

Example 1.3.7: Distances Between Two Points of the Line

Directly from the definition, $|5| = 5$ because $5 \geq 0$, and $|-7/2| = -(-7/2) = 7/2$ because $-7/2 < 0$. The distance from $-7/2$ to 5 is

$$\left|-\frac{7}{2} - 5\right| = \left|-\frac{17}{2}\right| = \frac{17}{2}$$

and computing it in the other order gives $|5 - (-7/2)| = |17/2| = 17/2$, the same number. Taking $y = 0$ recovers $|x - 0| = |x|$, so the absolute value of x is its distance from the origin.

Proposition 1.3.8: Comparison, Multiplicativity and the Triangle Inequality

Let $x, y \in \mathbb{R}$ and let $c \geq 0$.

1. $|x| \geq 0$, with $|x| = 0$ if and only if $x = 0$. Moreover $|-x| = |x|$ and $-|x| \leq x \leq |x|$.
2. $|x| \leq c$ if and only if $-c \leq x \leq c$, and $|x| < c$ if and only if $-c < x < c$.
3. $|xy| = |x|\,|y|$.
4. $|x + y| \leq |x| + |y|$.
5. $\big|\,|x| - |y|\,\big| \leq |x - y|$.

***Proof*:**

1. If $x \geq 0$ then $|x| = x \geq 0$; if $x < 0$ then $|x| = -x > 0$ by the sign rule of proposition 1.1.2(4). In both cases $|x| \geq 0$, and the second case shows $|x| = 0$ forces $x \geq 0$, whence $x = |x| = 0$. Conversely, $|0| = 0$. For $|-x| = |x|$: if $x > 0$ then $-x < 0$ and $|-x| = -(-x) = x = |x|$; if $x = 0$ both sides are 0; if $x < 0$ then $-x > 0$ and $|-x| = -x = |x|$. For the two-sided bound: if $x \geq 0$ then $x = |x|$ and $-|x| \leq 0 \leq x$, while if $x < 0$ then $x = -|x|$ and $x < 0 \leq |x|$.
2. Suppose $|x| \leq c$. By (1), $x \leq |x| \leq c$ and $-x \leq |-x| = |x| \leq c$, and the latter gives $-c \leq x$.

Conversely suppose $-c \le x \le c$. If $x \ge 0$ then $|x| = x \le c$; if $x < 0$ then $|x| = -x \le c$, this last from $-c \le x$. The same argument, with the two inequalities involving c read strictly, proves the second equivalence.

3. There are four cases, and each uses the sign rules of proposition 1.1.2(4) together with the observation that $|z| = -z$ holds for every $z \le 0$, the case $z = 0$ because both sides vanish. If $x \ge 0$ and $y \ge 0$ then $xy \ge 0$, so $|xy| = xy = |x|\,|y|$. If $x \ge 0$ and $y < 0$ then $xy \le 0$, so $|xy| = -(xy) = x(-y) = |x|\,|y|$. The case $x < 0$, $y \ge 0$ is the same with the roles of x and y exchanged. If $x < 0$ and $y < 0$ then $xy = (-x)(-y) > 0$, so $|xy| = xy = (-x)(-y) = |x|\,|y|$.

4. By (1), $x \le |x|$ and $y \le |y|$, so $x + y \le |x| + |y|$. Likewise $-x \le |x|$ and $-y \le |y|$, so $-(x + y) \le |x| + |y|$, that is $-(|x| + |y|) \le x + y$. The number $c = |x| + |y|$ is nonnegative by (1), so (2) applies and gives $|x + y| \le |x| + |y|$.

5. Writing $x = (x - y) + y$ and applying (4),

$$|x| \le |x - y| + |y|, \qquad \text{so} \qquad |x| - |y| \le |x - y|$$

Exchanging x and y gives $|y| - |x| \le |y - x| = |x - y|$, the last equality by (1) applied to $x - y$. So the number $|x| - |y|$ lies between $-|x - y|$ and $|x - y|$, and $|x - y| \ge 0$ by (1), so (2) gives $\big|\,|x| - |y|\,\big| \le |x - y|$, as we sought to show.

Part (2) is the form in which the absolute value is used: an inequality $|x - a| < \epsilon$ is a pair of inequalities placing x between $a - \epsilon$ and $a + \epsilon$, so a statement about distance converts into a statement about order and back at will. Part (5) is the version that estimates a difference of sizes by the size of a difference, and it is what later chapters use to show that passing to the absolute value cannot destroy an estimate already obtained.

Example 1.3.9: Solution Sets of Two Absolute Value Inequalities

By proposition 1.3.8(2) with $c = 1/2$, applied to the number $x - 2$, the condition $|x - 2| < 1/2$ holds if and only if $-1/2 < x - 2 < 1/2$, that is if and only if $3/2 < x < 5/2$. So

$$\{x \in \mathbb{R} \big| |x - 2| < 1/2\} = (3/2,\, 5/2)$$

an interval centred at 2 of total length 1. Likewise $|2x + 1| \le 3$ holds if and only if $-3 \le 2x + 1 \le 3$, which after subtracting 1 and multiplying by the positive number $1/2$ reads $-2 \le x \le 1$, so the solution set is $[-2, 1]$. Each computation is the same move: a single condition on a distance becomes two conditions on the order, and the order conditions are solved by the field operations.

Combining the two halves of the section gives the quantity that measures how far a function moves on a set. A bounded function has a supremum and an infimum there, and their difference bounds every difference of two of its values. The content of the next result is that this bound is the best one available, so the single number $\sup_D f - \inf_D f$ can replace the whole family of differences $|f(x) - f(y)|$ without loss.

Proposition 1.3.10: The Oscillation Bound

Let D be a nonempty set and let $f\colon D \to \mathbb{R}$ be bounded. Write $M = \sup_D f$ and $m = \inf_D f$. Then

1. $|f(x) - f(y)| \leq M - m$ for all $x, y \in D$;
2. $M - m = \sup\{|f(x) - f(y)| \big| x, y \in D\}$, so $M - m$ is the smallest constant C for which $|f(x) - f(y)| \leq C$ holds for all $x, y \in D$.

The number $M - m$ is called the *oscillation* of f on D.

***Proof*:**

1. Let $x, y \in D$. Then $f(x) \leq M$ and $m \leq f(y)$, so $f(x) - f(y) \leq M - m$. Exchanging x and y gives $f(y) - f(x) \leq M - m$, that is $-(M - m) \leq f(x) - f(y)$. Since D is nonempty there is at least one $z \in D$, and $m \leq f(z) \leq M$ gives $M - m \geq 0$, so proposition 1.3.8(2) applies with $c = M - m$ and yields $|f(x) - f(y)| \leq M - m$.
2. Let S denote the set $\{|f(x) - f(y)| \big| x, y \in D\}$, which is nonempty because D is, and which is bounded above by $M - m$ by part (1). By theorem 1.2.4 the supremum $\sup S$ exists, and $\sup S \leq M - m$ because $M - m$ is an upper bound.
   Suppose $\sup S < M - m$, and put $\epsilon = \big(M - m - \sup S\big)/2 > 0$. By proposition 1.2.6 applied to the set $f(D)$, there is an element of $f(D)$ exceeding $M - \epsilon$, that is a point $x \in D$ with $f(x) > M - \epsilon$. Likewise, by lemma 1.3.1 applied to $f(D)$, there is a point $y \in D$ with $f(y) < m + \epsilon$. Subtracting,
   $$f(x) - f(y) > (M - \epsilon) - (m + \epsilon) = M - m - 2\epsilon = \sup S$$
   and $|f(x) - f(y)| \geq f(x) - f(y)$ by proposition 1.3.8(1). So S contains an element strictly larger than $\sup S$, which contradicts that $\sup S$ is an upper bound of S. Hence $\sup S = M - m$.
   For the final clause, a constant C with $|f(x) - f(y)| \leq C$ for all $x, y \in D$ is exactly an upper bound of S, hence satisfies $C \geq \sup S = M - m$, and $C = M - m$ is such a constant by part (1), completing the proof.

The oscillation converts a two-point statement into a one-point one. A bound on $|f(x) - f(y)|$ for all pairs is a statement about $D \times D$, while $M - m$ is computed from two suprema over D, and the proposition says the two carry the same information. This is the form in which the quantity is used later, where a function is controlled on each piece of a subdivided interval by computing one supremum and one infimum per piece.

Example 1.3.11: The Oscillation of x^2 on $[-1, 2]$

Let $D = [-1, 2]$ and $f(x) = x^2$. For $x \in D$ the inequalities $-1 \leq x \leq 2$ give $-2 \leq x \leq 2$ and hence $|x| \leq 2$ by proposition 1.3.8(2). A square is nonnegative (proposition 1.1.2(6), together with $0 \cdot 0 = 0$), so $x^2 = |x^2|$ by definition 1.3.6, and $|x^2| = |x|^2$ by proposition 1.3.8(3). Multiplying $|x| \leq 2$ first by $|x| \geq 0$ and then by $2 > 0$ gives $|x|^2 \leq 2|x| \leq 4$, so $f(x) \leq 4$. Since $f(2) = 4$, the value 4 is attained and $M = \sup_D f = 4$. Every value $f(x) = x^2$ is nonnegative and $f(0) = 0$, so

$m = \inf_D f = 0$. The oscillation is therefore $M - m = 4$, and it is realized by an actual pair of points, since $|f(2) - f(0)| = 4$. On the smaller set $[0, 1]$ the same computation gives oscillation 1: the oscillation depends on the set as well as on the function.

The section closes with the theorem that puts the n-th roots into $\mathbb{R}$. Proposition 1.1.3 shows that no rational squares to 2, and the chapter opened by observing that the set of rationals with square less than 2 has upper bounds in $\mathbb{Q}$ but no least one. In $\mathbb{R}$ that set does have a least upper bound, by theorem 1.2.4, and the theorem below identifies it: it is a number whose square is exactly 2. The argument is written for a general exponent because later chapters need n-th roots of every order, and the hypothesis is $x \geq 0$, weaker than $x > 0$, because the convergence tests that use it are applied to sequences some of whose terms vanish. The case $x = 0$ is immediate and is treated separately in the proof.

Theorem 1.3.12: Existence and Uniqueness of n-th Roots

Let $x \in \mathbb{R}$ with $x \geq 0$ and let $n \geq 1$ be an integer. There is exactly one real $y \geq 0$ with $y^n = x$. It is written $y = x^{1/n}$, and $x^{1/2}$ is also written $\sqrt{x}$.

***Proof*:**
Step 1: powers are strictly increasing on the nonnegative reals. The claim is that $0 \leq a < b$ implies $a^n < b^n$ for every integer $n \geq 1$, and it is proved by induction on n. For $n = 1$ it is the hypothesis. Assume $a^n < b^n$. If $a = 0$ then $a^{n+1} = 0 < b^{n+1}$, since $b > 0$ makes b^{n+1} a product of positive factors. If $a > 0$ then multiplying $a^n < b^n$ by $a > 0$ gives $a^{n+1} < ab^n$, and multiplying $a < b$ by $b^n > 0$ gives $ab^n < b^{n+1}$, so that $a^{n+1} < b^{n+1}$. The same induction with $\leq$ in place of $<$ shows that $0 \leq a \leq b$ implies $a^n \leq b^n$, and this weak form is used for the exponent $n - 1$, where $n = 1$ makes both sides equal to 1.
Step 2: uniqueness. If $y_1, y_2 \geq 0$ both satisfy $y_i^n = x$ and $y_1 \neq y_2$, then one of them is smaller, say $y_1 < y_2$, and Step 1 gives $x = y_1^n < y_2^n = x$, which is impossible. So there is at most one such y, whatever $x \geq 0$ is.
Step 3: the difference bound. For $0 \leq a \leq b$ and $n \geq 1$,

$$b^n - a^n \leq n(b - a)b^{n-1} \tag{1.2}$$

Indeed, expanding the product and cancelling termwise,

$$(b - a)\sum_{k=0}^{n-1} b^{n-1-k}a^k = \sum_{k=0}^{n-1} b^{n-k}a^k - \sum_{k=0}^{n-1} b^{n-1-k}a^{k+1} = b^n - a^n$$

because the second sum is the first with the index shifted by one, leaving only the $k = 0$ term of the first and the $k = n - 1$ term of the second. In each summand $b^{n-1-k}a^k \leq b^{n-1-k}b^k = b^{n-1}$, using $a^k \leq b^k$ from Step 1 for $k \geq 1$ and the equality $a^0 = b^0 = 1$ for $k = 0$, and multiplying by $b^{n-1-k} \geq 0$. There are n summands, so the sum is at most nb^{n-1}, and multiplying by $b - a \geq 0$ gives (1.2).
Step 4: existence when $x = 0$. The number $y = 0$ satisfies $y^n = 0 = x$ and $y \geq 0$, and Step 2 makes it the only such number. Assume from now on that $x > 0$.
Step 5: the candidate. Let

$$E = \{t \in \mathbb{R} | t \geq 0 \text{ and } t^n < x\}$$

Put $t_0 = x/(1 + x)$. Then $1 + x > 0$, so $t_0 > 0$. Dividing $x < 1 + x$ by $1 + x$ gives $t_0 < 1$, and dividing $1 < 1 + x$ by $1 + x$ and then multiplying by $x > 0$ gives $t_0 < x$. From $0 < t_0 < 1$

and Step 1 one gets $t_0^n \leq t_0$ for every $n \geq 1$, by induction: the case $n = 1$ is an equality, and multiplying $t_0^n \leq t_0$ by $t_0 > 0$ gives $t_0^{n+1} \leq t_0^2 \leq t_0$, the last step multiplying $t_0 < 1$ by $t_0 > 0$. Hence $t_0^n \leq t_0 < x$ and $t_0 \in E$, so E is nonempty. Suppose instead that $t \geq 1 + x$, so that $t > 1$. Then $t^n \geq t$ for every $n \geq 1$, again by induction: the case $n = 1$ is an equality, and multiplying $t^n \geq t$ by $t > 0$ gives $t^{n+1} \geq t^2 \geq t$, the last step multiplying $t \geq 1$ by $t > 0$. So $t^n \geq t \geq 1 + x > x$ and $t \notin E$. Every element of E is therefore smaller than $1 + x$, and E is bounded above. By theorem 1.2.4 the number $y = \sup E$ exists, and $y \geq t_0 > 0$ because $t_0 \in E$.
Step 6: the case $y^n < x$ is impossible. Suppose $y^n < x$ and put

$$h = \min\left\{\frac{1}{2}, \frac{x - y^n}{2n(y+1)^{n-1}}\right\}$$

Both entries are positive, since $x - y^n > 0$, $n \geq 1$ and $(y+1)^{n-1} > 0$, so $0 < h < 1$ and $h < (x - y^n)/\big(n(y+1)^{n-1}\big)$. Apply (1.2) with $a = y$ and $b = y + h$, legitimate because $0 \leq y \leq y + h$:

$$(y+h)^n - y^n \leq nh(y+h)^{n-1} \leq nh(y+1)^{n-1} < x - y^n$$

where the middle inequality uses $y + h < y + 1$ together with the weak form of Step 1 at exponent $n - 1$, and the last uses the choice of h. Hence $(y+h)^n < x$ with $y + h \geq 0$, so $y + h \in E$. But $y + h > y = \sup E$, contradicting that y is an upper bound of E.
Step 7: the case $y^n > x$ is impossible. Suppose $y^n > x$ and put

$$k = \frac{y^n - x}{ny^{n-1}}$$

Then $k > 0$, since $y > 0$ makes $y^{n-1} > 0$ and $y^n - x > 0$. Also $k < y$: this is the inequality $y^n - x < ny^n$, which holds because $y^n - x < y^n \leq ny^n$, using $x > 0$ and $n \geq 1$. Let $t \in E$. If $t > y$ then Step 1 gives $t^n > y^n > x$, contradicting $t \in E$, so $t \leq y$. If moreover $t \geq y - k$, then $0 \leq y - t \leq k$ and (1.2) with $a = t$ and $b = y$ gives

$$y^n - t^n \leq n(y-t)y^{n-1} \leq nky^{n-1} = y^n - x$$

so that $t^n \geq x$, again contradicting $t \in E$. Every $t \in E$ therefore satisfies $t < y - k$, so $y - k$ is an upper bound of E smaller than y, contradicting that y is the *least* upper bound.
Step 8: conclusion. By trichotomy in the ordered field $\mathbb{R}$ (theorem 1.1.10) and Steps 6 and 7, the only remaining possibility is $y^n = x$, with $y > 0$. Uniqueness is Step 2, completing the proof.

No decimal expansion, no approximating sequence and no notion of limit appears in the argument. The number y is produced as a supremum, and Steps 6 and 7 show only that it cannot sit strictly on either side of the target. Where a later existence proof builds a real number directly, with no appeal to numbers already constructed, it has this same shape: describe the set of numbers below the one wanted and take the least upper bound.

Example 1.3.13: $\sqrt{2}$ as a Supremum
Take $x = 2$ and $n = 2$ in theorem 1.3.12. The set constructed in the proof is $E = \{t \geq 0 \,|\, t^2 < 2\}$, and $\sqrt{2} = \sup E$. Two rational computations locate it. First, $(7/5)^2 = 49/25 < 50/25 = 2$, so $7/5 \in E$ and hence $\sqrt{2} \geq 7/5$. Second, $(3/2)^2 = 9/4 > 8/4 = 2$, so if $t \geq 3/2$ then $t^2 \geq 9/4 > 2$

and $t \notin E$. Thus $3/2$ is an upper bound of E and $\sqrt{2} \leq 3/2$. So

$$\frac{7}{5} \leq \sqrt{2} \leq \frac{3}{2}$$

By proposition 1.1.3 the number $\sqrt{2}$ is not rational, so the supremum of E exists in $\mathbb{R}$ and lies outside the image of $\mathbb{Q}$ under the embedding of proposition 1.1.11. Inside $\mathbb{Q}$ the corresponding set of rationals has no rational least upper bound (example 1.2.3), which is the defect theorem 1.2.4 repairs.

Exercise 1.3.1

1. Let A and B be nonempty subsets of $\mathbb{R}$, both bounded above. Exercise 1.2.1 (4) gives $\sup(A \cup B)$. Show that if $A \cap B$ is nonempty then $\sup(A \cap B)$ is at most the smaller of $\sup A$ and $\sup B$, and give two sets for which this second inequality is strict.
2. Let A and B be nonempty subsets of $\mathbb{R}$ that are bounded below. Deduce from proposition 1.3.2 that $\inf(A + B) = \inf A + \inf B$, by applying the results there to $-A$ and $-B$.
3. Prove the identities

$$\max\{x, y\} = \frac{x + y + |x - y|}{2}, \qquad \min\{x, y\} = \frac{x + y - |x - y|}{2}$$

for all $x, y \in \mathbb{R}$, and deduce that $\max\{x, y\} + \min\{x, y\} = x + y$.
4. Let D be a nonempty set and let $f, g \colon D \to \mathbb{R}$ be bounded. Show that $|f - g|$ is bounded and that

$$\left|\sup_D f - \sup_D g\right| \leq \sup_D |f - g|$$

Show also that the corresponding statement with $\sup_D$ replaced by $\inf_D$ on the left holds as well.
5. Let D be a nonempty set, let $f \colon D \to \mathbb{R}$ be bounded, and let $E \subseteq D$ be nonempty. Show that the oscillation of f on E is at most the oscillation of f on D. Show also that the oscillation of f on D is 0 if and only if f is constant on D.
6. Let $x, y \geq 0$ and let $n \geq 1$ be an integer. Using the uniqueness clause of theorem 1.3.12, prove that $(xy)^{1/n} = x^{1/n} y^{1/n}$. Prove also that $x < y$ implies $x^{1/n} < y^{1/n}$, and conclude that the map $x \mapsto x^{1/n}$ is injective on the nonnegative reals.

1.4 Consequences of Completeness

Theorem 1.2.4 is a statement about sets: a nonempty set of reals with an upper bound has a least one. Later chapters use it through five statements about numbers and intervals, and this section proves those five. The natural numbers are not bounded above in $\mathbb{R}$; every real number has an integer part; $\mathbb{Q}$ and its complement both meet every interval; an inequality that holds with arbitrary slack holds with none; and a nested sequence of closed bounded intervals has a point lying in all of them. The last of these then gives a fact about $\mathbb{R}$ of a different kind from anything in sections 1.1 to 1.3: the real numbers cannot be arranged in a list indexed by $\mathbb{N}$.

Throughout, $\mathbb{Q}$ is identified with its image in $\mathbb{R}$ under the embedding of proposition 1.1.11, so that $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$ and a symbol such as n or m/n names a real number as well as a rational one. Induction and the well-ordering of $\mathbb{N}$ are used freely.

The first question to settle is whether $\mathbb{N}$, sitting inside $\mathbb{R}$ by that identification, exceeds every real number. Nothing in definition 1.1.9 makes this visible: a real number is a class of Cauchy sequences of rationals, and the definition says nothing about how far the integers extend among those classes. What hangs on the answer is whether $1/n$ can be brought below a prescribed positive tolerance by choosing n large, which is the shape of every estimate in the chapters that follow. The least upper bound property settles it.

Theorem 1.4.1: The Archimedean Property of $\mathbb{R}$

1. For all $x, y \in \mathbb{R}$ with $x > 0$ there is an $n \in \mathbb{N}$ with $nx > y$.
2. For every $y \in \mathbb{R}$ there is an $n \in \mathbb{N}$ with $n > y$.
3. For every $\epsilon > 0$ there is an $n \in \mathbb{N}$ with $0 < 1/n < \epsilon$.

***Proof*:**

1. Let $A = \{nx \big| n \in \mathbb{N}\}$, and suppose the conclusion fails, so that $nx \leq y$ for every $n \in \mathbb{N}$. Then A is nonempty, since $1 \cdot x = x$ lies in A, and y is an upper bound for A. By theorem 1.2.4, A has a least upper bound $\alpha = \sup A$.
   Since $x > 0$, the real number $\alpha - x$ is smaller than α. As α is the *least* upper bound of A, the smaller number $\alpha - x$ is not an upper bound of A, so some element of A exceeds it: there is an $m \in \mathbb{N}$ with $mx > \alpha - x$. Adding x to both sides gives $(m+1)x > \alpha$. But $m + 1 \in \mathbb{N}$, so $(m+1)x \in A$, and an element of A exceeds the upper bound α. This is impossible, so some $n \in \mathbb{N}$ satisfies $nx > y$.
2. Apply (1) with $x = 1$, which is positive: there is an $n \in \mathbb{N}$ with $n \cdot 1 = n > y$.
3. Apply (1) with ϵ in the role of x and 1 in the role of y: there is an $n \in \mathbb{N}$ with $n\epsilon > 1$. Now $n \geq 1 > 0$, so $1/n > 0$, and multiplying $n\epsilon > 1$ by the positive number $1/n$ preserves the inequality (proposition 1.1.2(5)) and gives $\epsilon > 1/n$, as we sought to show.

Form (3) is the one an ϵ-estimate uses: to make a quantity smaller than a prescribed ϵ, it is enough to make it smaller than $1/n$ for a single n chosen after ϵ. Form (2) says that $\mathbb{N}$ has no upper bound in $\mathbb{R}$, which is the same statement read as a fact about the set $\mathbb{N}$, where form (1) reads it as a fact about individual numbers.

The Archimedean property is not itself a form of completeness. Section 1.1 takes as one of its standing hypotheses that $\mathbb{Q}$ is Archimedean, while example 1.2.3 exhibits a set of rationals with no rational supremum, so the Archimedean property holds in a field that fails the least upper bound property. What the proof establishes is that completeness already contains it, so nothing further has to be assumed about $\mathbb{R}$.

Example 1.4.2: Suprema and Infima Governed by $\mathbb{N}$

1. $\mathbb{N}$ has no supremum in $\mathbb{R}$. By theorem 1.4.1(2), for every $y \in \mathbb{R}$ there is an $n \in \mathbb{N}$ with $n > y$, so no real number is an upper bound of $\mathbb{N}$. A set with no upper bound has no least upper bound, so $\sup \mathbb{N}$ does not exist. This is consistent with theorem 1.2.4, whose hypothesis is that the set be bounded above.

2. Let $E = \{1/n \big| n \in \mathbb{N}\}$. Then $\sup E = 1$ and $\inf E = 0$, and $0 \notin E$. For the supremum: every $n \in \mathbb{N}$ satisfies $n \geq 1 > 0$, so $1/n \leq 1$, and $1 = 1/1 \in E$, so 1 is an upper bound belonging to E and is therefore the least one. For the infimum: E is bounded below by 0, since $1/n > 0$ for every n, so $\inf E$ exists by corollary 1.2.5. If $t > 0$, then theorem 1.4.1(3) gives an $n \in \mathbb{N}$ with $1/n < t$, so t is not a lower bound of E. Hence 0 is the greatest lower bound. Finally $0 \notin E$ because every element of E is positive.

The second item shows an infimum that does not belong to the set, and theorem 1.4.1(3) is what makes it one: the elements of E come arbitrarily close to 0 without reaching it. An infimum is therefore not a smallest element, and definition 1.2.2 does not ask it to be.

The Archimedean property says that the integers eventually pass any real number. The sharper statement is that there is a *last* integer that does not pass it. That statement attaches a specific integer to a given real number, where theorem 1.4.1 only asserts that some large integer exists.

Proposition 1.4.3: The Integer Part of a Real Number

For every $x \in \mathbb{R}$ there is exactly one $m \in \mathbb{Z}$ with

$$m \leq x < m + 1$$

This integer is written $\lfloor x \rfloor$ and is called the *integer part* of x.

***Proof*:**
Step 1: Existence. By theorem 1.4.1(2) there is an $N \in \mathbb{N}$ with $N > |x|$. From proposition 1.3.8(1), both $x \leq |x|$ and $-x \leq |x|$, so $-N < x < N$.
Let

$$T = \{k \in \mathbb{N} \big| -N + k > x\}$$

This set is nonempty: $2N \in \mathbb{N}$ and $-N + 2N = N > x$, so $2N \in T$. By the well-ordering of $\mathbb{N}$, T has a least element k_0. Put $m = -N + k_0 - 1$, which lies in $\mathbb{Z}$ because N and k_0 do.
Then $m + 1 = -N + k_0 > x$, since $k_0 \in T$. For the other inequality there are two cases. If $k_0 = 1$ then $m = -N$, and $-N < x$ was established above, so $m \leq x$. If $k_0 > 1$ then $k_0 - 1 \in \mathbb{N}$ and $k_0 - 1 < k_0$, so $k_0 - 1 \notin T$ by minimality of k_0. Unwinding the definition of T, this reads $-N + (k_0 - 1) \leq x$, which is $m \leq x$. In both cases $m \leq x < m + 1$.
Step 2: Uniqueness. Suppose $m, m' \in \mathbb{Z}$ both satisfy the two inequalities. From $m \leq x$ and $x < m' + 1$ we get $m < m' + 1$, that is $m - m' < 1$. Exchanging the roles of m and m' gives $m' - m < 1$. So $m - m'$ is an integer with $-1 < m - m' < 1$.
No integer k satisfies $0 < k < 1$: such a k would be a positive integer, hence an element of $\mathbb{N}$, and 1 is the least element of $\mathbb{N}$, so $k \geq 1$. If an integer k satisfied $-1 < k < 0$, then $-k$ would be an integer with $0 < -k < 1$, which the previous sentence excludes. Hence $m - m' = 0$, so $m = m'$, completing the proof.

Writing $r = x - \lfloor x \rfloor$, the proposition splits every real number as $x = \lfloor x \rfloor + r$ with $\lfloor x \rfloor \in \mathbb{Z}$ and $0 \leq r < 1$, and the splitting is unique. For instance $\lfloor 3/2 \rfloor = 1$, since $1 \leq 3/2 < 2$, and $\lfloor 2 \rfloor = 2$. The negative case is the one to watch: $\lfloor -3/2 \rfloor = -2$, since $-2 \leq -3/2 < -1$. The integer part of x is the largest integer at or below x, not the integer nearest 0.

The integer part is what makes the next theorem a construction, with the rational produced explicitly. The question it answers is whether the rationals, which proposition 1.1.11 placed inside $\mathbb{R}$, are spread through $\mathbb{R}$ or clustered somewhere. The recipe is to multiply everything by an n large enough that nx and ny are more than 1 apart, and then to take the first integer above nx: it cannot have jumped past ny.

Theorem 1.4.4: $\mathbb{Q}$ Is Dense in $\mathbb{R}$

If $x, y \in \mathbb{R}$ and $x < y$, then there is a $p \in \mathbb{Q}$ with $x < p < y$.

***Proof*:**
Since $x < y$, the number $y - x$ is positive, so theorem 1.4.1(1), applied with $y - x$ in the role of x and with 1 in the role of y, gives an $n \in \mathbb{N}$ with

$$n(y - x) > 1$$

Put $m = \lfloor nx \rfloor + 1$, an integer by proposition 1.4.3. That proposition gives $\lfloor nx \rfloor \leq nx < \lfloor nx \rfloor + 1$, and the two halves read

$$nx < m \qquad \text{and} \qquad m \leq nx + 1$$

Combining the second with $1 < ny - nx$ gives

$$m \leq nx + 1 < nx + (ny - nx) = ny$$

so $nx < m < ny$. Since $n \geq 1 > 0$, the number $1/n$ is positive, and multiplying through by it preserves both inequalities (proposition 1.1.2(5)), giving $x < m/n < y$. The number $p = m/n$ is rational, as we sought to show.

The same interval has still to be shown to contain a number outside $\mathbb{Q}$, and nothing new is needed to produce one. Translation by a fixed irrational number preserves the order and carries rationals to irrationals, so the theorem just proved, applied to the translated interval, delivers the point.

Corollary 1.4.5: The Irrationals Are Dense in $\mathbb{R}$

If $x, y \in \mathbb{R}$ and $x < y$, then there is a $t \in \mathbb{R} \setminus \mathbb{Q}$ with $x < t < y$.

***Proof*:**
By theorem 1.3.12, applied with $n = 2$ and with 2 in the role of x, there is a unique real $s \geq 0$ with $s^2 = 2$. Write $s = \sqrt{2}$. This number is not rational. Suppose s were the image of some $p \in \mathbb{Q}$ under the embedding of proposition 1.1.11. That embedding is a field homomorphism, so the image of p^2 is $s^2 = 2$, which is also the image of the rational 2. The embedding is injective, so $p^2 = 2$ holds in $\mathbb{Q}$, contradicting proposition 1.1.3.
Adding $-s$ to both sides of $x < y$ gives $x - s < y - s$, so theorem 1.4.4 produces a $p \in \mathbb{Q}$ with $x - s < p < y - s$. Adding s throughout gives $x < p + s < y$. Put $t = p + s$. If t were rational,

then $s = t - p$ would be a difference of two rationals and hence rational, which has just been excluded. So $t \in \mathbb{R} \setminus \mathbb{Q}$ and $x < t < y$, completing the proof.

Between any two distinct reals there is a rational and also an irrational, so neither $\mathbb{Q}$ nor $\mathbb{R} \setminus \mathbb{Q}$ can be separated from $\mathbb{R}$ by looking at the order alone. Every interval meets both. The two sets are nonetheless of different sizes, and theorem 1.4.11 is where that difference appears.

Example 1.4.6: A Rational Between $\sqrt{2}$ and $\sqrt{3}$

The proof of theorem 1.4.4 is a recipe, and running it on $x = \sqrt{2}$ and $y = \sqrt{3}$ produces an explicit rational. Both roots exist by theorem 1.3.12. Throughout, use that squaring is strictly increasing on the nonnegative reals: if $0 \le u < v$ then $v^2 - u^2 = (v-u)(v+u) > 0$, since $v - u > 0$ and $v + u \ge v > 0$.

First, bracket the two roots by rationals. Compute $(7/5)^2 = 49/25 < 50/25 = 2$ and $(3/2)^2 = 9/4 > 8/4 = 2$. If $7/5 \ge \sqrt{2}$ held, then squaring would give $(7/5)^2 \ge 2$, so $7/5 < \sqrt{2}$; if $3/2 \le \sqrt{2}$ held, squaring would give $(3/2)^2 \le 2$, so $\sqrt{2} < 3/2$. The same two steps applied to $(17/10)^2 = 289/100 < 300/100 = 3$ and $(9/5)^2 = 81/25 > 75/25 = 3$ give $17/10 < \sqrt{3} < 9/5$.

Now follow the proof. Since $\sqrt{3} > 17/10$ and $\sqrt{2} < 3/2 = 15/10$,

$$\sqrt{3} - \sqrt{2} > \frac{17}{10} - \frac{15}{10} = \frac{1}{5}$$

so $n = 5$ satisfies $n(\sqrt{3} - \sqrt{2}) > 1$. Next, $5 \cdot (7/5) = 7 < 5\sqrt{2} < 5 \cdot (3/2) = 15/2 < 8$, so $7 \le 5\sqrt{2} < 8$ and the uniqueness clause of proposition 1.4.3 gives $\lfloor 5\sqrt{2} \rfloor = 7$. The proof takes $m = 8$. The rational it produces is $p = 8/5$, and the bracketing confirms the conclusion directly: $\sqrt{2} < 3/2 = 15/10 < 16/10 = 8/5$ and $8/5 = 16/10 < 17/10 < \sqrt{3}$.

An argument that compares two real numbers often produces not $a \le b$ but $a \le b + \epsilon$, with an ϵ that can be taken as small as desired and never reduced to 0 inside the argument itself. The next proposition removes the margin afterwards, and theorem 1.4.8 below is the first place it is applied.

Proposition 1.4.7: The ϵ-Principle

Let $a, b \in \mathbb{R}$.

1. If $a \le b + \epsilon$ for every $\epsilon > 0$, then $a \le b$.
2. If $|a - b| \le \epsilon$ for every $\epsilon > 0$, then $a = b$.

***Proof*:**

1. Suppose instead that $a > b$. Then $a - b > 0$, so $\epsilon_0 = (a-b)/2$ is positive, and the hypothesis applied to ϵ_0 gives
$$a \le b + \frac{a-b}{2} = \frac{a+b}{2}$$
On the other hand $b < a$ gives $a + b < 2a$, hence $(a+b)/2 < a$. Together these read $a < a$, which is false. Therefore $a \le b$.
2. Fix $\epsilon > 0$. If $a - b \ge 0$ then $a - b = |a - b| \le \epsilon$ by definition 1.3.6; if $a - b < 0$ then

$a - b < 0 < \epsilon$. Either way $a - b \leq \epsilon$, that is $a \leq b + \epsilon$. As $\epsilon > 0$ was arbitrary, part (1) gives $a \leq b$. Since $|b - a| = |a - b|$ (proposition 1.3.8(1)), the same argument with a and b exchanged gives $b \leq a$. Hence $a = b$, completing the proof.

Part (2) is the form used to prove that two numbers produced by different constructions are equal. A single computation showing $a = b$ is usually out of reach, and what one shows is that for each $\epsilon > 0$, a and b differ by less than ϵ, and the proposition converts the family of estimates into the equality.

Theorem 1.2.4 states completeness as a fact about bounded sets. A second way to say that $\mathbb{R}$ has no gaps is geometric: if a sequence of closed intervals is nested, each inside the last, then at least one point lies in every one of them. In $\mathbb{Q}$ this fails. Intervals with rational endpoints can close in on a point whose square is 2, and proposition 1.1.3 says no rational is available to be that point, so the intersection of the rational intervals is empty even though each of them is nonempty. Example 1.4.10 carries this out explicitly. Over $\mathbb{R}$ the conclusion holds, and completeness is what delivers it.

Theorem 1.4.8: The Nested Interval Theorem

For each $n \in \mathbb{N}$ let $a_n \leq b_n$ be real numbers, and put

$$I_n = \{x \in \mathbb{R} \big| a_n \leq x \leq b_n\}$$

Assume $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$. Then:

1. $\bigcap_{n\in\mathbb{N}} I_n$ is nonempty.
2. If, in addition, for every $\epsilon > 0$ there is an $n \in \mathbb{N}$ with $b_n - a_n < \epsilon$, then $\bigcap_{n\in\mathbb{N}} I_n$ has exactly one element.

***Proof*:**

1. First, the endpoints move monotonically. Fix n. Since $a_{n+1} \leq b_{n+1}$, the number a_{n+1} lies in I_{n+1}, hence in I_n, so $a_n \leq a_{n+1}$. The same argument applied to $b_{n+1} \in I_{n+1} \subseteq I_n$ gives $b_{n+1} \leq b_n$. By induction, $a_m \leq a_k$ and $b_k \leq b_m$ whenever $m \leq k$.
   Next, $a_m \leq b_n$ for *all* $m, n \in \mathbb{N}$, not only for $m = n$. Let k be the larger of m and n. Then $m \leq k$ and $n \leq k$, so the previous paragraph gives $a_m \leq a_k$ and $b_k \leq b_n$, and $a_k \leq b_k$ holds by hypothesis. Chaining the three, $a_m \leq a_k \leq b_k \leq b_n$.
   Let $A = \{a_n \big| n \in \mathbb{N}\}$. It is nonempty and, taking $n = 1$ above, bounded above by b_1, so theorem 1.2.4 gives $\alpha = \sup A$. Fix $n \in \mathbb{N}$. Then $a_n \leq \alpha$ because α is an upper bound of A, and $\alpha \leq b_n$ because b_n is an upper bound of A by the previous paragraph while α is the least such. So $a_n \leq \alpha \leq b_n$, that is $\alpha \in I_n$. As n was arbitrary, $\alpha \in \bigcap_{n\in\mathbb{N}} I_n$.
2. By part (1) the intersection contains at least one point, so it is enough to show it contains at most one. Let x and y both lie in $\bigcap_{n\in\mathbb{N}} I_n$ and fix $n \in \mathbb{N}$. From $a_n \leq x \leq b_n$ and $a_n \leq y \leq b_n$ it follows that $x - y \leq b_n - a_n$ and $y - x \leq b_n - a_n$, and since $|x - y|$ is one of $x - y$ and $y - x$ (definition 1.3.6),

$$|x - y| \leq b_n - a_n$$

   Now let $\epsilon > 0$. By hypothesis there is an n with $b_n - a_n < \epsilon$, and $|x - y| \leq b_n - a_n$ then gives $|x - y| \leq \epsilon$. Since $\epsilon > 0$ was arbitrary, proposition 1.4.7(2) gives $x = y$. The intersection therefore has exactly one element, completing the proof.

Both hypotheses on the intervals are needed, and theorem 1.4.1 is what produces the two failures.

Example 1.4.9: Two Failures of Nesting

1. *Closedness cannot be dropped.* Let $J_n = \{x \in \mathbb{R} \big| 0 < x < 1/n\}$. These are nested, since $1/(n+1) \leq 1/n$, and each is nonempty, since theorem 1.4.4 gives a point strictly between 0 and $1/n$. But $\bigcap_{n \in \mathbb{N}} J_n = \emptyset$: a point x of the intersection would satisfy $x > 0$ and $x < 1/n$ for every n, contradicting theorem 1.4.1(3), which produces an n with $1/n < x$.

2. *Boundedness cannot be dropped.* Let $K_n = \{x \in \mathbb{R} \big| x \geq n\}$. These are nested and nonempty, since $n \in K_n$. But $\bigcap_{n \in \mathbb{N}} K_n = \emptyset$: a point x of the intersection would satisfy $x \geq n$ for every $n \in \mathbb{N}$, contradicting theorem 1.4.1(2), which produces an n with $n > x$.

Example 1.4.10: Nested Rational Intervals with No Rational Point

Write $I_n = \{x \in \mathbb{R} \big| a_n \leq x \leq b_n\}$ throughout, and set $a_1 = 1$, $b_1 = 2$, so that $a_1, b_1 \in \mathbb{Q}$, $a_1 < b_1$, and $a_1^2 = 1 < 2 < 4 = b_1^2$. Given rationals $a_n < b_n$ with $a_n^2 < 2 < b_n^2$, let $c_n = (a_n + b_n)/2$, which is again rational and satisfies $a_n < c_n < b_n$, since $c_n - a_n = b_n - c_n = (b_n - a_n)/2 > 0$. By proposition 1.1.3 the rational c_n has $c_n^2 \neq 2$, so exactly one of $c_n^2 < 2$ and $c_n^2 > 2$ holds. Put

$$(a_{n+1}, b_{n+1}) = \begin{cases} (c_n, b_n) & \text{if } c_n^2 < 2 \\ (a_n, c_n) & \text{if } c_n^2 > 2 \end{cases}$$

In both cases a_{n+1} and b_{n+1} are rational, $a_{n+1} < b_{n+1}$, $a_{n+1}^2 < 2 < b_{n+1}^2$, the inclusion $I_{n+1} \subseteq I_n$ holds because $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$, and $b_{n+1} - a_{n+1} = (b_n - a_n)/2$. By induction $b_n - a_n = 1/2^{n-1}$.

These lengths become arbitrarily small. For every integer $k \geq 0$ one has $2^k \geq k+1$, by induction: $2^0 = 1$, and if $2^k \geq k+1$ then $2^{k+1} = 2 \cdot 2^k \geq 2k+2 \geq k+2$. Given $\epsilon > 0$, take $n \in \mathbb{N}$ with $1/n < \epsilon$ (theorem 1.4.1(3)). Then $2^{n-1} \geq n > 0$, so $1/2^{n-1} \leq 1/n < \epsilon$. Hence theorem 1.4.8(2) applies, and $\bigcap_{n \in \mathbb{N}} I_n$ consists of a single real number.

That number is $\sqrt{2}$. Let $s \geq 0$ be the real with $s^2 = 2$ given by theorem 1.3.12. Every a_n is positive, since $a_1 = 1$ and each a_{n+1} is either a_n or the larger number c_n, and $a_n^2 < 2 = s^2$. Were $a_n \geq s$, squaring would give $a_n^2 \geq s^2$, because $u \geq v \geq 0$ implies $u^2 - v^2 = (u-v)(u+v) \geq 0$. So $a_n < s$, and the same argument applied to $b_n > 0$ with $b_n^2 > s^2$ gives $s < b_n$. Thus $s \in I_n$ for every n, and by uniqueness $\bigcap_{n \in \mathbb{N}} I_n = \{s\}$.

Now read this inside $\mathbb{Q}$. The sets $I_n \cap \mathbb{Q}$ are nested, and each is nonempty, containing a_n. Their intersection is $\left(\bigcap_{n \in \mathbb{N}} I_n\right) \cap \mathbb{Q} = \{s\} \cap \mathbb{Q}$, which is empty: if s were a rational p, then $p^2 = 2$ in $\mathbb{Q}$, contradicting proposition 1.1.3. A nested sequence of nonempty closed intervals with rational endpoints therefore has empty intersection in $\mathbb{Q}$, which is the failure theorem 1.4.8 repairs.

Theorem 1.4.4 and corollary 1.4.5 left $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ indistinguishable by order: each meets every interval. They are distinguished by counting. Recall from *Preliminaries* [1] that a nonempty set is countable exactly when it is the range of some function defined on $\mathbb{N}$, and that $\mathbb{Q}$ is countable. The next theorem shows that $\mathbb{R}$ is not, so the set of classes built in definition 1.1.9 admits no listing while the set of rationals it was built from does. The proof produces a real number specified by nothing except the conditions it must avoid, and theorem 1.4.8 is what converts that list of conditions into a point.

Theorem 1.4.11: Uncountability of $\mathbb{R}$

Let $a, b \in \mathbb{R}$ with $a < b$, and write $[a,b] = \{x \in \mathbb{R} \big| a \leq x \leq b\}$. For every function $f : \mathbb{N} \to \mathbb{R}$ there is a point $x \in [a,b]$ with $x \neq f(n)$ for every $n \in \mathbb{N}$. Consequently the range of f does not contain $[a,b]$, and neither $\mathbb{R}$ nor $[a,b]$ is countable.

***Proof*:**
Step 1: One splitting step. Let $u < v$ be reals and let $z \in \mathbb{R}$. Put

$$c = u + \frac{v-u}{3}, \qquad d = u + \frac{2(v-u)}{3}$$

Then $u < c < d < v$, since each of $c-u$, $d-c$ and $v-d$ equals $(v-u)/3$, which is positive. The intervals $[u,c]$ and $[d,v]$ are therefore contained in $[u,v]$, each has distinct endpoints, and they are disjoint: a point of the first is at most c and a point of the second is at least d, while $c < d$. Consequently z fails to lie in at least one of them.
Define $\Phi([u,v],z)$ to be $[u,c]$ when $z \notin [u,c]$, and $[d,v]$ otherwise. In the second case $z \in [u,c]$, so $z \leq c < d$ and hence $z \notin [d,v]$. In both cases $\Phi([u,v],z)$ is an interval $[u',v']$ with $u' < v'$, contained in $[u,v]$, and not containing z.
Step 2: The recursion. Define $I_1 = \Phi([a,b], f(1))$ and $I_{n+1} = \Phi(I_n, f(n+1))$ for $n \in \mathbb{N}$. The rule Φ assigns a definite interval to each pair, so this defines the sequence (I_n) by recursion, with no choices made along the way. By induction on n, each I_n is an interval $[a_n, b_n]$ with $a_n < b_n$, satisfying $I_{n+1} \subseteq I_n$ and $I_1 \subseteq [a,b]$, and $f(n) \notin I_n$ by the last clause of Step 1.
Step 3: The point. The intervals I_n are closed, bounded and nested, so theorem 1.4.8(1) gives a point $x \in \bigcap_{n\in\mathbb{N}} I_n$. Fix $n \in \mathbb{N}$. Then $x \in I_n$ while $f(n) \notin I_n$, so $x \neq f(n)$. Moreover $x \in I_1 \subseteq [a,b]$, which establishes the first assertion.
Step 4: The consequences. If the range of f contained $[a,b]$, then the point x of Step 3 would be $f(n)$ for some n, which Step 3 excludes. So no function on $\mathbb{N}$ has $[a,b]$ inside its range. Both countability claims follow from this. Were $\mathbb{R}$ countable, some $h : \mathbb{N} \to \mathbb{R}$ would have range $\mathbb{R}$, and $\mathbb{R}$ contains $[a,b]$, so the range of h would contain $[a,b]$. Were $[a,b]$ countable, some $g : \mathbb{N} \to [a,b]$ would have range $[a,b]$, and g is in particular a function from $\mathbb{N}$ to $\mathbb{R}$ whose range contains $[a,b]$. Both are impossible, as we sought to show.

The theorem is stronger than the bare statement that $\mathbb{R}$ is uncountable: no list of real numbers can exhaust even a single interval, however short. Since $\mathbb{Q}$ is countable, this also settles that $\mathbb{R} \setminus \mathbb{Q}$ is not empty, which corollary 1.4.5 obtained by a different route, and, since a list of $\mathbb{Q}$ together with a list of $\mathbb{R} \setminus \mathbb{Q}$ could be interleaved into a single list of $\mathbb{R}$, that $\mathbb{R} \setminus \mathbb{Q}$ admits no list either. That interleaving is exercise 1.4.1 (6).[1]

Exercise 1.4.1

1. Let $x \in \mathbb{R}$ and put $A = \{q \in \mathbb{Q} \big| q < x\}$. Show that A is nonempty and bounded above, and that $\sup A = x$.

2. Show that $\lfloor x + m \rfloor = \lfloor x \rfloor + m$ for every $x \in \mathbb{R}$ and every $m \in \mathbb{Z}$. Then exhibit real numbers x

[1] The proof above is Cantor's first, published in 1874. He gave a second proof in 1891, the diagonal argument, which *Preliminaries* [1] develops in the form "$\{0,1\}^{\mathbb{N}}$ is uncountable". The 1874 argument is the one given here because it needs no decimal representation, which this chapter has not constructed, and rests only on theorem 1.4.8.

and y for which $\lfloor x+y \rfloor \neq \lfloor x \rfloor + \lfloor y \rfloor$.

3. Let $x < y$ be real numbers and let $k \in \mathbb{N}$. Show that there are rationals $p_1 < p_2 < \cdots < p_k$ with $x < p_1$ and $p_k < y$.

4. Let A and B be nonempty subsets of $\mathbb{R}$ such that $a \leq b$ for every $a \in A$ and every $b \in B$.

    1. Show that $\sup A$ and $\inf B$ exist and that $\sup A \leq \inf B$.
    2. Show that $\sup A = \inf B$ if and only if for every $\epsilon > 0$ there are $a \in A$ and $b \in B$ with $b - a < \epsilon$.

5. Let $I_n = \{x \in \mathbb{R} | a_n \leq x \leq b_n\}$ be as in theorem 1.4.8, with $a_n \leq b_n$ and $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$. Show that $\alpha = \sup\{a_n | n \in \mathbb{N}\}$ and $\beta = \inf\{b_n | n \in \mathbb{N}\}$ both exist, that $\alpha \leq \beta$, and that
$$\bigcap_{n\in\mathbb{N}} I_n = \{x \in \mathbb{R} | \alpha \leq x \leq \beta\}$$

6. Show that $\mathbb{R} \setminus \mathbb{Q}$ is uncountable. (Use that $\mathbb{Q}$ is countable, and build from a listing of $\mathbb{Q}$ and a hypothetical listing of $\mathbb{R} \setminus \mathbb{Q}$ a single function on $\mathbb{N}$ whose range is all of $\mathbb{R}$.)

7. Call a real number *dyadic* if it has the form $m/2^n$ with $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. Show that the dyadic numbers are dense in $\mathbb{R}$: if $x < y$ then there is a dyadic d with $x < d < y$. (Follow the proof of theorem 1.4.4, using theorem 1.4.1(1) together with the inequality $2^k \geq k+1$ established in example 1.4.10 to choose the denominator.)

8. Let $E \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s \in \mathbb{R}$. Show that $s = \sup E$ if and only if s is an upper bound of E and, for every $n \in \mathbb{N}$, there is an $x \in E$ with $x > s - 1/n$.

1.5⋆ Uniqueness of the Complete Ordered Field

⋆ *Optional. This section is an aside; nothing later in the book depends on it.*

Section 1.1 built one complete ordered field, and appendix A builds another by a different route. A reader is entitled to ask whether the two constructions produce the same object, and more generally whether the axioms pin $\mathbb{R}$ down at all or merely describe a family of fields among which the construction made an arbitrary choice. This section answers both: they do pin it down, so the choice was forced.

Call an ordered field *complete* if it satisfies the least upper bound property of theorem 1.2.4. One consequence comes first, because the uniqueness proof uses it and because it settles what the Archimedean property costs.

Corollary 1.5.1: Every Complete Ordered Field Is Archimedean

Let F be a complete ordered field. Then for every $x \in F$ there is a positive integer n with $x < n \cdot 1$.

***Proof*:**
Write N for the set of elements $n \cdot 1$ with n a positive integer, where $n \cdot 1$ means $1 + \cdots + 1$ with n summands. Suppose some $x \in F$ satisfies $n \cdot 1 \leq x$ for every n. Then N is nonempty and bounded above, so $s = \sup N$ exists in F by completeness. Since $s - 1 < s$ and s is the least upper bound, $s - 1$ is not an upper bound of N, so there is a positive integer m with $s - 1 < m \cdot 1$. Adding 1 gives $s < (m+1) \cdot 1$, and $(m+1) \cdot 1 \in N$, contradicting that s is an upper bound of N. So no such x exists, completing the proof.

For the field constructed in section 1.1 the Archimedean property was available before completeness, from proposition 1.1.12. Corollary 1.5.1 shows that it did not have to be assumed, since any complete ordered field has it. The two routes agree, and the second is the one available to a reader who meets $\mathbb{R}$ through its axioms, with no construction in hand.

An isomorphism of ordered fields is a bijection preserving addition, multiplication and the order. The theorem says there is exactly one between any two complete ordered fields, so that the phrase "the real numbers" is justified.

Theorem 1.5.2: Uniqueness of the Complete Ordered Field

Let F and F' be complete ordered fields. Then there is exactly one isomorphism of ordered fields $\varphi\colon F \to F'$.

***Proof*:**
Step 1: the rational subfield is forced. Any isomorphism sends 0 to 0 and 1 to 1, hence $n \cdot 1$ to $n \cdot 1$ for every positive integer n, hence $-(n \cdot 1)$ to $-(n \cdot 1)$, and hence $(m \cdot 1)(n \cdot 1)^{-1}$ to $(m \cdot 1)(n \cdot 1)^{-1}$ for integers m and $n \neq 0$. Write $\mathbb{Q}_F$ for the set of such elements of F, and $\mathbb{Q}_{F'}$ likewise. These are subfields, and the assignment just described is a bijection $\varphi_0\colon \mathbb{Q}_F \to \mathbb{Q}_{F'}$ preserving both operations and the order, since $(m \cdot 1)(n \cdot 1)^{-1} < (m' \cdot 1)(n' \cdot 1)^{-1}$ in either field reduces, after clearing denominators, to the same inequality between integers. So φ is determined on $\mathbb{Q}_F$ before anything is chosen.

Step 2: $\mathbb{Q}_F$ is dense in F. Let $x < y$ in F. By corollary 1.5.1 there is a positive integer n with $(y - x)^{-1} < n \cdot 1$, so $(n \cdot 1)^{-1} < y - x$ by proposition 1.1.2(7). Applying the corollary again gives a positive integer m_1 with $x \cdot n \cdot 1 < m_1 \cdot 1$ and a positive integer m_2 with $-(x \cdot n \cdot 1) < m_2 \cdot 1$, so that $-m_2 \cdot 1 < x \cdot n \cdot 1 < m_1 \cdot 1$. Among the finitely many integers k with $-m_2 \leq k \leq m_1$, those satisfying $x \cdot n \cdot 1 < k \cdot 1$ form a nonempty set, so there is a least such k_0, and it satisfies $(k_0 - 1) \cdot 1 \leq x \cdot n \cdot 1 < k_0 \cdot 1$. *(The integers must be bracketed on both sides first: taking k_0 least among the positive integers alone fails for $x < 0$, where $x \cdot n \cdot 1 < 0 < 1$ forces $k_0 = 1$.)* So

$$x < (k_0 \cdot 1)(n \cdot 1)^{-1} \leq x + (n \cdot 1)^{-1} < y$$

and the middle element lies in $\mathbb{Q}_F$.

Step 3: the extension. For $x \in F$ put

$$\varphi(x) = \sup\{\varphi_0(r) \big| r \in \mathbb{Q}_F \text{ and } r < x\}$$

The set on the right is nonempty and bounded above: nonempty by Step 2 applied to any element below x, and bounded above by $\varphi_0(r')$ for any $r' \in \mathbb{Q}_F$ with $x < r'$, again by Step 2.

So the supremum exists by completeness of F'. For $x \in \mathbb{Q}_F$ the value agrees with $\varphi_0(x)$, since φ_0 preserves the order and $\mathbb{Q}_F$ is dense, so no rational strictly below x has image above $\varphi_0(x)$ while every ϵ-band below $\varphi_0(x)$ is met.

Step 4: φ preserves the order and is bijective. If $x < y$, Step 2 gives $r, r' \in \mathbb{Q}_F$ with $x < r < r' < y$. Every rational below x is below r, so $\varphi(x) \leq \varphi_0(r) < \varphi_0(r') \leq \varphi(y)$, whence $\varphi(x) < \varphi(y)$. In particular φ is injective. The same construction with the roles of F and F' exchanged produces $\psi\colon F' \to F$, and $\psi \circ \varphi$ is an order-preserving map fixing $\mathbb{Q}_F$ pointwise. By Step 2 a map fixing a dense subset and preserving order is the identity, since $\psi(\varphi(x)) < x$ or $x < \psi(\varphi(x))$ would leave room for a rational between them, which is fixed and so cannot be moved across. Hence $\psi \circ \varphi$ and $\varphi \circ \psi$ are identities and φ is a bijection.

Step 5: φ preserves the operations. Let $x, y \in F$. If $r, s \in \mathbb{Q}_F$ satisfy $r < x$ and $s < y$ then $r + s < x + y$, so $\varphi_0(r) + \varphi_0(s) = \varphi_0(r + s) \leq \varphi(x + y)$. Taking suprema over r and then s gives $\varphi(x) + \varphi(y) \leq \varphi(x + y)$. For the reverse, if $t \in \mathbb{Q}_F$ satisfies $t < x + y$ then Step 2 gives $r, s \in \mathbb{Q}_F$ with $r < x$, $s < y$ and $t < r + s$, so $\varphi_0(t) < \varphi(x) + \varphi(y)$. Taking the supremum over t gives $\varphi(x + y) \leq \varphi(x) + \varphi(y)$. So φ preserves addition. For multiplication the same argument applies verbatim when $x, y > 0$, and the general case follows from $\varphi(-x) = -\varphi(x)$, which additivity gives, together with proposition 1.1.2(4).

Step 6: uniqueness. Any isomorphism agrees with φ_0 on $\mathbb{Q}_F$ by Step 1, and an order-preserving map is determined by its values on a dense subset, by the argument of Step 4. So φ is the only one, completing the proof.

Two consequences are worth stating. The field built in section 1.1 from Cauchy sequences and the field built in appendix A from cuts are both complete ordered fields, so the theorem identifies them: the appendix is an alternative route to the same object, not a second object. And the definite article in “the real numbers” is earned.

The theorem also disposes of the candidates a reader is most likely to raise.

1.5.3 Note: The p-adic Absolute Value Is Not an Order For a prime p, writing a nonzero rational as $p^k m/n$ with m and n prime to p and setting $|p^k m/n|_p = p^{-k}$ defines a second notion of size on $\mathbb{Q}$, and completing $\mathbb{Q}$ with respect to it produces the p-adic numbers. This is not a counterexample to theorem 1.5.2, and the reason is that $|\cdot|_p$ is an *absolute value*, not an order: it assigns a size to each element and does not say which of two elements is the larger. The theorem quantifies over ordered fields, so a field carrying only an absolute value is not a candidate for it.

Exercise 1.5.1

1. Show that $\mathbb{Q}$ is an Archimedean ordered field that is not complete, so that the converse of corollary 1.5.1 fails.
2. Show that an ordered field in which every nonempty set bounded below has an infimum satisfies the least upper bound property, so the two forms of completeness are interchangeable as an axiom.

3. Show that the only isomorphism of ordered fields $\mathbb{R} \to \mathbb{R}$ is the identity, and that the same is false for fields without an order by exhibiting a field with a nontrivial automorphism.
4. Where does the proof of theorem 1.5.2 use completeness of F', and where does it use completeness of F? Show that dropping either leaves the construction of φ incomplete.

2

Sequences and Series

Theorem 1.2.4 states completeness as a fact about sets: every nonempty set of reals with an upper bound has a least one. That is the form in which completeness was proved, and it is not the form in which it is usually spent. Almost every argument in the chapters that follow reaches instead for a statement about sequences, that a sequence whose terms crowd together has somewhere to converge to, and the first task here is to show that the two say the same thing.

The restatement is worth the work for a reason beyond convenience. The least upper bound property is about an order, and it can only be stated where there is one. The Cauchy condition is about distances, and it can be stated wherever distances can be measured. So the sequential form is what survives when the real line is replaced by a space of functions or a space of sequences, and it is the form the later volumes of this series inherit. Section 1.1 already used Cauchy sequences of rationals to build $\mathbb{R}$. This chapter turns the same idea on $\mathbb{R}$ itself.

The second half of the chapter is about infinite sums. A series is defined as a sequence of partial sums, so nothing new is needed to say what one means, and everything proved about sequences applies at once. What is genuinely new is that an infinite sum can depend on the order of its terms: reordering a convergent series can change its value, or destroy convergence altogether, and the condition separating the series where this happens from the series where it cannot is absolute convergence. That distinction is the reason the chapter treats convergence twice.

Section 2.1 defines the limit of a sequence and establishes the algebra of limits, the estimates that every later chapter spends without comment. Section 2.2 adds the monotone convergence theorem, the limit superior and inferior with the characterization that makes them usable, and the Bolzano-Weierstrass theorem. Section 2.3 proves the equivalence promised above and collects the forms of completeness into a single statement. Section 2.4 develops series and the tests that decide them, from the geometric and harmonic series through the comparison, ratio and root tests. Section 2.5 takes up the order dependence, proving that absolute convergence makes a sum rearrangement-invariant and that conditional convergence makes it arbitrary. Section 2.6 specializes to power series, computes the radius of convergence, and defines the exponential and trigonometric functions by the series that will later be shown to characterize them.

2.1 Convergence of Real Sequences

What section 1.1 never defines is what it means for a sequence of *real* numbers to approach a real number. This section gives that definition together with the facts that make it usable: a limit is unique, a convergent sequence is bounded, limits pass through sums, products and quotients, and limits respect weak inequalities.

The failure in $\mathbb{Q}$ is the one that forced the construction, restated in the vocabulary about to be defined. The sequence $1.4, 1.41, 1.414, 1.4142, \ldots$ of example 1.1.5(2) approaches nothing in $\mathbb{Q}$, and section 2.3 will show that it approaches a real number in $\mathbb{R}$. The last result here is the estimate $k^j r^k \to 0$ for $0 < r < 1$, which is what makes a geometric factor dominate a polynomial one.

Definition 2.1.1: [Real] Sequence

A *sequence* of real numbers is a function $a\colon \mathbb{N} \to \mathbb{R}$. The value $a(n)$ is written a_n and called the *n-th term*, and the function itself is written $(a_n)_{n\geq 1}$, or (a_n) when the index set is understood. A sequence indexed by the integers $n \geq n_0$ for a fixed $n_0 \in \mathbb{Z}$ is defined the same way, and every statement below holds for such a sequence with 1 replaced by n_0 throughout.

A sequence is presented by whatever rule produces its terms: a formula ($a_n = 1/n$), a recursion ($a_1 = 1$ and $a_{n+1} = a_n/2$, so that $a_n = 2^{1-n}$), or a description (a_n is the n-th decimal truncation of a square root of 2, as in example 1.1.5(2)). The notation (a_n) names the whole function and a_n names one of its values, and the two have to be kept apart as soon as sets of terms appear: the sequence $((-1)^n)$ has a term for every $n \geq 1$, while the set $\{(-1)^n \mid n \geq 1\} = \{-1, 1\}$ of its terms has two elements.

Convergence is the statement that the terms of a sequence eventually lie within any prescribed distance of one fixed number. Two words in that sentence need fixing before it becomes a definition. The *distance* from a_n to L is $|a_n - L|$, in the sense of definition 1.3.6, and *eventually* means from some index onwards, with the index allowed to depend on the prescribed distance and on nothing else.

Definition 2.1.2: Convergence and the Limit of a Sequence

A sequence (a_n) of real numbers *converges* to $L \in \mathbb{R}$ if for every $\epsilon > 0$ there is an $N \in \mathbb{N}$ such that

$$|a_n - L| < \epsilon \qquad \text{for every } n \geq N$$

Such an L is called a *limit* of (a_n), and one writes $a_n \to L$. A sequence converging to some real number is *convergent*, and one converging to no real number is *divergent*.

One convention changes here and is not mentioned again. Definition 1.1.4 quantified over *rational* ϵ, because that was all that existed when it was stated. From this point ϵ ranges over the positive reals. For a real sequence the two quantifications describe the same condition. A positive rational is a positive real under the embedding of proposition 1.1.11, so quantifying over all real $\epsilon > 0$ is at least as strong. Conversely, given a real $\epsilon > 0$, theorem 1.4.4 applied to $0 < \epsilon$ produces a rational q with $0 < q < \epsilon$, and an index N answering q also answers ϵ, since $|a_n - L| < q$ and $q < \epsilon$ give $|a_n - L| < \epsilon$.

Example 2.1.3: $1/n$, a Constant Sequence, $(n+1)/(2n+3)$, and $((-1)^n)$

1. $1/n \to 0$. Let $\epsilon > 0$. By theorem 1.4.1(3) there is an $N \in \mathbb{N}$ with $0 < 1/N < \epsilon$. Fix $n \geq N$. If $n > N$ then $0 < 1/n < 1/N$ by proposition 1.1.2(7), and if $n = N$ the two are equal, so in both cases $|1/n - 0| = 1/n \leq 1/N < \epsilon$.

2. A constant sequence $a_n = c$ converges to c. Every $\epsilon > 0$ is answered by $N = 1$, since $|a_n - c| = |0| = 0 < \epsilon$ for every n.

3. The sequence $\big((n+1)/(2n+3)\big)$ converges to $1/2$. Subtracting and clearing denominators,

$$\frac{n+1}{2n+3} - \frac{1}{2} = \frac{2(n+1) - (2n+3)}{2(2n+3)} = \frac{-1}{2(2n+3)}$$

so the distance to $1/2$ is $1/(2(2n+3))$, and this is smaller than $1/(4n)$ because $4n < 4n+6 = 2(2n+3)$. Given $\epsilon > 0$, apply theorem 1.4.1(3) to the positive real 4ϵ to get an $N \in \mathbb{N}$ with $0 < 1/N < 4\epsilon$. For $n \geq N$ the argument of (1) gives $1/n \leq 1/N$, hence

$$\left| \frac{n+1}{2n+3} - \frac{1}{2} \right| < \frac{1}{4n} \leq \frac{1}{4N} < \epsilon$$

4. $((-1)^n)$ is divergent. Suppose $(-1)^n \to L$ and take $\epsilon = 1$ in definition 2.1.2, producing an N with $|(-1)^n - L| < 1$ for all $n \geq N$. Among $2N$ and $2N+1$ the first is even and the second is odd, and both are at least N, so $|1 - L| < 1$ and $|-1 - L| < 1$. By proposition 1.3.8(4) and then (1),

$$2 = |1 - (-1)| = |(1 - L) + (L - (-1))| \leq |1 - L| + |-1 - L| < 2$$

which is false. So no such L exists.

Items (1) and (3) run the same route: bound $|a_n - L|$ above by an explicit expression in n, then use theorem 1.4.1 to choose an index past which that expression is below ϵ. Item (4) shows what a divergence proof has to do, since no candidate limit is given in advance and every one of them has to be excluded at once. Theorem 2.1.6 removes most of this labour by reducing a new limit to limits already known.

Definition 2.1.2 calls L *a* limit, since nothing said so far excludes a second one. The following excludes it. Two distinct real numbers are separated by a positive distance, while the terms of a convergent sequence eventually lie within every positive distance of each of its limits, and those two statements cannot both hold.

Proposition 2.1.4: A Sequence Has at Most One Limit

If $a_n \to L$ and $a_n \to L'$, then $L = L'$.

***Proof*:**
Let $\epsilon > 0$. Applying definition 2.1.2 to L with the positive real $\epsilon/2$ gives an N_1 with $|a_n - L| < \epsilon/2$ for all $n \geq N_1$, and applying it to L' with $\epsilon/2$ gives an N_2 with $|a_n - L'| < \epsilon/2$ for all $n \geq N_2$.

Fix an n at least as large as both. Then proposition 1.3.8(4) gives

$$|L - L'| = |(L - a_n) + (a_n - L')| \leq |L - a_n| + |a_n - L'| < \frac{\epsilon}{2} + \frac{\epsilon}{2}$$

where $|L - a_n| = |a_n - L|$ by proposition 1.3.8(1) applied to $x = a_n - L$. The right side is ϵ, so $|L - L'| \leq \epsilon$ holds for every $\epsilon > 0$, and proposition 1.4.7(2) gives $L = L'$, as we sought to show.

Uniqueness is what makes the notation well defined

$$\lim_{n\to\infty} a_n = L$$

for the limit of a convergent sequence, and with it the practice of treating $\lim_{n\to\infty} a_n$ as a number that can be added, multiplied and compared, which is what the next theorem makes legitimate.

Call a sequence (a_n) *bounded* if there is an $M \in \mathbb{R}$ with $|a_n| \leq M$ for every n. Such an M satisfies $M \geq |a_1| \geq 0$, so proposition 1.3.8(2) applies and says that $-M \leq a_n \leq M$ for every n, so that the set $\{a_n \big| n \geq 1\}$ of terms is bounded above by M and below by $-M$ in the sense of definition 1.2.1. A convergent sequence cannot fail this: past the index answering $\epsilon = 1$ every term is within 1 of the limit, and only finitely many terms are left unaccounted for. Part (3) of theorem 2.1.6 needs a bound of exactly this kind on one of its two factors.

Proposition 2.1.5: Convergent Sequences Are Bounded

Every convergent sequence of real numbers is bounded.

***Proof*:**
Let $a_n \to L$ and apply definition 2.1.2 with $\epsilon = 1$, producing an $N \in \mathbb{N}$ with $|a_n - L| < 1$ for all $n \geq N$. For such n, proposition 1.3.8(4) gives

$$|a_n| = |(a_n - L) + L| \leq |a_n - L| + |L| < 1 + |L|$$

If $N = 1$ this covers every term, and $M = 1 + |L|$ is a bound. If $N > 1$, the finitely many numbers $|a_1|, \ldots, |a_{N-1}|$ have a largest, since a finite nonempty set of reals has a largest element. Taking M to be the larger of that number and $1 + |L|$ bounds every term, completing the proof.

The converse fails, and example 2.1.3(4) is the witness: $((-1)^n)$ is bounded by 1 and has no limit. Boundedness is a necessary condition for convergence and nothing more, which is how it is used below, to control one factor of a product while the other factor is made small. Section 2.2 recovers a partial converse by weakening the conclusion from a limit of the sequence to a limit of one of its subsequences.

Computing a limit from definition 2.1.2 means producing an index for each ϵ, and doing that from scratch for each new sequence is avoidable whenever the sequence is assembled from ones whose limits are known. The following reduces the limit of an expression built from convergent sequences by the field operations to the limits of the sequences it is built from.

Theorem 2.1.6: The Algebra of Limits

Let (a_n) and (b_n) be real sequences with $a_n \to a$ and $b_n \to b$, and let $c \in \mathbb{R}$.

1. $a_n + b_n \to a + b$.
2. $c\, a_n \to c\, a$.
3. $a_n b_n \to ab$.
4. If $b \neq 0$ and $b_n \neq 0$ for every n, then $a_n/b_n \to a/b$.

***Proof*:**

1. Let $\epsilon > 0$. Choose N_1 with $|a_n - a| < \epsilon/2$ for $n \geq N_1$ and N_2 with $|b_n - b| < \epsilon/2$ for $n \geq N_2$, and let N be the larger. For $n \geq N$, proposition 1.3.8(4) gives
$$|(a_n + b_n) - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| < \epsilon$$

2. If $c = 0$ then $c\, a_n = 0$ for every n and $c\, a = 0$, by proposition 1.1.2(3), and the constant sequence 0 converges to 0 by example 2.1.3(2). Suppose $c \neq 0$, so that $|c| > 0$ by proposition 1.3.8(1). Let $\epsilon > 0$ and choose N with $|a_n - a| < \epsilon/|c|$ for $n \geq N$. For such n, proposition 1.3.8(3) gives
$$|c\, a_n - c\, a| = |c|\,|a_n - a| < |c| \cdot \frac{\epsilon}{|c|} = \epsilon$$

3. By proposition 2.1.5 there is an M with $|b_n| \leq M$ for every n, and $M \geq 0$ because $|b_1| \geq 0$. Put $K = M + |a| + 1$, so that $K \geq 1 > 0$, $K \geq M$ and $K \geq |a|$. For every n, proposition 1.3.8(4) and then (3) give
$$|a_n b_n - ab| = |(a_n - a)b_n + a(b_n - b)| \leq |a_n - a||b_n| + |a||b_n - b| \leq K\big(|a_n - a| + |b_n - b|\big)$$
the last step because $|b_n| \leq K$ and $|a| \leq K$, and multiplication by a nonnegative number preserves a weak inequality: a zero multiplier makes both products 0, and a positive multiplier is proposition 1.1.2(5) when the inequality is strict and an equality when it is not. Now let $\epsilon > 0$. Choose N_1 with $|a_n - a| < \epsilon/(2K)$ for $n \geq N_1$ and N_2 with $|b_n - b| < \epsilon/(2K)$ for $n \geq N_2$. For n at least as large as both, $|a_n b_n - ab| \leq K(|a_n - a| + |b_n - b|) < K(\epsilon/(2K) + \epsilon/(2K)) = \epsilon$.

4. It is enough to prove $1/b_n \to 1/b$, since part (3) applied to (a_n) and $(1/b_n)$ then gives $a_n/b_n = a_n \cdot (1/b_n) \to a \cdot (1/b) = a/b$.
Since $b \neq 0$, proposition 1.3.8(1) gives $|b| > 0$, so $|b|/2$ is a positive real and definition 2.1.2

produces an N_1 with $|b_n - b| < |b|/2$ for $n \geq N_1$. For such n, proposition 1.3.8(1) applied to $|b_n| - |b|$ and then proposition 1.3.8(5) give

$$|b| - |b_n| \leq \big||b_n| - |b|\big| \leq |b_n - b| < \frac{|b|}{2}$$

so $|b_n| > |b|/2 > 0$. Multiplying by $|b| > 0$ with proposition 1.1.2(5) and using proposition 1.3.8(3) gives $|b_n b| = |b_n||b| > |b|^2/2 > 0$, and inverting with proposition 1.1.2(7) gives $1/|b_n b| < 2/|b|^2$. Hence for $n \geq N_1$,

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| = \left|\frac{b - b_n}{b_n b}\right| = \frac{|b_n - b|}{|b_n b|} \leq \frac{2}{|b|^2}|b_n - b|$$

where the second equality uses $|b - b_n| = |b_n - b|$ from proposition 1.3.8(1) and proposition 1.3.8(3) in the form $|u/v| = |u|/|v|$, and the last step multiplies $1/|b_n b| < 2/|b|^2$ by the nonnegative number $|b_n - b|$, which gives an equality of zeros when $b_n = b$. Given $\epsilon > 0$, choose N_2 with $|b_n - b| < \epsilon|b|^2/2$ for $n \geq N_2$. For n at least as large as N_1 and N_2 the bound $(2/|b|^2)|b_n - b|$ on $|1/b_n - 1/b|$ is smaller than ϵ, so $1/b_n \to 1/b$, completing the proof.

With these four rules a limit is computed by rewriting the sequence in terms of sequences whose limits are known. Take

$$a_n = \frac{3n^2 + 2n}{n^2 - 5}$$

which is defined for $n \geq 3$, since $n^2 - 5 \geq 4$ there. Dividing numerator and denominator by n^2 turns it into

$$a_n = \frac{3 + 2/n}{1 - 5/n^2}$$

By example 2.1.3(1) and part (3), $1/n^2 = (1/n)(1/n) \to 0$. Parts (1) and (2), together with the constant sequences of example 2.1.3(2), give $3 + 2/n \to 3$ and $1 - 5/n^2 \to 1$. Part (4) applies because the limit 1 of the denominator is nonzero and each $1 - 5/n^2$ is nonzero for $n \geq 3$, where $5/n^2 \leq 5/9 < 1$. Therefore $a_n \to 3$.

Theorem 2.1.6 computes a limit from limits already known. A different situation is a sequence whose terms are not built from convergent sequences by the field operations, but which lies between two sequences having a common limit. That position alone forces convergence, and forces the limit to be the common one.

Theorem 2.1.7: The Squeeze Theorem

Let (a_n), (b_n) and (c_n) be real sequences and let $N_0 \in \mathbb{N}$ be an index with

$$a_n \leq b_n \leq c_n \qquad \text{for every } n \geq N_0$$

If $a_n \to L$ and $c_n \to L$, then $b_n \to L$.

***Proof*:**
Let $\epsilon > 0$. Choose N_1 with $|a_n - L| < \epsilon$ for $n \geq N_1$ and N_2 with $|c_n - L| < \epsilon$ for $n \geq N_2$, and let N be the largest of N_0, N_1 and N_2. Fix $n \geq N$. By proposition 1.3.8(2) with $c = \epsilon$, the bound $|a_n - L| < \epsilon$ reads $-\epsilon < a_n - L < \epsilon$ and so gives $L - \epsilon < a_n$, and $|c_n - L| < \epsilon$ gives $c_n < L + \epsilon$. Since $n \geq N_0$, the hypothesis inserts b_n between them:

$$L - \epsilon < a_n \leq b_n \leq c_n < L + \epsilon$$

By proposition 1.1.2(6), which reads $x < y$ as $0 < y - x$, the outer inequalities $L - \epsilon < b_n$ and $b_n < L + \epsilon$ say respectively that $0 < (b_n - L) + \epsilon$ and $0 < \epsilon - (b_n - L)$, that is $-\epsilon < b_n - L < \epsilon$. Then proposition 1.3.8(2), read in the other direction, turns this into $|b_n - L| < \epsilon$. As $\epsilon > 0$ was arbitrary, $b_n \to L$, as we sought to show.

The hypothesis is imposed only from the index N_0 onwards, which is what makes the theorem applicable: the two bounding sequences are typically produced by an estimate that is valid only for large n. For a first use, let (x_n) be any sequence with $|x_n| \leq 1$ for every n. Then $-1 \leq x_n \leq 1$ by proposition 1.3.8(2), and multiplying by the positive number $1/n$ gives $-1/n \leq x_n/n \leq 1/n$ for every n, by proposition 1.1.2(5) where the inequality is strict and by an equality where it is not. Both outer sequences converge to 0, by example 2.1.3(1) and theorem 2.1.6(2) with $c = -1$, so theorem 2.1.7 gives $x_n/n \to 0$ whatever the signs of the x_n.

Convergence interacts with the order of $\mathbb{R}$ as well as with its operations, and in a weakened form. An inequality between the terms of two convergent sequences passes to their limits, but a strict inequality between terms yields only a weak inequality between limits.

Proposition 2.1.8: Limits Preserve Weak Inequalities

Let (a_n) and (b_n) be real sequences with $a_n \to a$ and $b_n \to b$, and let $N_0 \in \mathbb{N}$.

1. If $a_n \leq b_n$ for every $n \geq N_0$, then $a \leq b$.
2. If $c \in \mathbb{R}$ satisfies $c \leq a_n$ for every $n \geq N_0$, then $c \leq a$. If $d \in \mathbb{R}$ satisfies $a_n \leq d$ for every $n \geq N_0$, then $a \leq d$.

***Proof*:**

1. Let $\epsilon > 0$. Choose N_1 with $|a_n - a| < \epsilon/2$ for $n \geq N_1$ and N_2 with $|b_n - b| < \epsilon/2$ for $n \geq N_2$, and fix an n at least as large as each of N_0, N_1 and N_2. Splitting $a - b$ across that index,

$$a - b = (a - a_n) + (a_n - b_n) + (b_n - b) \leq |a - a_n| + 0 + |b_n - b| < \frac{\epsilon}{2} + \frac{\epsilon}{2}$$

   where the first and third terms are bounded by proposition 1.3.8(1), which gives $x \leq |x|$, the middle term is at most 0 by the hypothesis $a_n \leq b_n$, and $|a - a_n| = |a_n - a|$ by proposition 1.3.8(1) again. So $a \leq b + \epsilon$ for every $\epsilon > 0$, and proposition 1.4.7(1) gives $a \leq b$.

2. The constant sequence with every term c converges to c by example 2.1.3(2), so part (1)

applied to it and to (a_n) gives $c \le a$. Applying part (1) to (a_n) and to the constant sequence with every term d gives $a \le d$, completing the proof.

Strict inequalities are not preserved. The constant sequence 0 and the sequence $(1/n)$ satisfy $0 < 1/n$ for every n, and their limits are both 0 by example 2.1.3(1) and (2). This is why part (2) is the form used to place a limit inside a closed interval: bounds $c \le a_n \le d$ on the terms give $c \le a \le d$ on the limit, and nothing sharper is available from the terms alone.

Every proof above reduced a claim about $a_n \to L$ to a claim about $|a_n - L|$ becoming small, which is convergence to 0. That case has a name, and it is the name already attached to rational sequences in definition 1.1.4.

Definition 2.1.9: Null Sequence

A real sequence (a_n) is *null* if $a_n \to 0$, that is, if for every $\epsilon > 0$ there is an $N \in \mathbb{N}$ with $|a_n| < \epsilon$ for every $n \ge N$.

The condition defining $a_n \to L$ is the inequality $|(a_n - L) - 0| < \epsilon$ for large n, which is the condition defining nullity of $(a_n - L)$. So $a_n \to L$ if and only if $(a_n - L)$ is null, and a convergence statement can always be restated as a statement about a null sequence. Example 2.1.3(1) says that $(1/n)$ is one, and example 2.1.3(4) shows $((-1)^n)$ is not, while $((-1)^n/n)$ is by the computation following theorem 2.1.7. For a sequence of rational numbers the definition above agrees with the one inside definition 1.1.4, by the same density argument that reconciled the two ranges of ϵ.

Three facts about null sequences are used throughout the chapters that follow. The first is that the null sequences are closed under sums and scalar multiples, the second turns any bound by a null sequence into nullity, and the third is a decay estimate: a fixed power of k multiplied by the k-th power of a number between 0 and 1 tends to zero, however large the fixed power and however close the number is to 1.

Proposition 2.1.10: Sums, Comparison, and the Decay of $k^j r^k$

1. If (a_n) and (b_n) are null and $c \in \mathbb{R}$, then $(a_n + b_n)$ and $(c\, a_n)$ are null.
2. If (b_n) is null and there is an $N_0 \in \mathbb{N}$ with $|a_n| \le b_n$ for every $n \ge N_0$, then (a_n) is null.
3. For every integer $j \ge 0$ and every real r with $0 < r < 1$, the sequence $(k^j r^k)_{k \ge 1}$ is null.

***Proof*:**

1. These are parts (1) and (2) of theorem 2.1.6 with $a = b = 0$, since $0 + 0 = 0$ and $c \cdot 0 = 0$ by proposition 1.1.2(3).

2. For $n \ge N_0$ the hypothesis gives $|a_n| \le b_n$, and $|a_n| \ge 0$ by proposition 1.3.8(1), so $b_n \ge 0$ and proposition 1.3.8(2), with b_n in the role of its constant c, turns the hypothesis into $-b_n \le a_n \le b_n$. The sequence $(-b_n)$ is null by part (1) with $c = -1$, so theorem 2.1.7 applies with $L = 0$ and the index N_0 and gives $a_n \to 0$.

3. Fix an integer $j \geq 0$ and a real r with $0 < r < 1$, and put $h = (1-r)/r$. Then $h > 0$, since $1 - r > 0$ and $r > 0$, and
$$1 + h = 1 + \frac{1-r}{r} = \frac{1}{r}$$
so that $r^k = (1+h)^{-k}$ for every $k \geq 1$. Write $m = j + 1$, an integer with $m \geq 1$, and let $k > 2m$.
By [1, the binomial theorem], $(1+h)^k$ expands as $\sum_{i=0}^{k} \binom{k}{i} h^i$, a sum of nonnegative terms containing $\binom{k}{m} h^m$ as one summand, the index m being admissible because $m < 2m < k$. By [1, the falling-factorial identity], $k(k-1)\cdots(k-m+1) = m!\binom{k}{m}$. Hence
$$(1+h)^k \geq \binom{k}{m} h^m = \frac{k(k-1)\cdots(k-m+1)}{m!}\, h^m$$
Each of the m factors in that product satisfies $k - i \geq k - m + 1 > k/2$ for $0 \leq i \leq m-1$, because $k > 2m$ gives $m < k/2$ and hence $k - m + 1 > k - k/2 = k/2$. Multiplying these m inequalities between positive numbers, by m applications of proposition 1.1.2(5), gives $k(k-1)\cdots(k-m+1) > (k/2)^m = k^m/2^m$. Multiplying that by the positive number $h^m/m!$ and combining with $(1+h)^k \geq k(k-1)\cdots(k-m+1)\, h^m/m!$,
$$(1+h)^k > \frac{k^m h^m}{2^m\, m!} > 0$$
Inverting this with proposition 1.1.2(7) and then multiplying by $k^j > 0$ with proposition 1.1.2(5) gives, for every $k > 2m$,
$$0 < k^j r^k = \frac{k^j}{(1+h)^k} < \frac{2^m\, m!}{h^m} \cdot \frac{k^j}{k^m} = \frac{C}{k}, \qquad C = \frac{2^m\, m!}{h^m}$$
the last equality because $m = j + 1$ makes $k^j/k^m = 1/k$. The sequence $(C/k)_{k \geq 1}$ is null by example 2.1.3(1) and part (1), and $|k^j r^k| = k^j r^k$ because these terms are positive. So part (2), applied with the index $N_0 = 2m + 1$, shows that $(k^j r^k)_{k \geq 1}$ is null, completing the proof.

Part (3) is stated with an arbitrary polynomial power because that is how it is used: a sequence bounded in absolute value by $Ck^j r^k$ is null, by parts (1) and (2), no matter how large j is, so long as $0 < r < 1$. For $-1 < r < 0$ the terms satisfy $|k^j r^k| = k^j |r|^k$ by repeated application of proposition 1.3.8(3), with $0 < |r| < 1$, so part (3) applied to $|r|$ and then part (2) give $k^j r^k \to 0$ there as well, and for $r = 0$ every term of $(k^j r^k)_{k \geq 1}$ is 0. So the conclusion holds for every r with $|r| < 1$, and how close $|r|$ is to 1 changes only the constant C in the proof, never the comparison index $2m + 1 = 2j + 3$ and never the conclusion.

Exercise 2.1.1

1. Show directly from definition 2.1.2, without using theorem 2.1.6, that
$$\frac{3n-1}{n+2} \to 3$$

2. Let (a_n) be a real sequence with $a_n \to a$. Show that $|a_n| \to |a|$. Then exhibit a sequence (a_n) for which $(|a_n|)$ converges while (a_n) does not, so that the converse fails.

3. Let (a_n) be a real sequence with $a_n \geq 0$ for every n and with $a_n \to a$. Show first that $a \geq 0$, so that $\sqrt{a}$ is defined by theorem 1.3.12, and then that $\sqrt{a_n} \to \sqrt{a}$. Treat the cases $a > 0$ and $a = 0$ separately.

4. Let $a_n \to a$ with $a > 0$. Show that there is an $N \in \mathbb{N}$ with $a_n > a/2$ for every $n \geq N$, and deduce that at most finitely many terms of the sequence are less than or equal to 0.

5. Show that $(n^5 2^{-n})_{n\geq 1}$ is null. Show also that $(n!/n^n)_{n\geq 1}$ is null, and explain why proposition 2.1.10(3) does not apply to the second sequence.

6. Let $x \in \mathbb{R}$ and write $\lfloor y \rfloor$ for the integer part of a real number y, as in proposition 1.4.3. Show that
$$\frac{\lfloor nx \rfloor}{n} \to x$$

2.2 Monotone Sequences, Limsup and Liminf

Every result in section 2.1 starts from a sequence already known to converge, because the definition of a limit names the limit in advance. This section produces convergence from hypotheses that never name the limit. Two hypotheses suffice: monotonicity together with boundedness gives a limit outright, and boundedness alone gives one along a subsequence.

Between those two statements sit the upper and lower limits, two numbers in the extended real line $[-\infty, +\infty]$ attached to every real sequence, convergent or not. The sequence has a limit in $[-\infty, +\infty]$ exactly when the two agree, and it converges exactly when they agree on a real value. The tests of section 2.4 and the radius of convergence of section 2.6 are both stated in terms of the upper limit, so the working properties proved here are what those sections use.

Monotonicity removes the obligation to name the limit first. An increasing sequence bounded above cannot pass the least of its upper bounds, and theorem 1.2.4 guarantees that this number exists.

Inside $\mathbb{Q}$ the guarantee fails, and one construction makes the failure explicit. By theorem 1.4.4 choose $q_1 \in \mathbb{Q}$ with $\sqrt{2} - 1 < q_1 < \sqrt{2}$, and, given q_n, choose $q_{n+1} \in \mathbb{Q}$ with

$$\max\Big\{q_n,\ \sqrt{2} - \tfrac{1}{n+1}\Big\} < q_{n+1} < \sqrt{2}$$

which is possible because both numbers inside the maximum are smaller than $\sqrt{2}$. Every term is rational, the sequence increases, and every term is smaller than $3/2$ because $q_n < \sqrt{2} \leq 3/2$ by example 1.3.13. Moreover $0 < \sqrt{2} - q_n < 1/n$ for every n. Now $1/n \to 0$: given $\epsilon > 0$, theorem 1.4.1 gives an N with $0 < 1/N < \epsilon$, and $0 < 1/n \leq 1/N < \epsilon$ for $n \geq N$ by proposition 1.1.2. So $\sqrt{2} - q_n \to 0$ by theorem 2.1.7, and $q_n \to \sqrt{2}$ by theorem 2.1.6. Since $\sqrt{2}$ is not rational (proposition 1.1.3) and a limit is unique (proposition 2.1.4), no rational number is the limit of (q_n). Boundedness and monotonicity produce nothing inside $\mathbb{Q}$. The theorem below is a statement about $\mathbb{R}$ and about nothing weaker.

Definition 2.2.1: Increasing, Decreasing and Monotone Sequences

A real sequence $(a_n)_{n\geq 1}$ is *increasing* if $a_n \leq a_{n+1}$ for every $n \geq 1$, and *decreasing* if $a_n \geq a_{n+1}$ for every $n \geq 1$. It is *monotone* if it is increasing or decreasing.

The rational sequence (q_n) built above is increasing, as is $a_n = 1 - 1/n$, while $a_n = 1/n$ is decreasing and $a_n = (-1)^n$ is neither. A constant sequence is both.

Theorem 2.2.2: Bounded Monotone Sequences Converge

Let $(a_n)_{n\geq 1}$ be a monotone real sequence. Then (a_n) converges if and only if it is bounded. If (a_n) is increasing and bounded above, then

$$\lim_{n\to\infty} a_n = \sup\{a_n \big| n \geq 1\}$$

and if (a_n) is decreasing and bounded below, then $\lim_{n\to\infty} a_n = \inf\{a_n \big| n \geq 1\}$.

***Proof*:**
If (a_n) converges then it is bounded by proposition 2.1.5, and this direction uses no monotonicity.

For the converse, suppose first that (a_n) is increasing and bounded above. The set $E = \{a_n \big| n \geq 1\}$ is nonempty and bounded above, so theorem 1.2.4 gives a real number $s = \sup E$. Let $\epsilon > 0$. Since s is an upper bound of E, proposition 1.2.6 produces an element of E larger than $s - \epsilon$, that is, an index N with $a_N > s - \epsilon$. Induction on n gives $a_n \geq a_N$ for every $n \geq N$: the case $n = N$ is an equality, and $a_{n+1} \geq a_n \geq a_N$ by definition 2.2.1. For every such n,

$$s - \epsilon < a_N \leq a_n \leq s$$

the last inequality because s is an upper bound of E. Hence $0 \leq s - a_n < \epsilon$, so $|a_n - s| = s - a_n < \epsilon$ by definition 1.3.6 and proposition 1.3.8. As ϵ was arbitrary, $a_n \to s$ in the sense of definition 2.1.2. Note that only the upper bound was used: an increasing sequence is bounded below by a_1, so for an increasing sequence boundedness and boundedness above are the same hypothesis.

Suppose next that (a_n) is decreasing and bounded below, and put $b_n = -a_n$. Then (b_n) is increasing, and it is bounded above: if $a_n \geq L$ for all n then $b_n \leq -L$ by proposition 1.1.2. The case already proved gives $b_n \to \sup\{b_n \big| n \geq 1\}$. Writing $A = \{a_n \big| n \geq 1\}$, the set $\{b_n \big| n \geq 1\}$ is $-A$, and A is bounded below, so proposition 1.3.2 gives $\sup(-A) = -\inf A$. Thus $b_n \to -\inf A$, and multiplying by the constant sequence with value -1 and applying theorem 2.1.6 gives $a_n \to \inf A$, as we sought to show.

The theorem converts the least upper bound property into a statement about sequences. What it gives is a limit whose value is not part of the hypothesis: the supremum comes from theorem 1.2.4, and identifying it afterwards is a separate problem. The next example is a sequence defined by a recursion in which no term is given in closed form, so that the limit exists well before any computation names it.

Example 2.2.3: The Nested Radicals $\sqrt{2}$, $\sqrt{2+\sqrt{2}}$, ...

Let $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2+a_n}$ for $n \geq 1$. Each term is defined and nonnegative: $a_1 \geq 0$, and if $a_n \geq 0$ then $2 + a_n \geq 2 > 0$, so theorem 1.3.12 produces the unique nonnegative square root a_{n+1}.

Every term is smaller than 2. First $a_1 < 2$, since $a_1 \geq 2$ would give $2 = a_1^2 \geq 4$ by proposition 1.1.2. If $a_n < 2$ then $2 + a_n < 4$, and $a_{n+1} \geq 2$ would give $2 + a_n = a_{n+1}^2 \geq 4$, so $a_{n+1} < 2$.

The sequence increases. For each n,

$$a_{n+1}^2 - a_n^2 = (2 + a_n) - a_n^2 = (2 - a_n)(1 + a_n)$$

and both factors are positive because $0 \leq a_n < 2$. If $a_{n+1} \leq a_n$ held, then $a_{n+1}^2 \leq a_{n+1}a_n \leq a_n^2$ by proposition 1.1.2, contradicting $a_{n+1}^2 - a_n^2 > 0$. So $a_{n+1} > a_n$.

By theorem 2.2.2 the sequence converges to a real number L. Since the sequence increases, $a_1 \leq a_n < 2$ for every n, so proposition 2.1.8, applied to (a_n) against the constant sequences with values a_1 and 2, gives $\sqrt{2} = a_1 \leq L \leq 2$. To identify L, note that the shifted sequence $b_n = a_{n+1}$ also converges to L: given $\epsilon > 0$ choose N with $|a_m - L| < \epsilon$ for all $m \geq N$, and then $n \geq N$ forces $n + 1 \geq N$, so $|b_n - L| < \epsilon$. Now $b_n^2 \to L^2$ by theorem 2.1.6 applied to the product of (b_n) with itself, while $b_n^2 = 2 + a_n \to 2 + L$ by theorem 2.1.6 applied to the sum with the constant sequence 2. Limits are unique (proposition 2.1.4), so $L^2 = 2 + L$, that is,

$$(L - 2)(L + 1) = 0$$

Since $L \geq \sqrt{2} > 0$, the factor $L + 1$ is not zero, so $L = 2$.

Boundedness alone does not give a limit. The sequence $(-1)^n$ does not converge: a limit a would give an N with $|(-1)^n - a| < 1$ for all $n \geq N$, and choosing one even and one odd such n gives $|1 - a| < 1$ and $|-1 - a| < 1$, whereas $2 = |1 - (-1)| \leq |1 - a| + |a + 1| < 2$ by proposition 1.3.8. What boundedness does give is a limit along part of the sequence, and to say this precisely takes two pieces of vocabulary. The first is a value system large enough to record unbounded behaviour, since the constructions below assign a value to every sequence and not only to the bounded ones.

Definition 2.2.4: The Extended Real Line

The *extended real line* is the set

$$[-\infty, +\infty] = \mathbb{R} \cup \{-\infty, +\infty\}$$

where $-\infty$ and $+\infty$ are two symbols not belonging to $\mathbb{R}$. Its order restricts to the order of $\mathbb{R}$ on $\mathbb{R}$, and in addition $-\infty < x < +\infty$ for every $x \in \mathbb{R}$, together with $-\infty < +\infty$. Negation extends by $-(+\infty) = -\infty$ and $-(-\infty) = +\infty$. Addition extends partially: $x + (+\infty) = (+\infty) + x = +\infty$ for every $x \neq -\infty$, and $x + (-\infty) = (-\infty) + x = -\infty$ for every $x \neq +\infty$, while $(+\infty) + (-\infty)$ is left undefined.

For $S \subseteq [-\infty, +\infty]$, an element u is an *upper bound* of S if $x \leq u$ for every $x \in S$, and $\sup S$ denotes the least upper bound of S when one exists. The terms *lower bound* and $\inf S$ are defined symmetrically. For a real sequence (a_n), write $a_n \to +\infty$ if for every $M \in \mathbb{R}$ there is an N with $a_n > M$ for all $n \geq N$, and $a_n \to -\infty$ if for every $M \in \mathbb{R}$ there is an N with $a_n < M$ for all $n \geq N$. The sequence (a_n) *has limit* $s \in [-\infty, +\infty]$ if s is real and $a_n \to s$ in the sense of definition 2.1.2, or if $s = \pm\infty$ and $a_n \to s$ in the sense just given.

The relation just described is a total order. For two real arguments it is the order of $\mathbb{R}$, for one real and one infinite argument it is one of the declared inequalities, and two infinite arguments are either equal or ordered by $-\infty < +\infty$. For transitivity, let $x < y < z$. If $x = -\infty$ then $z \neq -\infty$, since $y < z$, so $x < z$ by declaration; if $z = +\infty$ then $x \neq +\infty$, since $x < y$, so again $x < z$; and in the remaining case x, y, z are all real and transitivity is inherited from $\mathbb{R}$. Note that a sequence with limit $+\infty$ does not converge: the word *converges* stays reserved for a real limit, and $a_n \to +\infty$ records a specific way of failing to converge. As for the enlargement of suprema, no real number is an upper bound of $\mathbb{N}$ by theorem 1.4.1 and $-\infty$ is not one either, so $+\infty$ is the only upper bound and $\sup \mathbb{N} = +\infty$, while the supremum of $\{x \in \mathbb{R} | x^2 < 2\}$ is the real number $\sqrt{2}$ of example 1.3.13 and is unchanged.

The point of the enlargement is that suprema and infima now exist without any hypothesis whatever, which is what lets the definitions further down apply to every sequence.

Proposition 2.2.5: Suprema, Infima and Density in $[-\infty, +\infty]$

1. Every subset $S \subseteq [-\infty, +\infty]$ has a supremum and an infimum in $[-\infty, +\infty]$, and $\inf S = -\sup(-S)$, where $-S = \{-x | x \in S\}$. In particular $\sup \emptyset = -\infty$ and $\inf \emptyset = +\infty$.
2. If $u, v \in [-\infty, +\infty]$ and $u < v$, then there is a real number x with $u < x < v$.

Proof:

1. Four cases produce the supremum.

   If $S = \emptyset$, every element of $[-\infty, +\infty]$ is vacuously an upper bound, and $-\infty$ is the smallest element, so $\sup \emptyset = -\infty$.

   If $+\infty \in S$, then an upper bound u satisfies $u \geq +\infty$, so $u = +\infty$, and $+\infty$ is an upper bound. Hence $\sup S = +\infty$.

   If S is nonempty, $+\infty \notin S$ and $S \cap \mathbb{R} = \emptyset$, then $S = \{-\infty\}$, and $-\infty$ is an upper bound and is smaller than every other element, so $\sup S = -\infty$.

In the remaining case $S \cap \mathbb{R} \neq \emptyset$ and $+\infty \notin S$. If $S \cap \mathbb{R}$ is bounded above by some real number, theorem 1.2.4 gives $\sigma = \sup(S \cap \mathbb{R}) \in \mathbb{R}$. Then σ is an upper bound of S, since every real element of S is at most σ and $-\infty < \sigma$. If u is any upper bound of S, then $u \neq -\infty$ because $S \cap \mathbb{R} \neq \emptyset$, and if u is real then $u \geq \sigma$ because u bounds $S \cap \mathbb{R}$ above, while $u = +\infty$ certainly exceeds σ. So $\sup S = \sigma$. If instead $S \cap \mathbb{R}$ is not bounded above by any real number, then no real number is an upper bound of S, and neither is $-\infty$ because $S \cap \mathbb{R} \neq \emptyset$, so $+\infty$ is the only upper bound and $\sup S = +\infty$.

For the infimum, observe that $u < v$ holds if and only if $-v < -u$. For two real arguments this is proposition 1.1.2. If $u = -\infty$ and v is real, then $-v$ is real and $-u = +\infty$, so both sides hold. If u is real and $v = +\infty$, then $-v = -\infty < -u$ and both sides hold again. If $u = -\infty$ and $v = +\infty$, then $-v = -\infty < +\infty = -u$. No other case has $u < v$, and applying the equivalence to $-u$ and $-v$ gives the converse direction. Consequently u is a lower bound of S if and only if $-u$ is an upper bound of $-S$, and u is the greatest such if and only if $-u$ is the least such. Since $\sup(-S)$ exists by what was just proved, $-\sup(-S)$ is the greatest lower bound of S. Hence $\inf S$ exists and equals $-\sup(-S)$, and taking $S = \emptyset$ gives $\inf \emptyset = -\sup \emptyset = +\infty$.

2. If u and v are both real, proposition 1.1.2 gives $u < (u+v)/2 < v$. If $u = -\infty$ and v is real, take $x = v - 1$. If u is real and $v = +\infty$, take $x = u + 1$. If $u = -\infty$ and $v = +\infty$, take $x = 0$. No further case occurs, since $u < v$ excludes $u = +\infty$ and $v = -\infty$. This exhausts the possibilities, completing the proof.

The second piece of vocabulary is the operation of passing to part of a sequence. Keeping infinitely many terms of (a_n) in their original order gives a new sequence, and the rest of the section is about which limits such a sequence can have.

Definition 2.2.6: Subsequences and Subsequential Limits

Let $(a_n)_{n \geq 1}$ be a real sequence and let $n_1 < n_2 < n_3 < \cdots$ be a strictly increasing sequence of elements of $\mathbb{N}$. The sequence $k \mapsto a_{n_k}$ is a *subsequence* of (a_n), written $(a_{n_k})_{k \geq 1}$. An element $s \in [-\infty, +\infty]$ is a *subsequential limit* of (a_n) if some subsequence of (a_n) has limit s in the sense of definition 2.2.4.

For $a_n = (-1)^n$ the choice $n_k = 2k$ gives the constant subsequence with value 1, and $n_k = 2k - 1$ gives the constant subsequence with value -1, so both 1 and -1 are subsequential limits. The choice $n_k = k^2$ gives $a_{n_k} = (-1)^{k^2} = (-1)^k$, since k^2 and k have the same parity, so a subsequence need not converge. Taking $n_k = k$ shows that a sequence is a subsequence of itself. One inequality is used throughout: $n_k \geq k$ for every k, by induction, since $n_1 \geq 1$ because $n_1 \in \mathbb{N}$, and $n_{k+1} > n_k \geq k$ forces $n_{k+1} \geq k + 1$ because no element of $\mathbb{N}$ lies strictly between k and $k + 1$.

Proposition 2.2.7: Subsequences of a Sequence with a Limit

Let $(a_n)_{n\geq 1}$ be a real sequence and let $(a_{n_k})_{k\geq 1}$ be a subsequence.

1. If (a_n) has limit $s \in [-\infty, +\infty]$, then (a_{n_k}) has limit s.
2. If two subsequences of (a_n) have different limits in $[-\infty, +\infty]$, then (a_n) has no limit in $[-\infty, +\infty]$, and in particular it does not converge.
3. A subsequence of (a_{n_k}) is a subsequence of (a_n).

Proof:

1. Recall from the discussion after definition 2.2.6 that $n_k \geq k$ for every k.

   Suppose s is real and $\epsilon > 0$. Choose N with $|a_n - s| < \epsilon$ for all $n \geq N$. For $k \geq N$ we have $n_k \geq k \geq N$, so $|a_{n_k} - s| < \epsilon$. Suppose $s = +\infty$ and let $M \in \mathbb{R}$. Choose N with $a_n > M$ for all $n \geq N$, and then $a_{n_k} > M$ for all $k \geq N$ by the same inequality $n_k \geq k \geq N$. The case $s = -\infty$ is identical with the inequality reversed.

2. Suppose (a_n) had limit s. By part (1) every subsequence would have limit s, so it is enough to know that a sequence cannot have two different limits in $[-\infty, +\infty]$, which is the case. Two different real limits are excluded by proposition 2.1.4. A real limit s and the limit $+\infty$ are incompatible: a convergent sequence is bounded by proposition 2.1.5, say $a_n \leq M$ for all n, and then $a_n > M$ holds for no n at all. The same argument with a lower bound excludes a real limit together with $-\infty$. Finally $+\infty$ and $-\infty$ are incompatible: taking $M = 0$ in both definitions produces an N beyond which $a_n > 0$ and $a_n < 0$ simultaneously, which is impossible. So (a_n) has no limit, and since a convergent sequence has a real limit, it does not converge.

3. Let $(a_{n_{k_j}})_{j\geq 1}$ be a subsequence of (a_{n_k}), so that $k_1 < k_2 < \cdots$ in $\mathbb{N}$. Since (n_k) is strictly increasing, $k_j < k_{j+1}$ gives $n_{k_j} < n_{k_{j+1}}$. Thus $j \mapsto n_{k_j}$ is a strictly increasing sequence in $\mathbb{N}$, which is what definition 2.2.6 requires, completing the proof.

Part (2) is the standard way to prove that a specific sequence diverges, and $(-1)^n$ is the smallest instance: its even and odd subsequences are constant with values 1 and -1, so it has no limit. Parts (1) and (3) say that passing to a subsequence loses no limit and can be iterated, and both are used constantly below.

Deciding whether a given number is a subsequential limit by exhibiting a subsequence means constructing infinitely many indices, and the next result replaces that construction by a condition on the terms of the original sequence. It is the form in which subsequential limits are recognized in practice.

Proposition 2.2.8: Subsequential Limits Recognized Term by Term

Let $(a_n)_{n\geq 1}$ be a real sequence.

1. For $x \in \mathbb{R}$, the number x is a subsequential limit of (a_n) if and only if for every $\epsilon > 0$ the set $\{n \in \mathbb{N} \big| |a_n - x| < \epsilon\}$ is infinite.
2. The following three statements are equivalent: that $+\infty$ is a subsequential limit of (a_n), that the sequence (a_n) is not bounded above, and that for every $M \in \mathbb{R}$ the set $\{n \in \mathbb{N} \big| a_n > M\}$ is infinite. The three corresponding statements obtained by replacing $+\infty$ by $-\infty$, "bounded above" by "bounded below" and $a_n > M$ by $a_n < M$ are also equivalent.

***Proof*:**

1. Suppose $a_{n_k} \to x$ and let $\epsilon > 0$. There is a K with $|a_{n_k} - x| < \epsilon$ for all $k \geq K$. The indices $n_K < n_{K+1} < \cdots$ are pairwise distinct elements of $\mathbb{N}$ lying in the set in question, so that set is infinite.

   Conversely, suppose every such set is infinite. Define indices recursively, always taking the least admissible one so that no choice is made: let n_1 be the least n with $|a_n - x| < 1$, which exists because the set for $\epsilon = 1$ is infinite, hence nonempty, and $\mathbb{N}$ is well ordered. Given n_k, the set $\{n \big| |a_n - x| < 1/(k+1)\}$ is infinite, so it contains elements larger than n_k, and n_{k+1} is the least such element. Then $n_1 < n_2 < \cdots$ and $|a_{n_k} - x| < 1/k$ for every k. Given $\epsilon > 0$, theorem 1.4.1 gives a K with $0 < 1/K < \epsilon$, and for $k \geq K$, proposition 1.1.2 gives $|a_{n_k} - x| < 1/k \leq 1/K < \epsilon$. So $a_{n_k} \to x$ and x is a subsequential limit.

2. Suppose $+\infty$ is a subsequential limit, say $a_{n_k} \to +\infty$, and let $M \in \mathbb{R}$. There is a K with $a_{n_k} > M$ for all $k \geq K$, and the indices $n_K < n_{K+1} < \cdots$ are infinitely many, so $\{n \big| a_n > M\}$ is infinite. In particular it is nonempty for every M, so no real number bounds (a_n) above.

   Suppose next that (a_n) is not bounded above and let $M \in \mathbb{R}$. Write $F = \{n \big| a_n > M\}$ and suppose F were finite. If F is empty then M bounds (a_n) above. If F is nonempty, then $\{a_n \big| n \in F\}$ is a finite nonempty set of real numbers and so has a largest element c, and $a_n \leq \max\{M, c\}$ for every n, since $n \notin F$ gives $a_n \leq M$ and $n \in F$ gives $a_n \leq c$. Either way (a_n) would be bounded above, contrary to hypothesis. So F is infinite.

   Finally suppose $\{n \big| a_n > M\}$ is infinite for every $M \in \mathbb{R}$. Let n_1 be the least n with $a_n > 1$, and, given n_k, let n_{k+1} be the least element of $\{n \big| a_n > k+1\}$ exceeding n_k, which exists because that set is infinite. Then $a_{n_k} > k$ for every k. Given $M \in \mathbb{R}$, theorem 1.4.1 gives a $K \in \mathbb{N}$ with $K > M$, and for $k \geq K$ we get $a_{n_k} > k \geq K > M$. So $a_{n_k} \to +\infty$ and $+\infty$ is a subsequential limit. Together the implications just proved make the three statements equivalent.

   For the statements with $-\infty$, put $b_n = -a_n$. A sequence of indices makes (a_{n_k}) tend to $-\infty$ if and only if it makes (b_{n_k}) tend to $+\infty$, directly from definition 2.2.4, and (a_n)

is bounded below if and only if (b_n) is bounded above, by proposition 1.1.2. Likewise $a_n < M$ if and only if $b_n > -M$. So the three statements for (a_n) and $-\infty$ are the three statements for (b_n) and $+\infty$, which have just been shown equivalent, completing the proof.

The condition in part (1) is that (a_n) visits every neighbourhood of x infinitely often, and the content of the proposition is that this weak-looking condition already builds the subsequence. Nothing about the sequence is assumed: in particular a subsequential limit need not be a value of the sequence, and a value taken infinitely often is a subsequential limit.

Existence is the remaining question. For a bounded sequence, repeated bisection of an interval containing every term locates a point that the sequence visits infinitely often, and proposition 2.2.8 then converts that point into a convergent subsequence.

Theorem 2.2.9: Bolzano-Weierstrass

1. Every bounded real sequence has a convergent subsequence.
2. Every real sequence has at least one subsequential limit in $[-\infty, +\infty]$.

***Proof*:**

1. Let (a_n) be bounded, and choose real numbers $A \le B$ with $A \le a_n \le B$ for every n.

   Construct closed intervals $I_k = [A_k, B_k]$ recursively. Put $I_1 = [A, B]$, and note that $\{n | a_n \in I_1\} = \mathbb{N}$ is infinite. Given I_k with $\{n | a_n \in I_k\}$ infinite, let $C_k = (A_k + B_k)/2$ and consider the two halves $[A_k, C_k]$ and $[C_k, B_k]$. Their union is I_k, so

   $$\{n | a_n \in I_k\} = \{n | a_n \in [A_k, C_k]\} \cup \{n | a_n \in [C_k, B_k]\}$$

   and a union of two finite sets is finite, so at least one of the two sets on the right is infinite. Let I_{k+1} be the left half if the corresponding set is infinite, and the right half otherwise. This choice is canonical and keeps $\{n | a_n \in I_{k+1}\}$ infinite.

   By construction $I_{k+1} \subseteq I_k$, each half has $A_k \le C_k \le B_k$ so that $A_k \le B_k$ holds for every k, and $B_{k+1} - A_{k+1} = (B_k - A_k)/2$, so induction gives $B_k - A_k = (B - A)/2^{k-1} = 2(B - A) \cdot (1/2)^k$. The sequence $(1/2)^k$ is null by proposition 2.1.10, and a scalar multiple of a null sequence is null by the same result, so $B_k - A_k \to 0$. In particular, for every $\epsilon > 0$ there is a k with $B_k - A_k < \epsilon$. The hypotheses of theorem 1.4.8 are therefore met, and it gives a unique real number x with $x \in I_k$ for every k.

   Let $\epsilon > 0$ and choose k with $B_k - A_k < \epsilon$. For every one of the infinitely many indices n with $a_n \in I_k$, both a_n and x lie in $[A_k, B_k]$, so $a_n - x \le B_k - A_k$ and $x - a_n \le B_k - A_k$, whence $|a_n - x| \le B_k - A_k < \epsilon$ by proposition 1.3.8. So $\{n | |a_n - x| < \epsilon\}$ is infinite for every $\epsilon > 0$, and proposition 2.2.8 gives a subsequence converging to x.
2. If (a_n) is bounded, part (1) gives a real subsequential limit. If (a_n) is not bounded above, then $+\infty$ is a subsequential limit by proposition 2.2.8, and if (a_n) is not bounded below, then $-\infty$ is. Since a sequence that is bounded above and below is bounded, these three cases exhaust the possibilities, completing the proof.

The proof uses completeness twice, once through theorem 1.4.8 and once through the bisection that meets its hypotheses. Part (2) is what makes the following definition unconditional: the set of subsequential limits of a real sequence is never empty, so its supremum and infimum are attached to every sequence without exception.

Definition 2.2.10: Limit Superior and Limit Inferior of a Sequence

Let $(a_n)_{n\geq 1}$ be a real sequence and let $E \subseteq [-\infty, +\infty]$ be its set of subsequential limits. The *limit superior* and *limit inferior* of (a_n) are

$$\limsup_{n\to\infty} a_n = \sup E, \qquad \liminf_{n\to\infty} a_n = \inf E$$

the supremum and infimum being taken in $[-\infty, +\infty]$ as in definition 2.2.4.

Both exist by proposition 2.2.5, and $E \neq \emptyset$ by theorem 2.2.9, so any $y \in E$ satisfies $\inf E \leq y \leq \sup E$ and therefore $\liminf_n a_n \leq \limsup_n a_n$ for every real sequence. The two quantities are exchanged by negation:

$$\limsup_{n\to\infty}(-a_n) = -\liminf_{n\to\infty} a_n, \qquad \liminf_{n\to\infty}(-a_n) = -\limsup_{n\to\infty} a_n$$

To see this, write $E(a)$ for the set of subsequential limits of (a_n). A subsequence of $(-a_n)$ is the negative of a subsequence of (a_n) with the same indices, and $a_{n_k} \to y$ if and only if $-a_{n_k} \to -y$, by theorem 2.1.6 when y is real and directly from definition 2.2.4 when $y = \pm\infty$. Hence $E(-a) = -E(a)$, and proposition 2.2.5 gives $\sup(-E(a)) = -\inf E(a)$, which is the first identity. The second follows by applying the first to $(-a_n)$.

Example 2.2.11: $(-1)^n(1+1/n)$, an Unbounded Interlacing, and an Enumeration of $\mathbb{Q}$

1. Let $a_n = (-1)^n(1+1/n)$. The sequences $(1/(2k))_k$ and $(1/(2k-1))_k$ are subsequences of $(1/n)_n$, which tends to 0 by theorem 1.4.1, so both tend to 0 by proposition 2.2.7. The even terms are $a_{2k} = 1 + 1/(2k)$, which therefore tends to 1 by theorem 2.1.6, and the odd terms are $a_{2k-1} = -(1+1/(2k-1))$, which tends to -1 by the same result. So $\{-1, 1\} \subseteq E$. The sequence is bounded, since $|a_n| = 1 + 1/n \leq 2$, so neither $+\infty$ nor $-\infty$ lies in E by proposition 2.2.8. If $y \in E$ is real, say $a_{n_k} \to y$, then infinitely many of the indices n_k are even or infinitely many are odd. In the first case the corresponding sub-subsequence is a subsequence of (a_n) by proposition 2.2.7 and converges to y by the same result, while its indices have the form $2m_1 < 2m_2 < \cdots$, so it is also a subsequence of $(a_{2k})_k$ and converges to 1. Then $y = 1$ by proposition 2.1.4. The second case gives $y = -1$ in the same way. So $E = \{-1, 1\}$ and
$$\limsup_{n\to\infty} a_n = 1, \qquad \liminf_{n\to\infty} a_n = -1$$
The sequence does not converge, by proposition 2.2.7. Note that the value 1 is taken by no term of the sequence, so an upper limit need not be a value.

2. Let $a_n = n$ for even n and $a_n = 1/n$ for odd n. The even terms are not bounded above, so $+\infty \in E$ by proposition 2.2.8, and every term is positive, so $-\infty \notin E$ by the same result. The odd terms satisfy $0 < a_{2k-1} = 1/(2k-1) \leq 1/k$, so $a_{2k-1} \to 0$ by theorem 2.1.7 and $0 \in E$. Suppose $y \in E$ is real, say $a_{n_k} \to y$. The subsequence is bounded by proposition 2.1.5, say $|a_{n_k}| \leq M$, and theorem 1.4.1 gives an $m \in \mathbb{N}$ with $m > M$. An even index n_k has $a_{n_k} = n_k \leq M < m$, and there are at most m elements of $\mathbb{N}$ below m, so only finitely many

n_k are even. Beyond some K all indices are odd, and then $0 < a_{n_k} = 1/n_k \le 1/k$ using $n_k \ge k$, so $a_{n_k} \to 0$ by theorem 2.1.7 and $y = 0$ by proposition 2.1.4. Hence $E = \{0, +\infty\}$, and

$$\limsup_{n\to\infty} a_n = +\infty, \qquad \liminf_{n\to\infty} a_n = 0$$

3. Since $\mathbb{Q}$ is countable, fix a bijection $n \mapsto q_n$ from $\mathbb{N}$ onto $\mathbb{Q}$. Every element of $[-\infty, +\infty]$ is a subsequential limit of (q_n). For a real x and $\epsilon > 0$, the interval with endpoints x and $x + \epsilon$ contains infinitely many rationals: theorem 1.4.4 gives $p_1 \in \mathbb{Q}$ with $x < p_1 < x + \epsilon$, and, given p_j, gives $p_{j+1} \in \mathbb{Q}$ with $x < p_{j+1} < p_j$, so the p_j are pairwise distinct. As $n \mapsto q_n$ is injective, the set $\{n \big| |q_n - x| < \epsilon\}$ is infinite, and proposition 2.2.8 puts x in E. Moreover every element of $\mathbb{N}$ is a value of the sequence and $\mathbb{N}$ is not bounded above by theorem 1.4.1, so (q_n) is not bounded above, and the same argument with the negatives of the elements of $\mathbb{N}$ shows it is not bounded below. So $\pm\infty \in E$ as well, $E = [-\infty, +\infty]$, and

$$\limsup_{n\to\infty} q_n = +\infty, \qquad \liminf_{n\to\infty} q_n = -\infty$$

As defined, the upper limit is the supremum of a set that has to be determined in full before the supremum can be taken, which is what the example did three times over. The next theorem replaces the definition by two properties that refer only to the terms a_n: the supremum is attained, and every real number above it dominates the sequence eventually. Those two properties determine the upper limit, so they may be used as its definition wherever that is more convenient, and it is in this form that section 2.4 and section 2.6 use it.

Theorem 2.2.12: The Upper Limit Is Attained and Dominates Eventually

Let $(a_n)_{n\ge1}$ be a real sequence, let E be its set of subsequential limits, and put $s^* = \limsup_{n\to\infty} a_n$ and $s_* = \liminf_{n\to\infty} a_n$.

1. If $x \in \mathbb{R}$ and $x > s^*$, then there is an N with $a_n < x$ for every $n \ge N$.
2. $s^* \in E$ and $s_* \in E$.
3. If $x \in \mathbb{R}$ and $x < s^*$, then $a_n > x$ for infinitely many n.
4. s^* is the only $t \in [-\infty, +\infty]$ with both of the following properties: $t \in E$, and for every real $x > t$ there is an N with $a_n < x$ for all $n \ge N$.
5. For $s \in [-\infty, +\infty]$, the sequence (a_n) has limit s if and only if $s_* = s^* = s$.

***Proof*:**

1. Suppose, for contradiction, that no such N exists. Then for every N there is an $n \ge N$ with $a_n \ge x$, so $S = \{n \big| a_n \ge x\}$ is infinite. Let $n_1 < n_2 < \cdots$ enumerate S in increasing order, so that (a_{n_k}) is a subsequence with $a_{n_k} \ge x$ for every k. If (a_{n_k}) is bounded above, then it is bounded below by x as well, so theorem 2.2.9 gives a subsequence of it converging to some real y, and that subsequence is a subsequence of (a_n) by proposition 2.2.7, so $y \in E$ and $y \ge x$ by proposition 2.1.8 applied to it and the constant sequence with value x. If (a_{n_k}) is not bounded above, then proposition 2.2.8 gives a subsequence of it tending to $+\infty$, which is again a subsequence of (a_n) by proposition 2.2.7, so $y = +\infty$ lies in E and

$y \geq x$. Either way E contains an element $y \geq x > s^*$, contradicting $s^* = \sup E$.

2. Take s^* first, in three cases.

   Suppose $s^* = +\infty$. If (a_n) were bounded above by some real M, then $+\infty \notin E$ by proposition 2.2.8, while a real $y \in E$ would satisfy $y \leq M$ by proposition 2.1.8 applied to a subsequence converging to y and the constant sequence with value M, and $-\infty \leq M$ holds outright. Then M would be an upper bound of E and $s^* \leq M$, which is false. So (a_n) is not bounded above, and $+\infty \in E$ by proposition 2.2.8.

   Suppose $s^* = -\infty$. Every element of E is at most $\sup E = -\infty$, so $E \subseteq \{-\infty\}$, and $E \neq \emptyset$ by theorem 2.2.9. Hence $E = \{-\infty\}$ and $s^* \in E$.

   Suppose s^* is real, and let $\epsilon > 0$. Since $s^* - \epsilon/2 < s^*$, the number $s^* - \epsilon/2$ is not an upper bound of E, so there is $y \in E$ with $y > s^* - \epsilon/2$. That inequality rules out $y = -\infty$, and $y \leq \sup E = s^*$ rules out $y = +\infty$, so y is real. By proposition 2.2.8 there are infinitely many n with $|a_n - y| < \epsilon/2$, and each such n satisfies

$$a_n > y - \frac{\epsilon}{2} > s^* - \epsilon$$

   On the other hand part (1), applied to the real number $s^* + \epsilon > s^*$, gives an N with $a_n < s^* + \epsilon$ for all $n \geq N$. Removing the finitely many indices below N from an infinite set leaves an infinite set, so there are infinitely many n with $s^* - \epsilon < a_n < s^* + \epsilon$, that is, with $|a_n - s^*| < \epsilon$. As $\epsilon > 0$ was arbitrary, proposition 2.2.8 gives $s^* \in E$.

   For s_*, apply what has just been proved to $(-a_n)$: its upper limit is $-s_*$ by the identities following definition 2.2.10, so $-s_* \in E(-a) = -E(a)$, and hence $s_* \in E$.

3. Let $x \in \mathbb{R}$ with $x < s^*$. By part (2), $s^* \in E$. If s^* is real, apply proposition 2.2.8 with $\epsilon = s^* - x > 0$: infinitely many n satisfy $|a_n - s^*| < \epsilon$, and each of them satisfies $a_n > s^* - \epsilon = x$. If $s^* = +\infty$, then proposition 2.2.8 gives directly that $\{n | a_n > x\}$ is infinite. The case $s^* = -\infty$ does not arise, since no real x is smaller than $-\infty$.

4. That s^* has both properties is parts (2) and (1). Suppose t has both properties. Since $t \in E$, $t \leq \sup E = s^*$. If $t < s^*$, then proposition 2.2.5 gives a real x with $t < x < s^*$. The second property of t gives an N with $a_n < x$ for all $n \geq N$, while part (3) gives infinitely many n with $a_n > x$, and infinitely many of those n are at least N, which is a contradiction. Hence $t = s^*$.

5. Suppose (a_n) has limit s. Every subsequence has limit s by proposition 2.2.7, so every element of E equals s, and $s \in E$ because (a_n) is a subsequence of itself. Thus $E = \{s\}$ and $s_* = s^* = s$.

   Conversely suppose $s_* = s^* = s$. Consider first s real, and let $\epsilon > 0$. Part (1) applied to $s + \epsilon > s^*$ gives an N_1 with $a_n < s + \epsilon$ for $n \geq N_1$. Applying part (1) to the sequence $(-a_n)$, whose upper limit is $-s_* = -s$, and to the real number $-s + \epsilon > -s$, gives an N_2 with $-a_n < -s + \epsilon$, that is $a_n > s - \epsilon$, for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$ both hold, so $|a_n - s| < \epsilon$ and $a_n \to s$.

   Consider next $s = +\infty$. Every element of E is at least $\inf E = s_* = +\infty$, and $E \neq \emptyset$ by theorem 2.2.9, so $E = \{+\infty\}$. Suppose $a_n \to +\infty$ failed. Then there is an $M \in \mathbb{R}$

such that for every N some $n \geq N$ has $a_n \leq M$, so $\{n | a_n \leq M\}$ is infinite and the corresponding subsequence (a_{n_k}) satisfies $a_{n_k} \leq M$. If that subsequence is bounded below, theorem 2.2.9 gives a further subsequence converging to a real number, which lies in E by proposition 2.2.7, contradicting $E = \{+\infty\}$. If it is not bounded below, proposition 2.2.8 gives a further subsequence tending to $-\infty$, so $-\infty \in E$, again a contradiction. So $a_n \to +\infty$.

Finally, if $s = -\infty$ then the sequence $(-a_n)$ has $\limsup(-a_n) = -s_* = +\infty$ and $\liminf(-a_n) = -s^* = +\infty$, so $-a_n \to +\infty$ by the case just settled, and therefore $a_n \to -\infty$ by definition 2.2.4, completing the proof.

Parts (1) and (3) together say that s^* separates the real line into the numbers the sequence eventually stays below and the numbers it exceeds infinitely often, and part (4) says that this separation determines s^*. Part (5) turns a question about the existence of a limit in $[-\infty, +\infty]$, which quantifies over a bound and an index, into an equation between two numbers that exist for every sequence. The corresponding statements for s_* hold with the inequalities reversed, by applying the theorem to $(-a_n)$ and using the identities that follow definition 2.2.10.

The last two properties compare upper limits of two sequences. The first is an inequality that passes to the limit in both directions, while the second holds in one direction only.

Proposition 2.2.13: Comparison and Subadditivity for Upper Limits

Let $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ be real sequences.

1. If there is an N with $a_n \leq b_n$ for all $n \geq N$, then

$$\limsup_{n\to\infty} a_n \leq \limsup_{n\to\infty} b_n \qquad \text{and} \qquad \liminf_{n\to\infty} a_n \leq \liminf_{n\to\infty} b_n$$

2. If the sum $\limsup_n a_n + \limsup_n b_n$ is defined in $[-\infty, +\infty]$, then

$$\limsup_{n\to\infty}(a_n + b_n) \leq \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n$$

***Proof*:**
Write $s^* = \limsup_n a_n$ and $t^* = \limsup_n b_n$.

1. Suppose $s^* > t^*$. By proposition 2.2.5 there is a real x with $t^* < x < s^*$. Since $x > t^*$, theorem 2.2.12 gives an N_1 with $b_n < x$ for all $n \geq N_1$. Since $x < s^*$, theorem 2.2.12 gives infinitely many n with $a_n > x$, and infinitely many of those satisfy $n \geq \max\{N, N_1\}$. For such an n, $a_n > x > b_n$ while $a_n \leq b_n$, a contradiction. Hence $s^* \leq t^*$.

   For the lower limits, the hypothesis gives $-b_n \leq -a_n$ for $n \geq N$, so the case just proved applied to $(-b_n)$ and $(-a_n)$ gives $\limsup_n(-b_n) \leq \limsup_n(-a_n)$. By the identities following definition 2.2.10 this reads $-\liminf_n b_n \leq -\liminf_n a_n$, and negating reverses the inequality by proposition 1.1.2, giving $\liminf_n a_n \leq \liminf_n b_n$.

2. If $s^* + t^* = +\infty$ there is nothing to prove, since every element of $[-\infty, +\infty]$ is at most $+\infty$.

Suppose $s^* + t^* = -\infty$. Then at least one summand is $-\infty$ and, the sum being defined, neither is $+\infty$. The statement is symmetric in (a_n) and (b_n), so assume $s^* = -\infty$ and $t^* < +\infty$. Every element of the set of subsequential limits of (a_n) is at most $s^* = -\infty$, and that set is nonempty by theorem 2.2.9, so $\liminf_n a_n = -\infty$ as well, and $a_n \to -\infty$ by theorem 2.2.12. Since $t^* < +\infty$, proposition 2.2.5 gives a real x with $t^* < x < +\infty$, and theorem 2.2.12 gives an N_1 with $b_n < x$ for $n \geq N_1$. Let $M \in \mathbb{R}$. Since $a_n \to -\infty$ there is an N_2 with $a_n < M - x$ for $n \geq N_2$, and then $a_n + b_n < (M-x) + x = M$ for $n \geq \max\{N_1, N_2\}$. So $a_n + b_n \to -\infty$, and theorem 2.2.12 gives $\limsup_n (a_n + b_n) = -\infty = s^* + t^*$.

Suppose finally that $s^* + t^*$ is real, which forces both s^* and t^* to be real. Let $\epsilon > 0$. By theorem 2.2.12 applied to $s^* + \epsilon/2 > s^*$ there is an N_1 with $a_n < s^* + \epsilon/2$ for $n \geq N_1$, and applied to $t^* + \epsilon/2 > t^*$ there is an N_2 with $b_n < t^* + \epsilon/2$ for $n \geq N_2$. Put $u = s^* + t^* + \epsilon$, a real number. For $n \geq \max\{N_1, N_2\}$,

$$a_n + b_n < \left(s^* + \frac{\epsilon}{2}\right) + \left(t^* + \frac{\epsilon}{2}\right) = u$$

If $\limsup_n (a_n + b_n) > u$ held, then theorem 2.2.12 would give infinitely many n with $a_n + b_n > u$, and infinitely many of those would satisfy $n \geq \max\{N_1, N_2\}$, contradicting $a_n + b_n < u$. So $\limsup_n (a_n + b_n) \leq u$ for every $\epsilon > 0$. In particular $\limsup_n (a_n + b_n) \neq +\infty$. If it equals $-\infty$ the claimed inequality holds. Otherwise it is real, and $\limsup_n (a_n + b_n) \leq s^* + t^* + \epsilon$ for every $\epsilon > 0$ gives $\limsup_n (a_n + b_n) \leq s^* + t^*$ by proposition 1.4.7, completing the proof.

The inequality in part (2) can be strict, and the reason is that the two sequences may reach their upper limits along different sets of indices. Take $a_n = (-1)^n(1 + 1/n)$ and $b_n = -a_n$. By example 2.2.11, $\limsup_n a_n = 1$, and by the identities following definition 2.2.10, $\limsup_n b_n = -\liminf_n a_n = 1$, so the right-hand side of the inequality is 2. But $a_n + b_n = 0$ for every n, so every subsequence of the sum is constant with value 0, its set of subsequential limits is $\{0\}$, and $\limsup_n (a_n + b_n) = 0 < 2$. The same pair refutes the reverse inequality, which would read $0 \geq 2$.

Exercise 2.2.1

1. Let (a_n) be increasing and bounded above, with limit s. Show that $s = \sup\{a_n \mid n \geq 1\}$, and that (a_n) is eventually constant if and only if $s = a_N$ for some N.
2. Call an index n a *peak* of a real sequence (a_n) if $a_m \leq a_n$ for every $m > n$. Show that every real sequence has a monotone subsequence, by treating separately the case of infinitely many peaks and the case of finitely many. Deduce part (1) of theorem 2.2.9 from theorem 2.2.2.
3. For a real sequence (a_n) and $N \geq 1$ put $t_N = \sup\{a_n \mid n \geq N\}$, the supremum being taken in $[-\infty, +\infty]$. Show that
$$\limsup_{n\to\infty} a_n = \inf\{t_N \mid N \geq 1\}$$
and state and prove the corresponding formula for the lower limit. (Suggestion: verify that the right-hand side has the two properties in part (4) of theorem 2.2.12.)
4. Determine the set of subsequential limits, the upper limit and the lower limit of each of the sequences $a_n = (-1)^n + 1/n$ and $b_n = n\big(1 + (-1)^n\big)$, with proof.

5. Let (a_n) be a bounded real sequence with the property that every convergent subsequence of (a_n) has the same limit L. Show that $a_n \to L$. Show also that the boundedness hypothesis cannot be dropped, by exhibiting an unbounded sequence with no convergent subsequence at all.

6. Show that $\liminf_n (a_n + b_n) \geq \liminf_n a_n + \liminf_n b_n$ whenever the right-hand side is defined in $[-\infty, +\infty]$, and exhibit bounded sequences for which this inequality and the one in part (2) of proposition 2.2.13 are both strict.

7. Let (a_n) be a real sequence with $a_n > 0$ for every n. Show that $\liminf_n a_n \geq 0$, and that $\limsup_n a_n = 0$ forces $a_n \to 0$.

8. Show that an increasing real sequence that is not bounded above has limit $+\infty$, and conclude that every monotone real sequence has a limit in $[-\infty, +\infty]$. Show also that this fails without monotonicity, by exhibiting a sequence with no limit in $[-\infty, +\infty]$.

2.3 The Cauchy Criterion

The definition of convergence in definition 2.1.2 mentions the limit, so to apply it one must already know the number the sequence approaches. A sequence produced by a recursion, or by adding one more term at each stage, arrives without a candidate limit, and for such a sequence the definition cannot be used as a test. This section replaces it by a condition on the terms alone: past some index, any two terms differ by less than ϵ. The condition can be checked from the sequence itself, and in $\mathbb{R}$ it is equivalent to convergence.

In $\mathbb{Q}$ it is not. The decimal truncations of $\sqrt{2}$ come within any prescribed tolerance of one another and converge to no rational number, which is the failure section 1.1 was built to repair. So the equivalence proved below is a property of $\mathbb{R}$, carrying content beyond the definition, and it is the form in which the completeness of $\mathbb{R}$ is used for the rest of the book.

Definition 2.3.1: Cauchy Sequence in $\mathbb{R}$

A sequence $(a_n)_{n\geq 1}$ of real numbers is a *Cauchy sequence* if for every $\epsilon > 0$ there is an $N \in \mathbb{N}$ such that

$$|a_n - a_m| < \epsilon \qquad \text{for all } n, m \geq N$$

Definition 1.1.4 imposed the same condition on a sequence of rationals, with ϵ ranging over the positive rationals, because at that point no positive real was available. For a sequence of rationals the two conditions agree. If the real condition holds then so does the rational one, since every positive rational is a positive real. Conversely, suppose the rational condition holds and let $\epsilon > 0$ be real. By theorem 1.4.4 applied to $0 < \epsilon$ there is a rational p with $0 < p < \epsilon$, and the index N that works for p works for ϵ. Nothing below distinguishes the two, and the phrase *Cauchy sequence* refers to the real condition from here on.

Example 2.3.2: Decimal Truncations of $\sqrt{2}$, $((-1)^n)$ and $(\sqrt{n})$

Throughout, $\sqrt{x}$ for $x \geq 0$ is the number given by theorem 1.3.12. That theorem also gives the monotonicity used repeatedly below: if $0 \leq u < v$ then $\sqrt{u} < \sqrt{v}$, since $\sqrt{u} \geq \sqrt{v} \geq 0$ would force $u = (\sqrt{u})^2 \geq (\sqrt{v})^2 = v$.

1. Let $\alpha = \sqrt{2}$ and, for $n \geq 1$, let
$$a_n = \frac{\lfloor 10^n \alpha \rfloor}{10^n}$$
By proposition 1.4.3, $\lfloor 10^n \alpha \rfloor$ is the unique integer m with $m \leq 10^n \alpha < m+1$, so each a_n is rational and, dividing by $10^n > 0$,
$$0 \leq \alpha - a_n < 10^{-n}$$
For $m \geq n$ the triangle inequality proposition 1.3.8(4) gives
$$|a_m - a_n| \leq |a_m - \alpha| + |\alpha - a_n| < 10^{-m} + 10^{-n} \leq 2 \cdot 10^{-n}$$
By proposition 2.1.10 in the case $j = 0$ and $r = 1/10$, the sequence 10^{-n} is null, so given $\epsilon > 0$ there is an N with $10^{-N} < \epsilon/2$, and then $|a_m - a_n| < \epsilon$ for all $m, n \geq N$. So (a_n) is Cauchy. The same estimate with ϵ rational shows it is Cauchy in the sense of definition 1.1.4 as well.

   The bound $0 \leq \alpha - a_n < 10^{-n}$ also gives $a_n \to \alpha$. If some rational q satisfied $a_n \to q$, then $q = \alpha$ by proposition 2.1.4, so $q^2 = 2$ with $q \in \mathbb{Q}$, which contradicts proposition 1.1.3. This Cauchy sequence of rationals therefore has no rational limit. Note also that $0 \leq a_n \leq \alpha < 2$, the last inequality because $\alpha^2 = 2 < 4 = 2^2$, so the sequence is bounded.

2. Let $a_n = (-1)^n$, so that $|a_n| = 1$ for every n. Take $\epsilon = 1$. Given any N, the indices $n = N$ and $m = N+1$ satisfy $|a_N - a_{N+1}| = 2$, which is not less than 1. So no N works for $\epsilon = 1$, and (a_n) is a bounded sequence that is not Cauchy.

3. Let $a_n = \sqrt{n}$. Consecutive terms come together: from $(\sqrt{n+1} - \sqrt{n})(\sqrt{n+1} + \sqrt{n}) = (n+1) - n = 1$ and $\sqrt{n+1} + \sqrt{n} \geq 2\sqrt{n} > 0$,
$$0 < \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \leq \frac{1}{2\sqrt{n}}$$
Given $\epsilon > 0$, theorem 1.4.1(2) provides an integer $n_0 > 1/(4\epsilon^2)$, and for $n \geq n_0$ the monotonicity of the square root gives $\sqrt{n} > 1/(2\epsilon)$, hence $|a_{n+1} - a_n| < \epsilon$. The sequence is not Cauchy all the same. Take $\epsilon = 1$ and let N be given. By theorem 1.4.1(2) there is an integer $m > (\sqrt{N} + 1)^2$, and then $\sqrt{m} > \sqrt{N} + 1$, so $|a_m - a_N| > 1$.

Item (3) is why the definition quantifies over all pairs of large indices. Consecutive pairs give a weaker requirement, which $\sqrt{n}$ satisfies while being unbounded. Item (2) shows that boundedness is not sufficient either. Item (1) is the case the rest of the section is about: a sequence of rationals whose limit exists in $\mathbb{R}$ and not in $\mathbb{Q}$.

One of the two implications between convergence and the Cauchy condition follows from the triangle inequality alone.

Proposition 2.3.3: Convergent Sequences Are Cauchy

Every convergent sequence of real numbers is a Cauchy sequence.

***Proof*:**
Let $a_n \to a$ and let $\epsilon > 0$. Applying definition 2.1.2 with $\epsilon/2 > 0$ gives an $N \in \mathbb{N}$ with $|a_n - a| < \epsilon/2$ for all $n \geq N$. For $n, m \geq N$,

$$|a_n - a_m| = |(a_n - a) + (a - a_m)| \leq |a_n - a| + |a_m - a| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

using proposition 1.3.8(4) for the inequality and proposition 1.3.8(1) for $|a - a_m| = |a_m - a|$. The index N therefore satisfies the requirement of definition 2.3.1 for ϵ, as we sought to show.

The converse is the substantial direction, and it needs one preparation: a Cauchy sequence is bounded, which is what makes Bolzano-Weierstrass applicable to it. Lemma 1.1.6 is the same statement for sequences of rationals, proved there because the construction of $\mathbb{R}$ needed it before real ϵ existed.

Proposition 2.3.4: Cauchy Sequences in $\mathbb{R}$ Are Bounded

Every Cauchy sequence $(a_n)_{n\geq 1}$ of real numbers is bounded: there is an $M \geq 0$ with $|a_n| \leq M$ for all n.

***Proof*:**
Apply definition 2.3.1 with $\epsilon = 1$ to get an $N \in \mathbb{N}$ with $|a_n - a_m| < 1$ for all $n, m \geq N$. In particular, for $n \geq N$,

$$|a_n| = |(a_n - a_N) + a_N| \leq |a_n - a_N| + |a_N| < 1 + |a_N|$$

by proposition 1.3.8(4). Let M be the largest of the finitely many numbers

$$|a_1|,\ |a_2|,\ \ldots,\ |a_{N-1}|,\ 1 + |a_N|$$

Then $|a_n| \leq M$ for $n < N$ because $|a_n|$ is one of the listed numbers, and $|a_n| < 1 + |a_N| \leq M$ for $n \geq N$ by the displayed estimate. Since $M \geq |a_1| \geq 0$, this is the required bound, completing the proof.

The converse of proposition 2.3.4 fails, by example 2.3.2(2). What boundedness gives is a convergent subsequence, and the Cauchy condition is the extra information that forces the whole sequence to converge to the limit of that subsequence.

Theorem 2.3.5: The Cauchy Criterion

A sequence of real numbers converges if and only if it is a Cauchy sequence.

***Proof*:**
One direction is proposition 2.3.3. For the other, let $(a_n)_{n\geq 1}$ be a Cauchy sequence. By proposition 2.3.4 it is bounded, which is the hypothesis of theorem 2.2.9, so it has a convergent subsequence $(a_{n_k})_{k\geq 1}$, say $a_{n_k} \to a$. The indices satisfy $n_k \geq k$ for every k, by induction on k: the base case is $n_1 \geq 1$, and $n_{k+1} > n_k$ with both integers gives $n_{k+1} \geq n_k + 1 \geq k + 1$.

Let $\epsilon > 0$. The Cauchy condition gives an $N \in \mathbb{N}$ with $|a_n - a_m| < \epsilon/2$ for all $n, m \geq N$, and convergence of the subsequence gives a $K \in \mathbb{N}$ with $|a_{n_k} - a| < \epsilon/2$ for all $k \geq K$. Fix one index

$k = \max\{K, N\}$. Then $k \geq K$, and $n_k \geq k \geq N$, so for every $n \geq N$,

$$|a_n - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

using proposition 1.3.8(4). Hence $a_n \to a$, and the sequence converges, as we sought to show.

Theorem 2.3.5 makes convergence provable without producing the limit, and that is what later sections use it for: a sequence given by a recursion, or by adding one more term at each stage, can be tested on its terms alone. It is not the only result that gives a limit without naming it. Theorem 2.2.2 does the same for a bounded monotone sequence, and for the increasing partial sums of a series of nonnegative terms that is the route section 2.4 takes.

The condition in definition 2.3.1 refers to $|a_n - a_m|$, the distance between two terms, and to no other feature of $\mathbb{R}$. Replacing that distance by an abstract one gives a Cauchy condition on any set equipped with a notion of distance, and such a set is called *Cauchy-complete* when every Cauchy sequence in it converges. The qualifier matters, because section 1.5⋆ has already fixed *complete* for an ordered field to mean the least upper bound property, and the two senses are not equivalent in general: it is exactly the Archimedean hypothesis that closes the gap, which is why it appears in theorem 2.3.6. Theorem 2.3.5 says that $\mathbb{R}$ is *Cauchy*-complete. That it is complete in the order sense is theorem 1.2.4. The general theory belongs to the companion volume *Metric and Function Spaces*, and no argument in this book uses it.

Five statements have now been proved about $\mathbb{R}$, each of them a way of saying that $\mathbb{R}$ has no gaps: the least upper bound property theorem 1.2.4, monotone convergence theorem 2.2.2, the nested interval theorem theorem 1.4.8, Bolzano-Weierstrass theorem 2.2.9, and the Cauchy criterion. Inside $\mathbb{R}$ each of them is a theorem, so an implication among them is true for the trivial reason that its conclusion has already been proved. The content appears one level up, where the five are treated as conditions an Archimedean ordered field may or may not satisfy: for such a field each one implies the other four, and which to install as the axiom is a matter of preference. The theorem below is therefore stated for an arbitrary Archimedean ordered field, in the sense of definition 1.1.1, and that hypothesis is not removable.

Fix an Archimedean ordered field F. Absolute value in F is given by the formula in definition 1.1.1, and the definitions of convergence, of the Cauchy condition, of boundedness and of monotonicity read verbatim in F, with ϵ ranging over the positive elements of F and with a positive integer n read as $n \cdot 1 \in F$. For $a \leq b$ in F write $[a, b] = \{x \in F \big| a \leq x \leq b\}$.

Theorem 2.3.6: Equivalent Forms of Completeness

Let F be an Archimedean ordered field. The following five statements are equivalent.

1. Every nonempty subset of F that is bounded above has a supremum in F.
2. Every bounded monotone sequence in F converges.
3. If $[a_n, b_n]$ for $n \geq 1$ are intervals with $a_n \leq b_n$ and $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$ for every n, then $\bigcap_{n\geq 1}[a_n, b_n]$ is nonempty.
4. Every bounded sequence in F has a convergent subsequence.
5. Every Cauchy sequence in F converges.

***Proof*:**
The cycle (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (5) $\Rightarrow$ (1) is proved. Two observations are used more than once.

Observation A (halving). For every $\ell > 0$ and every $\epsilon > 0$ in F there is a positive integer k with $\ell \cdot (2^k)^{-1} < \epsilon$, where $2 = 1 + 1$. Indeed $2^k \geq k+1$ for every $k \geq 0$, by induction: the case $k = 0$ reads $1 \geq 1$, and $2^{k+1} = 2^k + 2^k \geq (k+1) + 1$ because $2^k \geq 1$. Next, $\epsilon^{-1} > 0$, since $\epsilon^{-1} < 0$ would give $1 = \epsilon \cdot \epsilon^{-1} < 0$ by proposition 1.1.2(5), contradicting $1 > 0$ from proposition 1.1.2(6). So $\ell\epsilon^{-1} > 0$, and the Archimedean property gives a positive integer k with $\ell\epsilon^{-1} < k$. Then $2^k \geq k + 1 > \ell\epsilon^{-1} > 0$, so $(2^k)^{-1} < (\ell\epsilon^{-1})^{-1} = \epsilon\ell^{-1}$ by proposition 1.1.2(7), and multiplying by $\ell > 0$ (proposition 1.1.2(5)) gives $\ell(2^k)^{-1} < \epsilon$.

Observation B (width). If $u, v \in [a, b]$ then $|u - v| \leq b - a$. Indeed $a \leq u \leq b$ and $a \leq v \leq b$ give both $u - v \leq b - a$ and $v - u \leq b - a$, and $|u - v|$ is one of these two numbers.

Step 1: (1) implies (2). Let (x_n) be increasing and bounded. The set $E = \{x_n \big| n \geq 1\}$ is nonempty and bounded above, so by (1) it has a supremum s. Let $\epsilon > 0$. Since $s - \epsilon < s$ and s is the least upper bound, $s - \epsilon$ is not an upper bound of E, so there is an N with $x_N > s - \epsilon$. For $n \geq N$ monotonicity gives $s - \epsilon < x_N \leq x_n \leq s$, hence $|x_n - s| = s - x_n < \epsilon$. So $x_n \to s$. If instead (x_n) is decreasing and bounded, then $(-x_n)$ is increasing and bounded, so $-x_n \to t$ for some $t \in F$. The elements $x_n - (-t) = x_n + t$ and $(-x_n) - t = -(x_n + t)$ have the same absolute value, so $|x_n - (-t)| < \epsilon$ whenever $|(-x_n) - t| < \epsilon$, and (x_n) converges to $-t$.

Step 2: (2) implies (3). Let the intervals $[a_n, b_n]$ be nested. Since a_{n+1} and b_{n+1} both lie in $[a_n, b_n]$, the sequence (a_n) is increasing and (b_n) is decreasing. Moreover $a_n \leq b_m$ for all n and m: for $n \geq m$ nesting gives $a_n \leq b_n \leq b_m$, and for $n < m$ it gives $a_n \leq a_m \leq b_m$. In particular (a_n) is bounded above by b_1 and below by a_1, so by (2) it converges, say $a_n \to a$. Fix m. If $a < a_m$, then putting $\epsilon = a_m - a > 0$, every $n \geq m$ has $a_n \geq a_m > a$ and hence $|a_n - a| = a_n - a \geq \epsilon$, contradicting convergence. If $a > b_m$, then putting $\epsilon = a - b_m > 0$, some n has $|a_n - a| < \epsilon$ and hence $a_n > a - \epsilon = b_m$, contradicting $a_n \leq b_m$. So $a_m \leq a \leq b_m$ for every m, and a lies in the intersection.

Step 3: (3) implies (4). Let (x_n) be bounded, say $|x_n| \leq M$ for all n, where $M > 0$ after replacing M by $M + 1$ if necessary. Then $x_n \in I_0 = [-M, M]$ for all n, by the definition of absolute value in definition 1.1.1. Intervals $I_k = [a_k, b_k]$ are built so that $a_k < b_k$, so that $I_{k+1} \subseteq I_k$, and so that $x_n \in I_k$ for infinitely many indices n. The set $\{n \big| x_n \in I_0\}$ is all of $\mathbb{N}$, so I_0 starts the construction. Given I_k, put $c = (a_k + b_k)(1 + 1)^{-1}$, so that $a_k < c < b_k$ by proposition 1.1.2(8). If $\{n \big| x_n \in [a_k, c]\}$ is infinite, set $I_{k+1} = [a_k, c]$, and otherwise set $I_{k+1} = [c, b_k]$. In the second case $\{n \big| x_n \in [c, b_k]\}$ is infinite, because $\{n \big| x_n \in I_k\}$ is the union of the two index sets and is infinite, while a union of two finite sets is finite. By construction $b_{k+1} - a_{k+1} = (b_k - a_k)(1 + 1)^{-1}$, so $b_k - a_k = (b_0 - a_0)(2^k)^{-1}$ for every k.

By (3) there is an $x \in \bigcap_{k \geq 0} I_k$. Define indices recursively: let n_1 be the least n with $x_n \in I_1$, and let n_{k+1} be the least $n > n_k$ with $x_n \in I_{k+1}$. Each exists because each index set is infinite, so the relevant set of positive integers is nonempty and has a least element. Both x_{n_k} and x lie in I_k, so Observation B gives $|x_{n_k} - x| \leq b_k - a_k = (b_0 - a_0)(2^k)^{-1}$. Let $\epsilon > 0$. Observation A gives a K with $(b_0 - a_0)(2^K)^{-1} < \epsilon$. For $k \geq K$ one has $2^k \geq 2^K > 0$, since each step from 2^j to $2^{j+1} = 2^j + 2^j$ adds a positive element, so $(2^k)^{-1} \leq (2^K)^{-1}$ by proposition 1.1.2(7) and therefore $|x_{n_k} - x| < \epsilon$. Hence $x_{n_k} \to x$.

Step 4: (4) implies (5). Let (x_n) be Cauchy. Taking $\epsilon = 1$ gives an N with $|x_n - x_N| < 1$ for $n \geq N$, so $|x_n| \leq |x_n - x_N| + |x_N| < 1 + |x_N|$ for those n by proposition 1.1.2(9), and (x_n) is bounded by the largest of $|x_1|, \ldots, |x_{N-1}|, 1 + |x_N|$. By (4) some subsequence converges, say $x_{n_k} \to x$, and $n_k \geq k$ for every k by induction on k, from $n_1 \geq 1$ and $n_{k+1} \geq n_k + 1 \geq k+1$. Let $\epsilon > 0$, choose N_1 with $|x_n - x_m| < \epsilon(1+1)^{-1}$ for $n, m \geq N_1$, choose K with $|x_{n_k} - x| < \epsilon(1+1)^{-1}$ for $k \geq K$, and fix $k = \max\{K, N_1\}$, so that $n_k \geq k \geq N_1$. For $n \geq N_1$,

$$|x_n - x| \leq |x_n - x_{n_k}| + |x_{n_k} - x| < \epsilon$$

by proposition 1.1.2(9). Hence $x_n \to x$.

Step 5: (5) implies (1). Let $E \subseteq F$ be nonempty and bounded above, say by b_0. Pick $x_0 \in E$ and put $a_0 = x_0 - 1$, so that $a_0 < x_0$ and a_0 is not an upper bound of E, while $a_0 < x_0 \leq b_0$. Build a_k and b_k recursively so that a_k is never an upper bound of E, b_k always is, and $a_k < b_k$: given the pair (a_k, b_k), put $c = (a_k + b_k)(1+1)^{-1}$, which satisfies $a_k < c < b_k$ by proposition 1.1.2(8), and set $(a_{k+1}, b_{k+1}) = (a_k, c)$ if c is an upper bound of E and $(a_{k+1}, b_{k+1}) = (c, b_k)$ otherwise. Then (a_k) is increasing, (b_k) is decreasing, and $b_k - a_k = (b_0 - a_0)(2^k)^{-1}$.

The sequence (b_k) is Cauchy. For $m \geq k$ one has $a_k \leq a_m < b_m \leq b_k$, so b_k and b_m both lie in $[a_k, b_k]$ and Observation B gives $|b_k - b_m| \leq b_k - a_k = (b_0 - a_0)(2^k)^{-1}$. Given $\epsilon > 0$, Observation A gives a K with $(b_0 - a_0)(2^K)^{-1} < \epsilon$, and for $m \geq k \geq K$ the monotonicity of $(2^k)^{-1}$ recorded in Step 3 gives $|b_k - b_m| \leq (b_0 - a_0)(2^k)^{-1} \leq (b_0 - a_0)(2^K)^{-1} < \epsilon$. By (5) the sequence converges, say $b_k \to s$.

The sequence (a_k) converges to s as well. Let $\epsilon > 0$. Observation A and the monotonicity just used give a K_1 with $b_k - a_k < \epsilon(1+1)^{-1}$ for all $k \geq K_1$, and convergence gives a K_2 with $|b_k - s| < \epsilon(1+1)^{-1}$ for all $k \geq K_2$. For $k \geq \max\{K_1, K_2\}$,

$$|a_k - s| \leq |a_k - b_k| + |b_k - s| < \epsilon$$

by proposition 1.1.2(9).

It remains to identify s as $\sup E$. Suppose some $x \in E$ had $x > s$. Put $\epsilon = x - s > 0$ and choose k with $|b_k - s| < \epsilon$. Then $b_k < s + \epsilon = x$, contradicting that b_k is an upper bound of E, so s is an upper bound. Now suppose some upper bound u had $u < s$. Put $\epsilon = s - u > 0$ and choose k with $|a_k - s| < \epsilon$. Then $a_k > s - \epsilon = u$, and since a_k is not an upper bound of E there is an $x \in E$ with $x > a_k > u$, contradicting that u is an upper bound. So s is the least upper bound of E, which is (1) and closes the cycle, completing the proof.

The Archimedean hypothesis enters the proof at exactly two places, in Step 3 and in Step 5, and at both through Observation A, which is what makes an interval of a fixed width halve below a prescribed ϵ after finitely many steps. It is a hypothesis and not a convenience of this particular argument. An ordered field satisfying (1) is Archimedean by corollary 1.5.1, so a version of the theorem for ordered fields at large would have to produce the Archimedean property out of (5) alone. That cannot be done, because there are non-Archimedean ordered fields in which every Cauchy sequence converges, the formal Laurent series in a positive infinitesimal being one such field. Statement (5) therefore does not imply statement (1) once the hypothesis is dropped.

A single failure in $\mathbb{Q}$ now accounts for five. The field $\mathbb{Q}$ is an ordered field and is Archimedean, these being among the hypotheses on $\mathbb{Q}$ recorded in section 1.1, so theorem 2.3.6 applies to it. By example 2.3.2(1) there is a Cauchy sequence of rationals with no rational limit, so $\mathbb{Q}$ fails (5), and therefore fails the other four as well. It has bounded sets of rationals with no rational supremum, which is what example 1.2.3(3) exhibited by hand, and it also has bounded monotone sequences that do not converge, nested rational intervals with empty intersection, and bounded sequences with no subsequence converging in $\mathbb{Q}$. These are four descriptions of the one defect, and theorem 2.3.6 is what identifies them.

Exercise 2.3.1

1. Let (a_n) and (b_n) be Cauchy sequences of real numbers. Prove directly from definition 2.3.1, without using theorem 2.3.5 or theorem 2.1.6, that $(a_n + b_n)$ and $(a_n b_n)$ are Cauchy sequences.
2. Let (a_n) be a Cauchy sequence of real numbers. Prove that $(|a_n|)$ is a Cauchy sequence, and show by an example that the converse fails: there is a sequence (a_n) such that $(|a_n|)$ is Cauchy and (a_n) is not.
3. Let $0 < r < 1$ be fixed and let $(a_n)_{n\geq 1}$ satisfy

$$|a_{n+1} - a_n| \leq r\,|a_n - a_{n-1}| \qquad \text{for all } n \geq 2$$

   Prove that (a_n) converges. (First bound $|a_{n+1} - a_n|$ by a multiple of r^{n-1}, then estimate $|a_m - a_n|$ for $m > n$ by summing the intermediate differences.)
4. For each of the following, either produce an example or prove that none exists.
    1. A Cauchy sequence of real numbers with a divergent subsequence.
    2. An unbounded sequence of real numbers with a Cauchy subsequence.
    3. A monotone sequence of real numbers that is not Cauchy.
5. Exhibit a bounded sequence of rational numbers no subsequence of which converges to a rational number, and prove that it has this property. (This is the failure of statement (4) of theorem 2.3.6 for $\mathbb{Q}$, produced directly, without passing through the cycle.)
6. For $n \geq 1$ let $J_n = \{x \in \mathbb{R} \big| 0 < x < 1/n\}$. Show that $J_{n+1} \subseteq J_n$ and $J_n \neq \emptyset$ for every n, and that $\bigcap_{n\geq 1} J_n = \emptyset$. Which hypothesis of theorem 1.4.8 does this family violate?

2.4 Series and the Comparison Tests

A series is rarely decided by computing its partial sums, since a closed form for s_N is the exception. The tests below decide convergence without one, and each of them works by measuring a series against another whose behaviour is already known. So the section first settles three reference series, the geometric series, the harmonic series and $\sum 1/n^p$, and then proves the comparison, ratio and root tests that reduce an arbitrary series to one of them.

The failure in $\mathbb{Q}$ sits at the criterion for nonnegative terms. Increasing bounded partial sums have a limit in $\mathbb{R}$ because of theorem 1.2.4, and over $\mathbb{Q}$ they need not: the partial sums of $\sum_{n\geq 0} 1/n!$ are rational and lie below 3, while proposition 2.4.19 shows that the number they approach is irrational. That example closes the section, alongside the decimal expansion of a real number, which is the geometric series applied to the digits produced by proposition 1.4.3.

Definition 2.4.1: [Real] Series and Partial Sums

Let $(a_n)_{n\geq 1}$ be a sequence of real numbers. For $N \geq 1$ the *N-th partial sum* is

$$s_N = \sum_{n=1}^{N} a_n$$

The *series* $\sum_{n\geq 1} a_n$ *converges* if the sequence $(s_N)_{N\geq 1}$ converges, and in that case the *sum* of the series is

$$\sum_{n=1}^{\infty} a_n := \lim_{N\to\infty} s_N$$

If (s_N) diverges, the series *diverges*. A series $\sum_{n\geq m} a_n$ indexed from an integer m other than 1 is treated identically, with $s_N = \sum_{n=m}^{N} a_n$ for $N \geq m$.
The symbol $\sum_{n\geq 1} a_n$ names the series, and the symbol $\sum_{n=1}^{\infty} a_n$ names its sum, which exists only when the series converges.

The sum of a convergent series is a limit and not an addition, so every statement about limits from sections 2.1 to 2.3 applies to it with no further work. Uniqueness of the sum is proposition 2.1.4 applied to (s_N). Two series arise immediately: a *finite* sum is the series whose terms vanish from some index on, whose partial sums are eventually constant, and the series $\sum_{n\geq 1} 1$, whose partial sums are $s_N = N$ and which therefore diverges by proposition 2.1.5, since $(N)_{N\geq 1}$ is unbounded by theorem 1.4.1(2).

Every result below is stated for a series indexed from $n = 1$ and holds verbatim for a series indexed from any integer m, since the two differ only in the bookkeeping of indices: the partial sums of $\sum_{n\geq m} a_n$ are those of $\sum_{n\geq 1} a_{n+m-1}$.

Convergence of (s_N) was characterized in theorem 2.3.5 without reference to a limit, and translating that characterization into a statement about the terms a_n gives the criterion every test in this section is proved from.

Proposition 2.4.2: The Cauchy Criterion for Series

The series $\sum_{n\geq 1} a_n$ converges if and only if for every $\epsilon > 0$ there is an $N \in \mathbb{N}$ such that

$$\left|\sum_{k=n}^{m} a_k\right| < \epsilon$$

for all integers $m \geq n \geq N$.

***Proof*:**
For $m \geq n \geq 2$ the identity $\sum_{k=n}^{m} a_k = s_m - s_{n-1}$ holds, since s_m and s_{n-1} share the terms $a_1, \ldots, a_{n-1}$, and for $n = 1$ the left side is s_m itself. Both directions are this identity together with theorem 2.3.5.
The condition is necessary. Suppose the series converges, so that (s_N) converges. By proposition 2.3.3, (s_N) is a Cauchy sequence: given $\epsilon > 0$ there is an $N_0 \in \mathbb{N}$ with $|s_p - s_q| < \epsilon$ for all

$p, q \geq N_0$. Put $N = N_0 + 1$. If $m \geq n \geq N$ then $m \geq N_0 + 1 \geq N_0$ and $n - 1 \geq N_0$, so

$$\left| \sum_{k=n}^{m} a_k \right| = |s_m - s_{n-1}| < \epsilon$$

which is the stated condition.
The condition is sufficient. Suppose the condition holds and let $\epsilon > 0$. Take N as in the condition and let $p > q \geq N$. Applying the condition with $n = q + 1$ and $m = p$, which satisfy $m \geq n \geq N + 1 \geq N$, gives

$$|s_p - s_q| = \left| \sum_{k=q+1}^{p} a_k \right| < \epsilon$$

For $p = q$ the difference is $0 < \epsilon$, and the case $q > p$ follows by symmetry. So (s_N) is a Cauchy sequence, and theorem 2.3.5 makes it convergent, completing the proof.

The criterion sees only the terms a_n with $n \geq N$, and N may be taken as large as one likes. Altering finitely many terms therefore changes whether a series converges not at all. Concretely, if $a_n = b_n$ for every $n > M$, then for $m \geq n > M$ the two sums $\sum_{k=n}^{m} a_k$ and $\sum_{k=n}^{m} b_k$ are the same number, so the condition of proposition 2.4.2 holds for one series exactly when it holds for the other. The two sums differ, by the amount $\sum_{n=1}^{M} (a_n - b_n)$.

Taking $m = n$ in the criterion leaves a single term, and the resulting necessary condition is the cheapest divergence test there is.

Proposition 2.4.3: The n-th Term Test

If $\sum_{n\geq 1} a_n$ converges, then $a_n \to 0$. Equivalently, if (a_n) does not converge to 0, then $\sum_{n\geq 1} a_n$ diverges.

***Proof*:**
Let $\epsilon > 0$ and take N as in proposition 2.4.2. For $n \geq N$ the choice $m = n$ is admissible, and the sum $\sum_{k=n}^{n} a_k$ is the single term a_n, so $|a_n| < \epsilon$. As ϵ was arbitrary, $a_n \to 0$ by definition 2.1.2. The second sentence is the contrapositive of the first, completing the proof.

The converse is false, and example 2.4.8 below is the standard example: $1/n \to 0$ while $\sum 1/n$ diverges. The test therefore establishes divergence and never convergence.

For a series of nonnegative terms the partial sums increase, and an increasing sequence has exactly one obstruction to convergence. This is where completeness enters the theory of series, and it is the criterion the condensation and comparison tests are proved from.

Proposition 2.4.4: Nonnegative Terms and Bounded Partial Sums

Let $a_n \geq 0$ for every $n \geq 1$. Then $\sum_{n\geq 1} a_n$ converges if and only if the set $\{s_N \big| N \geq 1\}$ of partial sums is bounded above, and in that case

$$\sum_{n=1}^{\infty} a_n = \sup_{N\geq 1} s_N$$

***Proof*:**
Since $s_{N+1} - s_N = a_{N+1} \geq 0$, the sequence (s_N) is increasing.
If the partial sums are bounded above, theorem 2.2.2 makes (s_N) convergent, so the series converges. Conversely, a convergent sequence is bounded by proposition 2.1.5, so the partial sums of a convergent series are bounded above.
Suppose now that the series converges and write $s = \lim_N s_N$. First, s is an upper bound for the partial sums: if $s_{N_0} > s$ for some N_0, then $s_N \geq s_{N_0}$ for every $N \geq N_0$ because (s_N) is increasing, and proposition 2.1.8 applied to the constant sequence s_{N_0} gives $s \geq s_{N_0} > s$, which is impossible. Second, let $\epsilon > 0$. Since $s_N \to s$ there is an N with $|s_N - s| < \epsilon$, hence $s_N > s - \epsilon$. So s is an upper bound that is approached to within every $\epsilon > 0$ from inside the set of partial sums, and proposition 1.2.6 identifies $s = \sup_N s_N$, as we sought to show.

Over $\mathbb{Q}$ the first half of this proposition fails. The partial sums of $\sum_{n\geq 0} 1/n!$ are rational and bounded above by 3, as example 2.4.18 computes, and no rational number is their limit, by proposition 2.4.19. The limit exists in $\mathbb{R}$ because theorem 2.2.2 produces it from theorem 1.2.4, and an ordered field without that property gives nothing in its place.

Sums and scalar multiples of convergent series behave as the algebra of limits predicts, and the triangle inequality passes from finite sums to infinite ones.

Proposition 2.4.5: The Algebra of Series and the Triangle Inequality

Let $\sum_{n\geq 1} a_n$ and $\sum_{n\geq 1} b_n$ converge, with sums A and B, and let $c \in \mathbb{R}$.

1. The series $\sum_{n\geq 1}(a_n + b_n)$ converges with sum $A + B$, and $\sum_{n\geq 1} ca_n$ converges with sum cA.

2. For all integers $m \geq n \geq 1$,
$$\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k|$$

3. If in addition $\sum_{n\geq 1}|a_n|$ converges, then
$$\left|\sum_{n=1}^{\infty} a_n\right| \leq \sum_{n=1}^{\infty} |a_n|$$

***Proof*:**

1. Write A_N and B_N for the partial sums of the two series. Rearranging a finite sum gives $\sum_{n=1}^{N}(a_n + b_n) = A_N + B_N$ and $\sum_{n=1}^{N} ca_n = cA_N$. Since $A_N \to A$ and $B_N \to B$, theorem 2.1.6 gives $A_N + B_N \to A + B$ and $cA_N \to cA$, which are the two assertions.

2. Fix n and induct on $m \geq n$. For $m = n$ both sides equal $|a_n|$. Assuming the inequality for m, the triangle inequality of proposition 1.3.8(4) gives

$$\left|\sum_{k=n}^{m+1} a_k\right| \leq \left|\sum_{k=n}^{m} a_k\right| + |a_{m+1}| \leq \sum_{k=n}^{m}|a_k| + |a_{m+1}| = \sum_{k=n}^{m+1}|a_k|$$

which closes the induction.

3. Write s_N for the partial sums of $\sum a_n$ and t_N for those of $\sum |a_n|$, and let $t = \lim_N t_N$. Part (2) with $n = 1$ and $m = N$ gives $|s_N| \leq t_N$ for every N. The sequence $(|s_N|)$ converges to $|A|$: by proposition 1.3.8(5), $\big||s_N| - |A|\big| \leq |s_N - A|$, and $(s_N - A)$ is a null sequence, so the comparison test for null sequences in proposition 2.1.10 makes $(|s_N| - |A|)$ null. Now proposition 2.1.8 applied to $|s_N| \leq t_N$ gives $|A| \leq t$, as we sought to show.

The three reference series come next. The first is the one case where the partial sums collapse, the second is the geometric series that every test in this section compares against, and the third is the harmonic series, which separates $a_n \to 0$ from convergence.

Example 2.4.6: The Telescoping Series $\sum 1/(n(n+1))$

Suppose $a_n = b_n - b_{n+1}$ for a sequence (b_n). Then

$$s_N = \sum_{n=1}^{N}(b_n - b_{n+1}) = b_1 - b_{N+1}$$

because every interior term $b_2, \ldots, b_N$ appears once with each sign, a cancellation confirmed by induction on N: the case $N = 1$ reads $b_1 - b_2$, and passing from N to $N + 1$ adds $b_{N+1} - b_{N+2}$ to $b_1 - b_{N+1}$. So $\sum_{n\geq 1} a_n$ converges precisely when $(b_{N+1})_{N\geq 1}$ converges, and then its sum is $b_1 - \lim_N b_{N+1}$.
Take $b_n = 1/n$. Then

$$b_n - b_{n+1} = \frac{1}{n} - \frac{1}{n+1} = \frac{(n+1) - n}{n(n+1)} = \frac{1}{n(n+1)}$$

and $b_{N+1} = 1/(N+1) \to 0$ by example 2.1.3(1). Hence

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$

This is the rare series whose sum is available directly from its partial sums.

Example 2.4.7: The Geometric Series

Let $x \in \mathbb{R}$. The series $\sum_{n\geq 0} x^n$ converges if and only if $|x| < 1$, and in that case

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$$

Suppose first that $|x| \geq 1$. An induction using proposition 1.3.8(3) gives $|x^n| = |x|^n$, and a second induction using proposition 1.1.2(5) gives $|x|^n \geq 1$ for every $n \geq 0$. So (x^n) does not converge to 0, and proposition 2.4.3 makes the series divergent.
Suppose now $|x| < 1$, so that $x \neq 1$. Multiplying out,

$$(1-x)\sum_{n=0}^{N} x^n = \sum_{n=0}^{N} x^n - \sum_{n=0}^{N} x^{n+1} = 1 - x^{N+1}$$

the middle step being the telescoping cancellation of example 2.4.6 applied to $b_n = x^n$. Dividing by $1 - x \neq 0$,

$$s_N = \sum_{n=0}^{N} x^n = \frac{1-x^{N+1}}{1-x}$$

If $x = 0$ then $x^{N+1} = 0$ for every N. If $x \neq 0$ then $r = |x|$ satisfies $0 < r < 1$, so $r^{N+1} \to 0$ by proposition 2.1.10 with $j = 0$, and since $|x^{N+1}| = r^{N+1}$ the comparison test for null sequences in the same proposition gives $x^{N+1} \to 0$. In both cases theorem 2.1.6 gives $s_N \to (1-0)/(1-x) = 1/(1-x)$.

Example 2.4.8: The Harmonic Series

The series $\sum_{n\geq 1} 1/n$ diverges, although its terms tend to 0.
The terms tend to 0 by example 2.1.3(1). For divergence, fix $n \geq 1$ and estimate the block of terms from $n+1$ to $2n$. It contains exactly n terms, and each satisfies $1/k \geq 1/(2n)$ because $k \leq 2n$ and proposition 1.1.2(7) reverses the inequality on taking reciprocals of positive numbers. Hence

$$\sum_{k=n+1}^{2n} \frac{1}{k} \geq n \cdot \frac{1}{2n} = \frac{1}{2}$$

Take $\epsilon = 1/2$ in proposition 2.4.2. Whatever N is proposed, the choice $n = N$ and $m = 2N$ satisfies $m \geq n \geq N$ and produces a block sum of at least $1/2$, so the condition of the criterion fails and the series diverges.

The harmonic series diverges slowly, and the block estimate says how slowly: applying it at $n = 2^j$ gives $s_{2^{j+1}} - s_{2^j} \geq 1/2$, so $s_{2^k} \geq s_1 + k/2 = 1 + k/2$ and the partial sums gain at least $1/2$ each time the index doubles. What decided the series was therefore the sequence of terms at the indices $1, 2, 4, 8, \ldots$, each counted once for every index in the block it heads. For a decreasing sequence of nonnegative terms that reduction is available in general.

Theorem 2.4.9: Cauchy's Condensation Test

Let $(a_n)_{n\geq 1}$ satisfy $a_1 \geq a_2 \geq a_3 \geq \cdots \geq 0$. Then $\sum_{n\geq 1} a_n$ converges if and only if

$$\sum_{k\geq 0} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots$$

converges.

***Proof*:**
Write $s_N = \sum_{n=1}^{N} a_n$ and $t_k = \sum_{j=0}^{k} 2^j a_{2^j}$. All terms of both series are nonnegative, so both sequences of partial sums are increasing and proposition 2.4.4 applies to each: the statement to be proved is that (s_N) is bounded above if and only if (t_k) is.
Step 1: $s_{2^{k+1}-1} \leq t_k$ for every integer $k \geq 0$. Split the index range $1 \leq n \leq 2^{k+1}-1$ into the blocks $2^j \leq n \leq 2^{j+1}-1$ for $j = 0, 1, \ldots, k$, which are disjoint and exhaust it. The j-th block contains $2^{j+1} - 2^j = 2^j$ indices, and each of its terms satisfies $a_n \leq a_{2^j}$ because $n \geq 2^j$ and (a_n) is decreasing. Hence

$$s_{2^{k+1}-1} = \sum_{j=0}^{k} \sum_{n=2^j}^{2^{j+1}-1} a_n \leq \sum_{j=0}^{k} 2^j a_{2^j} = t_k$$

Step 2: $t_k \leq 2s_{2^k}$ for every integer $k \geq 0$. Split the range $1 \leq n \leq 2^k$ into $\{1\}$ and the blocks $2^{j-1}+1 \leq n \leq 2^j$ for $j = 1, \ldots, k$. The j-th block contains $2^j - 2^{j-1} = 2^{j-1}$ indices, and each of its terms satisfies $a_n \geq a_{2^j}$ because $n \leq 2^j$. Hence

$$s_{2^k} = a_1 + \sum_{j=1}^{k} \sum_{n=2^{j-1}+1}^{2^j} a_n \geq a_1 + \sum_{j=1}^{k} 2^{j-1} a_{2^j} = a_1 + \frac{1}{2}(t_k - a_1) = \frac{a_1}{2} + \frac{t_k}{2}$$

Since $a_1 \geq 0$, this gives $s_{2^k} \geq t_k/2$, which is the claim.
Step 3: the two boundedness conditions agree. An induction gives $2^k \geq k+1$ for every integer $k \geq 0$: the case $k = 0$ reads $1 \geq 1$, and $2^{k+1} = 2 \cdot 2^k \geq 2k + 2 \geq k + 2$ since $k \geq 0$.
Suppose $\sum_{k\geq 0} 2^k a_{2^k}$ converges, so that $t_k \leq M$ for all k and some M, by proposition 2.4.4. Given $N \geq 1$, the inequality just proved gives $2^{N+1} - 1 \geq (N+2) - 1 > N$, so (s_N) increasing and Step 1 give

$$s_N \leq s_{2^{N+1}-1} \leq t_N \leq M$$

Hence (s_N) is bounded above and $\sum_{n\geq 1} a_n$ converges.
Conversely, suppose $\sum_{n\geq 1} a_n$ converges, so that $s_N \leq M'$ for all N and some M'. Step 2 gives $t_k \leq 2s_{2^k} \leq 2M'$ for every k, so (t_k) is bounded above and $\sum_{k\geq 0} 2^k a_{2^k}$ converges, completing the proof.

The test replaces a series by one whose terms are spaced geometrically, and that is what makes it decisive on the family the comparison test most often needs.

Example 2.4.10: The p-Series for Integer Exponents
Let $p \in \mathbb{Z}$. The series

$$\sum_{n\geq 1} \frac{1}{n^p}$$

converges if and only if $p \geq 2$.
Suppose first $p \leq 0$. Then $1/n^p = n^{-p}$ with $-p \geq 0$, and $n^{-p} \geq 1$ for every $n \geq 1$ by induction from proposition 1.1.2(5). So the terms do not converge to 0 and proposition 2.4.3 gives divergence.
Suppose $p \geq 1$. Then $a_n = 1/n^p$ is positive, and it is decreasing: if $1 \leq n < m$ then $n^p < m^p$ by induction from proposition 1.1.2(5), and proposition 1.1.2(7) inverts this to $1/m^p < 1/n^p$. So theorem 2.4.9 applies, and the condensed series has terms

$$2^k a_{2^k} = \frac{2^k}{(2^k)^p} = 2^k 2^{-kp} = \left(2^{1-p}\right)^k$$

This is the geometric series of example 2.4.7 with ratio $q = 2^{1-p} > 0$, so it converges if and only if $q < 1$. For $p = 1$ the ratio is $q = 1$. For $p \geq 2$ it is $q = 1/2^{p-1} \leq 1/2 < 1$, since $2^{p-1} \geq 2$. So the condensed series converges exactly when $p \geq 2$, and theorem 2.4.9 transfers that verdict to $\sum_{n\geq 1} 1/n^p$.

The case $p = 1$ recovers example 2.4.8 by a second route, since the condensed series of $\sum 1/n$ is $\sum_{k\geq 0} 1$. The three examples above are the whole stock of series whose convergence this section decides directly. Everything else is decided by comparison, and the tests that make that comparison usable are stated next.

Each of the three tests measures $|a_n|$, with the sign of a_n discarded, so each concludes something formally stronger than convergence.

Definition 2.4.11: Absolute Convergence

The series $\sum_{n\geq 1} a_n$ *converges absolutely* if the series $\sum_{n\geq 1} |a_n|$ converges.

Since $|a_n| \geq 0$, absolute convergence is decided by proposition 2.4.4: it holds exactly when the sums $\sum_{n=1}^{N} |a_n|$ are bounded above. For a concrete pair, the series $\sum_{n\geq 1} (-1)^{n+1}/n^2$ converges absolutely, since $\sum_{n\geq 1} 1/n^2$ converges by example 2.4.10, while $\sum_{n\geq 1} (-1)^{n+1}/n$ does not, since $\sum_{n\geq 1} 1/n$ diverges by example 2.4.8. The second series does nevertheless converge, which no test in this section can show. Section 2.5 proves it, and studies the series that converge without converging absolutely.

Theorem 2.4.12: Absolute Convergence Implies Convergence

If $\sum_{n\geq 1} a_n$ converges absolutely, then it converges.

***Proof*:**
Let $\epsilon > 0$. Applying proposition 2.4.2 to the convergent series $\sum_{n\geq 1} |a_n|$ produces an N such that

$$\sum_{k=n}^{m} |a_k| = \left| \sum_{k=n}^{m} |a_k| \right| < \epsilon$$

for all $m \geq n \geq N$, the first equality holding because a sum of nonnegative terms is nonnegative. For the same m and n, proposition 2.4.5(2) gives

$$\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k| < \epsilon$$

So $\sum_{n\geq 1} a_n$ satisfies the condition of proposition 2.4.2 and converges, as we sought to show.

Absolute convergence is what the criterion of proposition 2.4.4 can certify, and theorem 2.4.12 is what carries that certificate back to the original series. The comparison test is the first application, and it is the statement from which the ratio and root tests are both derived.

Theorem 2.4.13: The Comparison Test

Let (a_n), (c_n) and (d_n) be real sequences and let $N_0 \in \mathbb{N}$.

1. If $|a_n| \leq c_n$ for all $n \geq N_0$ and $\sum_{n\geq 1} c_n$ converges, then $\sum_{n\geq 1} a_n$ converges absolutely, hence converges.

2. If $a_n \geq d_n \geq 0$ for all $n \geq N_0$ and $\sum_{n\geq 1} d_n$ diverges, then $\sum_{n\geq 1} a_n$ diverges.

***Proof*:**

1. Let $\epsilon > 0$. Since $\sum c_n$ converges, proposition 2.4.2 gives an N_1 with $|\sum_{k=n}^{m} c_k| < \epsilon$ for all $m \geq n \geq N_1$. Put $N = \max\{N_0, N_1\}$ and let $m \geq n \geq N$. Every index k in that range satisfies $k \geq N_0$, so $c_k \geq |a_k| \geq 0$, and summing these inequalities gives

$$\sum_{k=n}^{m} |a_k| \leq \sum_{k=n}^{m} c_k = \left|\sum_{k=n}^{m} c_k\right| < \epsilon$$

where the middle equality holds because the c_k in the range are nonnegative. The left side is its own absolute value for the same reason, so $\sum_{n\geq 1}|a_n|$ satisfies the condition of proposition 2.4.2 and converges. That is absolute convergence of $\sum a_n$, and theorem 2.4.12 gives convergence.

2. Suppose, for a contradiction, that $\sum_{n\geq 1} a_n$ converges. Let $\epsilon > 0$ and take N_1 from proposition 2.4.2 for this series, and put $N = \max\{N_0, N_1\}$. For $m \geq n \geq N$ every index k in the range satisfies $a_k \geq d_k \geq 0$, so both $\sum_{k=n}^{m} d_k$ and $\sum_{k=n}^{m} a_k$ are nonnegative and

$$\left|\sum_{k=n}^{m} d_k\right| = \sum_{k=n}^{m} d_k \leq \sum_{k=n}^{m} a_k = \left|\sum_{k=n}^{m} a_k\right| < \epsilon$$

So $\sum_{n\geq 1} d_n$ satisfies the condition of proposition 2.4.2 and converges, contradicting the hypothesis. Hence $\sum_{n\geq 1} a_n$ diverges, completing the proof.

Both clauses tolerate a finite number of exceptions, which matters in practice: an inequality between two sequences almost never holds at the first few indices. Two computations show the shape of an application. The series $\sum_{n\geq 1}(n+1)/(n^3+2)$ converges, because $n+1 \leq 2n$ and $n^3+2 \geq n^3$ for $n \geq 1$ give

$$\frac{n+1}{n^3+2} \leq \frac{2n}{n^3} = \frac{2}{n^2}$$

while $\sum 2/n^2$ converges by example 2.4.10 and proposition 2.4.5(1). The series $\sum_{n\geq 1} 1/\sqrt{n}$ diverges, because $\sqrt{n} \geq 1$ for $n \geq 1$ gives $n = (\sqrt{n})^2 \geq \sqrt{n}$, hence $1/\sqrt{n} \geq 1/n > 0$ by proposition 1.1.2(7), and $\sum 1/n$ diverges by example 2.4.8. Here $\sqrt{n}$ is the nonnegative square root produced by theorem 1.3.12.

Applying the comparison test requires a candidate series to compare against, and the geometric series is almost always the candidate. The next two theorems automate that comparison: each extracts from (a_n) a number that decides whether $|a_n|$ is eventually dominated by β^n for some $\beta < 1$. They differ in how the number is extracted, and theorem 2.4.16 settles which extraction is sharper.

Theorem 2.4.14: The Ratio Test

Let $(a_n)_{n\geq 1}$ satisfy $a_n \neq 0$ for every n.

1. If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\sum_{n\geq 1} a_n$ converges absolutely.
2. If there is an n_0 with $|a_{n+1}/a_n| \geq 1$ for all $n \geq n_0$, then $\sum_{n\geq 1} a_n$ diverges.

***Proof*:**

1. Write α for the limit superior and choose a real β with $\alpha < \beta < 1$, which exists by proposition 1.1.2(8) applied to α and 1. Since $\beta > \alpha$, the eventual domination clause of theorem 2.2.12 gives an N with $|a_{n+1}/a_n| < \beta$ for all $n \geq N$, that is $|a_{n+1}| < \beta|a_n|$ for $n \geq N$ after multiplying by $|a_n| > 0$.
   An induction on $p \geq 0$ now gives $|a_{N+p}| \leq \beta^p|a_N|$: the case $p = 0$ is an equality, and the inductive step multiplies the hypothesis by $\beta > 0$ and applies $|a_{N+p+1}| < \beta|a_{N+p}|$. Setting $C = |a_N|\beta^{-N} > 0$ and writing $n = N + p$, this reads
   $$|a_n| \leq C\beta^n \qquad (n \geq N)$$
   Since $0 < \beta < 1$, the series $\sum_{n\geq 0} C\beta^n$ converges by example 2.4.7 and proposition 2.4.5(1). By theorem 2.4.13(1), with $N_0 = N$ and $c_n = C\beta^n$, the series $\sum_{n\geq 1} a_n$ converges absolutely.
2. The hypothesis gives $|a_{n+1}| \geq |a_n|$ for $n \geq n_0$, so an induction on $p \geq 0$ gives $|a_{n_0+p}| \geq |a_{n_0}|$. Because $a_{n_0} \neq 0$, the number $\delta = |a_{n_0}|$ is positive by proposition 1.3.8(1), and $|a_n| \geq \delta$ for all $n \geq n_0$. So no N has $|a_n - 0| < \delta$ for all $n \geq N$, and (a_n) does not converge to 0 by definition 2.1.2. By proposition 2.4.3 the series diverges, completing the proof.

Clause (2) is stated with a hypothesis on every large n, where clause (1) uses a limit superior, and the asymmetry is deliberate: a limit superior at least 1 is compatible with convergence, as $\sum(-1)^{n+1}/n^2$ shows, whose ratios have limit superior 1. The root test has no such asymmetry.

Theorem 2.4.15: The Root Test

Let $(a_n)_{n\geq 1}$ be a real sequence and put

$$\alpha = \limsup_{n\to\infty} |a_n|^{1/n} \in [0, +\infty]$$

where $|a_n|^{1/n}$ is the nonnegative n-th root of theorem 1.3.12.

1. If $\alpha < 1$, then $\sum_{n\geq 1} a_n$ converges absolutely.
2. If $\alpha > 1$, then $\sum_{n\geq 1} a_n$ diverges.
3. If $\alpha = 1$, both outcomes occur, so the test decides nothing.

***Proof*:**
Throughout, the inequality $0 \leq u < v$ implies $u^n < v^n$ for every $n \geq 1$, by induction on n: multiplying $u < v$ by $u \geq 0$ gives $u^{n+1} \leq u \cdot v^n$ once $u^n \leq v^n$ is known, and multiplying $u < v$ by $v^n > 0$ gives $uv^n < v^{n+1}$, both steps by proposition 1.1.2(5).

1. Choose a real β with $\alpha < \beta < 1$, by proposition 1.1.2(8). The eventual domination clause of theorem 2.2.12 gives an N with $|a_n|^{1/n} < \beta$ for all $n \geq N$. Raising both sides to the n-th power by the inequality above, and using $\left(|a_n|^{1/n}\right)^n = |a_n|$ from theorem 1.3.12, gives

$$|a_n| < \beta^n \qquad (n \geq N)$$

   The series $\sum_{n\geq 0} \beta^n$ converges by example 2.4.7 because $0 < \beta < 1$, so theorem 2.4.13(1) with $N_0 = N$ and $c_n = \beta^n$ gives absolute convergence.

2. Suppose $\alpha > 1$. Then $|a_n|^{1/n} > 1$ for infinitely many n. Otherwise there would be an N with $|a_n|^{1/n} \leq 1$ for every $n \geq N$, and the constant 1 would dominate the sequence from N on, forcing $\alpha \leq 1$ by theorem 2.2.12 and contradicting $\alpha > 1$. For each such n, raising $1 < |a_n|^{1/n}$ to the n-th power gives $|a_n| > 1$. So no N has $|a_n - 0| < 1$ for all $n \geq N$, hence (a_n) does not converge to 0 and proposition 2.4.3 gives divergence.

3. Take $a_n = 1$ for every n. Then $|a_n|^{1/n} = 1$ for every n, so $\alpha = 1$, and the series diverges by proposition 2.4.3. For a convergent example with $\alpha = 1$, take $a_n = 1/n^2$, which converges by example 2.4.10. Here $|a_n|^{1/n} = \left(n^{1/n}\right)^{-2}$, and the evaluation $\alpha = 1$ rests on $n^{1/n} \to 1$. That limit is proved as proposition 2.6.1 in section 2.6, and this one step of the present proof is deferred to it, completing the proof.

Both tests detect the same thing, domination of $|a_n|$ by a geometric series, and both fail in the same way, by returning the value 1. They are not equally sharp. The next theorem compares the two numbers they extract, and its right-hand inequality says both that whenever the ratio test establishes convergence the root test does too, and that whenever the root test is inconclusive so is the ratio test. Its left-hand inequality is needed for neither of those, and exercise 2.4.1 (8) is where it is used.

Theorem 2.4.16: The Root Test Is at Least as Sharp as the Ratio Test

For every sequence $(c_n)_{n\geq 1}$ of positive numbers,

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} c_n^{1/n} \leq \limsup_{n\to\infty} c_n^{1/n} \leq \limsup_{n\to\infty} \frac{c_{n+1}}{c_n}$$

***Proof*:**
The middle inequality holds for every sequence, since definition 2.2.10 takes both as an infimum and a supremum of the same nonempty set E of subsequential limits, and theorem 2.2.12(2) places s_* and s^* in E. So only the outer two are at issue. Both use two preliminary facts. First, if $0 < s < 1$ then $s^n \to 0$ by proposition 2.1.10 with $j = 0$, so for every $D > 0$ there is an index beyond which $s^n < D$. Second, $0 \leq u < v$ implies $u^n < v^n$ for every $n \geq 1$, by the induction recorded in the proof of theorem 2.4.15, so an inequality between $c_n^{1/n}$ and a nonnegative number may be raised to the n-th power, where $\left(c_n^{1/n}\right)^n = c_n$ by theorem 1.3.12.
Step 1: the right-hand inequality. Put $\alpha = \limsup_n c_{n+1}/c_n$. If $\alpha = +\infty$ there is nothing to prove, so assume α is a real number, necessarily $\alpha \geq 0$ because every ratio is positive. Let $\beta > \alpha$. Then $\beta > 0$. By the eventual domination clause of theorem 2.2.12 there is an N with $c_{n+1}/c_n < \beta$ for all $n \geq N$, hence $c_{n+1} < \beta c_n$ for $n \geq N$ after multiplying by $c_n > 0$. An induction on $p \geq 0$ gives $c_{N+p} \leq \beta^p c_N$, so with $C = c_N \beta^{-N} > 0$,

$$c_n \leq C\beta^n \qquad (n \geq N) \tag{2.1}$$

Suppose, for a contradiction, that $\gamma = \limsup_n c_n^{1/n}$ satisfies $\gamma > \beta$. Choose a real β' with $\beta < \beta' < \gamma$, taking the midpoint of β and γ when γ is finite (proposition 1.1.2(8)) and $\beta' = \beta + 1$ when $\gamma = +\infty$. If $c_n^{1/n} \leq \beta'$ held for every n beyond some index, then β' would dominate the sequence eventually and theorem 2.2.12 would give $\gamma \leq \beta'$, against the choice $\beta' < \gamma$. So $c_n^{1/n} > \beta'$ for infinitely many n, and raising to the n-th power gives $c_n > (\beta')^n$ for those n. Combining with (2.1), for infinitely many $n \geq N$,

$$(\beta')^n < C\beta^n, \qquad \text{that is} \qquad \left(\frac{\beta}{\beta'}\right)^n > \frac{1}{C}$$

But $s = \beta/\beta'$ lies in $(0, 1)$, so by the preliminary computation $s^n < 1/C$ for all large n, and only finitely many n can satisfy $(\beta/\beta')^n > 1/C$. This contradiction gives $\gamma \leq \beta$. As $\beta > \alpha$ was arbitrary, $\gamma \leq \alpha + \epsilon$ for every $\epsilon > 0$, and proposition 1.4.7(1) gives $\gamma \leq \alpha$.
Step 2: the left-hand inequality. Put $\alpha' = \liminf_n c_{n+1}/c_n$ and $\gamma' = \liminf_n c_n^{1/n}$, and note $\gamma' \geq 0$ since every $c_n^{1/n}$ is positive. If $\alpha' = 0$ then $\alpha' \leq \gamma'$ holds because $\gamma' \geq 0$, so let β be any real with $0 < \beta < \alpha'$. The eventual domination clause of theorem 2.2.12, in its form for the limit inferior, gives an N with $c_{n+1}/c_n > \beta$ for all $n \geq N$, hence $c_{n+1} > \beta c_n$. An induction on $p \geq 0$ gives $c_{N+p} \geq \beta^p c_N$, so with $C = c_N \beta^{-N} > 0$,

$$c_n \geq C\beta^n \qquad (n \geq N) \tag{2.2}$$

Suppose, for a contradiction, that $\gamma' < \beta$, and choose a real β' with $\gamma' < \beta' < \beta$ by proposition 1.1.2(8). If $c_n^{1/n} \geq \beta'$ held for every n beyond some index, then β' would bound the sequence from below eventually and theorem 2.2.12 would give $\gamma' \geq \beta'$, against the choice $\beta' > \gamma'$. So $c_n^{1/n} < \beta'$ for infinitely many n, and raising to the n-th power gives $c_n < (\beta')^n$ for those n. Combining with (2.2), for infinitely many $n \geq N$,

$$C\beta^n < (\beta')^n, \qquad \text{that is} \qquad \left(\frac{\beta'}{\beta}\right)^n > C$$

Here $s = \beta'/\beta$ lies in $(0,1)$, so $s^n < C$ for all large n and only finitely many n can satisfy $(\beta'/\beta)^n > C$. This contradiction gives $\gamma' \geq \beta$. As $\beta < \alpha'$ was arbitrary, $\gamma' \geq \alpha' - \epsilon$ for every $\epsilon \in (0, \alpha')$ when α' is finite, and proposition 1.4.7(1) gives $\gamma' \geq \alpha'$. When $\alpha' = +\infty$ the same conclusion $\gamma' \geq \beta$ for every real $\beta > 0$ makes γ' larger than every real number, so $\gamma' = +\infty = \alpha'$. This completes the proof.

Both consequences announced above follow from the right-hand inequality applied to $c_n = |a_n|$, which is positive because theorem 2.4.14 presupposes $a_n \neq 0$. If clause (1) of the ratio test applies, then $\limsup_n c_{n+1}/c_n < 1$, so $\limsup_n |a_n|^{1/n} < 1$ and theorem 2.4.15(1) applies as well. If the root test is inconclusive, so that $\limsup_n |a_n|^{1/n} = 1$, then $\limsup_n c_{n+1}/c_n \geq 1$ and clause (1) of the ratio test does not apply either. The reverse implications fail, and the following example is where.

Example 2.4.17: A Series the Root Test Decides and the Ratio Test Does Not

Define

$$a_n = \begin{cases} 2^{-n} & n \text{ odd} \\ 3^{-n} & n \text{ even} \end{cases}$$

so that the series is $\frac{1}{2} + \frac{1}{9} + \frac{1}{8} + \frac{1}{81} + \frac{1}{32} + \cdots$.

The root test decides it. By the uniqueness clause of theorem 1.3.12, $|a_n|^{1/n}$ equals $1/2$ for odd n and $1/3$ for even n, since $(1/2)^n = 2^{-n}$ and $(1/3)^n = 3^{-n}$. So the sequence $\left(|a_n|^{1/n}\right)$ takes only the two values $1/2$ and $1/3$, is dominated by $1/2$ everywhere, and equals $1/2$ for infinitely many n, so the constant subsequence indexed by the odd n converges to $1/2$. By theorem 2.2.12 these two facts identify $\alpha = 1/2 < 1$, and theorem 2.4.15(1) gives absolute convergence.

The ratio test does not. For n odd,

$$\frac{a_{n+1}}{a_n} = \frac{3^{-(n+1)}}{2^{-n}} = \frac{2^n}{3^{n+1}} = \frac{1}{3}\left(\frac{2}{3}\right)^n \leq \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$$

using that $(2/3)^n \leq 2/3$ for $n \geq 1$, which is an induction from proposition 1.1.2(5) since $0 < 2/3 < 1$, while for n even,

$$\frac{a_{n+1}}{a_n} = \frac{2^{-(n+1)}}{3^{-n}} = \frac{3^n}{2^{n+1}} = \frac{1}{2}\left(\frac{3}{2}\right)^n \geq \frac{1}{2} \cdot \frac{9}{4} = \frac{9}{8}$$

using that $(3/2)^n \geq (3/2)^2 = 9/4$ for $n \geq 2$, again an induction from proposition 1.1.2(5) since $3/2 > 1$. Clause (2) of theorem 2.4.14 fails, since ratios below 1 occur beyond every index. Clause (1) fails as well: if the limit superior of the ratios were some $\alpha < 1$, then choosing β with $\alpha < \beta < 1$ and applying the eventual domination clause of theorem 2.2.12 would put every ratio below $\beta < 1$ from some index on, contradicting the bound $a_{n+1}/a_n \geq 9/8 > 1$ just proved at every even index. So the ratio test yields no verdict on a series the root test settles at once.

The series is not itself difficult, since $a_n \leq 2^{-n}$ for every n and theorem 2.4.13(1) settles it.

The tests now produce the number e, together with an error estimate sharp enough to prove it irrational, and the decimal notation for a real number as a convergent series.

Example 2.4.18: The Series for e and Its Tail Estimate

The series $\sum_{n\geq 0} 1/n!$ converges. Its sum is written e, and it satisfies $2 < e < 3$. Writing $s_n = \sum_{k=0}^{n} 1/k!$ for its partial sums, every integer $n \geq 1$ satisfies

$$0 < e - s_n < \frac{1}{n!\,n} \tag{2.3}$$

Convergence and the bounds on e. An induction gives $n! \geq 2^{n-1}$ for every $n \geq 1$: the case $n = 1$ reads $1 \geq 1$, and $(n+1)! = (n+1)\cdot n! \geq 2\cdot 2^{n-1} = 2^n$ since $n+1 \geq 2$. Hence $1/n! \leq (1/2)^{n-1}$ for $n \geq 1$ by proposition 1.1.2(7), and theorem 2.4.13(1) with the convergent geometric series $\sum_{n\geq 1}(1/2)^{n-1}$ of example 2.4.7 gives convergence. The same comparison bounds the sum. The inequality $1/n! \leq (1/2)^{n-1}$ is strict at $n = 3$, where $1/6 < 1/4$, so for every $N \geq 3$,

$$\sum_{n=1}^{N} \frac{1}{n!} \leq \sum_{n=1}^{N} \left(\frac{1}{2}\right)^{n-1} - \left(\frac{1}{4} - \frac{1}{6}\right) \leq 2 - \frac{1}{12}$$

the last step using that the geometric partial sums are bounded above by their limit 2, by proposition 2.4.4. Letting $N \to \infty$ and applying proposition 2.1.8,

$$e = 1 + \sum_{n=1}^{\infty} \frac{1}{n!} \leq 3 - \frac{1}{12} < 3$$

For the lower bound, $e \geq s_2 = 1 + 1 + 1/2 = 5/2 > 2$, since proposition 2.4.4 identifies e as the supremum of the partial sums and s_2 is one of them.

The tail estimate. Fix $n \geq 1$ and let $m > n$. For $k = n+1+j$ with $0 \leq j \leq m-n-1$, the factorial $k!$ contains the factors $n+2, n+3, \ldots, k$ beyond $(n+1)!$, which are j factors each at least $n+2$, so $k! \geq (n+1)!\,(n+2)^j$ and hence $1/k! \leq (n+2)^{-j}/(n+1)!$ by proposition 1.1.2(7). Summing,

$$s_m - s_n = \sum_{k=n+1}^{m} \frac{1}{k!} \leq \frac{1}{(n+1)!} \sum_{j=0}^{m-n-1} \left(\frac{1}{n+2}\right)^j < \frac{1}{(n+1)!} \cdot \frac{1}{1 - \frac{1}{n+2}} = \frac{n+2}{(n+1)!\,(n+1)}$$

where the strict inequality holds because the finite geometric sum is smaller than the full geometric series of example 2.4.7, whose remaining terms are positive. Letting $m \to \infty$ and applying proposition 2.1.8,

$$e - s_n \leq \frac{n+2}{(n+1)!\,(n+1)}$$

Finally $n(n+2) = n^2 + 2n < n^2 + 2n + 1 = (n+1)^2$, so

$$\frac{n+2}{(n+1)!\,(n+1)} = \frac{1}{n!} \cdot \frac{n+2}{(n+1)^2} < \frac{1}{n!} \cdot \frac{1}{n}$$

which is the upper half of (2.3). The lower half holds because $e - s_n \geq s_{n+1} - s_n = 1/(n+1)! > 0$, again by proposition 2.1.8 and the fact that (s_m) is increasing.

Estimate (2.3) converges quickly enough to be used numerically, since s_{10} already approximates e

to within $1/(10! \cdot 10)$, a number below $3 \cdot 10^{-8}$. It also decides the arithmetic nature of e, because multiplying it by $n!$ produces a quantity confined between two consecutive integers.

Proposition 2.4.19: e Is Irrational

The number e is not rational.

***Proof*:**
Suppose $e = p/q$ with p and q positive integers, which is possible for a positive rational number. Apply (2.3) with $n = q \geq 1$ and multiply through by $q! > 0$, which preserves the inequalities by proposition 1.1.2(5):

$$0 < q!\,(e - s_q) < \frac{q!}{q!\,q} = \frac{1}{q} \leq 1$$

Both quantities inside are integers. First, $q!\,e = q!\,p/q = p \cdot (q-1)!$, an integer because $q \geq 1$. Second,

$$q!\,s_q = \sum_{k=0}^{q} \frac{q!}{k!}$$

and each summand is an integer, since $k \leq q$ makes $q!/k!$ the product of the integers $k+1, k+2, \ldots, q$, empty and equal to 1 when $k = q$. So $q!(e - s_q)$ is a difference of two integers, hence an integer, and it lies strictly between 0 and 1. No integer does, so the assumption is untenable and e is irrational, completing the proof.

This is the failure in $\mathbb{Q}$ promised at proposition 2.4.4, delivered in full: an increasing sequence of rational numbers, bounded above by 3, whose limit exists in $\mathbb{R}$ and is not itself rational.

The number e has a second description, as a limit, and the two are connected by the binomial theorem. The limit form is the one most applications of e use.

Theorem 2.4.20: e as a Limit

$$\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n = e$$

***Proof*:**
Write $t_n = (1 + 1/n)^n$ and $s_n = \sum_{k=0}^{n} 1/k!$ as in example 2.4.18. The binomial theorem ([1, the binomial theorem]) expands

$$t_n = \sum_{k=0}^{n} \binom{n}{k} \frac{1}{n^k}$$

and the k-th summand rewrites as

$$\binom{n}{k}\frac{1}{n^k} = \frac{1}{k!} \cdot \frac{n(n-1)\cdots(n-k+1)}{n^k} = \frac{1}{k!}\prod_{j=0}^{k-1}\left(1 - \frac{j}{n}\right) = \frac{1}{k!}\prod_{j=1}^{k-1}\left(1 - \frac{j}{n}\right)$$

the last step dropping the factor $j = 0$, which is 1. For $0 \leq j \leq k-1 \leq n-1$ each factor lies in $(0, 1]$, so the product lies in $(0, 1]$ as well.

Step 1: $t_n \le e$ for every n. Each summand is at most $1/k!$ by the previous paragraph, so $t_n \le s_n$, and $s_n < e$ by (2.3).
Step 2: a lower bound depending on a fixed truncation. Fix an integer $m \ge 1$. For $n \ge m$ all $n+1$ summands of t_n are nonnegative, so discarding those with $k > m$ gives

$$t_n \ge u_n^{(m)} := \sum_{k=0}^{m} \frac{1}{k!} \prod_{j=1}^{k-1} \left(1 - \frac{j}{n}\right)$$

Hold m fixed and let n grow. For each fixed j the sequence $(1 - j/n)_n$ converges to 1, since $j/n \to 0$ by example 2.1.3(1) and theorem 2.1.6. Each of the $m+1$ summands is a product of at most m such factors times a constant, so theorem 2.1.6 applied finitely many times gives $u_n^{(m)} \to s_m$ as $n \to \infty$.
Step 3: the two bounds close. Let $\epsilon > 0$. Since $s_m \to e$, there is an m with $s_m > e - \epsilon/2$. With that m fixed, Step 2 provides an $N \ge m$ with $|u_n^{(m)} - s_m| < \epsilon/2$, hence $u_n^{(m)} > s_m - \epsilon/2$, for all $n \ge N$. For such n,

$$e - \epsilon < s_m - \frac{\epsilon}{2} < u_n^{(m)} \le t_n \le e$$

so $|t_n - e| < \epsilon$. As $\epsilon > 0$ was arbitrary, $t_n \to e$ by definition 2.1.2, as we sought to show.

A decimal expansion is an infinite series, its convergence follows from the geometric series, and its digits are produced by proposition 1.4.3. Stating it as a theorem also settles the status of the identity $0.999\cdots = 1$.

Proposition 2.4.21: Decimal Expansions

Let $x \in \mathbb{R}$ with $x \ge 0$. Put $a_0 = \lfloor x \rfloor$ and, for $n \ge 1$,

$$a_n = \lfloor 10^n x \rfloor - 10\lfloor 10^{n-1} x \rfloor$$

Then each a_n with $n \ge 1$ is an integer with $0 \le a_n \le 9$, and

$$x = a_0 + \sum_{n=1}^{\infty} \frac{a_n}{10^n}$$

Distinct digit sequences can represent the same real number: the series $\sum_{n\ge1} 9\cdot 10^{-n}$ converges to 1.

***Proof*:**
Write $b_n = \lfloor 10^n x \rfloor$ for $n \ge 0$, so $a_0 = b_0$ and $a_n = b_n - 10b_{n-1}$ for $n \ge 1$. Each b_n is an integer and proposition 1.4.3 gives

$$b_n \le 10^n x < b_n + 1 \tag{2.4}$$

Step 1: the digits lie in $\{0, 1, \ldots, 9\}$. Multiplying (2.4) at index $n-1$ by $10 > 0$ gives $10b_{n-1} \le 10^n x < 10b_{n-1} + 10$. Combining the left half of this with the right half of (2.4) at index n gives $10b_{n-1} \le 10^n x < b_n + 1$, hence $10b_{n-1} < b_n + 1$ and so $a_n = b_n - 10b_{n-1} > -1$. Since a_n is an integer, $a_n \ge 0$. Combining the left half of (2.4) at index n with the right half of the multiplied inequality gives $b_n \le 10^n x < 10b_{n-1} + 10$, hence $a_n < 10$ and so $a_n \le 9$.

Step 2: the partial sums telescope. Dividing $a_n = b_n - 10b_{n-1}$ by 10^n gives

$$\frac{a_n}{10^n} = \frac{b_n}{10^n} - \frac{b_{n-1}}{10^{n-1}}$$

so the interior terms cancel in the partial sums and

$$a_0 + \sum_{n=1}^{N} \frac{a_n}{10^n} = \frac{b_N}{10^N} \qquad (N \geq 1)$$

This is an induction on N: for $N = 1$ the left side is $b_0 + (b_1/10 - b_0) = b_1/10$, and the step from N to $N+1$ adds $b_{N+1}/10^{N+1} - b_N/10^N$ to $b_N/10^N$.
Step 3: the partial sums converge to x. Dividing (2.4) by $10^N > 0$ gives

$$0 \leq x - \frac{b_N}{10^N} < \frac{1}{10^N}$$

and $(1/10)^N \to 0$ by proposition 2.1.10 with $j = 0$ and $r = 1/10$. By theorem 2.1.7 the sequence $(x - b_N 10^{-N})$ converges to 0, so by theorem 2.1.6 the partial sums $b_N 10^{-N}$ converge to x. That is the identity $x = a_0 + \sum_{n=1}^{\infty} a_n/10^n$.
Step 4: the nines. By example 2.4.7 and proposition 2.4.5(1),

$$\sum_{n=1}^{\infty} \frac{9}{10^n} = \frac{9}{10} \sum_{n=0}^{\infty} \left(\frac{1}{10}\right)^n = \frac{9}{10} \cdot \frac{1}{1 - \frac{1}{10}} = \frac{9}{10} \cdot \frac{10}{9} = 1$$

completing the proof.

Applied to $x = 1$ the recipe returns $a_0 = 1$ and $a_n = 0$ for $n \geq 1$, so the expansion it produces is the terminating one. Step 4 shows that the non-terminating sequence of nines represents the same number, which is the content of the identity $0.999\cdots = 1$: the two symbols are two convergent series with one sum, not two numbers. The proposition also gives a second proof that $\mathbb{Q}$ is dense in $\mathbb{R}$, established as theorem 1.4.4, since the truncation $a_0 + \sum_{n=1}^{N} a_n 10^{-n}$ is rational and Steps 2 and 3 together place it within 10^{-N} of x. The proposition asks for $x \geq 0$, and a negative x is reached by approximating $-x$ and negating.

Exercise 2.4.1

1. Show that $\sum_{n\geq 1} \dfrac{1}{n(n+2)}$ converges and compute its sum.

2. Compute $\displaystyle\sum_{n=1}^{\infty} \frac{2^n + 3^n}{6^n}$, justifying each step by a result of this section.

3. Show that $\sum_{n\geq 1} \dfrac{n}{2n+1}$ diverges.

4. Decide the convergence of each of the following, in each case naming the series used for comparison.

    1. $\displaystyle\sum_{n\geq 1} \frac{n+1}{n^4+5}$

2. $\displaystyle\sum_{n\geq 1} \frac{1}{2^n+n}$
3. $\displaystyle\sum_{n\geq 2} \frac{1}{n-1}$

5. Use theorem 2.4.9 to give a second proof that $\sum_{n\geq 1} 1/n$ diverges, and identify where the hypothesis that the terms decrease is used.
6. Let $a_1 \geq a_2 \geq \cdots \geq 0$ and suppose $\sum_{n\geq 1} a_n$ converges. Show that $na_{2n} \to 0$.
7. Apply theorem 2.4.14 to $\sum_{n\geq 1} 2^n/n!$ and to $\sum_{n\geq 1} n^2/2^n$, and in each case state the value of the limit superior of the ratios.
8. Let (c_n) be a sequence of positive numbers.
   1. Show that if $\lim_n c_{n+1}/c_n = L$ exists and is a real number, then $\lim_n c_n^{1/n} = L$.
   2. Deduce that $\displaystyle\lim_{n\to\infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}$.
9. Verify that $s_5 = 163/60$, and use (2.3) to deduce $163/60 < e < 1631/600$.
10. Show that $\displaystyle\frac{4}{10} + \sum_{n=2}^{\infty} \frac{9}{10^n} = \frac{1}{2}$, and determine which digit sequence the recipe of proposition 2.4.21 produces for $x = 1/2$.
11. Let $\sum_{n\geq 1} a_n$ converge absolutely and let (b_n) be a bounded sequence. Show that $\sum_{n\geq 1} a_n b_n$ converges absolutely, and show that boundedness of (b_n) cannot be dropped, by exhibiting an absolutely convergent $\sum_{n\geq 1} a_n$ and an unbounded (b_n) for which $\sum_{n\geq 1} a_n b_n$ diverges.

2.5 Conditional Convergence and Rearrangement

Every test proved in section 2.4 reaches its conclusion from a bound on $|a_n|$ alone, and so each of them establishes absolute convergence. A series whose terms change sign can converge for a different reason: not because its terms are small enough to be added with the signs discarded, but because consecutive terms cancel. This section proves the two tests that detect convergence of that second kind, names the series that converge for that reason and no other, and then measures the difference between the two kinds. An absolutely convergent series has the same sum in every order. A conditionally convergent one can be reordered so that its partial sums have any prescribed limit inferior and limit superior in $[-\infty, +\infty]$, subject only to the first not exceeding the second, and its Cauchy product with a convergent series can diverge.

Neither test below produces an expression for the sum whose existence it asserts. Each closes on the Cauchy criterion for series, proposition 2.4.2, which is theorem 2.3.5 read for the sequence of partial sums, so completeness is again what produces the number.

The tests of section 2.4 discard the signs of the terms at their first step, so none of them can see cancellation. What replaces them is an identity that exposes the cancellation directly: it trades the terms a_n for their partial sums and the terms b_n for their successive differences, so that a series whose partial sums merely stay bounded can still be handled. The identity is the discrete analogue of integration by parts.

Lemma 2.5.1: Summation by Parts

Let $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ be sequences of real numbers, and put

$$A_N = \sum_{n=1}^{N} a_n \quad (N \geq 1), \qquad A_0 = 0$$

Then for all integers $1 \leq p \leq q$,

$$\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q-1} A_n \left(b_n - b_{n+1}\right) + A_q b_q - A_{p-1} b_p$$

where the first sum on the right is empty, and therefore 0, when $p = q$.

***Proof*:**
For every $n \geq 1$, $a_n = A_n - A_{n-1}$, which for $n = 1$ reads $a_1 = A_1 - A_0 = A_1$. Hence

$$\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q} \left(A_n - A_{n-1}\right) b_n = \sum_{n=p}^{q} A_n b_n - \sum_{n=p}^{q} A_{n-1} b_n$$

Re-indexing the second sum by $m = n - 1$ turns it into $\sum_{m=p-1}^{q-1} A_m b_{m+1}$. Splitting the first sum at its top term and the second at its bottom term,

$$\sum_{n=p}^{q} A_n b_n - \sum_{n=p-1}^{q-1} A_n b_{n+1} = A_q b_q + \sum_{n=p}^{q-1} A_n b_n - A_{p-1} b_p - \sum_{n=p}^{q-1} A_n b_{n+1}$$

Collecting the two sums over the common range $p \leq n \leq q - 1$ gives $\sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p$, as we sought to show.

The right-hand side is small as soon as the numbers A_n are bounded and the increments $b_n - b_{n+1}$ have a small total over the range, and for a monotone (b_n) that total telescopes to $b_p - b_q$. This is what the next test uses.

Both the test and its corollary also use the following observation. A nonincreasing sequence with limit 0 has every term nonnegative: if $b_N < 0$ for some N, then $b_n \leq b_N$ for all $n \geq N$, and proposition 2.1.8 applied to the tail $(b_n)_{n\geq N}$ gives $0 = \lim_n b_n \leq b_N < 0$, which is false.

Theorem 2.5.2: Dirichlet's Test

Let $(a_n)_{n\geq 1}$ and $(b_n)_{n\geq 1}$ be sequences of real numbers such that

1. the partial sums $A_N = \sum_{n=1}^{N} a_n$ form a bounded sequence,
2. $b_1 \geq b_2 \geq b_3 \geq \cdots$, and
3. $b_n \to 0$.

Then $\sum_{n\geq 1} a_n b_n$ converges.

***Proof*:**
Choose $M > 0$ with $|A_n| \leq M$ for every $n \geq 0$, which is possible by (1) together with $A_0 = 0$. By the observation above, every b_n is nonnegative, and by (2) every difference $b_n - b_{n+1}$ is nonnegative as well.

Let $\epsilon > 0$. By (3) there is an N with $b_N \leq \epsilon/(2M)$. Fix integers $q \geq p \geq N$. Applying lemma 2.5.1 on the range $p \leq n \leq q$ and then the triangle inequality proposition 2.4.5(2) in its finite form,

$$\left|\sum_{n=p}^{q} a_n b_n\right| \leq \sum_{n=p}^{q-1} |A_n| \left(b_n - b_{n+1}\right) + |A_q|\, b_q + |A_{p-1}|\, b_p \leq M \left(\sum_{n=p}^{q-1} \left(b_n - b_{n+1}\right) + b_q + b_p\right)$$

where the nonnegativity of $b_n - b_{n+1}$, b_q and b_p is what allows the bound $|A_n| \leq M$ to be pulled out of each term. The remaining sum telescopes to $b_p - b_q$, so the right-hand side equals $M\left(b_p - b_q + b_q + b_p\right) = 2Mb_p$. Since $p \geq N$ and (b_n) is nonincreasing, $b_p \leq b_N \leq \epsilon/(2M)$, and therefore

$$\left|\sum_{n=p}^{q} a_n b_n\right| \leq 2Mb_N \leq \epsilon$$

for all $q \geq p \geq N$. This is the Cauchy criterion for series, proposition 2.4.2, so $\sum_{n\geq 1} a_n b_n$ converges, completing the proof.

Hypothesis (1) is much weaker than convergence of $\sum a_n$: it asks only that the partial sums stay in a bounded set, and permits them to oscillate forever. For a concrete case, let (a_n) repeat the block $1, 1, -2$ forever, so that $a_1 = a_2 = 1$, $a_3 = -2$, $a_4 = a_5 = 1$, $a_6 = -2$, and so on. The partial sums run $1, 2, 0, 1, 2, 0, \ldots$, so $|A_N| \leq 2$ for every N, while $\sum_{n\geq 1} a_n$ itself does not converge, its partial sums taking the value 2 and the value 0 infinitely often. Taking $b_n = 1/n$, which decreases to 0, theorem 2.5.2 gives the convergence of

$$\sum_{n\geq 1} \frac{a_n}{n} = 1 + \frac{1}{2} - \frac{2}{3} + \frac{1}{4} + \frac{1}{5} - \frac{2}{6} + \cdots$$

The same series with absolute values diverges: $|a_n| \geq 1$ for every n, so $|a_n|/n \geq 1/n$, and theorem 2.4.13 together with example 2.4.8 makes $\sum_n |a_n|/n$ divergent.

The case in which the block is $1, -1$ is the one that occurs most often, and it carries an estimate for the error committed by stopping at a finite stage. That estimate is what makes an alternating series usable for computation, and it is recorded here alongside the convergence statement.

Corollary 2.5.3: The Alternating Series Test

Let $b_1 \geq b_2 \geq b_3 \geq \cdots$ with $b_n \to 0$. Write $s_N = \sum_{n=1}^{N}(-1)^{n+1}b_n$. Then

1. $\sum_{n\geq 1}(-1)^{n+1}b_n$ converges. Call its sum s.
2. $s_{2k} \leq s \leq s_{2k-1}$ for every $k \geq 1$.
3. $|s - s_N| \leq b_{N+1}$ for every $N \geq 1$.

***Proof*:**

1. Put $a_n = (-1)^{n+1}$. Pairing consecutive terms gives $A_{2m} = \sum_{j=1}^{m}(1-1) = 0$ and $A_{2m+1} = A_{2m} + 1 = 1$, so $|A_N| \leq 1$ for every N. Hypotheses (2) and (3) of theorem 2.5.2 are the hypotheses assumed here, so that theorem applies and $\sum_{n\geq 1}(-1)^{n+1}b_n$ converges.
2. As noted before theorem 2.5.2, every b_n is nonnegative. Three computations from the definition of s_N record the shape of the partial sums:

   $$s_{2k+2} - s_{2k} = b_{2k+1} - b_{2k+2} \geq 0, \qquad s_{2k+1} - s_{2k-1} = b_{2k+1} - b_{2k} \leq 0, \qquad s_{2k+1} - s_{2k} = b_{2k+1} \geq 0$$

   So $(s_{2k})_{k\geq 1}$ is nondecreasing, and the second computation, read for $k \geq 1$, says that $(s_{2k-1})_{k\geq 1}$ is nonincreasing. Both are subsequences of the convergent sequence (s_N), so by proposition 2.2.7 both converge to s. Fix $k \geq 1$. For every $j \geq k$ the inequality $s_{2j} \geq s_{2k}$ holds, and letting $j \to \infty$ in it, proposition 2.1.8 gives $s \geq s_{2k}$. The same argument applied to the nonincreasing sequence $(s_{2j-1})_{j\geq k}$ gives $s \leq s_{2k-1}$.
3. Let $N \geq 1$. If $N = 2k$ is even, then part (2) read at $k+1$ gives $s \leq s_{2(k+1)-1} = s_{2k+1}$, so with the third computation above,

   $$0 \leq s - s_{2k} \leq s_{2k+1} - s_{2k} = b_{2k+1} = b_{N+1}$$

   If $N = 2k-1$ is odd with $k \geq 1$, then part (2) read at k gives $s_{2k} \leq s \leq s_{2k-1}$, while the definition of s_N gives $s_{2k} = s_{2k-1} + (-1)^{2k+1}b_{2k} = s_{2k-1} - b_{2k}$. Hence

   $$0 \leq s_{2k-1} - s \leq s_{2k-1} - s_{2k} = b_{2k} = b_{N+1}$$

   In each case $|s - s_N| \leq b_{N+1}$, as required.

Part (2) brackets the sum between two consecutive partial sums, and part (3) turns that bracketing into a bound on the truncation error: the error made by stopping after N terms is at most the first term omitted. The prototype is the alternating harmonic series $\sum_{n\geq 1}(-1)^{n+1}/n$, whose terms $1/n$ decrease to 0, so its sum s exists and satisfies $s_4 = 7/12 \leq s \leq 5/6 = s_3$.

That series converges while $\sum_{n\geq 1} 1/n$ diverges (example 2.4.8), so it converges for a reason that no test in section 2.4 could have detected. Series of this kind are the subject of the rest of the section.

Definition 2.5.4: Conditional Convergence and Rearrangement

Let $\sum_{n\geq 1} a_n$ be a series of real numbers.

1. The series *converges conditionally* if $\sum_{n\geq 1} a_n$ converges and $\sum_{n\geq 1}|a_n|$ diverges.
2. For a bijection $\sigma\colon \mathbb{N}\to\mathbb{N}$, the series $\sum_{n\geq 1} a_{\sigma(n)}$ is called a *rearrangement* of $\sum_{n\geq 1} a_n$.

The alternating harmonic series is conditionally convergent, by the two facts recorded above: it converges by corollary 2.5.3, and the series of absolute values is the harmonic series, which diverges by example 2.4.8. So is $\sum_{n\geq 1} a_n/n$ for the $1, 1, -2$ pattern computed after theorem 2.5.2. A convergent series is by definition either absolutely or conditionally convergent, and theorem 2.4.12 rules out the fourth possibility of a divergent series whose absolute values are summable, so every series of real numbers falls into exactly one of three classes: absolutely convergent, conditionally convergent, divergent.

A rearrangement changes nothing about which terms occur, only the order in which they are added, so a first guess is that it changes nothing about the sum. For absolutely convergent series that guess is correct.

Theorem 2.5.5: Absolutely Convergent Series Are Rearrangement Invariant

Let $\sum_{n\geq 1} a_n$ converge absolutely, with sum s. Then for every bijection $\sigma\colon \mathbb{N}\to\mathbb{N}$ the rearrangement $\sum_{n\geq 1} a_{\sigma(n)}$ converges absolutely, and its sum is s.

***Proof*:**

Write $t_N = \sum_{n=1}^{N}|a_n|$ and $t = \lim_N t_N$, which exists by the hypothesis of absolute convergence, and write $s_N = \sum_{n=1}^{N} a_n$.

Step 1: the rearranged series converges absolutely. Fix $M \geq 1$ and put $T = \{\sigma(1), \ldots, \sigma(M)\}$, a set of M distinct positive integers because σ is injective. Re-indexing the finite sum by the bijection σ from $\{1, \ldots, M\}$ onto T,

$$\sum_{n=1}^{M}|a_{\sigma(n)}| = \sum_{m\in T}|a_m| \leq \sum_{m=1}^{\max T}|a_m| = t_{\max T} \leq t$$

the middle inequality because $T \subseteq \{1, \ldots, \max T\}$ and every $|a_m|$ is nonnegative, and the last because (t_N) is nondecreasing with limit t, so proposition 2.1.8 gives $t_N \leq t$ for every N. Thus the partial sums of $\sum_n |a_{\sigma(n)}|$ are bounded above, and its terms are nonnegative, so it converges by proposition 2.4.4. By theorem 2.4.12 the rearrangement converges.

Step 2: the sum is unchanged. Let $\epsilon > 0$. Since $t_N \to t$ and $s_N \to s$, choose N with $t - t_N \leq \epsilon$ and $|s - s_N| \leq \epsilon$. Because σ is a bijection, each of $1, \ldots, N$ is $\sigma(i)$ for exactly one i, so

$$P = \max\left\{\sigma^{-1}(1), \ldots, \sigma^{-1}(N)\right\}$$

is a well-defined positive integer, and $\{1, \ldots, N\} \subseteq T$ whenever $M \geq P$, where $T = \{\sigma(1), \ldots, \sigma(M)\}$ as above.

Fix $M \geq P$ and set $S = T\setminus\{1, \ldots, N\}$, a finite set of integers all of which exceed N. Re-indexing as in Step 1 and cancelling the terms indexed by $\{1, \ldots, N\}$,

$$\sum_{n=1}^{M} a_{\sigma(n)} - s_N = \sum_{m \in T} a_m - \sum_{m=1}^{N} a_m = \sum_{m \in S} a_m$$

If S is empty this difference is 0. Otherwise the finite triangle inequality proposition 2.4.5(2) and the inclusion $S \subseteq \{N+1, \ldots, \max S\}$ give

$$\left|\sum_{m \in S} a_m\right| \leq \sum_{m \in S} |a_m| \leq \sum_{m=N+1}^{\max S} |a_m| = t_{\max S} - t_N \leq t - t_N \leq \epsilon$$

Combining the two displays,

$$\left|\sum_{n=1}^{M} a_{\sigma(n)} - s\right| \leq \left|\sum_{n=1}^{M} a_{\sigma(n)} - s_N\right| + |s_N - s| \leq 2\epsilon$$

for every $M \geq P$. As $\epsilon > 0$ was arbitrary, the partial sums of the rearranged series converge to s, completing the proof.

The hypothesis was used twice, and in both places it was used in the same way: the tail $t - t_N$ of the series of absolute values controls every finite sum of terms with index beyond N, no matter which such terms are chosen and no matter how many. A conditionally convergent series has no such control, since the partial sums of $\sum |a_n|$ are unbounded past every stage. The next theorem measures how completely that control is lost: for a conditionally convergent series, the pair consisting of the limit inferior and the limit superior of the rearranged partial sums can be prescribed arbitrarily.

Theorem 2.5.6: Riemann's Rearrangement Theorem

Let $\sum_{n \geq 1} a_n$ converge conditionally, and let $\alpha, \beta \in [-\infty, +\infty]$ with $\alpha \leq \beta$. Then there is a bijection $\sigma \colon \mathbb{N} \to \mathbb{N}$ whose partial sums $s'_N = \sum_{n=1}^{N} a_{\sigma(n)}$ satisfy

$$\liminf_{N \to \infty} s'_N = \alpha, \qquad \limsup_{N \to \infty} s'_N = \beta$$

***Proof*:**
Step 1: the positive and negative parts both diverge. For $n \geq 1$ put

$$p_n = \frac{|a_n| + a_n}{2}, \qquad q_n = \frac{|a_n| - a_n}{2}$$

so that $p_n - q_n = a_n$ and $p_n + q_n = |a_n|$, and both are nonnegative because $-|a_n| \leq a_n \leq |a_n|$ by proposition 1.3.8. Suppose first that $\sum p_n$ and $\sum q_n$ both converge. Then $\sum |a_n| = \sum (p_n + q_n)$ converges by proposition 2.4.5, contradicting conditional convergence. Suppose next that exactly one of them converges, say $\sum p_n$. Since $\sum a_n$ converges by hypothesis, proposition 2.4.5 makes $\sum (p_n - a_n) = \sum q_n$ convergent, again a contradiction, and the same argument with the roles exchanged rules out the case that only $\sum q_n$ converges. So both diverge, and since their terms are nonnegative, proposition 2.4.4 says that the partial sums of each are unbounded above.

Step 2: the two extracted sequences. Let $P_1, P_2, P_3, \ldots$ be the nonnegative terms of (a_n), listed in the order in which they occur, and let $Q_1, Q_2, Q_3, \ldots$ be the absolute values of the negative terms, again in their original order. Both lists are infinite: p_n is nonzero only where $a_n > 0$, so the divergence of $\sum p_n$ forces infinitely many a_n to be positive, and likewise the divergence of $\sum q_n$ forces infinitely many a_n to be negative. The partial sums of $\sum_j P_j$ are exactly the partial sums of $\sum_n p_n$ with the repetitions caused by the zero terms deleted, so the two sets of values coincide and the two suprema agree, and the same comparison holds for $\sum_j Q_j$ against $\sum_n q_n$. Hence the partial sums of $\sum_j P_j$ and of $\sum_j Q_j$ are unbounded above, and this remains true for the tail $\sum_{j>J} P_j$ for any fixed J, since its partial sums differ from those of $\sum_j P_j$ by the constant $P_1 + \cdots + P_J$. Finally, $\sum_n a_n$ converges, so $a_n \to 0$ by proposition 2.4.3, and (P_j) and (Q_j) are the subsequences of (a_n) and of $(-a_n)$ along the nonnegative and negative terms, so $P_j \to 0$ and $Q_j \to 0$ by proposition 2.2.7.

Step 3: two sequences of real targets. Define real numbers u_n and v_n for $n \geq 1$ by

$$u_n = \begin{cases} \alpha - 1/n, & \alpha \in \mathbb{R} \\ -n, & \alpha = -\infty \\ n, & \alpha = +\infty \end{cases} \qquad v_n = \begin{cases} \beta + 1/n, & \beta \in \mathbb{R} \\ n, & \beta = +\infty \\ -n, & \beta = -\infty \end{cases}$$

and put $u_0 = u_1$,

$$\alpha_n = u_n, \qquad \beta_n = \max\left\{v_n,\ u_n + \tfrac{1}{n},\ u_{n-1} + \tfrac{1}{n}\right\}$$

Two inequalities hold by construction, for every $n \geq 1$:

$$\alpha_n < \beta_n \quad (\text{because } \beta_n \geq u_n + \tfrac{1}{n}), \qquad \alpha_n < \beta_{n+1} \quad (\text{because } \beta_{n+1} \geq u_n + \tfrac{1}{n+1})$$

The behaviour of (β_n) is recorded case by case, and each case is a direct computation from the displayed formulas. If $\beta \in \mathbb{R}$ and $\alpha \in \mathbb{R}$, then for $n \geq 2$ one has $u_n + 1/n = \alpha \leq \beta < v_n$ and $u_{n-1} + 1/n < \alpha \leq \beta < v_n$, so $\beta_n = v_n = \beta + 1/n$ and $\beta_n \to \beta$. If $\beta \in \mathbb{R}$ and $\alpha = -\infty$, then $u_n + 1/n = -n + 1/n$ and $u_{n-1} + 1/n = -(n-1) + 1/n$ are both eventually smaller than $v_n = \beta + 1/n$, namely as soon as $n > 1 - \beta$, so again $\beta_n = v_n$ for all large n and $\beta_n \to \beta$. If $\beta = +\infty$, then $\beta_n \geq v_n = n$. If $\beta = -\infty$, then $\alpha = -\infty$ as well, and for $n \geq 2$ the largest of the three numbers is $u_{n-1} + 1/n$, so $\beta_n = -n + 1 + 1/n$. Correspondingly $\alpha_n = \alpha - 1/n \to \alpha$ when $\alpha \in \mathbb{R}$, while $\alpha_n = -n$ when $\alpha = -\infty$ and $\alpha_n = n$ when $\alpha = +\infty$.

Step 4: the rearrangement. Define integers $0 = m_0 < m_1 < m_2 < \cdots$ and $0 = k_0 < k_1 < k_2 < \cdots$ recursively. Given m_{n-1} and k_{n-1}, let m_n be the smallest integer greater than m_{n-1} for which

$$x_n := P_1 + \cdots + P_{m_1} - Q_1 - \cdots - Q_{k_1} + \cdots + P_{m_{n-1}+1} + \cdots + P_{m_n} > \beta_n$$

and then let k_n be the smallest integer greater than k_{n-1} for which

$$y_n := x_n - Q_{k_{n-1}+1} - \cdots - Q_{k_n} < \alpha_n$$

Both minima exist: by Step 2 the partial sums of the tails $\sum_{j>m_{n-1}} P_j$ and $\sum_{j>k_{n-1}} Q_j$ are unbounded above, so the running sum can be pushed above β_n by adding enough terms P_j and then below α_n by subtracting enough terms Q_j. Listing the terms in the order

$$P_1, \ldots, P_{m_1},\ -Q_1, \ldots, -Q_{k_1},\ P_{m_1+1}, \ldots, P_{m_2},\ -Q_{k_1+1}, \ldots, -Q_{k_2},\ \ldots$$

defines a map $\sigma\colon \mathbb{N} \to \mathbb{N}$ sending the position in this list to the original index of the term standing there. It is injective because each P_j and each Q_j is listed once, and surjective because $m_n \to \infty$ and $k_n \to \infty$, so every P_j and every Q_j eventually appears. Thus σ is a bijection, and

(s'_N) denotes the partial sums of the rearranged series. By construction $y_0 = 0$, and x_n and y_n are among the s'_N.

Step 5: the two estimates. Fix $n \geq 2$. The stopping rule gives $x_n > \beta_n$. For the reverse bound, note that the partial sum standing one place before x_n in the list is $x_n - P_{m_n}$, and that it is at most β_n: if $m_n > m_{n-1} + 1$ this is the minimality of m_n, and if $m_n = m_{n-1} + 1$ that partial sum is y_{n-1}, which satisfies $y_{n-1} < \alpha_{n-1} < \beta_n$ by Step 3. Hence

$$|x_n - \beta_n| = x_n - \beta_n \leq P_{m_n} \qquad (n \geq 2)$$

Similarly $y_n < \alpha_n$ for every $n \geq 1$, and the partial sum one place before y_n is $y_n + Q_{k_n}$, which is at least α_n: if $k_n > k_{n-1} + 1$ this is the minimality of k_n, and if $k_n = k_{n-1} + 1$ that partial sum is $x_n > \beta_n > \alpha_n$. Hence

$$|y_n - \alpha_n| = \alpha_n - y_n \leq Q_{k_n} \qquad (n \geq 1)$$

Step 6: the limits. Call the positions occupied by $P_{m_{n-1}+1}, \ldots, P_{m_n}, -Q_{k_{n-1}+1}, \ldots, -Q_{k_n}$ the n-th block. Every index $N \geq 1$ lies in exactly one block. Every P_j is nonnegative and every Q_j is nonnegative, so inside the P-part of the n-th block the partial sums are nondecreasing, running from y_{n-1} up to x_n, and inside the Q-part they are nonincreasing, running from x_n down to y_n. Hence

$$\min\{y_{n-1}, y_n\} \leq s'_N \leq x_n \qquad \text{for every } N \text{ in the } n\text{-th block}$$

The inequality $\limsup_N s'_N \leq \limsup_n x_n$. If $\limsup_n x_n = +\infty$ there is nothing to prove, so let $t \in \mathbb{R}$ satisfy $t > \limsup_n x_n$. By theorem 2.2.12(1) applied to (x_n) there is an n_0 with $x_n < t$ for every $n > n_0$. The blocks $1, \ldots, n_0$ occupy finitely many positions, so there is an N_0 with every $N \geq N_0$ lying in a block of index $n > n_0$, and for such N the display gives $s'_N \leq \sup_{n > n_0} x_n \leq t$. Comparing (s'_N) with the constant sequence t in proposition 2.2.13(1) gives $\limsup_N s'_N \leq t$. As t was an arbitrary real above $\limsup_n x_n$, the inequality follows.

The reverse inequality. This is immediate from definition 2.2.10 alone, with no tail formula: (x_n) is a subsequence of (s'_N), so every subsequential limit of (x_n) is a subsequential limit of (s'_N), whence the set E_x is contained in $E_{s'}$ and $\limsup_n x_n = \sup E_x \leq \sup E_{s'} = \limsup_N s'_N$.

The two arguments applied to (y_n), with the inequalities reversed and theorem 2.2.12(1) replaced by its lim inf counterpart, give $\liminf_N s'_N = \liminf_n y_n$. Therefore $\limsup_N s'_N = \limsup_n x_n$ and $\liminf_N s'_N = \liminf_n y_n$.

It remains to evaluate these two quantities, and Step 5 does it in each of the three cases for β. If $\beta \in \mathbb{R}$, then $\beta_n \to \beta$ by Step 3 and $x_n - \beta_n \to 0$ by theorem 2.1.7 applied to $0 \leq x_n - \beta_n \leq P_{m_n}$ for $n \geq 2$, so $x_n \to \beta$ by theorem 2.1.6, and $\limsup_n x_n = \beta$ by theorem 2.2.12. If $\beta = +\infty$, then $x_n > \beta_n \geq n$, so $\sup_{n>n_0} x_n = +\infty$ for every n_0 and $\limsup_n x_n = +\infty$. If $\beta = -\infty$, then (P_{m_n}) converges to 0 and is therefore bounded by some G (proposition 2.1.5), so $x_n \leq \beta_n + P_{m_n} \leq -n + 1 + 1/n + G$ for $n \geq 2$, and since $1/n \leq 1$ there, $\sup_{n>n_0} x_n \leq -n_0 + 1 + G$ for every $n_0 \geq 1$, whence $\limsup_n x_n = -\infty$. The three cases for α are identical with (y_n), (α_n) and (Q_{k_n}) in place of (x_n), (β_n) and (P_{m_n}), using $y_n < \alpha_n = -n$ when $\alpha = -\infty$ and $y_n \geq \alpha_n - Q_{k_n} \geq n - G'$ when $\alpha = +\infty$, where G' bounds (Q_{k_n}). In every case $\limsup_n x_n = \beta$ and $\liminf_n y_n = \alpha$, which completes the proof.

Taking $\alpha = \beta = s$ for a real number s makes the limit inferior and the limit superior of the rearranged partial sums both equal to s, so by theorem 2.2.12 the rearrangement converges to s: every real number is the sum of some rearrangement of a given conditionally convergent series. Taking $\alpha = \beta = +\infty$ produces a rearrangement whose partial sums tend to $+\infty$, and taking $\alpha = -\infty$, $\beta = +\infty$ produces one whose partial sums are unbounded in both directions. Together with theorem 2.5.5, this makes the distinction between absolute and conditional convergence exact: a convergent series has a sum independent of the order of its terms if and only if it converges absolutely.

The proof reorders the terms by an algorithm, and does not exhibit the reordering in closed form. For the alternating harmonic series there is a reordering with a simple description whose sum can be computed exactly, and computed without any information about what the original sum is.

Example 2.5.7: The Alternating Harmonic Series Rearranged Two Positive, One Negative

Let $s = \sum_{n\geq 1}(-1)^{n+1}/n$, which exists by corollary 2.5.3, and let s_N be its partial sums. Consider the rearrangement that takes two positive terms, then one negative term, and repeats:

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots$$

This is a rearrangement in the sense of definition 2.5.4: the reciprocals of the odd numbers appear once each, in order, and the negatives of the reciprocals of the even numbers appear once each, in order.

Write $H_n = \sum_{j=1}^{n} 1/j$ and let t_N denote the partial sums of the rearranged series. After k groups the terms used are $1, 1/3, \ldots, 1/(4k-1)$ and $-1/2, -1/4, \ldots, -1/(2k)$, so

$$t_{3k} = \sum_{j=1}^{2k} \frac{1}{2j-1} - \sum_{j=1}^{k} \frac{1}{2j}$$

The first sum is H_{4k} with the even-denominator terms removed, and those terms total $\sum_{j=1}^{2k} 1/(2j) = \frac{1}{2}H_{2k}$, so it equals $H_{4k} - \frac{1}{2}H_{2k}$. The second sum is $\frac{1}{2}H_k$. Hence

$$t_{3k} = H_{4k} - \tfrac{1}{2}H_{2k} - \tfrac{1}{2}H_k = (H_{4k} - H_{2k}) + \tfrac{1}{2}\left(H_{2k} - H_k\right)$$

Now $s_{2n} = H_{2n} - H_n$ for every n, because $s_{2n} = \sum_{m=1}^{2n} 1/m - 2\sum_{j=1}^{n} 1/(2j) = H_{2n} - H_n$. Applying this identity with $n = 2k$ to the first bracket of the previous display, and with $n = k$ to the second, gives

$$t_{3k} = s_{4k} + \tfrac{1}{2}s_{2k}$$

Both s_{4k} and s_{2k} converge to s by proposition 2.2.7, so $t_{3k} \to s + \frac{1}{2}s = \frac{3}{2}s$ by theorem 2.1.6. The two intermediate partial sums satisfy $t_{3k+1} = t_{3k} + 1/(4k+1)$ and $t_{3k+2} = t_{3k+1} + 1/(4k+3)$, and both correction terms converge to 0, so $t_{3k+1} \to \frac{3}{2}s$ and $t_{3k+2} \to \frac{3}{2}s$ as well. Every index $N \geq 1$ is of exactly one of the three forms $3k$, $3k+1$, $3k+2$, so given $\epsilon > 0$ each of the three subsequences lies within ϵ of $\frac{3}{2}s$ from some index onward, and beyond the largest of those three indices every t_N does. Hence $t_N \to \frac{3}{2}s$.

Finally $\frac{3}{2}s \neq s$, since corollary 2.5.3 gives $s \geq s_2 = 1/2 > 0$. The rearranged series therefore converges, and to a different number: reordering has multiplied the sum by $3/2$.

The rest of the section concerns two further interchanges, both of which are governed by absolute convergence in the same way theorem 2.5.5 is. The first is the interchange of two summations.

Theorem 2.5.8: Fubini's Theorem for Double Series

Let $a_{ij} \in \mathbb{R}$ be given for all integers $i, j \geq 1$.

1. For all $M, N \geq 1$, $\sum_{i=1}^{M} \sum_{j=1}^{N} a_{ij} = \sum_{j=1}^{N} \sum_{i=1}^{M} a_{ij}$.
2. Suppose that for each i the series $\sum_{j\geq 1} |a_{ij}|$ converges, with sum b_i, and that $\sum_{i\geq 1} b_i$ converges. Then every row sum $r_i = \sum_{j\geq 1} a_{ij}$ and every column sum $c_j = \sum_{i\geq 1} a_{ij}$ is defined, the series $\sum_{i\geq 1} r_i$ and $\sum_{j\geq 1} c_j$ both converge absolutely, and

$$\sum_{i\geq 1}\left(\sum_{j\geq 1} a_{ij}\right) = \sum_{j\geq 1}\left(\sum_{i\geq 1} a_{ij}\right)$$

***Proof*:**

1. Fix N and induct on M. For $M = 1$ both sides equal $\sum_{j=1}^{N} a_{1j}$. Assume the identity for M. Splitting each finite sum $\sum_{i=1}^{M+1}$ into $\sum_{i=1}^{M}$ plus the term $i = M + 1$, and using that a finite sum of finite sums may be split termwise,
$$\sum_{j=1}^{N}\sum_{i=1}^{M+1} a_{ij} = \sum_{j=1}^{N}\left(\sum_{i=1}^{M} a_{ij} + a_{M+1,j}\right) = \sum_{j=1}^{N}\sum_{i=1}^{M} a_{ij} + \sum_{j=1}^{N} a_{M+1,j}$$
By the inductive hypothesis the first term equals $\sum_{i=1}^{M}\sum_{j=1}^{N} a_{ij}$, and adding the second term gives $\sum_{i=1}^{M+1}\sum_{j=1}^{N} a_{ij}$. This is the identity for $M + 1$.
2. Write $S = \sum_{i\geq 1} b_i$ and record once a bound used several times below. If $\sum_{i\geq 1} u_i$ and $\sum_{i\geq 1}|u_i|$ both converge, then applying the finite triangle inequality proposition 2.4.5(2) in the form $-\sum_{i=1}^{M}|u_i| \leq \sum_{i=1}^{M} u_i \leq \sum_{i=1}^{M}|u_i|$ and letting $M \to \infty$ in both inequalities using proposition 2.1.8 gives $\left|\sum_{i\geq 1} u_i\right| \leq \sum_{i\geq 1}|u_i|$.

   The row sums are defined: $\sum_j |a_{ij}|$ converges by hypothesis, so $\sum_j a_{ij}$ converges by theorem 2.4.12, and the bound just recorded gives $|r_i| \leq b_i$. Since $\sum_i b_i$ converges, theorem 2.4.13 makes $\sum_i |r_i|$ convergent, so $\sum_i r_i$ converges absolutely. Write $R = \sum_{i\geq 1} r_i$.

   The column sums are defined: for fixed j, the number $|a_{ij}|$ is one of the nonnegative terms whose sum is b_i, so $|a_{ij}| \leq b_i$ and therefore $\sum_{i=1}^{M}|a_{ij}| \leq \sum_{i=1}^{M} b_i \leq S$ for every M, the last inequality because the partial sums of the nonnegative convergent series $\sum_i b_i$ are bounded by its sum (proposition 2.1.8). Hence $\sum_i |a_{ij}|$ converges by proposition 2.4.4 and $\sum_i a_{ij}$ converges. For absolute convergence of $\sum_j c_j$, fix N. Then $|c_j| \leq \sum_{i\geq 1}|a_{ij}|$ by the recorded bound, and by theorem 2.1.6 applied to the N convergent sequences involved, together with part (1),
$$\sum_{j=1}^{N}\sum_{i\geq 1}|a_{ij}| = \lim_{M\to\infty}\sum_{j=1}^{N}\sum_{i=1}^{M}|a_{ij}| = \lim_{M\to\infty}\sum_{i=1}^{M}\sum_{j=1}^{N}|a_{ij}| \leq S$$
the last step because each term of the sequence is at most $\sum_{i=1}^{M} b_i \leq S$, so proposition 2.1.8 bounds the limit by S. Hence $\sum_{j=1}^{N}|c_j| \leq S$ for every N, and $\sum_j c_j$ converges absolutely by proposition 2.4.4.

It remains to prove $\sum_{j\geq 1} c_j = R$. The same interchange of a finite sum with a limit, now without absolute values, gives for each N

$$\sum_{j=1}^{N} c_j = \lim_{M\to\infty} \sum_{i=1}^{M}\sum_{j=1}^{N} a_{ij} = \sum_{i\geq 1}\left(\sum_{j=1}^{N} a_{ij}\right)$$

the last series converging because the limit exists. Subtracting $R = \sum_i r_i$ term by term, which proposition 2.4.5 permits since both series converge,

$$\sum_{j=1}^{N} c_j - R = \sum_{i\geq 1}\left(\sum_{j=1}^{N} a_{ij} - r_i\right) = -\sum_{i\geq 1}\rho_i^{(N)}, \qquad \rho_i^{(N)} := \sum_{j>N} a_{ij}$$

where $\rho_i^{(N)}$ is the tail of a convergent series and satisfies $|\rho_i^{(N)}| \leq \sum_{j>N}|a_{ij}| \leq b_i$ by the recorded bound.

Let $\epsilon > 0$. Since $\sum_i b_i$ converges, choose I with $\sum_{i>I} b_i \leq \epsilon$. For each $i \leq I$ the series $\sum_j |a_{ij}|$ converges, so its tails tend to 0 and there is a J_i with $\sum_{j>J_i}|a_{ij}| \leq \epsilon/I$. Put $J = \max\{J_1, \ldots, J_I\}$ and fix $N \geq J$. Then for each $i \leq I$, $|\rho_i^{(N)}| \leq \sum_{j>N}|a_{ij}| \leq \sum_{j>J_i}|a_{ij}| \leq \epsilon/I$, while for $i > I$ only $|\rho_i^{(N)}| \leq b_i$ is used. Applying the recorded bound to $\sum_i \rho_i^{(N)}$, whose absolute values are dominated by the summable b_i, and splitting the resulting sum at $i = I$,

$$\left|\sum_{j=1}^{N} c_j - R\right| \leq \sum_{i=1}^{I}\left|\rho_i^{(N)}\right| + \sum_{i>I} b_i \leq I\cdot\frac{\epsilon}{I} + \epsilon = 2\epsilon$$

Since $\epsilon > 0$ was arbitrary, $\sum_{j\geq 1} c_j = R$, which is the asserted equality of the two iterated sums, completing the proof.

Part (1) is the statement that a finite rectangular array may be added by rows or by columns, and part (2) says that the same holds for an infinite array as soon as the row sums of absolute values are themselves summable. Without that hypothesis the two iterated sums can differ, and exercise 2.5.1 (8) asks for an array where they do.

The second interchange multiplies two series. Expanding the product of two finite sums and collecting the terms $a_i b_j$ by the value of $i + j$ produces the following series, and the question is whether it converges to the product of the two sums when the sums are infinite.

Theorem 2.5.9: Mertens' Theorem on Cauchy Products

Let $\sum_{n\geq 1} a_n$ converge absolutely with sum A, and let $\sum_{n\geq 1} b_n$ converge with sum B. Define the *Cauchy product* of the two series to be $\sum_{n\geq 1} c_n$, where

$$c_n = \sum_{k=1}^{n} a_k b_{n+1-k}$$

so that c_n collects exactly the products $a_i b_j$ with $i + j = n + 1$. Then $\sum_{n\geq 1} c_n$ converges, with sum AB.

***Proof*:**

Write $A_N = \sum_{n=1}^{N} a_n$, $B_N = \sum_{n=1}^{N} b_n$ and $C_N = \sum_{n=1}^{N} c_n$, and put $\gamma_N = B_N - B$, so that $\gamma_N \to 0$ by theorem 2.1.6.

Step 1: an expression for C_N. Define $d_{nk} = a_k b_{n+1-k}$ for $1 \leq k \leq n \leq N$ and $d_{nk} = 0$ for $1 \leq n < k \leq N$. Then $c_n = \sum_{k=1}^{N} d_{nk}$ for each $n \leq N$, so by theorem 2.5.8(1) applied to the square array (d_{nk}),

$$C_N = \sum_{n=1}^{N}\sum_{k=1}^{N} d_{nk} = \sum_{k=1}^{N}\sum_{n=1}^{N} d_{nk} = \sum_{k=1}^{N} a_k \sum_{n=k}^{N} b_{n+1-k} = \sum_{k=1}^{N} a_k B_{N+1-k}$$

where the inner sum was re-indexed by $j = n + 1 - k$, which runs over $1 \leq j \leq N + 1 - k$. Substituting $B_{N+1-k} = B + \gamma_{N+1-k}$ and separating the two parts,

$$C_N = BA_N + \delta_N, \qquad \delta_N := \sum_{k=1}^{N} a_k \gamma_{N+1-k}$$

Step 2: $\delta_N \to 0$. Let $\tau = \sum_{n\geq 1}|a_n|$, finite by hypothesis, and let $\tau_N = \sum_{n=1}^{N}|a_n|$. The sequence (γ_m) converges, hence is bounded (proposition 2.1.5), say $|\gamma_m| \leq G$ for all m.

Let $\epsilon > 0$. Choose M with $|\gamma_m| \leq \epsilon$ for all $m > M$. Since $\tau_N \to \tau$, choose $N_0 > M$ with $\tau_N - \tau_{N-M} \leq \epsilon$ for all $N \geq N_0$, which is possible because $\tau_N - \tau_{N-M} \to \tau - \tau = 0$ by theorem 2.1.6. Fix $N \geq N_0$ and split the sum defining δ_N at $k = N - M$. For $1 \leq k \leq N - M$ the index $N + 1 - k$ exceeds M, so $|\gamma_{N+1-k}| \leq \epsilon$, while for $N - M < k \leq N$ only the crude bound G is used. The finite triangle inequality proposition 2.4.5(2) then gives

$$|\delta_N| \leq \epsilon \sum_{k=1}^{N-M} |a_k| + G \sum_{k=N-M+1}^{N} |a_k| \leq \epsilon\tau + G\,(\tau_N - \tau_{N-M}) \leq \epsilon\,(\tau + G)$$

As $\tau + G$ does not depend on ϵ and $\epsilon > 0$ was arbitrary, $\delta_N \to 0$.

By Step 1 and theorem 2.1.6, $C_N = BA_N + \delta_N \to BA + 0 = AB$, completing the proof.

Absolute convergence is assumed of one factor only, and the theorem is silent when neither factor converges absolutely. It has to be: two conditionally convergent series can have a Cauchy product whose terms do not even tend to 0.

Example 2.5.10: A Divergent Cauchy Product of Two Convergent Series

Let $a_n = b_n = (-1)^{n+1}/\sqrt{n}$ for $n \geq 1$, where $\sqrt{n}$ is the nonnegative square root given by theorem 1.3.12. Square roots preserve the order on $[0,\infty)$: if $0 \leq x < y$ then $x^2 < y^2$ by proposition 1.1.2, so $\sqrt{u} > \sqrt{v} \geq 0$ would force $u > v$, and therefore $0 \leq u \leq v$ implies $\sqrt{u} \leq \sqrt{v}$. The sequence $1/\sqrt{n}$ decreases to 0, so $\sum_{n\geq 1} a_n$ converges by corollary 2.5.3. It does not converge absolutely: $n \leq n^2$ for $n \geq 1$ gives $\sqrt{n} \leq n$ by the order fact just recorded, hence $1/\sqrt{n} \geq 1/n$ by proposition 1.1.2, and theorem 2.4.13 with example 2.4.8 makes $\sum_n 1/\sqrt{n}$ divergent. So both factors are conditionally convergent.

Their Cauchy product has terms

$$c_n = \sum_{k=1}^{n} \frac{(-1)^{k+1}}{\sqrt{k}} \cdot \frac{(-1)^{n+2-k}}{\sqrt{n+1-k}} = (-1)^{n+1} \sum_{k=1}^{n} \frac{1}{\sqrt{k}\sqrt{n+1-k}}$$

since $(-1)^{k+1}(-1)^{n+2-k} = (-1)^{n+3} = (-1)^{n+1}$. For $1 \leq k \leq n$, the uniqueness clause of theorem 1.3.12 identifies $\sqrt{k}\sqrt{n+1-k}$ with $\sqrt{k(n+1-k)}$, because the left-hand side is nonnegative and its square is $k(n+1-k)$. Writing $m = n+1$, the algebraic identity

$$\frac{m^2}{4} - k(m-k) = \frac{m^2 - 4km + 4k^2}{4} = \frac{(m-2k)^2}{4} \geq 0$$

gives $k(n+1-k) \leq \left(\frac{n+1}{2}\right)^2$. By the order fact recorded at the start, $\sqrt{k(n+1-k)} \leq \frac{n+1}{2}$, so each of the n summands is at least $\frac{2}{n+1}$ and

$$|c_n| \geq n \cdot \frac{2}{n+1} = \frac{2n}{n+1} \geq 1$$

for every $n \geq 1$. The terms c_n therefore do not converge to 0, and $\sum_{n\geq 1} c_n$ diverges by proposition 2.4.3.

Exercise 2.5.1

1. Let $s = \sum_{n\geq 1}(-1)^{n+1}/n$. Using only corollary 2.5.3, prove the bounds $1/2 \leq s \leq 1$ and then the sharper bounds $7/12 \leq s \leq 5/6$. Show more generally that for a sequence $b_1 \geq b_2 \geq \cdots$ with $b_n \to 0$, the sum s of $\sum_{n\geq 1}(-1)^{n+1}b_n$ satisfies $b_1 - b_2 \leq s \leq b_1$.

2. Let $p \geq 1$ be an integer and let $(a_n)_{n\geq 1}$ be periodic with period p, meaning $a_{n+p} = a_n$ for every n, and suppose $a_1 + a_2 + \cdots + a_p = 0$. Prove that $\sum_{n\geq 1} a_n/n$ converges. Prove furthermore that if $a_n \neq 0$ for at least one n, then the convergence is conditional in the sense of definition 2.5.4.

3. Show that the monotonicity hypothesis (2) of theorem 2.5.2 cannot be dropped, even when the remaining hypotheses are strengthened to $b_n > 0$ for all n. Specifically, let $a_n = (-1)^{n+1}$ and let $b_n = 1/n$ for odd n and $b_n = 1/n^2$ for even n. Verify that $b_n \to 0$ and that (A_N) is bounded, and prove that $\sum_{n\geq 1} a_n b_n$ diverges.

4. With s as in exercise 2.5.1 (1), consider the rearrangement of $\sum_{n\geq 1}(-1)^{n+1}/n$ that takes one

positive term followed by two negative terms and repeats:

$$1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \frac{1}{5} - \frac{1}{10} - \frac{1}{12} + \cdots$$

Show that it converges, and that its sum is $\frac{1}{2}s$. (Follow the pattern of example 2.5.7, computing the partial sum after k groups in terms of $H_n = \sum_{j=1}^{n} 1/j$.)

5. Prove *Abel's test*: if $\sum_{n\ge1} a_n$ converges and (b_n) is monotone and bounded, then $\sum_{n\ge1} a_n b_n$ converges. (Treat the nonincreasing and nondecreasing cases separately. For a nonincreasing (b_n) write $b_n = b + c_n$ with $b = \lim_n b_n$, so that (c_n) is nonincreasing with limit 0. For a nondecreasing (b_n) put $c_n = b - b_n$ instead. Either way (c_n) is nonnegative, nonincreasing and tends to 0.)

6. Let $|x| < 1$. Compute the Cauchy product of $\sum_{n\ge1} x^{n-1}$ with itself, and use theorem 2.5.9 to conclude that $\sum_{n\ge1} nx^{n-1}$ converges with sum $1/(1-x)^2$. Explain why theorem 2.5.9 applies.

7. Suppose $\sum_{n\ge1} a_n$ converges and every rearrangement of it converges as well. Prove that $\sum_{n\ge1} a_n$ converges absolutely. (The converse is theorem 2.5.5.)

8. Define $a_{ij} = 1$ if $i = j$, $a_{ij} = -1$ if $i = j+1$, and $a_{ij} = 0$ otherwise. Compute every row sum $r_i = \sum_{j\ge1} a_{ij}$ and every column sum $c_j = \sum_{i\ge1} a_{ij}$, and deduce that $\sum_i \sum_j a_{ij} \ne \sum_j \sum_i a_{ij}$. Identify precisely which hypothesis of theorem 2.5.8(2) fails.

2.6 Power Series and the Radius of Convergence

The tests of section 2.4 decide one series at a time. A power series presents a whole family at once, one series for each real x, and the question is which members of the family converge. The answer is a dichotomy with a single exception: there is a number R computed from the coefficients alone, such that the series converges absolutely at every x within distance R of the centre and diverges at every x farther away. Only the two points at distance exactly R are left undecided, and example 2.6.4 exhibits all three of the behaviours that can occur there.

The same computation is what makes the exponential and the trigonometric functions available. Each is defined below by a power series whose radius is infinite, and the identity $\exp(x+y) = \exp(x)\exp(y)$, which is what makes exp an exponential, is the Cauchy product of section 2.5 evaluated with the binomial theorem.

The coefficient whose n-th root is wanted is usually a constant, a power of n, or a product of the two. Two limits cover both cases, and both of their proofs use one fact about powers: for $s, t \ge 0$ and an integer $n \ge 1$,

$$s < t \quad \Rightarrow \quad s^n < t^n \tag{2.5}$$

For $n = 1$ this is the hypothesis itself. If it holds for n, then $t > s \ge 0$ gives $t > 0$, so multiplying $s^n < t^n$ by $s \ge 0$ gives $s^{n+1} \le s\,t^n$ and multiplying $s < t$ by $t^n > 0$ gives $s\,t^n < t^{n+1}$, both by proposition 1.1.2(5), whence $s^{n+1} < t^{n+1}$. Taking contrapositives, $s^n \le t^n$ implies $s \le t$ for $s, t \ge 0$. Together with theorem 1.3.12, which produces a unique nonnegative n-th root of every nonnegative real, (2.5) says that raising to the n-th power and extracting the nonnegative n-th root both preserve the order on the nonnegative reals.

Proposition 2.6.1: Two Special Limits

1. $\lim_{n\to\infty} n^{1/n} = 1$.
2. $\lim_{n\to\infty} c^{1/n} = 1$ for every real $c > 0$.

***Proof*:**

1. For $n \geq 1$ put $x_n = n^{1/n} - 1$. If $n^{1/n} < 1$ then (2.5), together with the identity $(n^{1/n})^n = n$ of theorem 1.3.12, gives $n = (n^{1/n})^n < 1^n = 1$, which contradicts $n \geq 1$, so $x_n \geq 0$. Let $n \geq 2$. The binomial theorem [1, the binomial theorem] expands

   $$n = (1 + x_n)^n = \sum_{k=0}^{n} \binom{n}{k} x_n^k \;\geq\; \binom{n}{2} x_n^2 = \frac{n(n-1)}{2}\, x_n^2$$

   where the inequality discards terms that are all nonnegative, since $x_n \geq 0$. Dividing by the positive number $n(n-1)/2$ gives

   $$0 \;\leq\; x_n^2 \;\leq\; \frac{2}{n-1}$$

   Now let $\epsilon > 0$. By theorem 1.4.1(2) there is an $N \in \mathbb{N}$ with $N > 1 + 2/\epsilon^2$, and $N \geq 2$ because $1 + 2/\epsilon^2 > 1$. For $n \geq N$ one has $n - 1 > 2/\epsilon^2$ and therefore $x_n^2 \leq 2/(n-1) < \epsilon^2$. Were $x_n \geq \epsilon$, then (2.5) with exponent 2 would give $x_n^2 \geq \epsilon^2$, so in fact $x_n < \epsilon$. Thus $|n^{1/n} - 1| = x_n < \epsilon$ for all $n \geq N$, which is definition 2.1.2 for the limit 1.

2. For $c = 1$ the sequence is constant, since $1^n = 1$ and n-th roots are unique by theorem 1.3.12.

   Let $c > 1$ and put $y_n = c^{1/n} - 1$. If $c^{1/n} \leq 1$ then the contrapositive form of (2.5) gives $c = (c^{1/n})^n \leq 1^n = 1$, contradicting $c > 1$, so $y_n > 0$. The binomial expansion of $(1 + y_n)^n$ has nonnegative terms, and discarding all but the first two of them gives $c = (1 + y_n)^n \geq 1 + ny_n$, so

   $$0 \;<\; y_n \;\leq\; \frac{c-1}{n}$$

   Given $\epsilon > 0$, choose $N \in \mathbb{N}$ with $N > (c-1)/\epsilon$ by theorem 1.4.1(2). For $n \geq N$ one has $1/n \leq 1/N$ by proposition 1.1.2(7), so $y_n \leq (c-1)/N < \epsilon$, the last inequality being $N > (c-1)/\epsilon$ multiplied by the positive number ϵ/N. Hence $c^{1/n} \to 1$.

   Finally let $0 < c < 1$ and put $d = 1/c$, so that $d > 1$ by proposition 1.1.2(7). The number $1/d^{1/n}$ is positive and satisfies $(1/d^{1/n})^n = 1/(d^{1/n})^n = 1/d = c$, so uniqueness in theorem 1.3.12 identifies it with $c^{1/n}$. The previous case gives $d^{1/n} \to 1$, and since the limit 1 is not 0 the quotient rule of theorem 2.1.6 gives $c^{1/n} = 1/d^{1/n} \to 1$, as we sought to show.

The two limits together control every coefficient sequence built from a constant and a power of n. For $c > 0$ and an integer k, the number $c^{1/n}\,(n^{1/n})^k$ is positive and its n-th power is $c\,n^k$, so uniqueness in theorem 1.3.12 gives $(c\,n^k)^{1/n} = c^{1/n}\,(n^{1/n})^k$, and theorem 2.1.6 then sends the right-hand side to 1. Coefficients of that shape all produce the same radius below, whatever c and k are.

A power series is written down by naming its coefficients, and everything about its convergence is computable from them. The definition accordingly attaches a number to the coefficient sequence first, and the theorem afterwards says what that number does.

Definition 2.6.2: Power Series and its Radius of Convergence

Let $a \in \mathbb{R}$ and let $(c_n)_{n\geq 0}$ be real numbers. The *power series with coefficients* (c_n) *centred at* a assigns to each $x \in \mathbb{R}$ the series

$$\sum_{n\geq 0} c_n(x-a)^n$$

in which $(x-a)^0$ is read as 1 for every x, including $x = a$. Put

$$\alpha = \limsup_{n\to\infty} |c_n|^{1/n}$$

the limit superior being taken over $n \geq 1$, where the root is defined. Every term of that sequence is nonnegative, so $\alpha \in [0, +\infty]$. The *radius of convergence* of the power series is

$$R = \begin{cases} +\infty & \text{if } \alpha = 0 \\ 1/\alpha & \text{if } 0 < \alpha < +\infty \\ 0 & \text{if } \alpha = +\infty \end{cases}$$

Two conventions apply below. The coefficients are indexed from $n = 0$, where section 2.4 started at $n = 1$, and every statement of section 2.4 and section 2.5 about a series $\sum_{n\geq 1} a_n$ applies after the relabelling $b_n = a_{n-1}$, which leaves the sequence of partial sums unchanged apart from its indexing. That relabelling is left implicit below. The extended values 0 and $+\infty$ for α are the ones given by definition 2.2.4, and both of them occur.

Take $a = 0$ and $c_n = 1$ for every n, so that the power series is $\sum_{n\geq 0} x^n$. The sequence $|c_n|^{1/n}$ is constantly 1, and a convergent sequence has its limit for a limit superior: by proposition 2.2.7 every subsequence of a convergent sequence has the same limit, while by theorem 2.2.12 the limit superior is the limit of some subsequence. So $\alpha = 1$ and $R = 1$. The geometric series of example 2.4.7 converges for $|x| < 1$, and for $|x| \geq 1$ the contrapositive form of (2.5) gives $|x^n| = |x|^n \geq 1^n = 1$, so the terms do not tend to 0 and proposition 2.4.3 makes the series divergent. The set of convergence is exactly $(-1, 1)$, which is the pattern the next theorem asserts in general.

Theorem 2.6.3: Cauchy-Hadamard: Convergence Inside the Radius, Divergence Outside

Let $\sum_{n\geq 0} c_n(x-a)^n$ be a power series with radius of convergence R as in definition 2.6.2, and let $x \in \mathbb{R}$.

1. If $|x-a| < R$, the series converges absolutely.
2. If $|x-a| > R$, the series diverges.

***Proof*:**
Write $\alpha = \limsup_{n\to\infty} |c_n|^{1/n}$, so that R is the value attached to α in definition 2.6.2.

1. Suppose $|x-a| < R$. Then $R > 0$, so $\alpha < +\infty$. The first task is to produce a real ρ with

$$|x-a| < \rho \qquad \text{and} \qquad \frac{1}{\rho} > \alpha$$

If $\alpha = 0$, take $\rho = |x-a| + 1$, which is positive, so that $1/\rho > 0 = \alpha$. If $0 < \alpha < +\infty$ then $R = 1/\alpha$ and $|x-a| < 1/\alpha$, so $\rho = (|x-a| + 1/\alpha)(1+1)^{-1}$ satisfies $|x-a| < \rho < 1/\alpha$ by proposition 1.1.2(8), and then $0 < \rho < 1/\alpha$ gives $1/\rho > \alpha$ by proposition 1.1.2(7).
Since $1/\rho > \alpha$, the eventual domination clause of theorem 2.2.12 gives an $N \in \mathbb{N}$ with $|c_n|^{1/n} < 1/\rho$ for all $n \geq N$. Raising this to the n-th power with (2.5), and using theorem 1.3.12 to write $|c_n| = (|c_n|^{1/n})^n$, gives $|c_n| < \rho^{-n}$, that is $|c_n|\rho^n < 1$, for all $n \geq N$.
Put $r = |x-a|/\rho$, so that $0 \leq r < 1$, and put

$$M = 1 + \sum_{n=0}^{N-1} |c_n|\rho^n$$

a finite sum of nonnegative numbers increased by 1. For $n < N$ the number $|c_n|\rho^n$ is one of the summands, so $|c_n|\rho^n \leq M$, and for $n \geq N$ the previous paragraph gives $|c_n|\rho^n < 1 \leq M$. Since $|x-a| = \rho r$, applying the multiplicativity of proposition 1.3.8(3) n times yields, for every $n \geq 0$,

$$|c_n(x-a)^n| = |c_n|\,|x-a|^n = \big(|c_n|\rho^n\big) r^n \ \leq \ M r^n$$

The geometric series $\sum_{n\geq 0} r^n$ converges because $0 \leq r < 1$ (example 2.4.7), so $\sum_{n\geq 0} M r^n$ converges by the scalar rule of proposition 2.4.5, and theorem 2.4.13 then makes $\sum_{n\geq 0} |c_n(x-a)^n|$ convergent. That is absolute convergence in the sense of definition 2.4.11.

2. Suppose $|x-a| > R$. Then $|x-a| > 0$, so $\beta = 1/|x-a|$ is a positive real, and $\beta < \alpha$ in each of the three cases of definition 2.6.2: if $\alpha = 0$ then $R = +\infty$ and no x satisfies the hypothesis, so there is nothing to prove; if $0 < \alpha < +\infty$ then $|x-a| > R = 1/\alpha > 0$, which proposition 1.1.2(7) inverts to $\beta < \alpha$; if $\alpha = +\infty$ then $\beta < \alpha$ because β is a real number.
By theorem 2.2.12 the value α is the limit of a subsequence $(|c_{n_k}|^{1/n_k})_{k\geq 1}$, in the extended sense that this subsequence exceeds every bound when $\alpha = +\infty$. In either case there is a K with $|c_{n_k}|^{1/n_k} > \beta$ for all $k \geq K$: for finite α apply definition 2.1.2 with $\epsilon = \alpha - \beta > 0$, and for $\alpha = +\infty$ apply the unboundedness with the bound β.
Raising this to the n_k-th power with (2.5) gives $|c_{n_k}| > \beta^{n_k}$, and multiplying by the positive number $|x-a|^{n_k}$ (proposition 1.1.2(5)) gives

$$|c_{n_k}(x-a)^{n_k}| > \beta^{n_k}|x-a|^{n_k} = \big(\beta|x-a|\big)^{n_k} = 1$$

for every $k \geq K$. So infinitely many terms of the series have absolute value larger than 1. Were the terms to tend to 0, all but finitely many would have absolute value below 1, so the terms do not tend to 0, and proposition 2.4.3 makes the series divergent, completing the proof.

The theorem justifies the word radius. Write S for the set of $|x-a|$ with x a point at which the series converges. Part (1) gives $[0,R) \subseteq S$, part (2) gives $S \subseteq [0,R]$, and $0 \in S$ because at $x=a$ every term after the first vanishes and the partial sums are constantly c_0. When $R=+\infty$ the first inclusion makes S unbounded. When R is finite, the second inclusion gives $\sup S \le R$, and the reverse inequality holds as well: an upper bound b of S satisfies $b \ge 0$ because $0 \in S$, and $b < R$ would be contradicted by the number $(b+R)(1+1)^{-1}$, which lies in $[0,R) \subseteq S$ and exceeds b by proposition 1.1.2(8). So $\sup S = R$. When $0 < R < +\infty$ the set of points of convergence therefore contains the open interval $(a-R, a+R)$ and is contained in the closed interval $[a-R, a+R]$, and the two endpoints are the only points theorem 2.6.3 leaves undecided. The degenerate cases were settled above and leave nothing undecided: for $R=+\infty$ the series converges at every real, and for $R=0$ it converges at $x=a$ alone.

Example 2.6.4: Five Radii, and Three Behaviours at an Endpoint
All five power series below are centred at $a=0$.

1. $\sum_{n\ge 0} n!\,x^n$ has $R=0$. Fix $x \ne 0$ and use theorem 1.4.1(2) to choose $N \in \mathbb{N}$ with $N > 2/|x|$, so that $j|x| > N|x| > 2$ for every integer $j > N$. Passing from n to $n+1$ multiplies $|n!\,x^n|$ by $(n+1)|x|$, which exceeds $2 > 1$ once $n \ge N$, so induction gives $|n!\,x^n| \ge |N!\,x^N|$ for every $n \ge N$. Since $|N!\,x^N| > 0$, the terms do not tend to 0, and proposition 2.4.3 gives divergence at every $x \ne 0$. If R were positive there would be an $x \ne 0$ with $|x| < R$, namely $x = R/2$ for finite R and $x=1$ for $R=+\infty$, and part (1) of theorem 2.6.3 would make the series converge there. So $R=0$. The formula of definition 2.6.2 is not evaluated directly here, because the root $(n!)^{1/n}$ is not among the limits available.

2. $\sum_{n\ge 0} x^n/n!$ has $R=+\infty$. At $x=0$ every term after the first vanishes. For $x \ne 0$ every term is nonzero and the ratios are
$$\frac{|x^{n+1}/(n+1)!|}{|x^n/n!|} = \frac{|x|}{n+1}$$
which tend to 0: given $\epsilon > 0$, theorem 1.4.1(2) provides an N with $N > |x|/\epsilon$, and then $|x|/(n+1) < |x|/N < \epsilon$ for $n \ge N$. So theorem 2.4.14 gives absolute convergence at every $x \ne 0$. The series therefore converges at every real point, and a finite R would contradict part (2) of theorem 2.6.3 at any x with $|x| > R$, one of which exists by theorem 1.4.1(2).

3. $\sum_{n\ge 0} x^n$ has $R=1$, since every $|c_n|^{1/n}$ equals 1, a constant sequence converges to its value, and a convergent sequence has its limit for a limit superior (proposition 2.2.7 and theorem 2.2.12). It diverges at both endpoints: the terms at $x=1$ are constantly 1 and the terms at $x=-1$ have absolute value 1, so in neither case do they tend to 0.

4. $\sum_{n\ge 1} x^n/n$ has $R=1$. Here $c_0=0$ and $c_n = 1/n$ for $n \ge 1$, and $1/n^{1/n}$ is positive with n-th power $1/n$, so $|c_n|^{1/n} = 1/n^{1/n}$ by theorem 1.3.12. By proposition 2.6.1(1) and the quotient rule of theorem 2.1.6 this tends to 1, and a convergent sequence has its limit for a limit superior (proposition 2.2.7 and theorem 2.2.12), so $\alpha = 1$ and $R=1$. At $x=1$ the series is the harmonic series and diverges (example 2.4.8). At $x=-1$ it is $\sum_{n\ge 1}(-1)^n/n$, which converges by corollary 2.5.3 because $1/n$ decreases to 0, while $\sum_{n\ge 1} 1/n$ diverges. So the convergence at $x=-1$ is conditional in the sense of definition 2.5.4.

5. $\sum_{n\ge 1} x^n/n^2$ has $R=1$. The number $1/(n^{1/n})^2$ is positive with n-th power $1/n^2$, so it is $|c_n|^{1/n}$ by theorem 1.3.12, and it tends to 1 by proposition 2.6.1(1) and theorem 2.1.6. Hence

$\alpha = 1$ and $R = 1$. At $x = 1$ and at $x = -1$ the series of absolute values is $\sum_{n\geq 1} 1/n^2$, which converges by example 2.4.10, so both endpoints give absolute convergence.

Items (3), (4) and (5) have the same radius and differ at the boundary in all three possible ways: divergence at both endpoints, convergence at one and divergence at the other, and absolute convergence at both. So no statement phrased in terms of R alone can decide an endpoint, and each such point is settled by applying the tests of section 2.4 and section 2.5 to the single numerical series that the endpoint produces.

A power series of infinite radius defines a function on all of $\mathbb{R}$, and the exponential and trigonometric functions are introduced in exactly that way. The three series below are their definitions in this book, and each converges absolutely at every real point.

For exp that is item (2) of example 2.6.4. For the other two, fix $x \in \mathbb{R}$ and consider the cosine series, whose m-th coefficient is $(-1)^n/(2n)!$ when $m = 2n$ is even and 0 when m is odd. Writing $u_n = (-1)^n x^{2n}/(2n)!$ for the terms that can be nonzero, the same ratio computation as in item (2) gives

$$\frac{|u_{n+1}|}{|u_n|} = \frac{x^2}{(2n+1)(2n+2)} \leq \frac{x^2}{2n+1}$$

for $x \neq 0$, and the right-hand side falls below any prescribed $\epsilon > 0$ once $n \geq N$, where $N > x^2/\epsilon$ is chosen by theorem 1.4.1(2), since then $2n + 1 > n \geq N$. The ratios are nonnegative, so they converge to 0, and $\sum_{n\geq 0}|u_n|$ converges by theorem 2.4.14. The partial sum $\sum_{m=0}^{M}|c_m x^m|$ omits the odd terms and so equals $\sum_{n\,:\,2n\leq M}|u_n|$, which is at most $\sum_{n=0}^{M}|u_n|$, and this in turn is at most $\sum_{n\geq 0}|u_n|$, because the partial sums of a series of nonnegative terms increase and proposition 2.1.8, applied to the tail of that increasing sequence, bounds each of them by the limit. Bounded partial sums and nonnegative terms give convergence by proposition 2.4.4, so the cosine series converges absolutely at x, and theorem 2.4.12 makes it convergent. The case $x = 0$ leaves only the term $n = 0$, and the sine series is treated identically with $v_n = (-1)^n x^{2n+1}/(2n+1)!$, whose ratios are $x^2/((2n+2)(2n+3))$. Since all three series converge at every real point, a finite radius R would contradict part (2) of theorem 2.6.3 at any x with $|x| > R$, one of which exists by theorem 1.4.1(2). So all three radii are $+\infty$.

Definition 2.6.5: The Exponential, Cosine and Sine Functions

For $x \in \mathbb{R}$ set

$$\exp(x) = \sum_{n\geq 0} \frac{x^n}{n!}, \qquad \cos x = \sum_{n\geq 0} \frac{(-1)^n x^{2n}}{(2n)!}, \qquad \sin x = \sum_{n\geq 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

with the conventions $0! = 1$ and $x^0 = 1$ for every x. Each of the three series converges absolutely at every real x and has radius of convergence $+\infty$, so each defines a function $\mathbb{R} \to \mathbb{R}$.

Setting $x = 1$ in the first series gives the series of example 2.4.18 term by term, so $\exp(1) = e$. Every coefficient $1/n!$ lies in $\mathbb{Q}$ and so does the point $x = 1$, and yet the value does not, since proposition 2.4.19 shows $e \notin \mathbb{Q}$. The value of a power series is a limit of rationals whenever its coefficients and its argument are rational, and the limit of a sequence of rationals need not be rational, which is the failure that section 1.1 built $\mathbb{R}$ to repair. The values at $x = 0$ come straight from the series: only the term $n = 0$ is nonzero in the first two and every term of the third vanishes, so $\exp(0) = 1$, $\cos 0 = 1$

and $\sin 0 = 0$.

Nothing in definition 2.6.5 explains the name of exp. What does explain it is the identity below, which converts addition of arguments into multiplication of values, and its proof is the Cauchy product of section 2.5 followed by one application of the binomial theorem.

Theorem 2.6.6: The Functional Equation of the Exponential

For all $x, y \in \mathbb{R}$,

$$\exp(x + y) = \exp(x)\exp(y)$$

***Proof*:**
Fix $x, y \in \mathbb{R}$. By definition 2.6.5 the series $\sum_{n\geq 0} x^n/n!$ converges absolutely and the series $\sum_{n\geq 0} y^n/n!$ converges, which are the hypotheses of theorem 2.5.9. That theorem therefore applies: the Cauchy product of the two, whose n-th term is

$$d_n = \sum_{k=0}^{n} \frac{x^k}{k!} \cdot \frac{y^{n-k}}{(n-k)!}$$

converges, and $\sum_{n\geq 0} d_n = \exp(x)\exp(y)$.
For $0 \leq k \leq n$ the binomial coefficient is $\binom{n}{k} = n!/(k!\,(n-k)!)$, so $1/(k!\,(n-k)!) = \binom{n}{k}/n!$ and

$$d_n = \frac{1}{n!}\sum_{k=0}^{n}\binom{n}{k}x^k y^{n-k} = \frac{(x+y)^n}{n!}$$

the second equality being the binomial theorem [1, the binomial theorem] applied to $(x+y)^n$. Hence $\sum_{n\geq 0} d_n$ is the series defining $\exp(x+y)$, and comparing the two evaluations of that one convergent series gives $\exp(x+y) = \exp(x)\exp(y)$, as we sought to show.

Two consequences follow at once. Taking $y = -x$ in theorem 2.6.6 gives $\exp(x)\exp(-x) = \exp(0) = 1$, so $\exp(x)$ is never 0 and $\exp(-x) = 1/\exp(x)$. And induction on n, with theorem 2.6.6 giving the step $\exp(n+1) = \exp(n)\exp(1)$, gives $\exp(n) = e^n$ for every $n \in \mathbb{N}$. The functional equation is what makes the alternative notation e^x for $\exp(x)$ consistent with integer powers of e.

Exercise 2.6.1

1. Compute the radius of convergence of $\sum_{n\geq 1} \frac{2^n}{n} x^n$, and decide the behaviour of the series at each of the two endpoints $x = 1/2$ and $x = -1/2$, saying in the convergent case whether the convergence is absolute or conditional.
2. Let $c_n = 2^n$ for even n and $c_n = 3^n$ for odd n. Show that the power series $\sum_{n\geq 0} c_n x^n$ has radius of convergence $1/3$. Show also that the sequence of ratios $|c_{n+1}/c_n|$ does not converge, so that no test phrased in terms of a limit of consecutive ratios reaches this radius.
3. Show that $\sum_{n\geq 1} x^n/n^n$ has radius of convergence $+\infty$, and that $\sum_{n\geq 1} n^k x^n$ has radius of convergence 1 for every fixed integer $k \geq 1$.
4. Prove that $\exp(x) \geq 1 + x$ for every $x \geq 0$, and deduce that $\exp(x) > 0$ for every $x \in \mathbb{R}$. (For $x < 0$, use theorem 2.6.6.)

5. Working directly from definition 2.6.5, prove that $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$ for every $x \in \mathbb{R}$.

6. Let $c_m = 1$ when m is a perfect square and $c_m = 0$ otherwise. Show that $\sum_{m\geq 0} c_m x^m$ has radius of convergence 1, and that it diverges at $x = 1$ and at $x = -1$. (For the radius, verify both clauses of theorem 2.2.12 for the candidate value 1.)

3

Limits and Continuity

Chapter 2 spent completeness on sequences. This chapter spends it on functions, and almost every theorem below is one of its results read through a single translation.

That translation is the sequential criterion of Section 3.1: a function has limit L at a point exactly when it carries every sequence approaching that point to a sequence converging to L. Once it is proved, the algebra of limits, the squeeze and theorem 2.2.9 transfer with no further work, and the counterexamples that settle the delicate questions become sequences chosen by hand.

Stating the definition at all takes vocabulary this book has not yet needed. A limit at c says nothing about $f(c)$, and c need not lie in the domain, so what the definition does require is that c be approachable from the domain. Naming that condition means naming limit points, closure, and the closed subsets of $\mathbb{R}$. Those notions belong to a general theory the later volumes develop, and only the part the real line needs is built here.

The last two sections are where completeness is spent. On a closed bounded interval a continuous function attains every intermediate value, stays bounded, attains its bounds, and is uniformly continuous. All four fail over $\mathbb{Q}$, and the first fails visibly, since $x^2 - 2$ is negative at 0 and positive at 2 with no rational root between them. Every proof runs through theorem 2.2.9 or through theorem 1.2.4 directly.

Section 3.1 builds the point-set vocabulary, defines the functional limit, proves the sequential criterion, and constructs the Cantor set as the worked example of a closed set containing no interval. Section 3.2 defines continuity, establishes the algebra and the composition rule, and separates the standard pathologies: the Dirichlet function, continuous nowhere; Thomae's function, whose discontinuity set is exactly $\mathbb{Q}$; and the monotone functions, whose discontinuity sets can be no larger than that. Section 3.3 proves the intermediate value and extreme value theorems, with the counterexamples showing that neither half of the hypothesis can be dropped. Section 3.4 strengthens continuity by making the modulus independent of the point, shows that a closed bounded interval forces the strengthening at no cost, and records the three consequences the later chapters spend.

3.1 Limits of Functions

A sequence has a single limiting direction. Its index runs through $\mathbb{N}$, and the only question is the behaviour of the terms for large n. A function $f\colon E \to \mathbb{R}$ with $E \subseteq \mathbb{R}$ can be examined near any point c that has other points of E arbitrarily close to it, so before the limit can be defined at all, the points c at which the question makes sense have to be identified. This section fixes that vocabulary for subsets of the line, defines $\lim_{x\to c} f(x)$, and proves the characterization that turns a statement about convergent sequences into a statement about functions.

An interval is a subset of $\mathbb{R}$ with no gaps in it. A betweenness condition says this without naming endpoints, so that the unbounded and the degenerate cases fall under the same words, and the nine standard notations are fixed alongside it.

Definition 3.1.1: [Real] Interval

A set $I \subseteq \mathbb{R}$ is an *interval* if for all $x, y \in I$ and all $z \in \mathbb{R}$,

$$x < z < y \qquad \text{implies} \qquad z \in I$$

For $a, b \in \mathbb{R}$ the following nine sets carry the names shown, and each is an interval:

$$\begin{aligned}
(a,b) &= \{x \in \mathbb{R} \mid a < x < b\}, & [a,b] &= \{x \in \mathbb{R} \mid a \le x \le b\},\\
[a,b) &= \{x \in \mathbb{R} \mid a \le x < b\}, & (a,b] &= \{x \in \mathbb{R} \mid a < x \le b\},\\
(a,\infty) &= \{x \in \mathbb{R} \mid x > a\}, & [a,\infty) &= \{x \in \mathbb{R} \mid x \ge a\},\\
(-\infty,b) &= \{x \in \mathbb{R} \mid x < b\}, & (-\infty,b] &= \{x \in \mathbb{R} \mid x \le b\},
\end{aligned}$$

together with $(-\infty, \infty) = \mathbb{R}$. An interval of the form $[a, b]$ with $a \le b$ is called *closed and bounded*.

The symbols ∞ and $-\infty$ in these names record that an end is unbounded and are not elements of $\mathbb{R}$. They are the two points adjoined in definition 2.2.4. Two degenerate sets satisfy the definition, and later statements quantify over intervals without excluding them. The empty set is an interval, since the condition on x, y, z is never triggered, and so is a one-point set $\{a\} = [a, a]$.

Example 3.1.2: Intervals and Three Sets That Are Not Intervals

1. Each of the nine sets named in definition 3.1.1 is an interval. For $(a, b]$, suppose $x, y \in (a, b]$ and $x < z < y$. Then $a < x < z$ and $z < y \le b$, so $a < z \le b$ and $z \in (a, b]$. The other eight are identical, with the strict and weak inequalities changed to match.
2. $\mathbb{Q} \cap [0, 1]$ is not an interval. Both 0 and 1 lie in it, and corollary 1.4.5 gives a $t \in \mathbb{R} \setminus \mathbb{Q}$ with $0 < t < 1$. This t lies between two points of the set and is not in it.
3. $[0, 1] \cup [2, 3]$ is not an interval. It contains 0 and 3, and $3/2$ lies strictly between them, but $3/2 > 1$ and $3/2 < 2$, so $3/2$ belongs to neither piece.
4. $\mathbb{R} \setminus \{0\}$ is not an interval, since -1 and 1 lie in it and 0 does not.

Four definitions now separate the points that a set approaches from the points that it merely contains.

Three of them are stated with the absolute value of definition 1.3.6, so that $|y - x| < \delta$ says that y is within δ of x, as computed in example 1.3.7.

Definition 3.1.3: Adherent Point

Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is *adherent* to E if for every real $\delta > 0$ there is a $y \in E$ with $|y - x| < \delta$.

Definition 3.1.4: [Real] Closure

Let $E \subseteq \mathbb{R}$. The *closure* of E, written $\overline{E}$, is the set of all points of $\mathbb{R}$ adherent to E.

Definition 3.1.5: [Real] Limit Point

Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. Then x is a *limit point* of E if for every real $\delta > 0$ there is a $y \in E$ with

$$0 < |y - x| < \delta$$

The set of limit points of E is written E'.

Definition 3.1.6: [Real] Isolated Point

Let $E \subseteq \mathbb{R}$ and $x \in E$. Then x is an *isolated point* of E if there is a real $\delta > 0$ such that the only $y \in E$ with $|y - x| < \delta$ is $y = x$.

The single character of difference between definition 3.1.3 and definition 3.1.5 is the strict inequality $0 < |y - x|$, which forbids the witness y from being x itself. Every point of E is adherent to E for the trivial reason that it can serve as its own witness, so adherence records nothing new at points of E. Being a limit point does, because the witnesses must be other points. The terminology of chapter 2 is not reused here: a *subsequential limit* of a sequence is the notion of definition 2.2.6, and a limit point in the present sense is always a limit point of a set.

Example 3.1.7: Limit Points and Closures of Four Subsets of the Line

1. $E = (0, 1)$. Every $x \in [0, 1]$ is a limit point of E. Let $\delta > 0$ and put $r = \min(\delta, 1)$, so $0 < r \leq 1$. If $x < 1$, set $y = \min(x + r/2, (x + 1)/2)$. Both entries exceed x, the first because $r/2 > 0$ and the second by proposition 1.1.2(8), so $y > x \geq 0$. The same part (8) gives $(x + 1)/2 < 1$, and $y \leq (x + 1)/2$, so $y < 1$ and $y \in (0, 1)$. Since $y \leq x + r/2$, also $0 < y - x \leq r/2 < \delta$. If $x = 1$, set $y = \max(1 - r/2, 1/2)$, so $1/2 \leq y < 1$ and $y \in (0, 1)$, while $0 < 1 - y \leq r/2 < \delta$. No point outside $[0, 1]$ is adherent to E: if $x < 0$ then every $y \in (0, 1)$ has $|y - x| = y - x > -x$, and if $x > 1$ then every such y has $|y - x| = x - y > x - 1$. So $E' = [0, 1]$ and $\overline{E} = [0, 1]$, and E has no isolated points.

2. $E = \mathbb{Z}$. Distinct integers differ by at least 1, so for $m \in \mathbb{Z}$ the value $\delta = 1$ shows m is isolated in $\mathbb{Z}$ and shows that m is not a limit point of $\mathbb{Z}$. A point $x \in \mathbb{R} \setminus \mathbb{Z}$ is not adherent either: with $m = \lfloor x \rfloor$ from proposition 1.4.3 one has $m < x < m + 1$, and $\delta = \min(x - m,\ m + 1 - x) > 0$ leaves no integer within δ of x, since an integer $k \leq m$ has $x - k \geq x - m \geq \delta$ and an integer $k \geq m + 1$ has $k - x \geq m + 1 - x \geq \delta$. So $\mathbb{Z}' = \emptyset$ and $\overline{\mathbb{Z}} = \mathbb{Z}$.

3. $E = \mathbb{Q}$. Every $x \in \mathbb{R}$ is a limit point. Given $\delta > 0$, theorem 1.4.4 applied to $x < x + \delta$ produces a $p \in \mathbb{Q}$ with $x < p < x + \delta$, so $0 < |p - x| < \delta$. Hence $\mathbb{Q}' = \overline{\mathbb{Q}} = \mathbb{R}$, and $\mathbb{Q}$ has no isolated points. This is the failure that section 1.1 was built to repair: $\mathbb{Q}$ is a proper subset of the line and is adherent to all of it.

4. $E = \{0, 1, 2\}$. Every point of E is isolated, by the same computation as in (2) with $\delta = 1$. No $x \in \mathbb{R}$ is a limit point of E. If $x \notin E$, the number $\delta = \min\{|y - x| \big| y \in E\}$ is a minimum over a finite nonempty set of positive reals, so $\delta > 0$, and no $y \in E$ has $|y - x| < \delta$, so x is not even adherent to E. If $x \in E$, then $\delta = 1$ admits no $y \in E$ with $0 < |y - x| < 1$, since the only $y \in E$ within 1 of x is x itself. Hence $E' = \emptyset$ and $\overline{E} = E$.

The four computations above already show the pattern that proposition 3.1.8 records: the closure of E is E with its limit points added, and inside E the two possibilities for a point are to be isolated or to be a limit point. Part (3) disposes of every finite set at once, and part (5) is what example 3.1.14 uses to see that a union of finitely many closed sets is closed.

Proposition 3.1.8: Adherent Points, Limit Points and Isolated Points

Let $E, A, B \subseteq \mathbb{R}$ and let E' be the set of limit points of E.

1. $E \subseteq \overline{E}$ and $E' \subseteq \overline{E}$, and $\overline{E} = E \cup E'$.
2. Every point of E is either an isolated point of E or a limit point of E, and no point of E is both.
3. If $x \in E'$, then for every $\delta > 0$ the set $\{y \in E \big| |y - x| < \delta\}$ is infinite. In particular a finite subset of $\mathbb{R}$ has no limit points.
4. If $A \subseteq B$ then $\overline{A} \subseteq \overline{B}$.
5. $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

***Proof*:**

1. If $x \in E$ then for every $\delta > 0$ the point $y = x$ lies in E and satisfies $|y - x| = 0 < \delta$, using proposition 1.3.8(1). Hence $E \subseteq \overline{E}$. If $x \in E'$ then a y with $0 < |y - x| < \delta$ in particular has $|y - x| < \delta$, so $E' \subseteq \overline{E}$, and therefore $E \cup E' \subseteq \overline{E}$. For the reverse inclusion let $x \in \overline{E}$ with $x \notin E$, and let $\delta > 0$. There is a $y \in E$ with $|y - x| < \delta$, and $y \neq x$ because $y \in E$ while $x \notin E$, so $|y - x| > 0$ by proposition 1.3.8(1). Hence $x \in E'$.

2. Let $x \in E$ and suppose x is not isolated in E. Then for every $\delta > 0$ the statement defining isolation fails, so there is a $y \in E$ with $|y - x| < \delta$ and $y \neq x$. As in (1) this gives $0 < |y - x| < \delta$, so $x \in E'$. Conversely, if x is isolated with witness δ, then no $y \in E$ satisfies $0 < |y - x| < \delta$, because such a y would be a point of E within δ of x and different from x, so $x \notin E'$.

3. Let $x \in E'$, let $\delta > 0$, and put $F = \{y \in E \big| 0 < |y - x| < \delta\}$. Since $x \in E'$, the set F is nonempty. Suppose F were finite. A finite nonempty set of positive real numbers has a least element, so $r = \min\{|y - x| \big| y \in F\}$ exists and $r > 0$. Applying the definition of a limit point with r in place of δ produces a $y_0 \in E$ with $0 < |y_0 - x| < r$. Then $|y_0 - x| < r \leq \delta$, so $y_0 \in F$ and hence $r \leq |y_0 - x|$, contradicting $|y_0 - x| < r$. So F is

infinite, and so is the larger set $\{y \in E \big| |y - x| < \delta\}$. If E is finite and $x \in E'$, then taking $\delta = 1$ exhibits an infinite subset of E, which is impossible, so $E' = \emptyset$.

4. If $x \in \overline{A}$ and $\delta > 0$, a witness $y \in A$ with $|y - x| < \delta$ is also a point of B.

5. The inclusion $\overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$ is (4) applied twice. For the reverse, let $x \in \overline{A \cup B}$ and suppose $x \notin \overline{A}$ and $x \notin \overline{B}$. Then there are $\delta_1 > 0$ with no $y \in A$ satisfying $|y - x| < \delta_1$, and $\delta_2 > 0$ with no $y \in B$ satisfying $|y - x| < \delta_2$. Put $\delta = \min(\delta_1, \delta_2) > 0$. Then no $y \in A \cup B$ satisfies $|y - x| < \delta$, contradicting $x \in \overline{A \cup B}$, and completing the proof.

The closure has a second description that will be used more often than the definition itself, because it converts a question about all $\delta > 0$ into a question about a single sequence. Section 3.4 extends a uniformly continuous function from a set to its closure, and it is proposition 3.1.9 that produces the point at which the extended function must be evaluated.

Proposition 3.1.9: The Closure Described by Sequences

Let $E \subseteq \mathbb{R}$ and $x \in \mathbb{R}$.

1. $x \in \overline{E}$ if and only if there is a sequence (x_n) with $x_n \in E$ for every n and $x_n \to x$.

2. x is a limit point of E if and only if there is a sequence (x_n) with $x_n \in E$ and $x_n \neq x$ for every n, and $x_n \to x$.

***Proof*:**

1. Suppose $x \in \overline{E}$. For each $n \in \mathbb{N}$ the number $1/n$ is positive, so there is a point of E within $1/n$ of x. Choose one and call it x_n.[a] Let $\epsilon > 0$. By theorem 1.4.1(3) there is an $N \in \mathbb{N}$ with $0 < 1/N < \epsilon$, and for $n \geq N$ one has $1/n \leq 1/N$ by proposition 1.1.2(7), so $|x_n - x| < 1/n \leq 1/N < \epsilon$. Hence $x_n \to x$. Conversely, suppose $x_n \in E$ and $x_n \to x$, and let $\delta > 0$. Definition 2.1.2 gives an N with $|x_N - x| < \delta$, and $x_N \in E$, so $x \in \overline{E}$.

2. The same argument with $0 < |x_n - x| < 1/n$ in place of $|x_n - x| < 1/n$ proves the forward direction, the choice being available because x is a limit point. For the converse, let $\delta > 0$ and take N with $|x_N - x| < \delta$. Since $x_N \neq x$, proposition 1.3.8(1) gives $|x_N - x| > 0$, so $0 < |x_N - x| < \delta$ with $x_N \in E$, as we sought to show.

[a] Choosing one such x_n for every n at once is an appeal to countable choice.

A set that contains every point adherent to it is one on which limits of sequences cannot escape, and that is the property the extreme value theorem of section 3.3 will need of $[a, b]$.

Definition 3.1.10: [Real] Closed Set

A set $E \subseteq \mathbb{R}$ is *closed* if $\overline{E} = E$.

Since $E \subseteq \overline{E}$ always holds by proposition 3.1.8(1), a set is closed exactly when $\overline{E} \subseteq E$, that is, when every point adherent to E already lies in E.

Proposition 3.1.11: Closed Sets Contain Their Limit Points

A set $E \subseteq \mathbb{R}$ is closed if and only if every limit point of E belongs to E.

***Proof*:**
By proposition 3.1.8(1), $\overline{E} = E \cup E'$. If $E' \subseteq E$ then $E \cup E' = E$, so $\overline{E} = E$ and E is closed. If E is closed then $E' \subseteq E \cup E' = \overline{E} = E$, completing the proof.

Proposition 3.1.12: Closed Sets Are Exactly the Sequentially Closed Sets

A set $E \subseteq \mathbb{R}$ is closed if and only if the following holds: whenever (x_n) is a sequence with $x_n \in E$ for every n and $x_n \to x$ for some $x \in \mathbb{R}$, the limit x lies in E.

***Proof*:**
Suppose E is closed and let (x_n) be a sequence in E with $x_n \to x$. By proposition 3.1.9(1), $x \in \overline{E} = E$. Conversely, suppose the stated condition holds and let $x \in \overline{E}$. By proposition 3.1.9(1) there is a sequence (x_n) in E with $x_n \to x$, so the condition gives $x \in E$. Hence $\overline{E} \subseteq E$, and with proposition 3.1.8(1) this gives $\overline{E} = E$, as we sought to show.

Example 3.1.13: Closed and Non-Closed Subsets of the Line

1. $\emptyset$ is closed, since it has no adherent points at all: the definition asks for a $y \in \emptyset$, and there is none. And $\mathbb{R}$ is closed, since $\overline{\mathbb{R}} \subseteq \mathbb{R}$ by definition.

2. For $a \leq b$ the interval $[a, b]$ is closed. Let (x_n) be a sequence in $[a, b]$ with $x_n \to x$. Then $a \leq x_n \leq b$ for every n, so $a \leq x \leq b$ by proposition 2.1.8(2), and proposition 3.1.12 applies. The same argument with only one of the two inequalities shows that $[a, \infty)$ and $(-\infty, b]$ are closed.

3. $(0, 1)$ is not closed, and neither is $[0, 1)$. By example 3.1.7(1) the closure of $(0, 1)$ is $[0, 1]$, which is not $(0, 1)$. The point 1 is adherent to $[0, 1)$ by the computation there, while $1 \notin [0, 1)$.

4. $\mathbb{Z}$ is closed and $\mathbb{Q}$ is not, by example 3.1.7(2) and (3).

5. Every finite subset of $\mathbb{R}$ is closed. It has no limit points by proposition 3.1.8(3), so proposition 3.1.11 applies.

6. A union $F_1 \cup \cdots \cup F_m$ of finitely many closed sets is closed. For $m = 1$ this is the hypothesis, and if it holds for m then proposition 3.1.8(5) with $A = F_1 \cup \cdots \cup F_m$ and $B = F_{m+1}$ gives

$$\overline{F_1 \cup \cdots \cup F_{m+1}} = \overline{F_1 \cup \cdots \cup F_m} \cup \overline{F_{m+1}} = (F_1 \cup \cdots \cup F_m) \cup F_{m+1}$$

so the union of $m + 1$ of them is closed. No such statement holds for infinite unions: each $\{1/n\}$ is closed by (5), while $\bigcup_{n \in \mathbb{N}} \{1/n\}$ has 0 as an adherent point, since $|1/n - 0| < \delta$ for any n with $1/n < \delta$, which theorem 1.4.1(3) gives.

The closed sets met so far are unions of intervals, or so sparse that they have no limit points at all. The following construction produces one that is neither. It is built from theorem 1.4.8 and the geometric series of example 2.4.7, and nothing else.

Example 3.1.14: The Cantor Set

For $A \subseteq \mathbb{R}$, $\lambda \in \mathbb{R}$ and $t \in \mathbb{R}$ write $\lambda A = \{\lambda a \big| a \in A\}$ and $t + A = \{t + a \big| a \in A\}$. Define sets $C_n \subseteq \mathbb{R}$ by

$$C_0 = [0,1], \qquad C_{n+1} = \tfrac{1}{3}C_n \cup \left(\tfrac{2}{3} + \tfrac{1}{3}C_n\right)$$

and put $C = \bigcap_{n \geq 0} C_n$, the *Cantor set.* Then:

1. $C_n \subseteq [0,1]$ and $C_{n+1} \subseteq C_n$ for every n, and C_n is a union of 2^n closed intervals each of length 3^{-n} and each with both endpoints of the form $k/3^n$ for an integer k.

2. C is nonempty. More precisely, for every sequence $(\sigma_n)_{n\geq 1}$ with $\sigma_n \in \{0,2\}$ for each n there is a point of C obtained from it, and 0, 1 and 1/4 all lie in C.

3. C is closed.

4. If $I \subseteq C$ is an interval, then I has at most one point.

5. For every $\epsilon > 0$ there are finitely many closed intervals whose union contains C and the sum of whose lengths is less than ϵ.

Proof of (1). The map $x \mapsto x/3$ preserves the order by proposition 1.1.2(5), and so does $x \mapsto 2/3 + x/3$. Hence $\frac{1}{3}[\alpha,\beta] = [\alpha/3, \beta/3]$ and $\frac{2}{3} + \frac{1}{3}[\alpha,\beta] = [2/3+\alpha/3,\ 2/3+\beta/3]$ for $\alpha \leq \beta$, the inclusion in each direction being the observation that $x \mapsto x/3$ and $x \mapsto 2/3 + x/3$ are bijections of $\mathbb{R}$ with inverses $y \mapsto 3y$ and $y \mapsto 3y - 2$. Now argue by induction. Certainly $C_0 = [0,1]$ is one closed interval of length 1, and $1 = 2^0$, $1 = 3^{-0}$. If $C_n \subseteq [0,1]$, then $\frac{1}{3}C_n \subseteq [0,1/3]$ and $\frac{2}{3} + \frac{1}{3}C_n \subseteq [2/3,1]$, so $C_{n+1} \subseteq [0,1]$. If C_n is the union of the 2^n closed intervals $[\alpha_i, \alpha_i + 3^{-n}]$ with each $\alpha_i = k_i/3^n$ for an integer k_i, then $\frac{1}{3}C_n$ is the union of the 2^n closed intervals $[\alpha_i/3,\ \alpha_i/3 + 3^{-n-1}]$ and $\frac{2}{3} + \frac{1}{3}C_n$ is the union of the 2^n closed intervals $[2/3 + \alpha_i/3,\ 2/3 + \alpha_i/3 + 3^{-n-1}]$, so C_{n+1} is a union of 2^{n+1} closed intervals of length 3^{-n-1}. The new left endpoints are $k_i/3^{n+1}$ and $(2 \cdot 3^n + k_i)/3^{n+1}$, and the right endpoints exceed them by $1/3^{n+1}$, so all are of the required form. For the nesting, first $C_1 = [0,1/3] \cup [2/3,1] \subseteq [0,1] = C_0$. If $C_{n+1} \subseteq C_n$, then $\frac{1}{3}C_{n+1} \subseteq \frac{1}{3}C_n$ and $\frac{2}{3} + \frac{1}{3}C_{n+1} \subseteq \frac{2}{3} + \frac{1}{3}C_n$, whence $C_{n+2} \subseteq C_{n+1}$. Iterating the nesting gives $C_n \subseteq C_m$ whenever $m \leq n$.

Proof of (2). Write $T_0(x) = x/3$ and $T_2(x) = (x+2)/3$, so that $C_{n+1} = T_0(C_n) \cup T_2(C_n)$ and, by the computation in (1), $T_0([\alpha,\beta]) = [\alpha/3, \beta/3]$ and $T_2([\alpha,\beta]) = [(\alpha+2)/3, (\beta+2)/3]$, each of length $(\beta - \alpha)/3$. Given $\sigma_1, \sigma_2, \ldots$ in $\{0,2\}$, set $J_0 = [0,1]$ and

$$J_n = T_{\sigma_1} \circ T_{\sigma_2} \circ \cdots \circ T_{\sigma_n}([0,1])$$

Each J_n is a closed interval of length 3^{-n}, by induction on n from the length computation just made. Since $T_0([0,1]) = [0,1/3]$ and $T_2([0,1]) = [2/3,1]$ are both contained in $[0,1]$, and each T_σ carries a subset into the image of any set containing it, applying $T_{\sigma_1} \circ \cdots \circ T_{\sigma_n}$ to $T_{\sigma_{n+1}}([0,1]) \subseteq [0,1]$ gives $J_{n+1} \subseteq J_n$. Next, $J_n \subseteq C_n$. Prove instead the stronger statement that $T_{\tau_1} \circ \cdots \circ T_{\tau_n}([0,1]) \subseteq C_n$ for *every* $\tau_1, \ldots, \tau_n$ in $\{0,2\}$, by induction on n. For $n = 0$ this is $[0,1] = C_0$. For the step, write $T_{\tau_1} \circ \cdots \circ T_{\tau_{n+1}}([0,1]) = T_{\tau_1}(K)$ where $K = T_{\tau_2} \circ \cdots \circ T_{\tau_{n+1}}([0,1])$ is built from a list of length n, so $K \subseteq C_n$ by the inductive hypothesis, and then $T_{\tau_1}(K) \subseteq T_{\tau_1}(C_n) \subseteq C_{n+1}$.

The lengths 3^{-n} form a null sequence by proposition 2.1.10(3) with $j = 0$ and $r = 1/3$, so for every $\epsilon > 0$ there is an n with $3^{-n} < \epsilon$. The hypotheses of theorem 1.4.8(2) are therefore met by $(J_n)_{n\geq 0}$, reindexed to start at 1, and it gives a point x lying in every J_n. Then $x \in C_n$ for every n, so $x \in C$. Taking every $\sigma_n = 0$ gives $J_n = [0, 3^{-n}]$ and $x = 0$, and taking every $\sigma_n = 2$ gives $x = 1$. For $1/4$, argue instead that $1/4 \in C_n$ and $3/4 \in C_n$ for every n, by simultaneous induction: both lie in $C_0 = [0,1]$, and if both lie in C_n then $1/4 = \frac{1}{3} \cdot \frac{3}{4} \in \frac{1}{3}C_n \subseteq C_{n+1}$ and $3/4 = \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{4} \in \frac{2}{3} + \frac{1}{3}C_n \subseteq C_{n+1}$, the second because $2/3 + 1/12 = 8/12 + 1/12 = 3/4$. So $1/4 \in C$. Together with 0 and 1 this proves (2).

The point $1/4$ is not an endpoint of any of the intervals of any C_n: by (1) such an endpoint is $k/3^n$ for an integer k, and $1/4 = k/3^n$ would give $3^n = 4k$, which is impossible because 3^n is odd and $4k$ is even. It is the point that the argument above produces from the sequence with $\sigma_n = 0$ for odd n and $\sigma_n = 2$ for even n. Indeed $T_\sigma(x) = (x+\sigma)/3$ for both values of σ, so applying T_{σ_1} to a closed interval with left endpoint α returns one with left endpoint $(\alpha + \sigma_1)/3$, and induction on the length of the list makes the left endpoint of J_n equal to $\sum_{i=1}^{n} \sigma_i 3^{-i}$. That endpoint is within the length 3^{-n} of the point of C lying in every J_n, so that point is the limit of these endpoints. Dropping the term $k = 0$, whose value is 1, from the geometric series of example 2.4.7 with $x = 1/9$ evaluates the limit as

$$2\sum_{k=1}^{\infty} 9^{-k} = 2\left(\frac{1}{1-1/9} - 1\right) = 2\left(\frac{9}{8} - 1\right) = \frac{1}{4}$$

so the sequence alternating 0 and 2 produces $1/4$.

Proof of (3). Each of the 2^n closed intervals composing C_n is closed by example 3.1.13(2), so C_n is closed by example 3.1.13(6). Now let (x_k) be a sequence in C with $x_k \to x$. Fix n. Every x_k lies in C and hence in C_n, so proposition 3.1.12 applied to the closed set C_n gives $x \in C_n$. As n was arbitrary, $x \in C$, and proposition 3.1.12 applied to C shows C is closed.

Proof of (4). First a claim: for every $n \geq 0$, if $a \leq b$ and $[a,b] \subseteq C_n$, then $b - a \leq 3^{-n}$. For $n = 0$ this says $[a,b] \subseteq [0,1]$ forces $b - a \leq 1$, which holds since $0 \leq a \leq b \leq 1$. Assume the claim for n and let $[a,b] \subseteq C_{n+1}$ with $a \leq b$. If $a = b$ there is nothing to prove, so assume $a < b$. By (1), $C_{n+1} \subseteq [0,1/3] \cup [2/3,1]$, since $\frac{1}{3}C_n \subseteq [0,1/3]$ and $\frac{2}{3} + \frac{1}{3}C_n \subseteq [2/3,1]$. Suppose $b > 1/3$ and $a < 2/3$, and put $u = \max(a, 1/3)$ and $v = \min(b, 2/3)$. Each of the four comparisons $a < b$, $a < 2/3$, $1/3 < b$ and $1/3 < 2/3$ holds, so $u < v$. By proposition 1.1.2(8) the point $w = (u+v)/2$ satisfies $u < w < v$, hence $a \leq u < w < v \leq b$ and $w \in [a,b] \subseteq C_{n+1}$, while $1/3 \leq u < w < v \leq 2/3$ puts w outside $[0,1/3] \cup [2/3,1]$. That is a contradiction, so $b \leq 1/3$ or $a \geq 2/3$. In the first case every $x \in [a,b]$ satisfies $x \leq 1/3 < 2/3$, so $x \notin [2/3,1]$ and therefore $x \notin \frac{2}{3} + \frac{1}{3}C_n$. Being in C_{n+1}, such an x lies in $\frac{1}{3}C_n$, that is $3x \in C_n$. Every $y \in [3a, 3b]$ is $3x$ for the point $x = y/3$ of $[a,b]$, so $[3a,3b] \subseteq C_n$, and the inductive hypothesis gives $3b - 3a \leq 3^{-n}$, that is $b - a \leq 3^{-n-1}$. In the second case every $x \in [a,b]$ satisfies $x \geq 2/3 > 1/3$, so $x \notin \frac{1}{3}C_n$ and hence $3x - 2 \in C_n$. The same reasoning gives $[3a-2, 3b-2] \subseteq C_n$ and again $b - a \leq 3^{-n-1}$. This proves the claim. Now let $I \subseteq C$ be an interval with two distinct points $a < b$. Every z with $a < z < b$ lies in I by definition 3.1.1, and a, b lie in I, so $[a,b] \subseteq I \subseteq C \subseteq C_n$ for every n. The claim gives $0 < b - a \leq 3^{-n}$ for every n, and since $3^{-n} \to 0$ by proposition 2.1.10(3), proposition 2.1.8(2) forces $b - a \leq 0$, a contradiction.

Proof of (5). The sequence $(2/3)^n$ is null by proposition 2.1.10(3) with $j = 0$ and $r = 2/3$, so there is an n with $(2/3)^n < \epsilon$. By (1), $C \subseteq C_n$ and C_n is a union of 2^n closed intervals of length 3^{-n}, whose lengths sum to $2^n \cdot 3^{-n} = (2/3)^n < \epsilon$.

So C is a closed subset of $[0,1]$ containing no interval of positive length, and yet it is not thin in the way a finite set is. Each of the 2^n intervals from which C_n is assembled is the interval J_n of some sequence (σ_k), and the proof of (2) puts a point of C inside every such J_n, so C meets all 2^n of them.

It also contains points such as $1/4$ that are not endpoints of any of them. The vocabulary is now complete enough to define the limit.

Definition 3.1.15: Limit of a Function

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$, let c be a limit point of E, and let $L \in \mathbb{R}$. Then f *has limit L at c*, written

$$\lim_{x\to c} f(x) = L$$

if for every real $\epsilon > 0$ there is a real $\delta > 0$ such that

$$|f(x) - L| < \epsilon \quad \text{for every } x \in E \text{ with } 0 < |x - c| < \delta$$

3.1.16 Convention: The Deleted Limit Two features of definition 3.1.15 hold throughout the book and are not restated later.

1. The point c need not belong to E, and when it does, the value $f(c)$ plays no part. The condition $0 < |x - c|$ excludes $x = c$ from every requirement the definition makes. A limit of this kind is called *deleted*.
2. The point c must be a limit point of E. Without that hypothesis the definition is empty: if c is not a limit point, some $\delta > 0$ admits no $x \in E$ with $0 < |x - c| < \delta$, so the displayed condition holds for every $L \in \mathbb{R}$ at once and f would have every real number as a limit at c.

The two work together. Deleting c is what allows a limit to be taken at a point where f is undefined or takes an unrelated value, and requiring c to be a limit point is what stops that deletion from removing all the information at once.

Example 3.1.17: Four Functional Limits

1. $E = \mathbb{R}$, $f(x) = 3x - 1$, $c \in \mathbb{R}$. Then $\lim_{x\to c} f(x) = 3c - 1$. Every $c \in \mathbb{R}$ is a limit point of $\mathbb{R}$, since for $\delta > 0$ the point $y = c + \delta/2$ satisfies $0 < |y - c| = \delta/2 < \delta$. Given $\epsilon > 0$ put $\delta = \epsilon/3$. If $0 < |x - c| < \delta$, then $|f(x) - (3c - 1)| = |3(x - c)| = 3|x - c| < 3\delta = \epsilon$, using proposition 1.3.8(3).
2. $E = \mathbb{R} \setminus \{1\}$, $f(x) = (x^2 - 1)/(x - 1)$, $c = 1$. Here $c \notin E$. For $x \in E$ one has $x - 1 \neq 0$, so $f(x) = (x - 1)(x + 1)/(x - 1) = x + 1$. Also 1 is a limit point of E, since for $\delta > 0$ the point $x = 1 + \min(\delta, 1)/2$ lies in E and satisfies $0 < |x - 1| < \delta$. Given $\epsilon > 0$ put $\delta = \epsilon$. If $x \in E$ and $0 < |x - 1| < \delta$, then $|f(x) - 2| = |x - 1| < \epsilon$. So $\lim_{x\to 1} f(x) = 2$ although $f(1)$ is not defined.
3. $E = \mathbb{R}$, $g(x) = x + 1$ for $x \neq 1$ and $g(1) = 17$. Here 1 is a limit point of $\mathbb{R}$ by (1). The points at which definition 3.1.15 tests g are those with $0 < |x - 1| < \delta$, and $x = 1$ is not among them, so g agrees there with the f of (2) and the same choice $\delta = \epsilon$ works. Hence $\lim_{x\to 1} g(x) = 2$ while $g(1) = 17$.
4. $E = (0, 1) \cup \{2\}$ and $c = 2$. Every $x \in (0, 1)$ has $|x - 2| = 2 - x > 1$, so $\delta = 1/2$ admits no $x \in E$ with $0 < |x - 2| < \delta$. Hence 2 is not a limit point of E, and definition 3.1.15 assigns no limit at 2 to any function on E. This is the situation box 3.1.16(2) excludes.

Proposition 3.1.18: A Function Has at Most One Limit at a Point

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$, and let c be a limit point of E. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} f(x) = L'$, then $L = L'$.

***Proof*:**
Suppose $L \neq L'$ and put $\epsilon = |L - L'|/2$. Then $\epsilon > 0$ by proposition 1.3.8(1). The definition gives a $\delta > 0$ with $|f(x) - L| < \epsilon$ for every $x \in E$ with $0 < |x - c| < \delta$, and a $\delta' > 0$ with $|f(x) - L'| < \epsilon$ for every $x \in E$ with $0 < |x - c| < \delta'$. Since c is a limit point of E and $\min(\delta, \delta') > 0$, there is an $x_0 \in E$ with $0 < |x_0 - c| < \min(\delta, \delta')$, and this x_0 satisfies both estimates. By proposition 1.3.8(4),

$$|L - L'| \le |L - f(x_0)| + |f(x_0) - L'| < \epsilon + \epsilon = |L - L'|$$

which is impossible. So $L = L'$, completing the proof.

The limit-point hypothesis is what made that proof possible: it produced the point x_0 at which the two estimates could be compared. Uniqueness is what entitles the notation $\lim_{x\to c} f(x)$ to name a single real number, and from here on the phrase "the limit of f at c exists" means that some $L \in \mathbb{R}$ satisfies definition 3.1.15.

Everything chapter 2 proved about convergent sequences now becomes available, through a single translation. Theorem 3.1.19 is the one result of this section that later sections use most often. The algebra of limits, the squeeze theorem and the counterexamples of section 3.2 are corollaries of it together with theorem 2.1.6.

Theorem 3.1.19: The Sequential Criterion for Functional Limits

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$, let c be a limit point of E, and let $L \in \mathbb{R}$.

1. $\lim_{x\to c} f(x) = L$ if and only if $f(x_n) \to L$ for every sequence (x_n) with $x_n \in E$ and $x_n \neq c$ for all n and $x_n \to c$.

2. If two such sequences (x_n) and (y_n) have $f(x_n) \to L_1$ and $f(y_n) \to L_2$ with $L_1 \neq L_2$, or if one such sequence has $(f(x_n))$ divergent, then f has no limit at c.

***Proof*:**

1. Sequences of the stated kind exist, by proposition 3.1.9(2) applied to the limit point c. Suppose first that $\lim_{x\to c} f(x) = L$, and let (x_n) be such a sequence. Let $\epsilon > 0$ and take $\delta > 0$ from definition 3.1.15. Since $x_n \to c$, there is an N with $|x_n - c| < \delta$ for every $n \ge N$, and $x_n \neq c$ gives $|x_n - c| > 0$ by proposition 1.3.8(1). So $0 < |x_n - c| < \delta$ and hence $|f(x_n) - L| < \epsilon$ for every $n \ge N$, which is $f(x_n) \to L$.
For the converse, suppose $\lim_{x\to c} f(x) = L$ fails. Then there is an $\epsilon_0 > 0$ such that no $\delta > 0$ works, that is, such that for every $\delta > 0$ there is an $x \in E$ with $0 < |x - c| < \delta$ and $|f(x) - L| \ge \epsilon_0$. For each $n \in \mathbb{N}$ apply this with $\delta = 1/n$ and call the resulting point x_n. Then $x_n \in E$ and $x_n \neq c$, and $|x_n - c| < 1/n$ gives $x_n \to c$ by the argument used in proposition 3.1.9(1). But $|f(x_n) - L| \ge \epsilon_0$ for every n, so $(f(x_n))$ does not converge to L.

So the sequential condition fails as well, which is the contrapositive of the claim.

2. Suppose f had a limit L at c. In the first case part (1) gives $f(x_n) \to L$ and $f(y_n) \to L$, so $L = L_1$ and $L = L_2$ by proposition 2.1.4, contradicting $L_1 \neq L_2$. In the second case part (1) gives $f(x_n) \to L$, so $(f(x_n))$ converges, contrary to hypothesis. Hence no such L exists, completing the proof.

Part (2) is how a limit is shown not to exist: produce two sequences approaching c along which f does two different things. Parts (1) and (2) together say that the functional limit carries exactly as much information as the collection of all sequential limits along $E \setminus \{c\}$, which is why corollary 3.1.20 and corollary 3.1.21 need no new estimates.

Corollary 3.1.20: The Algebra of Functional Limits

Let $E \subseteq \mathbb{R}$, let c be a limit point of E, let $f, g\colon E \to \mathbb{R}$ with $\lim_{x\to c} f(x) = A$ and $\lim_{x\to c} g(x) = B$, and let $\lambda \in \mathbb{R}$.

1. $\lim_{x\to c}\big(f(x) + g(x)\big) = A + B$.
2. $\lim_{x\to c} \lambda f(x) = \lambda A$.
3. $\lim_{x\to c} f(x)g(x) = AB$.
4. If $B \neq 0$, then c is a limit point of $E^* = \{x \in E \big| g(x) \neq 0\}$, and the function $f/g\colon E^* \to \mathbb{R}$ satisfies $\lim_{x\to c} f(x)/g(x) = A/B$.

Proof:

1. Let (x_n) be a sequence in E with $x_n \neq c$ and $x_n \to c$. By theorem 3.1.19(1), $f(x_n) \to A$ and $g(x_n) \to B$, so $f(x_n) + g(x_n) \to A + B$ by theorem 2.1.6(1). Since (x_n) was arbitrary, theorem 3.1.19(1) applied to $f + g$ gives the claim.
2. The same argument with theorem 2.1.6(2).
3. The same argument with theorem 2.1.6(3).
4. Apply definition 3.1.15 to g with $\epsilon = |B|/2$, which is positive by proposition 1.3.8(1): there is a $\delta_0 > 0$ such that $|g(x) - B| < |B|/2$ for every $x \in E$ with $0 < |x - c| < \delta_0$. For such an x, proposition 1.3.8(1) and (5) give
$$|B| - |g(x)| \leq \big|\,|B| - |g(x)|\,\big| \leq |B - g(x)| = |g(x) - B| < \frac{|B|}{2}$$
so $|g(x)| > |B|/2 > 0$ and hence $g(x) \neq 0$, that is $x \in E^*$. To see that c is a limit point of E^*, let $\delta > 0$ and put $\delta' = \min(\delta, \delta_0) > 0$. Since c is a limit point of E there is an $x \in E$ with $0 < |x - c| < \delta'$, and the previous sentence puts this x in E^* while $0 < |x - c| < \delta$. Now let (x_n) be a sequence in E^* with $x_n \neq c$ and $x_n \to c$. It is in particular a sequence in E, so theorem 3.1.19(1) gives $f(x_n) \to A$ and $g(x_n) \to B$, and $g(x_n) \neq 0$ for every n because $x_n \in E^*$. Theorem 2.1.6(4) gives $f(x_n)/g(x_n) \to A/B$. Since (x_n) was arbitrary, theorem 3.1.19(1) applied to f/g on E^* completes the proof.

Corollary 3.1.21: The Squeeze Theorem for Functions

Let $E \subseteq \mathbb{R}$, let c be a limit point of E, and let $f, g, h \colon E \to \mathbb{R}$. Suppose there is an $r > 0$ with

$$f(x) \leq g(x) \leq h(x) \quad \text{for every } x \in E \text{ with } 0 < |x - c| < r$$

If $\lim_{x \to c} f(x) = L$ and $\lim_{x \to c} h(x) = L$, then $\lim_{x \to c} g(x) = L$.

***Proof*:**
Let (x_n) be a sequence in E with $x_n \neq c$ for every n and $x_n \to c$. Since $r > 0$, there is an N_0 with $|x_n - c| < r$ for every $n \geq N_0$, and $|x_n - c| > 0$ because $x_n \neq c$. So $f(x_n) \leq g(x_n) \leq h(x_n)$ for every $n \geq N_0$. By theorem 3.1.19(1), $f(x_n) \to L$ and $h(x_n) \to L$, so theorem 2.1.7 gives $g(x_n) \to L$. Since (x_n) was arbitrary, theorem 3.1.19(1) gives $\lim_{x \to c} g(x) = L$, as we sought to show.

The limit at c is determined by the values of f arbitrarily close to c and by nothing else. Proposition 3.1.22 makes that precise, and one-sided limits are then instances of definition 3.1.15 applied to a restricted domain.

Proposition 3.1.22: The Limit Depends Only on the Values Near the Point

Let $E \subseteq \mathbb{R}$, let $f \colon E \to \mathbb{R}$, let c be a limit point of E, let $r > 0$, and put $E_r = E \cap (c - r,\ c + r)$.

1. c is a limit point of E_r.
2. For $L \in \mathbb{R}$, $\lim_{x \to c} f(x) = L$ if and only if $\lim_{x \to c} \big(f|_{E_r}\big)(x) = L$.
3. If $g \colon E \to \mathbb{R}$ satisfies $g(x) = f(x)$ for every $x \in E_r$ with $x \neq c$, then f has a limit at c exactly when g does, and the two limits are equal.

***Proof*:**

1. Let $\delta > 0$ and put $\delta' = \min(\delta, r) > 0$. Since c is a limit point of E there is a $y \in E$ with $0 < |y - c| < \delta'$. Then $|y - c| < r$ gives $c - r < y < c + r$ by proposition 1.3.8(2), so $y \in E_r$, and $0 < |y - c| < \delta$.
2. Suppose $\lim_{x \to c} f(x) = L$ and let $\epsilon > 0$. The δ furnished by definition 3.1.15 for f also works for $f|_{E_r}$, since every $x \in E_r$ with $0 < |x - c| < \delta$ is a point of E with the same property. Conversely, suppose $\lim_{x \to c} (f|_{E_r})(x) = L$ and let $\epsilon > 0$. Take $\delta > 0$ from the definition for $f|_{E_r}$ and put $\delta'' = \min(\delta, r) > 0$. If $x \in E$ and $0 < |x - c| < \delta''$, then $|x - c| < r$ puts x in E_r, and $0 < |x - c| < \delta$ gives $|f(x) - L| < \epsilon$.
3. The functions $f|_{E_r}$ and $g|_{E_r}$ agree at every point of E_r other than c, and definition 3.1.15 never evaluates either at c, so the two satisfy exactly the same instances of the definition and have the same limits at c. Now apply (2) to f and then to g, completing the proof.

Proposition 3.1.23: Limits Along a Subset

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$, let $c \in \mathbb{R}$ and $L \in \mathbb{R}$.

1. Let $D \subseteq E$ and suppose c is a limit point of D. Then c is a limit point of E, and if $\lim_{x\to c} f(x) = L$ then $\lim_{x\to c}\big(f|_D\big)(x) = L$.
2. Let $D_1, D_2 \subseteq E$ with $E \setminus \{c\} \subseteq D_1 \cup D_2$, suppose c is a limit point of both D_1 and D_2, and suppose $\lim_{x\to c}(f|_{D_1})(x) = L$ and $\lim_{x\to c}(f|_{D_2})(x) = L$. Then $\lim_{x\to c} f(x) = L$.

***Proof*:**

1. For each $\delta > 0$, a point $y \in D$ with $0 < |y - c| < \delta$ is also a point of E, so c is a limit point of E. Given $\epsilon > 0$, take $\delta > 0$ from definition 3.1.15 for f. Every $x \in D$ with $0 < |x - c| < \delta$ lies in E, so $|f|_D(x) - L| = |f(x) - L| < \epsilon$.
2. By (1) applied to D_1, the point c is a limit point of E, so $\lim_{x\to c} f(x)$ is a meaningful assertion. Given $\epsilon > 0$, take $\delta_1 > 0$ for $f|_{D_1}$ and $\delta_2 > 0$ for $f|_{D_2}$, and put $\delta = \min(\delta_1, \delta_2) > 0$. Let $x \in E$ with $0 < |x - c| < \delta$. Then $x \neq c$ by proposition 1.3.8(1), so $x \in D_1 \cup D_2$. If $x \in D_1$ then $0 < |x - c| < \delta_1$ gives $|f(x) - L| < \epsilon$, and if $x \in D_2$ the same holds with δ_2, completing the proof.

Definition 3.1.24: One-Sided Limits

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ and let $c \in \mathbb{R}$. Put

$$E_c^+ = E \cap (c, \infty), \qquad E_c^- = E \cap (-\infty, c)$$

If c is a limit point of E_c^+, the *right-hand limit* of f at c is

$$\lim_{x\to c^+} f(x) := \lim_{x\to c}\big(f|_{E_c^+}\big)(x)$$

whenever the right-hand side exists. If c is a limit point of E_c^-, the *left-hand limit* $\lim_{x\to c^-} f(x)$ is defined the same way with E_c^- in place of E_c^+.

For $E = \mathbb{R} \setminus \{0\}$, $f(x) = |x|/x$ and $c = 0$, the two sets are $E_0^+ = (0, \infty)$ and $E_0^- = (-\infty, 0)$, and f takes the constant value 1 on the first and the constant value -1 on the second. The right-hand limit at 0 is then 1 and the left-hand limit is -1, and example 3.1.29(1) carries out the estimates.

A one-sided limit is therefore the limit of a restriction, and it needs no theory of its own. What it does need is a direct description, so that a computation does not have to pass through the restricted domain each time.

Proposition 3.1.25: The Direct Description of a One-Sided Limit

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$, let $c \in \mathbb{R}$ and $L \in \mathbb{R}$.

1. c is a limit point of E_c^+ if and only if for every $\delta > 0$ there is an $x \in E$ with $c < x < c+\delta$.
2. Assume c is a limit point of E_c^+. Then $\lim_{x\to c^+} f(x) = L$ if and only if for every $\epsilon > 0$ there is a $\delta > 0$ with $|f(x) - L| < \epsilon$ for every $x \in E$ satisfying $c < x < c + \delta$.
3. Assume c is a limit point of E_c^+. Then $\lim_{x\to c^+} f(x) = L$ if and only if $f(x_n) \to L$ for every sequence (x_n) with $x_n \in E$ and $x_n > c$ for all n and $x_n \to c$.

The three statements hold with E_c^-, $c - \delta < x < c$ and $x_n < c$ in place of E_c^+, $c < x < c + \delta$ and $x_n > c$.

***Proof*:**
For $x \in E$ the two conditions

$$x \in E_c^+ \text{ and } 0 < |x - c| < \delta, \qquad c < x < c + \delta$$

are equivalent. Indeed, if $x > c$ then $x - c > 0$, so $|x - c| = x - c$ by definition 1.3.6, and $0 < x - c < \delta$ is exactly $c < x < c + \delta$. Conversely $c < x < c + \delta$ gives $x > c$ and $0 < x - c < \delta$. Parts (1) and (2) are now definition 3.1.5 and definition 3.1.15 for the function $f|_{E_c^+}$, with the left-hand condition of the display replaced by the right-hand one. Part (3) is theorem 3.1.19(1) applied to $f|_{E_c^+}$, once one notes that a sequence in E_c^+ with no term equal to c is the same thing as a sequence in E with every term greater than c, the case $x_n = c$ being impossible inside E_c^+. The left-handed statements follow by the same computation with $|x - c| = c - x$ for $x < c$, completing the proof.

Proposition 3.1.26: A Limit Exists Exactly When the One-Sided Limits Agree

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$, let c be a limit point of E and let $L \in \mathbb{R}$. Then c is a limit point of at least one of E_c^+ and E_c^-, and:

1. If c is a limit point of both, then $\lim_{x\to c} f(x) = L$ if and only if both one-sided limits at c exist and are equal to L.
2. If c is a limit point of E_c^+ and not of E_c^-, then $\lim_{x\to c} f(x) = L$ if and only if $\lim_{x\to c^+} f(x) = L$. Symmetrically with the two sides exchanged.

***Proof*:**
Suppose c were a limit point of neither. Then there are $\delta_+ > 0$ admitting no $y \in E_c^+$ with $0 < |y - c| < \delta_+$, and $\delta_- > 0$ admitting no such $y \in E_c^-$. Put $\delta = \min(\delta_+, \delta_-) > 0$. Since c is a limit point of E there is a $y \in E$ with $0 < |y - c| < \delta$. This y satisfies $y \neq c$ by proposition 1.3.8(1), so $y > c$ or $y < c$, placing y in E_c^+ or in E_c^- and contradicting the choice of δ_+ or of δ_-.

1. If $\lim_{x\to c} f(x) = L$, then proposition 3.1.23(1) applied to $D = E_c^+$ and to $D = E_c^-$ gives both one-sided limits, equal to L. Conversely, every $x \in E$ with $x \neq c$ satisfies $x > c$ or

$x < c$, so $E \setminus \{c\} \subseteq E_c^+ \cup E_c^-$, and proposition 3.1.23(2) with $D_1 = E_c^+$ and $D_2 = E_c^-$ gives $\lim_{x\to c} f(x) = L$.

2. The forward implication is proposition 3.1.23(1) with $D = E_c^+$. For the converse, since c is not a limit point of E_c^- there is an $r > 0$ admitting no $y \in E_c^-$ with $0 < |y - c| < r$. Let $\epsilon > 0$ and take $\delta_+ > 0$ from proposition 3.1.25(2). Put $\delta = \min(\delta_+, r) > 0$ and let $x \in E$ with $0 < |x - c| < \delta$. Then $x \neq c$. If $x < c$ then $x \in E_c^-$ and $0 < |x - c| < r$, which is impossible, so $x > c$, and then $c < x < c + \delta_+$ gives $|f(x) - L| < \epsilon$. Hence $\lim_{x\to c} f(x) = L$, completing the proof.

Two ends of the definition remain. A domain may be unbounded above, and the question is then what f does at large x. The values of f may also grow beyond every bound. Both cases are recorded by relaxing one half of definition 3.1.15 while keeping the other.

Definition 3.1.27: Limits at Infinity

Let $E \subseteq \mathbb{R}$ be unbounded above, let $f\colon E \to \mathbb{R}$ and let $L \in \mathbb{R}$. Write $\lim_{x\to\infty} f(x) = L$ if for every real $\epsilon > 0$ there is an $M \in \mathbb{R}$ such that

$$|f(x) - L| < \epsilon \quad \text{for every } x \in E \text{ with } x > M$$

For E unbounded below, $\lim_{x\to-\infty} f(x) = L$ is defined the same way with $x < M$ in place of $x > M$.

These limits are unique as well. If $\lim_{x\to\infty} f(x)$ were both L and L' with $L \neq L'$, then taking $\epsilon = |L - L'|/2$ produces M and M'. Since E is unbounded above there is an $x_0 \in E$ with $x_0 > \max(M, M')$, and proposition 1.3.8(4) gives $|L - L'| \leq |L - f(x_0)| + |f(x_0) - L'| < |L - L'|$, which is impossible. The unboundedness of E here does the work that the limit-point hypothesis does in proposition 3.1.18.

Definition 3.1.28: Infinite Limits

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ and let c be a limit point of E. Write $\lim_{x\to c} f(x) = +\infty$ if for every $M \in \mathbb{R}$ there is a real $\delta > 0$ such that

$$f(x) > M \quad \text{for every } x \in E \text{ with } 0 < |x - c| < \delta$$

and $\lim_{x\to c} f(x) = -\infty$ if for every $M \in \mathbb{R}$ there is a $\delta > 0$ with $f(x) < M$ for every such x. The one-sided forms $\lim_{x\to c^\pm} f(x) = \pm\infty$ are the corresponding statements for the restrictions of definition 3.1.24. For E unbounded above, $\lim_{x\to\infty} f(x) = +\infty$ means that for every $M \in \mathbb{R}$ there is an $N \in \mathbb{R}$ with $f(x) > M$ for every $x \in E$ with $x > N$, and the remaining three combinations of signs are defined analogously.

The symbols $+\infty$ and $-\infty$ are those of definition 2.2.4 and are not real numbers, so "$\lim_{x\to c} f(x) = +\infty$" is a statement about divergence and not the assertion that some limit exists. The two situations never overlap: if $\lim_{x\to c} f(x) = L$ for a real L, then taking $\epsilon = 1$ gives a δ with $|f(x) - L| < 1$, hence $f(x) < L + 1$ by proposition 1.3.8(2), for every $x \in E$ with $0 < |x - c| < \delta$, while $\lim_{x\to c} f(x) = +\infty$ with $M = L + 1$ gives a δ' with $f(x) > L + 1$ for every such x with δ' in place of δ. Since c is a limit point there is a point of E satisfying both restrictions, and it would give $f(x) < L + 1 < f(x)$.

Example 3.1.29: Three One-Sided and Infinite Limits

1. $E = \mathbb{R} \setminus \{0\}$ and $f(x) = |x|/x$, with $c = 0$. Every $\delta > 0$ admits $x = \min(\delta, 1)/2 \in E$ with $0 < x < \delta$ and admits $-x$, so 0 is a limit point of both E_0^+ and E_0^-. For $x > 0$ one has $|x| = x$ and $f(x) = 1$, so proposition 3.1.25(2) with any δ gives $\lim_{x\to 0^+} f(x) = 1$; for $x < 0$ one has $|x| = -x$ and $f(x) = -1$, giving $\lim_{x\to 0^-} f(x) = -1$. Since $1 \neq -1$, proposition 3.1.26(1) shows that f has no limit at 0.

2. $E = \mathbb{R} \setminus \{0\}$ and $f(x) = 1/x^2$, with $c = 0$. Then $\lim_{x\to 0} f(x) = +\infty$. Every $x \in E$ has $x \neq 0$, so $x^2 > 0$ by proposition 1.1.2(6), and applying proposition 1.1.2(7) to $0 < x^2 < x^2 + 1$ gives $1/x^2 > 0$. Let $M \in \mathbb{R}$. If $M \le 0$, take $\delta = 1$, and the previous sentence gives $1/x^2 > 0 \ge M$ for every $x \in E$. If $M > 0$, then $1/M > 0$ by proposition 1.1.2(7), and theorem 1.3.12 with $n = 2$ gives a $\delta \ge 0$ with $\delta^2 = 1/M$. Here $\delta \neq 0$, since $\delta = 0$ would give $1/M = 0$. For $x \in E$ with $0 < |x - 0| < \delta$ one has $0 < |x| < \delta$. Two applications of proposition 1.1.2(5), first multiplying by $|x| > 0$ and then by $\delta > 0$, give $|x|^2 < |x|\delta < \delta^2$, and $|x|^2 = |x^2| = x^2$ by proposition 1.3.8(3) and definition 1.3.6. So $0 < x^2 < \delta^2 = 1/M$, and proposition 1.1.2(7) applied to this gives $1/x^2 > M$.

3. $E = (3, \infty)$ and $f(x) = (2x + 1)/(x - 3)$. Then E is unbounded above: given $y \in \mathbb{R}$, theorem 1.4.1(2) produces an $n \in \mathbb{N}$ with $n > \max(y, 3)$, and such an n lies in E and exceeds y. Moreover $\lim_{x\to\infty} f(x) = 2$. For $x \in E$ one has $x - 3 > 0$, and

$$f(x) - 2 = \frac{2x + 1 - 2(x - 3)}{x - 3} = \frac{7}{x - 3}$$

Given $\epsilon > 0$ put $M = 3 + 7/\epsilon$. If $x \in E$ and $x > M$ then $x - 3 > 7/\epsilon > 0$, so $0 < 7/(x - 3) < \epsilon$ by proposition 1.1.2(7) and (5), and $|f(x) - 2| < \epsilon$.

The last two definitions inherit no algebra. Nothing like corollary 3.1.20 holds for them, and the difference of two functions tending to $+\infty$ can do anything. On $E = \mathbb{R} \setminus \{0\}$ with $c = 0$, the function $1/x^2$ tends to $+\infty$ by example 3.1.29(2), and so does $2/x^2$, since $2/x^2 > 1/x^2 > M$ wherever the estimate there holds. Yet $2/x^2 - 1/x^2 = 1/x^2$ tends to $+\infty$, the difference of $1/x^2$ with itself is the constant 0, and $1/x^2 - 2/x^2 = -1/x^2$ tends to $-\infty$, because $1/x^2 > -M$ gives $-1/x^2 < M$. What these definitions give instead is a vocabulary precise enough to say what a function does at a point where it has no limit.

Exercise 3.1.1

1. Let $E = \{1/n \mid n \in \mathbb{N}\}$. Determine the set of limit points of E, determine which points of E are isolated, and compute $\overline{E}$. Prove each of the three answers.

2. Let $E \subseteq \mathbb{R}$. Prove that $\overline{\overline{E}} = \overline{E}$, and conclude that $\overline{E}$ is closed for every E.

3. Let $A, B \subseteq \mathbb{R}$. Prove that $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$, and give sets A and B for which the inclusion is strict. Compare with proposition 3.1.8(5), where the corresponding statement for unions is an equality.

4. Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = x^2$ and let $c \in \mathbb{R}$. Given $\epsilon > 0$, exhibit an explicit $\delta > 0$, depending on ϵ and c, that witnesses $\lim_{x\to c} f(x) = c^2$ in definition 3.1.15. Then show that no single $\delta > 0$ works for $\epsilon = 1$ at every $c \in \mathbb{R}$ simultaneously.

5. Let $E \subseteq \mathbb{R}$, let c be a limit point of E and let $f\colon E \to \mathbb{R}$ with $\lim_{x\to c} f(x) = L$. Prove that $\lim_{x\to c}|f(x)| = |L|$. Then give an E, a c and an f for which $\lim_{x\to c}|f(x)|$ exists and $\lim_{x\to c} f(x)$ does not.
6. Let $E \subseteq \mathbb{R}$ be nonempty and bounded above. Prove that $\sup E \in \overline{E}$, and deduce that a nonempty closed set that is bounded above contains its supremum.
7. Let $f\colon (0,\infty) \to \mathbb{R}$ be $f(x) = 1/x$. Prove directly from definition 3.1.27 that $\lim_{x\to\infty} f(x) = 0$, and use proposition 3.1.25(3) to show that there is no $L \in \mathbb{R}$ with $\lim_{x\to 0^+} f(x) = L$.
8. Let $E \subseteq \mathbb{R}$, let c be a limit point of E, and let $f, g\colon E \to \mathbb{R}$ satisfy $f(x) \le g(x)$ for every $x \in E$. Suppose both limits at c exist. Prove that $\lim_{x\to c} f(x) \le \lim_{x\to c} g(x)$. Then give an example with $f(x) < g(x)$ for every $x \in E$ in which the two limits are equal.

3.2 Continuous Functions

Section 3.1 attached a limit to a point its domain approaches and read nothing from the function's value there. Continuity at a point is the condition that the value and the limit agree, and the definition below states it by dropping the restriction $x \ne c$ from the ϵ-δ sentence. Two things change with it. The point c must now lie in the domain, since $f(c)$ occurs in the inequality, and c need no longer be a limit point of the domain, since the condition can be satisfied with no nearby domain points at all.

Definition 3.2.1: [Real] Continuity at a Point

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ and let $c \in E$. Then f is *continuous at c* if for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(c)| < \epsilon \qquad \text{for every } x \in E \text{ with } |x - c| < \delta$$

If f is not continuous at c, then f is *discontinuous at c*, and c is a *discontinuity* of f.

Definition 3.2.2: Continuous Function

Let $E \subseteq \mathbb{R}$ and let $f\colon E \to \mathbb{R}$. Then f is *continuous* if it is continuous at every point of E. For $A \subseteq E$, the function f is *continuous on A* if the restriction of f to A is continuous in this sense.

The clause about a subset is not idle. The restriction of f to A is a function on the smaller domain A, so its continuity at $c \in A$ quantifies only over the points of A within δ of c, which is a weaker condition than continuity of f at c. The Dirichlet function of example 3.2.10 is continuous on $\mathbb{Q}$ and continuous at no point of $\mathbb{R}$.

A constant function $f(x) = \lambda$ on E is continuous at every $c \in E$, since $|f(x) - f(c)| = |\lambda - \lambda| = 0 < \epsilon$ for every $x \in E$, so any $\delta > 0$ answers any ϵ. For $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = 3x - 4$ and any $c \in \mathbb{R}$,

$$|f(x) - f(c)| = |3x - 3c| = 3|x - c|$$

the last equality by proposition 1.3.8(3). Given $\epsilon > 0$, put $\delta = \epsilon/3$, which is positive by proposition 1.1.2(7). Then $|x - c| < \delta$ gives $|f(x) - f(c)| = 3|x - c| < 3\delta = \epsilon$, using proposition 1.1.2(5) to multiply the inequality by $3 > 0$. So f is continuous.

The definition of the functional limit required c to be a limit point of E, and box 3.1.16(2) showed why: at an isolated point every candidate value satisfies the ϵ-δ condition vacuously, so no limit is determined. Definition 3.2.1 carries no such hypothesis, and the vacuous case does no harm here, because $f(c)$ is already fixed by f and nothing has to be determined. The next proposition records both halves.

Proposition 3.2.3: Continuity Against the Functional Limit

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ and let $c \in E$.

1. If c is a limit point of E, then f is continuous at c if and only if $\lim_{x\to c} f(x)$ exists and equals $f(c)$.
2. If c is an isolated point of E, then f is continuous at c.

***Proof*:**

1. Suppose f is continuous at c and let $\epsilon > 0$. Take $\delta > 0$ as in definition 3.2.1. Every $x \in E$ with $0 < |x - c| < \delta$ satisfies $|x - c| < \delta$, so $|f(x) - f(c)| < \epsilon$. Since c is a limit point of E, this is the condition of definition 3.1.15 for the value $f(c)$, so $\lim_{x\to c} f(x) = f(c)$.

   Conversely suppose $\lim_{x\to c} f(x) = f(c)$ and let $\epsilon > 0$. Definition 3.1.15 gives a $\delta > 0$ with $|f(x) - f(c)| < \epsilon$ for every $x \in E$ satisfying $0 < |x - c| < \delta$. The only point of E with $|x - c| < \delta$ not covered by that statement is $x = c$ itself, and there $|f(c) - f(c)| = 0 < \epsilon$ by proposition 1.3.8(1). So the same δ answers ϵ in definition 3.2.1.

2. By definition 3.1.6 the point c lies in E and is not a limit point of E. Negating definition 3.1.5, there is a $\delta_0 > 0$ such that no $x \in E$ satisfies $0 < |x - c| < \delta_0$. Let $\epsilon > 0$ and take $\delta = \delta_0$. Suppose $x \in E$ and $|x - c| < \delta_0$. The quantity $|x - c|$ is nonnegative by proposition 1.3.8(1) and cannot be positive by the previous sentence, so it is 0 and $x = c$, again by proposition 1.3.8(1). Hence $|f(x) - f(c)| = 0 < \epsilon$, and f is continuous at c, completing the proof.

Part (2) says that continuity detects nothing at an isolated point. A function on $\mathbb{Z}$, or on $\{1/n \big| n \in \mathbb{N}\}$, is continuous no matter how its values are assigned, since each point of either set is isolated. Continuity is a statement about a function together with the way its domain approaches each point, and where the domain offers no approach the statement is empty.

The characterization by sequences is the form in which continuity is applied below. It holds at every point of the domain, with no limit-point hypothesis, which theorem 3.1.19 cannot dispense with.

Proposition 3.2.4: The Sequential Characterization of Continuity

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ and let $c \in E$. Then f is continuous at c if and only if $f(x_n) \to f(c)$ for every sequence $(x_n)_{n\geq 1}$ of points of E with $x_n \to c$.

***Proof*:**
Step 1: Continuity gives the sequential condition. Suppose f is continuous at c and let (x_n) be a sequence in E with $x_n \to c$. Let $\epsilon > 0$ and take $\delta > 0$ as in definition 3.2.1. By definition 2.1.2 applied to (x_n) with the positive real δ, there is an $N \in \mathbb{N}$ with $|x_n - c| < \delta$ for every $n \geq N$. Each such x_n lies in E, so $|f(x_n) - f(c)| < \epsilon$ for every $n \geq N$. Since $\epsilon > 0$ was arbitrary, $f(x_n) \to f(c)$.

Step 2: The sequential condition gives continuity. The argument is by contraposition. Suppose f is not continuous at c. Negating definition 3.2.1, there is an $\epsilon_0 > 0$ such that for every $\delta > 0$ some $x \in E$ satisfies $|x - c| < \delta$ and $|f(x) - f(c)| \geq \epsilon_0$. For each $n \in \mathbb{N}$ apply this with $\delta = 1/n$, positive by proposition 1.1.2(7), and let $x_n \in E$ be a point it produces.[a]

The sequence (x_n) converges to c. Given $\epsilon > 0$, theorem 1.4.1(3) gives an $N \in \mathbb{N}$ with $0 < 1/N < \epsilon$, and for $n \geq N$ the bound $|x_n - c| < 1/n \leq 1/N < \epsilon$ holds, the middle inequality by proposition 1.1.2(7). But $|f(x_n) - f(c)| \geq \epsilon_0$ for every n, so no index can force $|f(x_n) - f(c)| < \epsilon_0$, and $(f(x_n))$ does not converge to $f(c)$. So the sequential condition fails, completing the proof.

[a]Selecting one such x_n for every n at once is an appeal to countable choice. Unlike the least-index choices used for subsequences in theorem 2.2.9, the sets drawn from here are sets of real numbers and carry no canonical element.

The results of chapter 2 about sequences now transfer to functions, and the first case is the algebra. The proof does not repeat the ϵ-δ estimates of theorem 2.1.6, which were carried out once there and enter here through proposition 3.2.4.

Theorem 3.2.5: The Algebra of Continuous Functions

Let $E \subseteq \mathbb{R}$, let $f, g\colon E \to \mathbb{R}$ be continuous at $c \in E$ and let $\lambda \in \mathbb{R}$. Then

1. $f + g$ is continuous at c;
2. λf is continuous at c;
3. fg is continuous at c;
4. if $g(c) \neq 0$, then f/g, viewed as a function on $E^* = \{x \in E \,|\, g(x) \neq 0\}$, is continuous at c.

Consequently, if f and g are continuous on E, then so are $f + g$, λf and fg, and f/g is continuous on E^*.

***Proof*:**

1. Let (x_n) be a sequence in E with $x_n \to c$. By proposition 3.2.4 applied to f and then to g, $f(x_n) \to f(c)$ and $g(x_n) \to g(c)$. By theorem 2.1.6(1), $(f+g)(x_n) = f(x_n) + g(x_n) \to f(c) + g(c) = (f+g)(c)$. Since (x_n) was an arbitrary sequence in E converging to c, proposition 3.2.4 makes $f+g$ continuous at c.

2. The same sequence argument with theorem 2.1.6(2) gives $\lambda f(x_n) \to \lambda f(c)$.

3. The same sequence argument with theorem 2.1.6(3) gives $f(x_n)g(x_n) \to f(c)g(c)$.

4. Since $g(c) \neq 0$, the point c lies in E^*, so f/g is defined at c and the statement has content. Let (x_n) be a sequence in E^* with $x_n \to c$. Every x_n lies in E, so $f(x_n) \to f(c)$ and $g(x_n) \to g(c)$ as above. The hypotheses of theorem 2.1.6(4) hold: the limit $g(c)$ is nonzero, and $g(x_n) \neq 0$ for every n because $x_n \in E^*$. So $f(x_n)/g(x_n) \to f(c)/g(c)$, and proposition 3.2.4, applied to the function f/g on the domain E^*, makes f/g continuous at c.

The final sentence is each part applied at every point of the relevant domain, completing the proof.

Taking $\lambda = -1$ in part (2) shows that f is continuous at c exactly when $-f$ is, since applying (2) twice returns f. That symmetry transfers every statement below about increasing functions to decreasing ones.

Example 3.2.6: Polynomials and Rational Functions

1. The identity function $f(x) = x$ on $\mathbb{R}$ is continuous: given $c \in \mathbb{R}$ and $\epsilon > 0$, the choice $\delta = \epsilon$ gives $|f(x) - f(c)| = |x - c| < \epsilon$ whenever $|x - c| < \delta$. Constant functions are continuous by the computation following definition 3.2.2. By induction on n, the function $x \mapsto x^n$ is continuous on $\mathbb{R}$ for every integer $n \geq 0$: the case $n = 0$ is the constant 1, and if $x \mapsto x^n$ is continuous then so is $x \mapsto x^{n+1} = x \cdot x^n$ by theorem 3.2.5(3). Applying theorem 3.2.5(2) to each $a_k x^k$ and then theorem 3.2.5(1) repeatedly, every polynomial

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n$$

with real coefficients is continuous on $\mathbb{R}$. If p and q are polynomials, theorem 3.2.5(4) makes the rational function p/q continuous on $\{x \in \mathbb{R} \big| q(x) \neq 0\}$.

2. The general argument hides the estimate, so here is one case run in full. Fix $c \in \mathbb{R}$ and let

$f(x) = x^2$. For any x,

$$|f(x) - f(c)| = |x^2 - c^2| = |x - c|\,|x + c|$$

by proposition 1.3.8(3), since $x^2 - c^2 = (x - c)(x + c)$. Write $x + c = (x - c) + 2c$, so that $|x + c| \leq |x - c| + 2|c|$ by parts (3) and (4) of the same proposition. Given $\epsilon > 0$, put

$$\delta = \min\left\{1, \; \frac{\epsilon}{1 + 2|c|}\right\}$$

which is positive because $1 + 2|c| \geq 1 > 0$ by proposition 1.3.8(1). If $|x - c| < \delta$ then $|x - c| < 1$, so $|x + c| < 1 + 2|c|$, and therefore

$$|f(x) - f(c)| < \delta\,(1 + 2|c|) \leq \frac{\epsilon}{1 + 2|c|}\,(1 + 2|c|) = \epsilon$$

So $x \mapsto x^2$ is continuous at c. The δ depends on c as well as on ϵ, and it shrinks as $|c|$ grows. Section 3.4 takes up the question of when that dependence can be removed.

Composition is the operation the functional limit handles badly and continuity handles cleanly.

Theorem 3.2.7: Compositions of Continuous Functions

Let $E, F \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ satisfy $f(E) \subseteq F$ and let $g\colon F \to \mathbb{R}$. If f is continuous at $c \in E$ and g is continuous at $f(c)$, then $g \circ f\colon E \to \mathbb{R}$ is continuous at c. If f is continuous on E and g is continuous on F, then $g \circ f$ is continuous on E.

***Proof*:**
Write $b = f(c)$, a point of F because $f(E) \subseteq F$. Let $\epsilon > 0$. Since g is continuous at b, definition 3.2.1 gives an $\eta > 0$ such that

$$|g(y) - g(b)| < \epsilon \qquad \text{for every } y \in F \text{ with } |y - b| < \eta$$

Since f is continuous at c and $\eta > 0$, definition 3.2.1 gives a $\delta > 0$ such that $|f(x) - b| < \eta$ for every $x \in E$ with $|x - c| < \delta$. Fix such an x and put $y = f(x)$. Then $y \in F$ and $|y - b| < \eta$, so the displayed inequality applies and gives

$$|(g \circ f)(x) - (g \circ f)(c)| = |g(y) - g(b)| < \epsilon$$

So $g \circ f$ is continuous at c. The second sentence is the first applied at every $c \in E$, using that $f(c) \in F$ for each such c, completing the proof.

The step that carries the proof is the one where $y = f(x)$ is fed to the inequality for g without asking whether $y = b$. A functional limit forbids exactly that question, since $\lim_{y \to b} g(y)$ is computed with $y = b$ excluded, and the next example shows that the corresponding statement for limits is false.

Example 3.2.8: The Composition of Two Functional Limits

Define $g\colon \mathbb{R} \to \mathbb{R}$ by

$$g(y) = \begin{cases} 1 & \text{if } y = 0 \\ 0 & \text{if } y \neq 0 \end{cases}$$

Then $\lim_{y\to 0} g(y) = 0$: given $\epsilon > 0$, every y with $0 < |y - 0| < 1$ satisfies $y \neq 0$ and hence $|g(y) - 0| = 0 < \epsilon$, so $\delta = 1$ answers every ϵ.

1. Let $f\colon \mathbb{R} \to \mathbb{R}$ be the constant function $f(x) = 0$. Then $\lim_{x\to 0} f(x) = 0$ by the same argument, and $g(f(x)) = g(0) = 1$ for every x, so $\lim_{x\to 0} g(f(x)) = 1$. Both inner and outer limits are 0 while the composite has limit 1.

2. The failure does not depend on f being constant. Define $h\colon \mathbb{R} \to \mathbb{R}$ by $h(x) = 0$ for $x \in \mathbb{Q}$ and $h(x) = x$ for $x \notin \mathbb{Q}$. Then $\lim_{x\to 0} h(x) = 0$, since $|h(x) - 0| \leq |x|$ for every x, so $\delta = \epsilon$ answers ϵ. The composite $g \circ h$ has no limit at 0 at all. For each $n \in \mathbb{N}$, theorem 1.4.4 applied to $0 < 1/n$ gives a rational p_n with $0 < p_n < 1/n$, and corollary 1.4.5 applied to the same pair gives an irrational t_n with $0 < t_n < 1/n$. Both sequences converge to 0: given $\epsilon > 0$, theorem 1.4.1(3) produces N with $0 < 1/N < \epsilon$, and for $n \geq N$ both $|p_n| < 1/n \leq 1/N < \epsilon$ and $|t_n| < 1/n \leq 1/N < \epsilon$. Both sequences also avoid 0. But $g(h(p_n)) = g(0) = 1$ for every n, while $g(h(t_n)) = g(t_n) = 0$ for every n, since $t_n \neq 0$. Two sequences approaching 0 from within $\mathbb{R} \setminus \{0\}$ produce composite values with different limits, so theorem 3.1.19 rules out any limit for $g \circ h$ at 0.

What breaks in both cases is that the inner function attains the value 0, and 0 is the one point deleted in the computation of the outer limit. Redefining $g(0) = 0$ makes g continuous at 0 and removes both pathologies, which is theorem 3.2.7 at work.

Example 3.2.9: The Absolute Value and the Pointwise Maximum

The function $x \mapsto |x|$ is continuous on $\mathbb{R}$. Indeed proposition 1.3.8(5) gives

$$\big|\, |x| - |c| \,\big| \leq |x - c|$$

for all $x, c \in \mathbb{R}$, so $\delta = \epsilon$ answers every $\epsilon > 0$ at every c. Combining this with theorem 3.2.7, if $f\colon E \to \mathbb{R}$ is continuous then so is $|f|$, the function $x \mapsto |f(x)|$.

For $f, g\colon E \to \mathbb{R}$ write $\max(f, g)$ for the function $x \mapsto \max\{f(x), g(x)\}$ and $\min(f, g)$ for $x \mapsto \min\{f(x), g(x)\}$. These are continuous whenever f and g are, because of the identities

$$\max(f, g) = \frac{f + g + |f - g|}{2}, \qquad \min(f, g) = \frac{f + g - |f - g|}{2}$$

which hold pointwise. At a point where $f(x) \geq g(x)$ the right-hand side of the first is $(f(x)+g(x)+f(x)-g(x))/2 = f(x)$, and at a point where $f(x) < g(x)$ it is $(f(x)+g(x)+g(x)-f(x))/2 = g(x)$. The second identity is checked the same way. The right-hand sides are built from f, g and $|f - g|$ by the operations of theorem 3.2.5, so both are continuous.

That is the whole catalogue available at this point: polynomials, rational functions, the absolute value, and everything obtained from them by the algebra and by composition. The exponential, cosine and sine functions were defined by their power series in definition 2.6.5, but nothing proved so far says that a power series defines a continuous function, and their continuity is established in **??**. The n-th root function is absent for a different reason: its continuity follows from a theorem about inverses of continuous *strictly* monotone functions, which is theorem 3.3.12 in the next section.

The catalogue is short, and the three functions that follow show why a longer one would still leave open the question of which subsets of $\mathbb{R}$ occur as sets of discontinuities. Each treats rational and irrational arguments differently, and between them they realize $\mathbb{R}$, $\mathbb{R} \setminus \{0\}$ and $\mathbb{Q}$.

Example 3.2.10: The Dirichlet Function

The *Dirichlet function* $\mathbf{1}_{\mathbb{Q}} \colon \mathbb{R} \to \mathbb{R}$ is the indicator of the rationals,

$$\mathbf{1}_{\mathbb{Q}}(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

It is discontinuous at every point of $\mathbb{R}$.

Let $c \in \mathbb{R}$. For each $n \in \mathbb{N}$, theorem 1.4.4 applied to $c < c + 1/n$ gives a rational p_n with $c < p_n < c + 1/n$, and corollary 1.4.5 applied to the same pair gives an irrational t_n with $c < t_n < c + 1/n$. Both sequences converge to c: given $\epsilon > 0$, theorem 1.4.1(3) produces N with $0 < 1/N < \epsilon$, and for $n \geq N$ both $|p_n - c| < 1/n \leq 1/N < \epsilon$ and $|t_n - c| < 1/n \leq 1/N < \epsilon$, the middle inequality by proposition 1.1.2(7).

Now $\mathbf{1}_{\mathbb{Q}}(p_n) = 1$ for every n and $\mathbf{1}_{\mathbb{Q}}(t_n) = 0$ for every n, so the two image sequences converge to 1 and to 0. At most one of these equals $\mathbf{1}_{\mathbb{Q}}(c)$, since $1 \neq 0$, so proposition 3.2.4 fails for one of the two sequences and $\mathbf{1}_{\mathbb{Q}}$ is discontinuous at c.

The restriction of $\mathbf{1}_{\mathbb{Q}}$ to $\mathbb{Q}$ is the constant 1 and its restriction to $\mathbb{R} \setminus \mathbb{Q}$ is the constant 0, so $\mathbf{1}_{\mathbb{Q}}$ is continuous on each of those two sets in the sense of definition 3.2.2 while being continuous at no point of $\mathbb{R}$. It is also bounded and not Riemann integrable.

Discontinuity everywhere is one extreme. The next function differs from $\mathbf{1}_{\mathbb{Q}}$ by a single factor of x and stops one point short of that extreme, being continuous at exactly one point and discontinuous at every other.

Example 3.2.11: The Dirichlet Function Multiplied by x

Define $u \colon \mathbb{R} \to \mathbb{R}$ by $u(x) = x\,\mathbf{1}_{\mathbb{Q}}(x)$, so that $u(x) = x$ for rational x and $u(x) = 0$ for irrational x. Then u is continuous at 0 and discontinuous at every other point.

Continuity at 0*.* For every $x \in \mathbb{R}$ the value $u(x)$ is either x or 0, so $|u(x)| \leq |x|$ by proposition 1.3.8(1). Since $u(0) = 0$, given $\epsilon > 0$ the choice $\delta = \epsilon$ gives $|u(x) - u(0)| = |u(x)| \leq |x| < \epsilon$ whenever $|x - 0| < \delta$.

Discontinuity elsewhere. Let $c \neq 0$ and take the sequences (p_n) of rationals and (t_n) of irrationals converging to c constructed in example 3.2.10. Then $u(p_n) = p_n \to c$ and $u(t_n) = 0$ for every n, so $u(t_n) \to 0$. Since $c \neq 0$, the two limits differ, so at most one equals $u(c)$ and proposition 3.2.4 fails for one of the two sequences. So u is discontinuous at c.

Both functions so far are discontinuous at all but at most one point, and neither settles whether a set whose complement is itself dense, such as $\mathbb{Q}$, can be exactly the set of discontinuities of a function. Thomae answered that in 1875 with the function below, whose value at a rational records how large a denominator that rational needs. The argument proves a single statement, that the limit of t is 0 at every real number, and both the continuity and the discontinuity follow from it.

Example 3.2.12: Thomae's Function

Every rational number has exactly one representation p/q with $p \in \mathbb{Z}$, q a positive integer and p/q in lowest terms, which is among the facts about $\mathbb{Z}$ and $\mathbb{Q}$ taken as given in section 1.1. *Thomae's function* $t\colon \mathbb{R} \to \mathbb{R}$ is

$$t(x) = \begin{cases} 1/q & \text{if } x \in \mathbb{Q} \text{ and } x = p/q \text{ in lowest terms with } q \geq 1 \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

In particular $t(0) = 1$, since $0 = 0/1$ in lowest terms, and $t(50/100) = t(1/2) = 1/2$, the reduction to lowest terms being what makes the value depend on the number alone, whatever fraction is written for it. Then t is continuous at every irrational number and discontinuous at every rational number.

Step 1: The rationals within 1 *of* c *whose reduced denominator is at most* N. Fix $c \in \mathbb{R}$ and let $\epsilon > 0$. By theorem 1.4.1(3) choose $N \in \mathbb{N}$ with $0 < 1/N < \epsilon$, and set

$$F = \{p/q \big| q \in \{1, \ldots, N\},\ p \in \mathbb{Z},\ \lfloor q(c-1) \rfloor \leq p \leq \lfloor q(c+1) \rfloor\}$$

the integer parts being those of proposition 1.4.3. For each of the N values of q the integer p is confined between two fixed integers, so only finitely many p occur, and F is a finite set of rational numbers.

Step 2: F catches every rational near c with a small denominator. Let $x \in \mathbb{Q}$ satisfy $|x - c| \leq 1$ and write $x = p/q$ in lowest terms with $1 \leq q \leq N$. By proposition 1.3.8(2) the hypothesis reads $c - 1 \leq x \leq c + 1$, and multiplying through by $q > 0$ using proposition 1.1.2(5) gives $q(c-1) \leq p \leq q(c+1)$. Now proposition 1.4.3 gives $\lfloor q(c-1) \rfloor \leq q(c-1)$, so $\lfloor q(c-1) \rfloor \leq p$. And if $p > \lfloor q(c+1) \rfloor$ then $p \geq \lfloor q(c+1) \rfloor + 1 > q(c+1)$, again by proposition 1.4.3, contradicting $p \leq q(c+1)$. So $p \leq \lfloor q(c+1) \rfloor$, and hence $x \in F$.

Step 3: The choice of δ. The set $F \setminus \{c\}$ is finite, and $|c - r| > 0$ for each of its elements r by proposition 1.3.8(1). Let δ be the least of the number 1 and the numbers $|c - r|$ for $r \in F \setminus \{c\}$, a minimum over a finite set of positive numbers, so $\delta > 0$.

Step 4: The estimate. Let $x \in \mathbb{R}$ with $0 < |x - c| < \delta$. If $x \notin \mathbb{Q}$ then $|t(x) - 0| = 0 < \epsilon$. If $x \in \mathbb{Q}$, write $x = p/q$ in lowest terms with $q \geq 1$, and suppose $q \leq N$. Since $|x - c| < \delta \leq 1$, Step 2 puts x in F, and $|x - c| > 0$ gives $x \neq c$, so $x \in F \setminus \{c\}$ and therefore $|c - x| \geq \delta$ by the choice of δ. That contradicts $|x - c| < \delta$. So $q > N$, and proposition 1.1.2(7) gives

$$|t(x) - 0| = \frac{1}{q} < \frac{1}{N} < \epsilon$$

Since $\epsilon > 0$ was arbitrary and c is a limit point of $\mathbb{R}$, definition 3.1.15 gives $\lim_{x\to c} t(x) = 0$.

Step 5: Continuity and its failure. Every $c \in \mathbb{R}$ is a limit point of $\mathbb{R}$, so proposition 3.2.3(1) makes t continuous at c exactly when $t(c) = \lim_{x\to c} t(x) = 0$. If $c \notin \mathbb{Q}$ then $t(c) = 0$ and t is continuous at c. If $c \in \mathbb{Q}$, write $c = p/q$ in lowest terms, so that $t(c) = 1/q > 0$ by proposition 1.1.2(7) and t is discontinuous at c.

So the set of discontinuities of a function can be all of $\mathbb{R}$, all of $\mathbb{R}$ with one point removed, or exactly $\mathbb{Q}$. Determining which subsets of $\mathbb{R}$ occur in general is beyond what has been built here. For one class of functions the constraint is severe and provable now, and reaching it takes a way of separating failures of continuity into kinds, since the three examples above fail in visibly different ways.

Definition 3.2.13: Discontinuities of the First and Second Kind

Let $a < b$ be real numbers, let $f\colon (a,b) \to \mathbb{R}$ and let $c \in (a,b)$ be a discontinuity of f. The discontinuity is *simple*, or *of the first kind*, if both one-sided limits $\lim_{x\to c^-} f(x)$ and $\lim_{x\to c^+} f(x)$ exist in $\mathbb{R}$, and *of the second kind* otherwise. A simple discontinuity is *removable* if the two one-sided limits are equal, and a *jump* if they are not.

The two names for a simple discontinuity describe the two ways it arises. If the one-sided limits agree, then $\lim_{x\to c} f(x)$ exists by proposition 3.1.26, and the only obstruction to continuity is the value $f(c)$, which can be redefined to that common limit. If they disagree, no value assigned at c can help.

Each of the three named functions is now classified. Thomae's function has a removable discontinuity at every rational: example 3.2.12 gives $\lim_{x\to c} t(x) = 0$, so proposition 3.1.26 makes both one-sided limits exist and equal 0, while $t(c) > 0$. The Dirichlet function has a discontinuity of the second kind at every $c \in (a,b)$, since neither one-sided limit exists. For the right-hand side, apply theorem 1.4.4 and corollary 1.4.5 inside the interval with endpoints c and $\min\{b, c + 1/n\}$ to get a rational p_n and an irrational t_n in (c,b) converging to c. Their images are the constant sequences 1 and 0, so theorem 3.1.19, applied to the restriction of $\mathbf{1}_{\mathbb{Q}}$ to (c,b) as proposition 3.1.25 permits, rules out a right-hand limit. The left-hand side is the same argument inside the interval with endpoints $\max\{a, c - 1/n\}$ and c. A jump is easiest to exhibit directly: let $f\colon (-1,1) \to \mathbb{R}$ be $f(x) = 0$ for $x < 0$ and $f(x) = 1$ for $x \geq 0$. Then $\lim_{x\to 0^-} f(x) = 0$ and $\lim_{x\to 0^+} f(x) = 1$, each because f is constant on the relevant side, so f has a jump at 0. It is continuous at every other point of $(-1,1)$, since every such point has a neighbourhood inside $(-1,0)$ or inside $(0,1)$ on which f is constant. The arithmetic of $\mathbb{Q}$ is not the only source of a second-kind discontinuity. Oscillation produces one too, in the function $x \mapsto \sin(1/x)$ extended by the value 0 at 0, treated in **??** once sin is available.

Monotone functions are the class for which the second kind is impossible, and the reason is that a supremum can be produced whenever a value is needed. The definition applies to a function on any subset of $\mathbb{R}$, while the results that follow are stated on an open interval, where both sides of every point carry domain points.

Definition 3.2.14: [Real] Monotone Function

Let $E \subseteq \mathbb{R}$ and let $f\colon E \to \mathbb{R}$. Then f is

1. *increasing* if $f(x) \leq f(y)$ whenever $x, y \in E$ satisfy $x \leq y$, and *strictly increasing* if $f(x) < f(y)$ whenever $x, y \in E$ satisfy $x < y$;
2. *decreasing* if $f(x) \geq f(y)$ whenever $x, y \in E$ satisfy $x \leq y$, and *strictly decreasing* if $f(x) > f(y)$ whenever $x, y \in E$ satisfy $x < y$.

It is *monotone* if it is increasing or decreasing, and *strictly monotone* if it is strictly increasing or strictly decreasing.

The affine function $f(x) = 2x + 5$ on $\mathbb{R}$ is strictly increasing, since $x < y$ gives $2x < 2y$ by proposition 1.1.2(5) and then $2x+5 < 2y+5$ by the third ordered-field axiom of definition 1.1.1. The same two steps with the factor -2 make $x \mapsto -2x+5$ strictly decreasing, using proposition 1.1.2(5) in its second form. A constant function is both increasing and decreasing and neither strictly. The Dirichlet function is neither on any interval containing more than one point, since theorem 1.4.4 and corollary 1.4.5 place a rational and an irrational between any two of its points. Definition 2.2.1 attached the same words to sequences. The two definitions agree when a sequence is read as a function on $\mathbb{N}$, and the one above is what the rest of this chapter uses.

Proposition 3.2.15: One-Sided Limits of an Increasing Function

Let $a < b$ be real numbers and let $f\colon (a,b) \to \mathbb{R}$ be increasing. Then for every $c \in (a,b)$ both one-sided limits of f at c exist in $\mathbb{R}$, and

$$\sup\{f(x) \big| a < x < c\} = \lim_{x\to c^-} f(x) \leq f(c) \leq \lim_{x\to c^+} f(x) = \inf\{f(x) \big| c < x < b\}$$

Moreover, if $a < c < d < b$, then $\lim_{x\to c^+} f(x) \leq \lim_{x \to d^-} f(x)$.

***Proof*:**
Step 1: The left-hand limit. Let $A = \{f(x) \big| a < x < c\}$. This set is nonempty, since $a < c$ gives a point of (a,c) by proposition 1.1.2(8), and it is bounded above by $f(c)$, since $a < x < c$ gives $f(x) \leq f(c)$ by definition 3.2.14. By theorem 1.2.4 the number $L = \sup A$ exists, and $L \leq f(c)$ because $f(c)$ is an upper bound of A and L is the least one.

Let $\epsilon > 0$. By proposition 1.2.6 there is a point of A exceeding $L - \epsilon$, that is, an $x_0 \in (a,c)$ with $L - \epsilon < f(x_0)$. Put $\delta = c - x_0 > 0$. If $x \in (a,c)$ satisfies $|x - c| < \delta$, then $c - x < \delta = c - x_0$ gives $x_0 < x < c$, so $f(x_0) \leq f(x)$ by definition 3.2.14 and $f(x) \leq L$ because L is an upper bound of A. Combining, $L - \epsilon < f(x) \leq L$, which gives $-\epsilon < f(x) - L \leq 0$ and so $|f(x) - L| < \epsilon$ by proposition 1.3.8(2). Since c is a limit point of (a,c), this is the condition of definition 3.1.24, and $\lim_{x\to c^-} f(x) = L$.

Step 2: The right-hand limit. Let $B = \{f(x) \big| c < x < b\}$, nonempty by proposition 1.1.2(8) and bounded below by $f(c)$, since $c < x < b$ gives $f(c) \leq f(x)$. By corollary 1.2.5 the number $M = \inf B$ exists, and $f(c) \leq M$ because $f(c)$ is a lower bound and M is the greatest one. Let

$\epsilon > 0$. By lemma 1.3.1 there is an $x_1 \in (c, b)$ with $M \leq f(x_1) < M + \epsilon$. Put $\delta = x_1 - c > 0$. If $x \in (c, b)$ satisfies $|x - c| < \delta$, then $c < x < x_1$, so $M \leq f(x) \leq f(x_1) < M + \epsilon$ and $|f(x) - M| < \epsilon$ by proposition 1.3.8(2). So $\lim_{x \to c^+} f(x) = M$.

Step 3: Comparison across two points. Let $a < c < d < b$. By proposition 1.1.2(8) there is a w with $c < w < d$. Then $w \in (c, b)$, so $f(w) \in B$ and $\lim_{x \to c^+} f(x) = \inf B \leq f(w)$. Also $w \in (a, d)$, so $f(w)$ belongs to the set whose supremum Step 1 identifies with $\lim_{x \to d^-} f(x)$, giving $f(w) \leq \lim_{x \to d^-} f(x)$. Chaining the two inequalities gives $\lim_{x \to c^+} f(x) \leq \lim_{x \to d^-} f(x)$, completing the proof.

Applying the proposition to $-f$, which is increasing exactly when f is decreasing, and using corollary 3.1.20 to move the sign outside each one-sided limit, gives the decreasing case with every inequality reversed. So a monotone function has both one-sided limits at every interior point of its interval, and by definition 3.2.13 it has no discontinuity of the second kind. The gap between the two one-sided limits is the whole of the failure, and the next theorem bounds how often that gap can be nonzero.

Theorem 3.2.16: The Discontinuity Set of a Monotone Function

Let $a < b$ be real numbers and let $f \colon (a, b) \to \mathbb{R}$ be monotone. Then the set of points of (a, b) at which f is discontinuous is finite or countable.

***Proof*:**
Step 1: Reduction to the increasing case. If f is decreasing then $-f$ is increasing, and f is continuous at a point exactly when $-f$ is, by theorem 3.2.5(2) applied with $\lambda = -1$ in both directions. So the two functions have the same set of discontinuities and it is enough to treat an increasing f. Write D for that set.

Step 2: A discontinuity is a strict gap. Let $c \in (a, b)$ and write $L(c) = \lim_{x \to c^-} f(x)$ and $M(c) = \lim_{x \to c^+} f(x)$, which exist by proposition 3.2.15 and satisfy $L(c) \leq f(c) \leq M(c)$. Suppose $L(c) = M(c)$. Then both numbers equal $f(c)$, and c is a limit point of (a, c) and of (c, b), so proposition 3.1.26 gives $\lim_{x \to c} f(x) = f(c)$. Since c is a limit point of (a, b), proposition 3.2.3(1) makes f continuous at c. Contrapositively, $c \in D$ forces $L(c) < M(c)$.

Step 3: An injection into $\mathbb{N}$. By [1, countable sets and the countability of $\mathbb{Q}$] the set $\mathbb{Q}$ is countable, so there is a function $r \colon \mathbb{N} \to \mathbb{Q}$ whose range is all of $\mathbb{Q}$. Let $c \in D$. Theorem 1.4.4 applied to $L(c) < M(c)$ gives a rational strictly between them, and that rational equals $r(n)$ for some n, so

$$S(c) = \{n \in \mathbb{N} \big| L(c) < r(n) < M(c)\}$$

is a nonempty set of positive integers. Let $n(c)$ be its least element, which exists by the well-ordering of $\mathbb{N}$ recorded in section 1.1. No choice is made, since $n(c)$ is determined by c.

Let $c_1, c_2 \in D$ with $c_1 \neq c_2$, say $c_1 < c_2$. The last clause of proposition 3.2.15 gives $M(c_1) \leq L(c_2)$, so

$$r(n(c_1)) < M(c_1) \leq L(c_2) < r(n(c_2))$$

and in particular $r(n(c_1)) \neq r(n(c_2))$, which forces $n(c_1) \neq n(c_2)$. So $c \mapsto n(c)$ is an injection of D into $\mathbb{N}$.

Step 4: Conclusion. The injection identifies D with a subset of $\mathbb{N}$, and $\mathbb{N}$ is countable, being the range of the identity function on $\mathbb{N}$. A subset of a finite or countable set is finite or countable, by [1, countable sets], so D is finite or countable, completing the proof.

The bound is exact. Every nonempty finite or countable subset of an interval is realized as a set of discontinuities, by a function built from a convergent series of positive numbers whose n-th term is deposited at the n-th point of the set.

Example 3.2.17: A Monotone Function with Prescribed Discontinuities
Let $a < b$ be real and let $S \subseteq (a, b)$ be nonempty, finite or countable. Then there is an increasing $f : (a, b) \to \mathbb{R}$ whose set of discontinuities is exactly S.

The construction. By [1, countable sets] there is a function $\sigma : \mathbb{N} \to S$ whose range is all of S. For $n \in \mathbb{N}$ define $u_n : (a, b) \to \mathbb{R}$ by $u_n(x) = 1$ if $\sigma(n) < x$ and $u_n(x) = 0$ otherwise, and set

$$f(x) = \sum_{n=1}^{\infty} 2^{-n} u_n(x)$$

The terms are nonnegative, and the partial sums satisfy $\sum_{n=1}^{M} 2^{-n} u_n(x) \leq \sum_{n=1}^{M} 2^{-n} = 1-2^{-M} \leq 1$, the middle equality by summing the identity $2^{-n} = 2^{-(n-1)}-2^{-n}$ from $n = 1$ to M and cancelling as in example 2.4.6. So the partial sums are bounded above, proposition 2.4.4 makes the series convergent, and f is defined on all of (a, b).

f is increasing. Let $x \leq y$ in (a, b) and let $n \in \mathbb{N}$. If $u_n(x) = 1$ then $\sigma(n) < x \leq y$, so $u_n(y) = 1$. Hence $u_n(x) \leq u_n(y)$ for every n, so the partial sums of the two series compare term by term, and proposition 2.1.8(1) gives $f(x) \leq f(y)$.

f is discontinuous at every point of S. Let $s \in S$ and choose $m \in \mathbb{N}$ with $\sigma(m) = s$, which exists because σ has range S. Let $x \in (a, b)$ with $x > s$. For every n, if $u_n(s) = 1$ then $\sigma(n) < s < x$ and $u_n(x) = 1$, so $u_n(x) - u_n(s) \geq 0$. Also $u_m(x) - u_m(s) = 1 - 0 = 1$, since $\sigma(m) = s < x$ while $\sigma(m) = s$ is not less than s. By proposition 2.4.5(1), applied with the scalar -1 and then with the sum, the series $\sum_{n\geq 1} 2^{-n}(u_n(x) - u_n(s))$ converges with sum $f(x) - f(s)$, and each of its partial sums with $M \geq m$ is at least 2^{-m}, because every term is nonnegative and the m-th equals 2^{-m}. By proposition 2.1.8(2), $f(x) - f(s) \geq 2^{-m}$.

Now take $\epsilon_0 = 2^{-m} > 0$ and let $\delta > 0$ be arbitrary. Since $s < b$ and $s < s + \delta$, proposition 1.1.2(8) gives a point x strictly between s and $\min\{b,\ s + \delta\}$, so $x \in (a, b)$ and $|x - s| < \delta$ while $|f(x) - f(s)| \geq \epsilon_0$. So no δ answers ϵ_0 in definition 3.2.1, and f is discontinuous at s.

f is continuous at every other point. Let $c \in (a, b)$ with $c \notin S$ and let $\epsilon > 0$. By proposition 2.1.10(3), applied with $j = 0$ and $r = 1/2$, the sequence $(2^{-k})_{k\geq 1}$ is null, so there is an $N \in \mathbb{N}$ with $2^{-N} < \epsilon$. For each $n \in \{1, \dots, N\}$ the point $\sigma(n)$ lies in S and so differs from c, whence $|c - \sigma(n)| > 0$ by proposition 1.3.8(1). Let δ be the least of those N numbers, a minimum over a

finite set of positive numbers, so $\delta > 0$.

Let $x \in (a, b)$ with $|x - c| < \delta$ and let $n \leq N$. If $\sigma(n) < c$ then $c - \sigma(n) \geq \delta$, so $\sigma(n) \leq c - \delta < x$ and $u_n(x) = 1 = u_n(c)$. If $\sigma(n) > c$ then $\sigma(n) - c \geq \delta$, so $\sigma(n) \geq c + \delta > x$ and $u_n(x) = 0 = u_n(c)$. The case $\sigma(n) = c$ does not arise. So the first N terms of the series for $f(x) - f(c)$ vanish, and for every $M > N$, using proposition 2.4.5(2) and $|u_n(x) - u_n(c)| \leq 1$,

$$\left|\sum_{n=1}^{M} 2^{-n}\big(u_n(x) - u_n(c)\big)\right| = \left|\sum_{n=N+1}^{M} 2^{-n}\big(u_n(x) - u_n(c)\big)\right| \leq \sum_{n=N+1}^{M} 2^{-n} = 2^{-N} - 2^{-M} \leq 2^{-N}$$

the last equality again by summing $2^{-n} = 2^{-(n-1)} - 2^{-n}$, now from $n = N+1$ to M. These partial sums converge to $f(x) - f(c)$ by proposition 2.4.5(1), and every one of them lies between -2^{-N} and 2^{-N}, so proposition 2.1.8(2) gives $-2^{-N} \leq f(x) - f(c) \leq 2^{-N}$ and hence $|f(x) - f(c)| \leq 2^{-N} < \epsilon$ by proposition 1.3.8(2). So f is continuous at c.

Taking $(a, b) = (0, 1)$ and $S = \mathbb{Q} \cap (0, 1)$ produces an increasing function on $(0, 1)$ whose discontinuities are exactly the rationals, the same set that example 3.2.12 produced by an entirely different construction.

Exercise 3.2.1

1. Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ be continuous at $c \in E$ and suppose $f(c) > 0$. Show that there is a $\delta > 0$ such that $f(x) > f(c)/2$ for every $x \in E$ with $|x - c| < \delta$. State and prove the corresponding statement when $f(c) < 0$.
2. Let I be an interval containing more than one point and let $f, g\colon I \to \mathbb{R}$ be continuous with $f(x) = g(x)$ for every $x \in I \cap \mathbb{Q}$. Show that $f = g$. Where does the argument use that I contains more than one point?
3. Let $f\colon \mathbb{R} \to \mathbb{R}$ satisfy $f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$. Show that if f is continuous at 0, then f is continuous at every point of $\mathbb{R}$.
4. Let t be Thomae's function of example 3.2.12. Show that t is monotone on no interval containing more than one point.
5. Give a continuous $f\colon \mathbb{R} \to \mathbb{R}$ and a discontinuous $g\colon \mathbb{R} \to \mathbb{R}$ with $g \circ f$ continuous at every point of $\mathbb{R}$, and a discontinuous f and a continuous g with the same property.
6. Let $f, g\colon \mathbb{R} \to \mathbb{R}$ be continuous. Show that $\{x \in \mathbb{R} \,|\, f(x) = g(x)\}$ is a closed subset of $\mathbb{R}$, and deduce that a continuous function agreeing with a polynomial at every rational number agrees with it at every real number.
7. Let $a < b$ be real, let $f\colon (a, b) \to \mathbb{R}$ be increasing and for $c \in (a, b)$ write $J(c) = \lim_{x \to c^+} f(x) - \lim_{x \to c^-} f(x)$. Show that $J(c) \geq 0$, and that for any points $u < c_1 < c_2 < \cdots < c_k < v$ of (a, b),
$$\sum_{i=1}^{k} J(c_i) \leq f(v) - f(u)$$

8. Let $f\colon \mathbb{R} \to \mathbb{R}$ be $f(x) = \lfloor x \rfloor$, the integer part of proposition 1.4.3. Show that f is increasing and continuous at every $c \notin \mathbb{Z}$. Fix $m \in \mathbb{Z}$ and restrict f to $(m-1, m+1)$, which puts it in the setting of definition 3.2.13; show that the restriction has a jump discontinuity at m, computing both one-sided limits there.

3.3 Continuous Functions on a Closed Bounded Interval

Section 3.1 and section 3.2 placed no condition on the domain: a continuous function was $f\colon E \to \mathbb{R}$ for an arbitrary $E \subseteq \mathbb{R}$, and the algebra and composition rules hold at that generality. This section imposes conditions on E and records what each one gives. That E is an interval produces the intermediate value theorem. That E is a closed bounded interval produces the boundedness of f, the attainment of its bounds, and the description of its image. Adding strict monotonicity to the first of these produces a continuous inverse.

Every hypothesis below constrains the domain, so the first thing to record is that shrinking a domain costs nothing.

Proposition 3.3.1: Restriction of a Continuous Function

Let $E \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ be continuous and let $D \subseteq E$. Then the restriction of f to D is continuous.

***Proof*:**
Let $x \in D$ and $\epsilon > 0$. The continuity of f at x gives a $\delta > 0$ such that every $t \in E$ with $|t - x| < \delta$ satisfies $|f(t) - f(x)| < \epsilon$ (definition 3.2.1). The points $t \in D$ with $|t - x| < \delta$ are among those t, so the same δ answers ϵ for the restriction at x.

In particular a polynomial function, regarded as a function on any subset of $\mathbb{R}$, is continuous, by proposition 3.3.1 together with example 3.2.6.

The intermediate value theorem fails over $\mathbb{Q}$, and the failure is concrete. Let $E_0 = \mathbb{Q} \cap [0, 2]$ and let $g\colon E_0 \to \mathbb{R}$ be $g(q) = q^2 - 2$. It is continuous, being a polynomial restricted to E_0, and $g(0) = -2 < 0 < 2 = g(2)$, yet $g(q) = 0$ has no solution in E_0 because $q^2 = 2$ has no rational solution (proposition 1.1.3). What E_0 lacks is the betweenness condition of definition 3.1.1. The number $\sqrt{2}$ given by theorem 1.3.12 satisfies $1 < \sqrt{2} < 2$. It is positive, since $(\sqrt{2})^2 = 2 \neq 0$ forces $\sqrt{2} \neq 0$ while $\sqrt{2} \geq 0$. If $\sqrt{2} = 1$ then $2 = 1$, and if $\sqrt{2} < 1$ then multiplying by the positive number $\sqrt{2}$ gives $2 = (\sqrt{2})^2 < \sqrt{2} < 1$ (proposition 1.1.2(5)), so $\sqrt{2} > 1$. If $\sqrt{2} = 2$ then $2 = 4$, and if $\sqrt{2} > 2$ then the same multiplication gives $2 = (\sqrt{2})^2 > 2\sqrt{2}$ while $2\sqrt{2} > 4$, so $\sqrt{2} < 2$. Hence 1 and 2 lie in E_0 while the number $\sqrt{2}$ between them does not.

Theorem 3.3.2: The Intermediate Value Theorem

1. Let $a < b$ and let $f\colon [a, b] \to \mathbb{R}$ be continuous. If $y \in \mathbb{R}$ satisfies $\min\{f(a), f(b)\} < y < \max\{f(a), f(b)\}$, then $f(c) = y$ for some $c \in (a, b)$.
2. Let $I \subseteq \mathbb{R}$ be an interval, let $f\colon I \to \mathbb{R}$ be continuous, and let $u, v \in I$ with $u < v$. If $y \in \mathbb{R}$ satisfies $\min\{f(u), f(v)\} < y < \max\{f(u), f(v)\}$, then $f(c) = y$ for some $c \in (u, v)$.

***Proof*:**

1. *Reduction to a sign change at* 0. If $f(a) < y < f(b)$, put $g = f - y$. If $f(b) < y < f(a)$, put $g = y - f$. In both cases $g\colon [a, b] \to \mathbb{R}$ is continuous by theorem 3.2.5, applied to f and to the constant function with value y, which is a polynomial restricted to $[a, b]$. In the first case $g(a) = f(a) - y < 0$ and $g(b) = f(b) - y > 0$; in the second, $g(a) = y - f(a) < 0$ and $g(b) = y - f(b) > 0$. In both cases $g(c) = 0$ holds exactly when $f(c) = y$. So it suffices to produce a $c \in (a, b)$ with $g(c) = 0$, given a continuous $g\colon [a, b] \to \mathbb{R}$ with $g(a) < 0 < g(b)$.

   The supremum. Let
   $$S = \{x \in [a, b] \big| g(x) < 0\}$$
   Then $a \in S$, so S is nonempty, and b is an upper bound of S. By theorem 1.2.4 the number $c = \sup S$ exists, and $a \le c \le b$, since $a \in S$ forces $a \le c$ and b is an upper bound.

   Step 1: $a < c$. Apply definition 3.2.1 at a with $\epsilon = -g(a) > 0$: there is a $\delta_1 > 0$ such that every $x \in [a, b]$ with $|x - a| < \delta_1$ satisfies $|g(x) - g(a)| < -g(a)$, hence $g(a) < g(x) - g(a) < -g(a)$ by proposition 1.3.8(2), hence $g(x) < 0$. Put $x_0 = \min\{b,\ a + \delta_1/2\}$. Then $a < x_0 \le b$, because $a < b$ and $a < a + \delta_1/2$, and $|x_0 - a| = x_0 - a \le \delta_1/2 < \delta_1$. So $g(x_0) < 0$, that is $x_0 \in S$, and $c \ge x_0 > a$.

   Step 2: $c < b$. Apply definition 3.2.1 at b with $\epsilon = g(b) > 0$: there is a $\delta_2 > 0$ such that every $x \in [a, b]$ with $|x - b| < \delta_2$ satisfies $|g(x) - g(b)| < g(b)$, hence $g(x) > 0$ by proposition 1.3.8(2). Put $x_1 = \max\{a,\ b - \delta_2/2\}$, so that $a \le x_1 < b$. Every $x \in [a, b]$ with $x > x_1$ satisfies $b - x < b - (b - \delta_2/2) = \delta_2/2 < \delta_2$, so $g(x) > 0$ and $x \notin S$. Every element of S is therefore at most x_1, and $c \le x_1 < b$.

   Step 3: $g(c) \le 0$. Suppose $g(c) > 0$. Apply definition 3.2.1 at c with $\epsilon = g(c)$: there is a $\delta > 0$ such that every $x \in [a, b]$ with $|x - c| < \delta$ satisfies $g(x) > 0$, again by proposition 1.3.8(2). Put $t = \max\{a,\ c - \delta/2\}$. Then $t < c$, since $a < c$ by Step 1 and $c - \delta/2 < c$. If $x \in S$ satisfied $x > t$, then $x \le c$ and $c - x < c - (c - \delta/2) = \delta/2 < \delta$, so $|x - c| < \delta$ and $g(x) > 0$, contradicting $x \in S$. So t is an upper bound of S strictly smaller than c, which contradicts the definition of c as the least upper bound (definition 1.2.2).

   Step 4: $g(c) \ge 0$. Suppose $g(c) < 0$. Apply definition 3.2.1 at c with $\epsilon = -g(c)$: there is a $\delta > 0$ such that every $x \in [a, b]$ with $|x - c| < \delta$ satisfies $g(x) < 0$. Put $x_2 = \min\{b,\ c + \delta/2\}$. By Step 2, $c < b$, so $c < x_2 \le b$, and $|x_2 - c| = x_2 - c \le \delta/2 < \delta$. Hence $g(x_2) < 0$, that is $x_2 \in S$, and $x_2 > c = \sup S$ contradicts that c is an upper bound.

Steps 3 and 4 give $g(c) = 0$, and Steps 1 and 2 give $c \in (a, b)$.

2. Since I is an interval and $u, v \in I$ with $u < v$, every z with $u < z < v$ lies in I (definition 3.1.1), so $[u, v] \subseteq I$. The restriction of a continuous function to a subset of its domain is continuous, so f restricted to $[u, v]$ is continuous. Part (1) applied to that restriction on $[u, v]$ produces the required $c \in (u, v)$, completing the proof.

The theorem asserts existence and nothing else. It does not say how many solutions $f(c) = y$ has, and it gives no procedure for finding one. The two hypotheses enter at separate places. Continuity is what the four applications of definition 3.2.1 in Steps 1 to 4 use, and completeness is what the single line producing $c = \sup S$ uses. Over $\mathbb{Q}$ the second is unavailable, and the function $g(q) = q^2 - 2$ on $\mathbb{Q} \cap [0, 2]$ is what its absence looks like.

One immediate application settles a question about polynomials that the algebra of section 3.2 cannot reach.

Corollary 3.3.3: Odd-Degree Polynomials Have Real Roots

Let $n \geq 1$ be an odd integer, let $c_0, \ldots, c_n \in \mathbb{R}$ with $c_n \neq 0$, and let

$$p(x) = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0$$

Then $p(c) = 0$ for some $c \in \mathbb{R}$.

***Proof*:**
The polynomial $-p$ has the same roots as p and leading coefficient $-c_n$, so it may be assumed that $c_n > 0$.

Step 1: a radius beyond which the leading term dominates. Put $A = |c_0| + \cdots + |c_{n-1}|$ and $M = 1 + A/c_n$, so that $M \geq 1$ and $c_n M = c_n + A$. For every integer k with $0 \leq k \leq n - 1$,

$$M^k \leq M^{n-1}$$

To see it, note that $M \geq 1 > 0$ makes every power M^j positive (proposition 1.1.2(3) and (5)), and multiplying $1 \leq M$ by M^j gives $M^j \leq M^{j+1}$, with equality when $M = 1$ and strict inequality by proposition 1.1.2(5) when $M > 1$. These inequalities chain from k up to $n - 1$. Hence, using $|c_k M^k| = |c_k| M^k$ (proposition 1.3.8(3)) and applying proposition 1.3.8(4) $n - 1$ times,

$$\left| \sum_{k=0}^{n-1} c_k M^k \right| \leq \sum_{k=0}^{n-1} |c_k| M^k \leq A M^{n-1}$$

The same estimate holds with M replaced by $-M$, since $|(-M)^k| = M^k$ by proposition 1.3.8(3) applied k times.

Step 2: the two sign evaluations. From $c_n M^n = (c_n M) M^{n-1} = (c_n + A) M^{n-1}$ and $M^{n-1} > 0$,

$$p(M) = c_n M^n + \sum_{k=0}^{n-1} c_k M^k \geq c_n M^n - A M^{n-1} = c_n M^{n-1} > 0$$

where the first inequality uses $s \geq -|s|$ (proposition 1.3.8(1)) with $s = \sum_{k<n} c_k M^k$. Since n is odd, $(-M)^n = -M^n$, so the same computation gives

$$p(-M) = -c_n M^n + \sum_{k=0}^{n-1} c_k (-M)^k \leq -c_n M^n + AM^{n-1} = -c_n M^{n-1} < 0$$

Step 3: the intermediate value theorem. The function p is continuous on $\mathbb{R}$ (example 3.2.6), the set $\mathbb{R}$ is an interval, and $-M < M$ because $M \geq 1 > 0$. Since $\min\{p(-M), p(M)\} = p(-M) < 0 < p(M) = \max\{p(-M), p(M)\}$, the value 0 lies strictly between $p(-M)$ and $p(M)$, and theorem 3.3.2(2) produces a $c \in (-M, M)$ with $p(c) = 0$, as we sought to show.

The hypothesis that n be odd is what makes the two evaluations have opposite signs, and it cannot be dropped: $p(x) = x^2+1$ satisfies $p(x) \geq 1$ for every real x, since $x^2 > 0$ for $x \neq 0$ by proposition 1.1.2(6) and $0 \cdot 0 = 0$ by proposition 1.1.2(3).

Theorem 3.3.2(2) says that a continuous function on an interval attains every value between any two of its values. That conclusion is a property a function may have on its own, without reference to continuity, and it is given a name because the two conditions are not equivalent.

Definition 3.3.4: Intermediate Value Property

Let $I \subseteq \mathbb{R}$ be an interval. A function $f\colon I \to \mathbb{R}$ has the *intermediate value property* if for all $u, v \in I$ with $u < v$ and every $y \in \mathbb{R}$ with $\min\{f(u), f(v)\} < y < \max\{f(u), f(v)\}$ there is a $c \in (u, v)$ with $f(c) = y$.

Every continuous function on an interval has the property, which is exactly theorem 3.3.2(2). The polynomial $p(x) = x^3 - x$ on $\mathbb{R}$ has it for that reason. Not every function does: define $f\colon [-1, 1] \to \mathbb{R}$ by $f(x) = -1$ for $x < 0$ and $f(x) = 1$ for $x \geq 0$. Then $f(-1) = -1$ and $f(1) = 1$, and 0 lies strictly between them, but the only values f takes are -1 and 1, so no c satisfies $f(c) = 0$.

The converse also fails: a function can have the intermediate value property at every pair of points and still be discontinuous. The standard example is $x \mapsto \sin(1/x)$ on $[-1, 1]$, extended by the value 0 at the origin, which oscillates between -1 and 1 on every interval around 0 and therefore omits no intermediate value, while failing to have a limit at 0. Constructing sin and establishing both facts is the work of **??**. The intermediate value property is therefore strictly weaker than continuity, and theorem 3.3.2 cannot be reversed.

Theorem 3.3.5 and theorem 3.3.7 replace "interval" by "closed bounded interval", and both are false without the strengthening. They also fail over $\mathbb{Q}$, and again it is the absence of $\sqrt{2}$ that breaks them. Let $h\colon E_0 \to \mathbb{R}$ be $h(q) = 1/(2 - q^2)$, again with $E_0 = \mathbb{Q} \cap [0, 2]$. It is defined at every point of E_0, since $q^2 = 2$ has no rational solution (proposition 1.1.3), and it is continuous by theorem 3.2.5 applied to the polynomials 1 and $2 - q^2$, the second of which does not vanish on E_0. It is unbounded. Given $n \in \mathbb{N}$, theorem 1.4.4 produces a rational q with $\sqrt{2} - 1/n < q < \sqrt{2}$. Then $q > 0$ because $\sqrt{2} - 1/n \geq \sqrt{2} - 1 > 0$, and $q < 2$ because $\sqrt{2} < 2$, so $q \in E_0$. Factoring,

$$0 < 2 - q^2 = (\sqrt{2} - q)(\sqrt{2} + q) < 4/n$$

using $0 < \sqrt{2} - q < 1/n$ and $0 < \sqrt{2} + q < 4$, so $h(q) > n/4$ by proposition 1.1.2(7). Since $\{n/4 \mid n \in \mathbb{N}\}$ has no upper bound in $\mathbb{R}$ (theorem 1.4.1(2) applied to $4M$ for a proposed bound M), neither does the set of values of h.

Theorem 3.3.5: A Continuous Function on a Closed Bounded Interval Is Bounded

Let $a \leq b$ and let $f\colon [a,b] \to \mathbb{R}$ be continuous. Then there is an $M \in \mathbb{R}$ with $|f(x)| \leq M$ for every $x \in [a,b]$.

***Proof*:**
Suppose no such M exists. Then for every $n \in \mathbb{N}$ the number n is not such an M, so there is an $x_n \in [a,b]$ with $|f(x_n)| > n$.

Step 1: a convergent subsequence. The sequence $(x_n)_{n\geq 1}$ satisfies $a \leq x_n \leq b$ for every n, so it is bounded. By theorem 2.2.9(1) it has a convergent subsequence $(x_{n_k})_{k\geq 1}$, say $x_{n_k} \to x$. Applying proposition 2.1.8(2) to the constant bounds $a \leq x_{n_k} \leq b$ gives $a \leq x \leq b$, so $x \in [a,b]$ and $f(x)$ is defined.

Step 2: the contradiction. By proposition 3.2.4, applied to f at the point $x \in [a,b]$ and to the sequence (x_{n_k}) in $[a,b]$ converging to x, the sequence $(f(x_{n_k}))_{k\geq 1}$ converges to $f(x)$. A convergent sequence is bounded (proposition 2.1.5), so there is a $C \in \mathbb{R}$ with $|f(x_{n_k})| \leq C$ for every k. Because $n_1 < n_2 < \cdots$ in $\mathbb{N}$, an induction gives $n_k \geq k$ for every k: indeed $n_1 \geq 1$, and $n_{k+1} > n_k \geq k$ forces $n_{k+1} \geq k+1$, since no element of $\mathbb{N}$ lies strictly between k and $k+1$. Choose $k_0 \in \mathbb{N}$ with $k_0 > C$, which theorem 1.4.1(2) provides. Then

$$C \geq |f(x_{n_{k_0}})| > n_{k_0} \geq k_0 > C$$

which is impossible. So some M works, completing the proof.

Both halves of "closed and bounded" are used, and the proof shows where. That the interval is bounded is what makes (x_n) a bounded sequence, so that theorem 2.2.9 applies. That it is closed is what puts the limit x back inside the domain, so that $f(x)$ exists and proposition 3.2.4 can be invoked. Dropping either one breaks the argument, and the next example shows that each failure is real.

Example 3.3.6: Unbounded Continuous Functions on $(0,1]$ and on $[0,\infty)$

1. Let $f\colon (0,1] \to \mathbb{R}$ be $f(x) = 1/x$. It is continuous by theorem 3.2.5, applied to the polynomials 1 and x, the second of which does not vanish on $(0,1]$. It is unbounded: given $M \in \mathbb{R}$, theorem 1.4.1(2) gives an $n \in \mathbb{N}$ with $n > M$, and $1/n \in (0,1]$ has $f(1/n) = n > M$. The domain is a bounded interval, and it is not closed, since 0 is a limit point of $(0,1]$ that does not belong to it. That 0 is a limit point is theorem 1.4.1(3): every $\delta > 0$ admits an n with $1/n < \delta$, and $1/n$ is a point of $(0,1]$ other than 0 within δ of 0. So $(0,1]$ is not closed by proposition 3.1.11.

2. Let $f\colon [0,\infty) \to \mathbb{R}$ be $f(x) = x$, continuous as a polynomial restricted to $[0,\infty)$. Given $M \in \mathbb{R}$, theorem 1.4.1(2) gives an $n \in \mathbb{N}$ with $n > M$, and $f(n) = n > M$, so f is unbounded. The domain is a closed interval and it is not bounded.

With boundedness in hand, the supremum of the set of values exists. The substantial question is whether it is attained, and the answer uses theorem 2.2.9 a second time.

Theorem 3.3.7: The Extreme Value Theorem

Let $a \leq b$ and let $f\colon [a,b] \to \mathbb{R}$ be continuous. Then there are $p, q \in [a,b]$ with

$$f(p) \leq f(x) \leq f(q) \qquad \text{for every } x \in [a,b]$$

Consequently the set $f([a,b]) = \{f(x) \big| x \in [a,b]\}$ has both a supremum and an infimum, and both belong to it: $\sup f([a,b]) = f(q)$ and $\inf f([a,b]) = f(p)$.

***Proof*:**
Write $E = f([a,b])$.

Step 1: the supremum exists. The set E is nonempty, since $a \in [a,b]$. By theorem 3.3.5 there is an M with $|f(x)| \leq M$ for every $x \in [a,b]$, hence $f(x) \leq M$ by proposition 1.3.8(1), so M is an upper bound of E. By theorem 1.2.4 the number $s = \sup E$ exists.

Step 2: a sequence approaching the supremum. Let $n \in \mathbb{N}$. By proposition 1.2.6, applied with $\epsilon = 1/n$, there is an element of E exceeding $s - 1/n$, that is, there is an $x_n \in [a,b]$ with $s - 1/n < f(x_n)$. Since s is an upper bound of E,

$$s - \frac{1}{n} < f(x_n) \leq s$$

Step 3: passing to a subsequence. The sequence (x_n) lies in $[a,b]$ and is therefore bounded, so theorem 2.2.9(1) gives a convergent subsequence $x_{n_k} \to q$, and $q \in [a,b]$ by proposition 2.1.8(2) applied to $a \leq x_{n_k} \leq b$. By proposition 3.2.4, $f(x_{n_k}) \to f(q)$.

Step 4: identifying the limit. The sequence $(1/n)_{n\geq 1}$ converges to 0 by example 2.1.3(1), so its subsequence $(1/n_k)_{k\geq 1}$ converges to 0 by proposition 2.2.7(1). The constant sequence with value s converges to s by example 2.1.3(2), so $s - 1/n_k \to s$ by theorem 2.1.6(1) and (2). The estimate of Step 2 reads $s - 1/n_k < f(x_{n_k}) \leq s$ for every k, so theorem 2.1.7, applied with the lower sequence $(s - 1/n_k)_{k\geq 1}$ and the upper sequence constantly equal to s, gives $f(x_{n_k}) \to s$. A sequence has at most one limit (proposition 2.1.4), so $f(q) = s$. Since s is an upper bound of E, this gives $f(x) \leq f(q)$ for every $x \in [a,b]$.

Step 5: the minimum. The function $-f$ is continuous on $[a,b]$ by theorem 3.2.5, so Steps 1 to 4 applied to it produce a $p \in [a,b]$ with $-f(x) \leq -f(p)$ for every $x \in [a,b]$. Each such inequality gives $f(p) \leq f(x)$: if $-f(x) = -f(p)$ then $f(x) = f(p)$, and if $-f(x) < -f(p)$ then $f(p) < f(x)$ by proposition 1.1.2(6).

Step 6: the two extrema are the supremum and the infimum. The number $f(q)$ is an upper bound of E and belongs to E, so any upper bound u of E satisfies $f(q) \leq u$, and $f(q) = \sup E$ by definition 1.2.2. The number $f(p)$ is a lower bound of E and belongs to E, so any lower bound ℓ satisfies $\ell \leq f(p)$, and $f(p) = \inf E$, completing the proof.

The content is the last clause: the supremum of the set of values is one of the values. For an arbitrary bounded set of reals the supremum need not belong to the set (example 1.2.3(2)), and the theorem says that a continuous function on a closed bounded interval never produces such a set. Both hypotheses on the domain are again needed, and this time the failure is subtler than in example 3.3.6, since the functions of example 3.3.8 are bounded.

Example 3.3.8: Bounded Continuous Functions That Do Not Attain a Bound

1. Let $f\colon (0,1) \to \mathbb{R}$ be $f(x) = x$, continuous as a polynomial restricted to $(0,1)$, with values in $(0,1)$ and therefore bounded. Its set of values is $(0,1)$, whose supremum is 1. Indeed 1 is an upper bound, and no $u < 1$ is one. If $u < 1/2$, then $1/2 \in (0,1)$ exceeds u. If $1/2 \le u$, then proposition 1.1.2(8) gives a v with $u < v < 1$, and $v > u \ge 1/2 > 0$ puts v in $(0,1)$. Since $1 \notin (0,1)$, no point of the domain has $f(x) = 1$, so the supremum is not attained. The same argument with the order reversed gives $\inf f((0,1)) = 0$. The number 0 is a lower bound, and no $\ell > 0$ is one: if $\ell > 1/2$ then $1/2 \in (0,1)$ is smaller than ℓ, and if $\ell \le 1/2$ then proposition 1.1.2(8) applied to $0 < \ell$ gives a v with $0 < v < \ell \le 1/2 < 1$, so $v \in (0,1)$ is smaller than ℓ. Again $0 \notin (0,1)$. The domain is a bounded interval that is not closed.
2. Let $f\colon [0,\infty) \to \mathbb{R}$ be $f(x) = 1/(1+x^2)$. For every real x the number x^2 is nonnegative (proposition 1.1.2(3) and (6)), so $1 + x^2 \ge 1 > 0$ and f is continuous by theorem 3.2.5. Its values satisfy $0 < f(x) \le 1$: for $x = 0$ the value is 1, and for $x \ne 0$ the inequality $0 < 1 < 1 + x^2$ and proposition 1.1.2(7) give $0 < f(x) < 1$. So f is bounded and $f(0) = 1$ is a maximum. There is no minimum. The number 0 is a lower bound that no value attains, and it is the infimum: given $\epsilon > 0$, theorem 1.4.1(3) gives an $n \in \mathbb{N}$ with $1/n < \epsilon$, and multiplying $1 \le n$ by $n > 0$ gives $n \le n^2 < 1 + n^2$, so proposition 1.1.2(7) gives $f(n) = 1/(1+n^2) < 1/n < \epsilon$. The domain is a closed interval that is not bounded.

Theorem 3.3.2 and theorem 3.3.7 combine into a single statement about the image, and it is this packaged form that later chapters quote.

Corollary 3.3.9: The Continuous Image of a Closed Bounded Interval

Let $a \le b$, let $f\colon [a,b] \to \mathbb{R}$ be continuous, and put $m = \inf f([a,b])$ and $M = \sup f([a,b])$, both of which exist by theorem 3.3.7. Then

$$f([a,b]) = [m, M]$$

In particular the image is again a closed bounded interval, degenerating to the single point $\{m\}$ when $m = M$.

***Proof*:**
By theorem 3.3.7 there are $p, q \in [a,b]$ with $f(p) = m$, $f(q) = M$ and $m \le f(x) \le M$ for every $x \in [a,b]$. The last inequality is the inclusion $f([a,b]) \subseteq [m, M]$.

For the reverse inclusion let $y \in [m, M]$. If $y = m$ then $y = f(p)$, and if $y = M$ then $y = f(q)$, so assume $m < y < M$. Then $f(p) \ne f(q)$, so $p \ne q$. Put $u = \min\{p,q\}$ and $v = \max\{p,q\}$, so that $u < v$ and both lie in $[a,b]$. The two values $f(u)$ and $f(v)$ are m and M in one order or the other, so $\min\{f(u), f(v)\} = m < y < M = \max\{f(u), f(v)\}$. Since $[a, b]$ is an interval (definition 3.1.1), theorem 3.3.2(2) gives a $c \in (u, v)$ with $f(c) = y$, and $c \in [a, b]$. Hence $[m, M] \subseteq f([a, b])$, as we sought to show.

The corollary is the reason the two theorems are usually quoted together: it converts a continuous function on $[a, b]$ into a surjection onto a closed bounded interval whose endpoints are values of the function. One consequence is a statement about self-maps that requires no further hypothesis at all.

Theorem 3.3.10: Fixed Points of a Continuous Self-Map of $[0, 1]$

Let $f : [0, 1] \to [0, 1]$ be continuous. Then $f(c) = c$ for some $c \in [0, 1]$.

***Proof*:**
Let $g : [0, 1] \to \mathbb{R}$ be $g(x) = f(x) - x$. It is continuous by theorem 3.2.5, applied to f and to the polynomial x restricted to $[0, 1]$. Since f takes values in $[0, 1]$, its values satisfy $0 \le f(x) \le 1$, so

$$g(0) = f(0) \ge 0 \qquad \text{and} \qquad g(1) = f(1) - 1 \le 0$$

If $g(0) = 0$ then $f(0) = 0$ and $c = 0$ works. If $g(1) = 0$ then $f(1) = 1$ and $c = 1$ works. Otherwise $g(1) < 0 < g(0)$, so $\min\{g(0), g(1)\} = g(1) < 0 < g(0) = \max\{g(0), g(1)\}$, and theorem 3.3.2(1) gives a $c \in (0, 1)$ with $g(c) = 0$, that is $f(c) = c$. In every case a fixed point exists, completing the proof.

The hypothesis that f map $[0, 1]$ into itself is what forces the two sign conditions, and dropping it destroys the conclusion: $f(x) = x + 1$ is continuous on $[0, 1]$ and $f(c) = c$ would give $1 = 0$.

Theorem 3.3.12 concerns inverses. A function is invertible onto its image exactly when it is injective, and a strictly monotone function (definition 3.2.14) is injective. The question is whether the inverse of a continuous strictly monotone function on an interval is again continuous. It is, and the interval hypothesis carries most of the argument: the continuity of f is used only to see that the image is again an interval.

Proposition 3.3.11: Powers Are Strictly Increasing on the Nonnegative Reals

Let $n \ge 1$ be an integer. If $0 \le s < t$ then $s^n < t^n$. So $x \mapsto x^n$ is strictly increasing on $[0, \infty)$ in the sense of definition 3.2.14.

***Proof*:**
Induction on n. For $n = 1$ the claim is the hypothesis $s < t$. Suppose it holds for n, and let $0 \le s < t$, so that $t > 0$. Every power t^j is then positive, by proposition 1.1.2(3) and (5) applied repeatedly to $0 < t$. If $s = 0$ then $s^{n+1} = 0 < t^{n+1}$. If $s > 0$, then multiplying $s^n < t^n$ by $s > 0$ gives $s^{n+1} < st^n$, and multiplying $s < t$ by $t^n > 0$ gives $st^n < t^{n+1}$, both by proposition 1.1.2(5). Transitivity of the order gives $s^{n+1} < t^{n+1}$, completing the proof.

Theorem 3.3.12: Continuous Strictly Monotone Functions Have Continuous Inverses

Let $I \subseteq \mathbb{R}$ be an interval and let $f\colon I \to \mathbb{R}$ be continuous and strictly monotone (definition 3.2.14). Put $J = f(I) = \{f(x) \big| x \in I\}$.

1. J is an interval.

2. The map $f\colon I \to J$ is a bijection.

3. The inverse $f^{-1}\colon J \to I$ is strictly monotone in the same direction as f, and it is continuous.

***Proof*:**
Assume first that f is strictly increasing. The strictly decreasing case is reduced to this one at the end.

1. Let $y_1, y_2 \in J$ and let $z \in \mathbb{R}$ satisfy $y_1 < z < y_2$. Write $y_1 = f(u_1)$ and $y_2 = f(u_2)$ with $u_1, u_2 \in I$. Then $u_1 < u_2$: the case $u_1 = u_2$ would give $y_1 = y_2$, and the case $u_1 > u_2$ would give $f(u_1) > f(u_2)$ by strict increase, both contradicting $y_1 < y_2$. Now $\min\{f(u_1), f(u_2)\} = y_1 < z < y_2 = \max\{f(u_1), f(u_2)\}$, so theorem 3.3.2(2), applied to the interval I and the points $u_1 < u_2$, gives a $c \in (u_1, u_2)$ with $f(c) = z$. Hence $z \in J$. This is the betweenness condition of definition 3.1.1, so J is an interval.

2. The map $f\colon I \to J$ is surjective by the definition of J. It is injective: if $u \neq v$ in I, say $u < v$, then $f(u) < f(v)$ by strict increase, so $f(u) \neq f(v)$.

3. Write $g = f^{-1}\colon J \to I$, which exists by part (2).

   Step 1: g is strictly increasing. Let $y_1 < y_2$ in J and put $u_i = g(y_i)$, so $f(u_i) = y_i$. The argument in part (1) gives $u_1 < u_2$, which is $g(y_1) < g(y_2)$.

   Step 2: a one-sided approximation inside I. Fix $y_0 \in J$, put $u_0 = g(y_0) \in I$, and let $\epsilon > 0$. Claim: if I contains a point greater than u_0, then it contains a point u_+ with $u_0 < u_+ \leq u_0 + \epsilon$. Indeed, let $w \in I$ with $w > u_0$. If $w \leq u_0 + \epsilon$, take $u_+ = w$. Otherwise $u_0 < u_0 + \epsilon < w$ with $u_0, w \in I$, so $u_0 + \epsilon \in I$ by definition 3.1.1, and $u_+ = u_0 + \epsilon$ serves. Symmetrically, if I contains a point smaller than u_0, it contains a point u_- with $u_0 - \epsilon \leq u_- < u_0$.

   Step 3: the choice of δ. Strict increase gives $f(u_-) < f(u_0) = y_0 < f(u_+)$ whenever the relevant point exists. Set

$$\delta = \begin{cases} \min\{f(u_+) - y_0,\ y_0 - f(u_-)\} & \text{if both } u_+ \text{ and } u_- \text{ exist} \\ f(u_+) - y_0 & \text{if only } u_+ \text{ exists} \\ y_0 - f(u_-) & \text{if only } u_- \text{ exists} \\ 1 & \text{if neither exists} \end{cases}$$

   In each case $\delta > 0$.

Step 4: δ works. Let $y \in J$ with $|y - y_0| < \delta$ and put $u = g(y)$. If $u = u_0$ then $|g(y) - g(y_0)| = 0 < \epsilon$. Suppose $u > u_0$. Then I contains a point greater than u_0, namely u, so u_+ exists and $\delta \leq f(u_+) - y_0$. From $|y - y_0| < \delta$ and proposition 1.3.8(2), $y < y_0 + \delta \leq f(u_+)$. If $u \geq u_+$ held, then $f(u) \geq f(u_+)$, since $u = u_+$ gives equality and $u > u_+$ gives strict inequality, so $y \geq f(u_+)$, which is false. Hence $u < u_+ \leq u_0 + \epsilon$, and $|g(y) - g(y_0)| = u - u_0 < \epsilon$. The case $u < u_0$ is the mirror image: u_- exists, $\delta \leq y_0 - f(u_-)$, $y > y_0 - \delta \geq f(u_-)$ forces $u > u_- \geq u_0 - \epsilon$, and $|g(y) - g(y_0)| = u_0 - u < \epsilon$. In all three cases $|g(y) - g(y_0)| < \epsilon$, which is the condition of definition 3.2.1 for g at y_0.

The strictly decreasing case. Let f be continuous and strictly decreasing on I and put $h = -f$, which is continuous by theorem 3.2.5 and strictly increasing, since $u < v$ gives $f(u) > f(v)$ and hence $-f(u) < -f(v)$ by proposition 1.1.2(6). By the case already proved, $K = h(I)$ is an interval, $h\colon I \to K$ is a bijection, and $h^{-1}\colon K \to I$ is continuous and strictly increasing. Now $J = f(I) = \{-z | z \in K\}$. If $y_1 < z < y_2$ with $y_1, y_2 \in J$, then $-y_2 < -z < -y_1$ with $-y_1, -y_2 \in K$ (proposition 1.1.2(6)), so $-z \in K$ and $z \in J$, and J is an interval. The map $f\colon I \to J$ is a bijection, being surjective by the definition of J and injective because $u < v$ gives $f(u) > f(v)$. For $y \in J$ the element $x = h^{-1}(-y) \in I$ satisfies $-f(x) = -y$, so $f(x) = y$ and $f^{-1}(y) = h^{-1}(-y)$. The map $\nu(y) = -y$ is a polynomial, hence continuous on J, and it carries J into K, so $f^{-1} = h^{-1} \circ \nu$ is continuous by theorem 3.2.7. Finally $y_1 < y_2$ in J gives $-y_1 > -y_2$, hence $h^{-1}(-y_1) > h^{-1}(-y_2)$, that is $f^{-1}(y_1) > f^{-1}(y_2)$, so f^{-1} is strictly decreasing, completing the proof.

The interval hypothesis is used twice. Part (1) uses it through theorem 3.3.2(2), and its conclusion fails without it: the identity map on $[0, 1] \cup [2, 3]$ is continuous and strictly increasing, and its image is that same set, which is not an interval (example 3.1.2(3)). Step 2 of part (3) uses it again, and that step is where the continuity of the inverse would break. On a domain that is not an interval, the points above u_0 can be separated from u_0 by a gap, so no u_+ within ϵ of u_0 need exist. A strictly increasing continuous bijection from a set that is not an interval onto an interval can then have a discontinuous inverse, and exercise 3.3.1 (7) exhibits one. Continuity of f, by contrast, entered only in part (1), through theorem 3.3.2: parts (2) and (3) hold for any strictly monotone f on an interval, with no continuity assumed.

The immediate application extends the catalogue of continuous functions of section 3.2, which stopped at polynomials, rational functions and $|x|$.

Corollary 3.3.13: The n-th Root Function Is Continuous

Let $n \geq 1$ be an integer, and for $x \geq 0$ write $x^{1/n}$ for the unique $y \geq 0$ with $y^n = x$ furnished by theorem 1.3.12. The function $r\colon [0, \infty) \to \mathbb{R}$ given by $r(x) = x^{1/n}$ is a strictly increasing continuous bijection from $[0, \infty)$ onto $[0, \infty)$.

***Proof*:**
Let $p\colon [0, \infty) \to \mathbb{R}$ be $p(x) = x^n$. It is continuous, being a polynomial restricted to $[0, \infty)$, and strictly increasing there by proposition 3.3.11. The set $[0, \infty)$ is an interval.

Its image is $[0, \infty)$. Every value $p(y) = y^n$ with $y \geq 0$ is nonnegative: for $y = 0$ it is 0, and for $y > 0$ repeated use of proposition 1.1.2(3) and (5) gives $y^n > 0$. Conversely, given $x \geq 0$, theorem 1.3.12 produces exactly one $y \geq 0$ with $y^n = x$, namely $y = x^{1/n}$, so $x = p(x^{1/n})$ lies in the image.

By theorem 3.3.12, applied to p on the interval $[0, \infty)$, the map $p\colon [0, \infty) \to [0, \infty)$ is a bijection and its inverse is continuous and strictly increasing. That inverse sends x to the unique $y \geq 0$ with $y^n = x$, which is $x^{1/n}$ by the uniqueness clause of theorem 1.3.12. So $r = p^{-1}$ has the stated properties, as we sought to show.

Combined with theorem 3.2.5 and theorem 3.2.7, this shows that any function built from polynomials, quotients with nonvanishing denominators and n-th roots is continuous on the set where those operations are defined. The function $x \mapsto (1 + x^2)^{1/2}$ on $\mathbb{R}$ is continuous for this reason: it is r with $n = 2$ composed with the polynomial $1 + x^2$, whose values lie in $[1, \infty) \subseteq [0, \infty)$.

Exercise 3.3.1

1. Let $f\colon [0, 1] \to \mathbb{R}$ be continuous and suppose that $f(x) \in \mathbb{Q}$ for every $x \in [0, 1]$. Show that f is constant.

2. Let $f\colon [0, 2] \to \mathbb{R}$ be continuous with $f(0) = f(2)$. Show that there is a $c \in [0, 1]$ with $f(c) = f(c + 1)$.

3. Show that the polynomial $p(x) = x^5 - 3x + 1$ has at least three distinct real roots, and name an interval of length at most 2 containing each of them.

4. Let $a \leq b$ and let $f\colon [a, b] \to \mathbb{R}$ be continuous with $f(x) > 0$ for every $x \in [a, b]$. Show that there is an $\eta > 0$ with $f(x) \geq \eta$ for every $x \in [a, b]$. Then give a continuous $g\colon (0, 1] \to \mathbb{R}$ with $g(x) > 0$ for every x and $\inf g((0, 1]) = 0$, and say which hypothesis of theorem 3.3.7 the domain $(0, 1]$ fails.

5. Let $a < b$ and let $f\colon [a, b] \to \mathbb{R}$ be continuous and injective. Show that f is strictly monotone. (Begin by showing that for $u < v < w$ in $[a, b]$ the value $f(v)$ lies strictly between $f(u)$ and $f(w)$, using theorem 3.3.2 on $[u, v]$ and on $[v, w]$ to contradict injectivity otherwise.)

6. Prove that $x \mapsto x^{1/2}$ is continuous on $[0, \infty)$ directly from definition 3.2.1, without appealing to theorem 3.3.12, by factoring $s - t = (s^{1/2} - t^{1/2})(s^{1/2} + t^{1/2})$. Treat $t > 0$ and $t = 0$ separately, and say in each case which δ answers a given ϵ. (The δ you produce at $t > 0$ depends on t. That dependence is the subject of section 3.4.)

7. Let $E = [0, 1) \cup [2, 3]$ and define $f\colon E \to \mathbb{R}$ by $f(x) = x$ for $x \in [0, 1)$ and $f(x) = x - 1$ for $x \in [2, 3]$. Show that f is continuous and strictly increasing, that $f(E) = [0, 2]$, and that the inverse map $[0, 2] \to E$ is not continuous at 1. Which hypothesis of theorem 3.3.12 does E fail?

8. Show that $\{x^2 | x \in (-1, 1)\} = [0, 1)$. Conclude that corollary 3.3.9 becomes false if the domain is a bounded interval that is not closed, even though the image is still an interval.

3.4 Uniform Continuity

Continuity at a point is a statement about that point. The δ answering a given ϵ in definition 3.2.1 is produced after the point has been named, and for most functions it has to be, since the same ϵ needs a smaller δ wherever the function moves faster. This section names the condition that holds when a single δ answers ϵ at every point of the domain at once, exhibits continuous functions that fail it, and shows that on a closed bounded interval continuity already implies it. Three consequences follow: such a function carries Cauchy sequences to Cauchy sequences, extends continuously to the closure of its domain, and carries bounded sets to bounded sets.

Definition 3.4.1: [Real] Uniform Continuity

Let $E \subseteq \mathbb{R}$ and let $f\colon E \to \mathbb{R}$. Then f is *uniformly continuous* on E if for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon \qquad \text{for all } x, y \in E \text{ with } |x - y| < \delta$$

The condition constrains the pair (x, y), no point of E is distinguished, and E itself is unrestricted. The function $x \mapsto 2x + 1$ on $\mathbb{R}$ satisfies it: given $\epsilon > 0$, the choice $\delta = \epsilon/2$ gives $|(2x+1) - (2y+1)| = 2|x - y| < \epsilon$ whenever $|x - y| < \delta$.

3.4.2 Note: The Order of the Quantifiers Written out, continuity of f on E says

$$\forall c \in E \quad \forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall x \in E \quad \big(|x - c| < \delta \;\Rightarrow\; |f(x) - f(c)| < \epsilon\big)$$

and uniform continuity of f on E says

$$\forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall c \in E \quad \forall x \in E \quad \big(|x - c| < \delta \;\Rightarrow\; |f(x) - f(c)| < \epsilon\big)$$

The two strings differ only in the position of $\exists\delta$, which has moved to the left of $\forall c$. In the first, δ may depend on c as well as on ϵ; in the second it may depend on ϵ only.
One implication holds with no hypothesis on E: a uniformly continuous function is continuous. Given $c \in E$ and $\epsilon > 0$, the δ produced by definition 3.4.1 satisfies $|f(x) - f(c)| < \epsilon$ for every $x \in E$ with $|x - c| < \delta$, which is what definition 3.2.1 asks for at c. The converse is false, and example 3.4.3 gives three witnesses.

Negating definition 3.4.1, a function f fails to be uniformly continuous on E exactly when there is an $\epsilon_0 > 0$ such that for every $\delta > 0$ there are $x, y \in E$ with $|x - y| < \delta$ and $|f(x) - f(y)| \geq \epsilon_0$. Such an ϵ_0 is a tolerance that no single δ meets across all of E, and for each of the three functions below, uniform continuity is refuted by exhibiting one.

Example 3.4.3: $1/x$ on $(0, 1)$, x^2 on $\mathbb{R}$, and $1/(2 - q^2)$ on the Rationals in $[0, 2]$

1. Let $f(x) = 1/x$ on $E = (0, 1)$. The function $x \mapsto x$ is continuous and vanishes nowhere on E, so f is continuous on E by theorem 3.2.5. Take $\epsilon_0 = 1$ and let $\delta > 0$. By theorem 1.4.1(2) there is an $n \in \mathbb{N}$ with $n > \max\{2, 1/\delta\}$, and from $0 < 1/\delta < n$ and proposition 1.1.2(7) it follows that $0 < 1/n < \delta$. Put $x = 1/n$ and $y = 1/(2n)$, both of which lie in $(0, 1)$ because

$n > 2$. Then

$$|x - y| = \frac{1}{n} - \frac{1}{2n} = \frac{1}{2n} < \frac{1}{n} < \delta$$

while $|f(x) - f(y)| = |n - 2n| = n > 2 > \epsilon_0$. So no δ answers ϵ_0, and f is not uniformly continuous on $(0, 1)$.

2. Let $g(x) = x^2$ on $\mathbb{R}$, which is continuous by example 3.2.6. Take $\epsilon_0 = 1$ and let $\delta > 0$. By theorem 1.4.1(2) choose $n \in \mathbb{N}$ with $n > 1/\delta$, and multiply this inequality by $\delta > 0$ using proposition 1.1.2(5) to get $\delta n > 1$. Put $x = n$ and $y = n + \delta/2$, so that $0 < x < y$ and $|x - y| = \delta/2 < \delta$. Then

$$|g(y) - g(x)| = y^2 - x^2 = (y - x)(y + x) = \frac{\delta}{2}\left(2n + \frac{\delta}{2}\right) = \delta n + \frac{\delta^2}{4} > \delta n > 1 = \epsilon_0$$

So g is not uniformly continuous on $\mathbb{R}$. Here the domain is all of $\mathbb{R}$ and the function is bounded on every bounded set, so the obstruction is neither of the ones that defeat (1) and (3).

3. Let $E_0 = \mathbb{Q} \cap [0, 2]$ and let $h\colon E_0 \to \mathbb{R}$ be $h(q) = 1/(2 - q^2)$. This is the function Section 3.3 sets against theorem 3.3.5, and it is defined, continuous and unbounded above on E_0 for the reasons given there. Throughout, $\sqrt{2}$ is the number given by theorem 1.3.12, and $1 < \sqrt{2} < 2$ is proved in Section 3.3.
For each $n \in \mathbb{N}$, theorem 1.4.4 produces a rational q_n with $\sqrt{2} - 1/n < q_n < \sqrt{2}$. Then $q_n > 0$, since $\sqrt{2} - 1/n \geq \sqrt{2} - 1 > 0$, and $q_n < 2$, so $q_n \in E_0$. From $|q_n - \sqrt{2}| < 1/n$, together with example 2.1.3(1) and definition 2.1.2, it follows that $q_n \to \sqrt{2}$, and then theorem 2.1.6(3) and (1) give $2 - q_n^2 \to 0$. Each $2 - q_n^2$ is positive, since $0 \leq q_n < \sqrt{2}$ forces $q_n^2 < (\sqrt{2})^2 = 2$ by proposition 3.3.11. Now let $M > 0$. Applying definition 2.1.2 to $(2 - q_n^2)$ with $\epsilon = 1/M$ produces an N' such that $0 < 2 - q_n^2 < 1/M$ for every $n \geq N'$, and inverting with proposition 1.1.2(7) gives $h(q_n) > M$ for every $n \geq N'$.
Now take $\epsilon_0 = 1$ and let $\delta > 0$. Since $q_n \to \sqrt{2}$, there is an $N \in \mathbb{N}$ with $|q_n - \sqrt{2}| < \delta/2$ for all $n \geq N$, so proposition 1.3.8(4) gives $|q_n - q_m| < \delta$ for all $n, m \geq N$. The value $h(q_N)$ is positive, so the previous paragraph applies to $M = h(q_N) + 1$ and returns an N' with $h(q_n) > h(q_N) + 1$ for every $n \geq N'$. Put $m = \max\{N, N'\}$. Then $|h(q_m) - h(q_N)| > 1 = \epsilon_0$ while $|q_m - q_N| < \delta$. So h is not uniformly continuous on E_0.

Two of these three failures occur on a bounded domain, and in both the function is unbounded near a point the domain does not contain: 0 in item (1) and $\sqrt{2}$ in item (3). Item (3) is the failure over $\mathbb{Q}$. Its domain $E_0 = \mathbb{Q} \cap [0, 2]$ is bounded and contains both of its bounds, and it is nonetheless neither an interval in the sense of definition 3.1.1 nor a closed set in the sense of definition 3.1.10. Both failures occur at $\sqrt{2}$, which lies outside E_0 by proposition 1.1.3 and satisfies $1 < \sqrt{2} < 2$, as Section 3.3 proves. The betweenness condition fails for the pair $0, 2 \in E_0$, and the rationals q_n of item (3) lie in E_0 and converge to $\sqrt{2}$, so E_0 is not closed by proposition 3.1.12. The theorem below therefore cannot be proved from order and boundedness alone, and its proof goes through theorem 2.2.9.

Theorem 3.4.4: Continuity on a Closed Bounded Interval Is Uniform

Let $a < b$ be real numbers and let $f\colon [a,b] \to \mathbb{R}$ be continuous. Then f is uniformly continuous on $[a,b]$.

***Proof*:**
Suppose f is not uniformly continuous on $[a,b]$. Negating definition 3.4.1, there is an $\epsilon_0 > 0$ such that every $\delta > 0$ admits $x, y \in [a,b]$ with $|x-y| < \delta$ and $|f(x) - f(y)| \geq \epsilon_0$. Applying this to $\delta = 1/n$ for each $n \in \mathbb{N}$ produces points $x_n, y_n \in [a,b]$ with

$$|x_n - y_n| < \frac{1}{n} \qquad \text{and} \qquad |f(x_n) - f(y_n)| \geq \epsilon_0 \tag{3.1}$$

Step 1: a convergent subsequence of (x_n). Every term of (x_n) lies in $[a,b]$, so the sequence is bounded, and theorem 2.2.9(1) gives a subsequence $(x_{n_k})_{k\geq 1}$ converging to some $x \in \mathbb{R}$. Since $a \leq x_{n_k} \leq b$ for every k, two applications of proposition 2.1.8(2) give $a \leq x \leq b$. So $x \in [a,b]$ and f is continuous at x.
Step 2: the matching subsequence of (y_n) *has the same limit.* The sequence $(1/n)_{n\geq 1}$ converges to 0 by example 2.1.3(1), and $(1/n_k)_{k\geq 1}$ is a subsequence of it because $k \mapsto n_k$ is strictly increasing, so $1/n_k \to 0$ by proposition 2.2.7(1). Let $\epsilon > 0$ and take K with $1/n_k < \epsilon$ for every $k \geq K$. For such k the first half of (3.1) gives

$$\big|(y_{n_k} - x_{n_k}) - 0\big| = |y_{n_k} - x_{n_k}| < \frac{1}{n_k} < \epsilon$$

so $y_{n_k} - x_{n_k} \to 0$ by definition 2.1.2. Writing $y_{n_k} = x_{n_k} + (y_{n_k} - x_{n_k})$ and applying theorem 2.1.6(1) gives $y_{n_k} \to x$.
Step 3: the contradiction. Both (x_{n_k}) and (y_{n_k}) are sequences in $[a,b]$ converging to x, and f is continuous at x by Step 1, so proposition 3.2.4 gives $f(x_{n_k}) \to f(x)$ and $f(y_{n_k}) \to f(x)$. By theorem 2.1.6(2) with $c = -1$ and then theorem 2.1.6(1), the sequence $d_k = f(x_{n_k}) - f(y_{n_k})$ converges to $f(x) - f(x) = 0$. Applying definition 2.1.2 to (d_k) with $\epsilon = \epsilon_0$ produces a k with $|d_k| < \epsilon_0$, while the second half of (3.1) gives $|d_k| \geq \epsilon_0$ for every k. This contradiction shows that f is uniformly continuous on $[a,b]$, as we sought to show.

On a closed bounded interval the two conditions therefore coincide, and nothing is gained by checking the stronger one. This is one of the statements about a continuous function on $[a,b]$ that section 3.3 and this section prove from completeness, and it is the one whose failure over $\mathbb{Q}$ is visible in example 3.4.3(3). The proof applied continuity at the single point x, and everything else was theorem 2.2.9 and the algebra of limits.

The three results that follow hold for an arbitrary domain $E \subseteq \mathbb{R}$. The first is what the other two rest on.

Proposition 3.4.5: Uniform Continuity Carries Cauchy Sequences to Cauchy Sequences

Let $E \subseteq \mathbb{R}$, let $f: E \to \mathbb{R}$ be uniformly continuous, and let $(x_n)_{n\geq 1}$ be a Cauchy sequence with $x_n \in E$ for every n. Then $(f(x_n))_{n\geq 1}$ is a Cauchy sequence.

***Proof*:**
Let $\epsilon > 0$. By definition 3.4.1 there is a $\delta > 0$ with $|f(x) - f(y)| < \epsilon$ for all $x, y \in E$ satisfying $|x - y| < \delta$. Since (x_n) is Cauchy, definition 2.3.1 applied to δ gives an $N \in \mathbb{N}$ with $|x_n - x_m| < \delta$ for all $n, m \geq N$. For such n and m both x_n and x_m lie in E, so the choice of δ gives $|f(x_n) - f(x_m)| < \epsilon$. As ϵ was arbitrary, $(f(x_n))$ satisfies definition 2.3.1, completing the proof.

Continuity alone does not give this. Take $f(x) = 1/x$ on $(0, 1)$ and $x_n = 1/(n + 1)$. Every x_n lies in $(0, 1)$, and (x_n) is a subsequence of $(1/n)_{n\geq 1}$, hence converges to 0 by example 2.1.3(1) and proposition 2.2.7(1), hence is Cauchy by proposition 2.3.3. But $f(x_n) = n+1$, and no $M \geq 0$ bounds these values, since theorem 1.4.1(2) produces an $n \in \mathbb{N}$ with $n > M$. So $(f(x_n))$ is unbounded, and proposition 2.3.4 then says it is not Cauchy. This settles example 3.4.3(1) a second time, without producing an explicit pair of nearby points.

Once Cauchy sequences are preserved, a uniformly continuous function has a value forced on it at every point its domain approaches, because the images of the approaching sequences converge and the limit does not depend on which sequence is used. Proposition 3.1.9 is what turns a point of the closure into such a sequence.

Theorem 3.4.6: Uniformly Continuous Functions Extend to the Closure

Let $E \subseteq \mathbb{R}$ be nonempty and let $f: E \to \mathbb{R}$ be uniformly continuous. Then there is exactly one continuous function $g: \overline{E} \to \mathbb{R}$ with $g(x) = f(x)$ for every $x \in E$, and this g is uniformly continuous on $\overline{E}$.

***Proof*:**
Step 1: a candidate value at each point of the closure. Let $a \in \overline{E}$, the closure of definition 3.1.4. By proposition 3.1.9 there is a sequence (x_n) with $x_n \in E$ for every n and $x_n \to a$. By proposition 2.3.3 the sequence (x_n) is Cauchy, so $(f(x_n))$ is Cauchy by proposition 3.4.5, and therefore converges by theorem 2.3.5.
Step 2: the value does not depend on the sequence. Let (x_n) and (y_n) be sequences in E with $x_n \to a$ and $y_n \to a$, and let L and L' be the limits of $(f(x_n))$ and $(f(y_n))$ produced in Step 1. Define $z_{2n-1} = x_n$ and $z_{2n} = y_n$ for $n \geq 1$, so that every z_m lies in E. Then $z_m \to a$. Indeed, given $\epsilon > 0$, pick N_1 with $|x_n - a| < \epsilon$ for $n \geq N_1$ and N_2 with $|y_n - a| < \epsilon$ for $n \geq N_2$, and set $N = 2\max\{N_1, N_2\}$. If $m \geq N$ and $m = 2n - 1$, then $2n = m + 1 > N$, so $n > N/2 = \max\{N_1, N_2\} \geq N_1$. If $m \geq N$ and $m = 2n$, then $n = m/2 \geq N/2 \geq N_2$. In both cases $|z_m - a| < \epsilon$.
By Step 1 applied to (z_n), the sequence $(f(z_n))$ converges, to M say. The index maps $n \mapsto 2n-1$ and $n \mapsto 2n$ are strictly increasing, so $(f(x_n))$ and $(f(y_n))$ are subsequences of $(f(z_n))$, and proposition 2.2.7(1) gives $f(x_n) \to M$ and $f(y_n) \to M$. By proposition 2.1.4, $L = M = L'$. Setting $g(a) = L$ therefore defines a function $g: \overline{E} \to \mathbb{R}$.
Step 3: g extends f. Let $a \in E$. The constant sequence $x_n = a$ lies in E and converges to a

by example 2.1.3(2), so $a \in \overline{E}$ by proposition 3.1.9, and in particular $E \subseteq \overline{E}$. By Step 2 the value $g(a)$ may be computed from this sequence, and $f(x_n) = f(a)$ is constant, so $g(a) = f(a)$ by example 2.1.3(2) and proposition 2.1.4.
Step 4: g is uniformly continuous. Let $\epsilon > 0$. Apply definition 3.4.1 to f and the positive number $\epsilon/3$ to get a $\delta > 0$ with $|f(x) - f(y)| < \epsilon/3$ for all $x, y \in E$ with $|x - y| < \delta$. Let $a, b \in \overline{E}$ satisfy $|a - b| < \delta/3$, and choose sequences (x_n) and (y_n) in E with $x_n \to a$ and $y_n \to b$, so that $f(x_n) \to g(a)$ and $f(y_n) \to g(b)$ by Steps 1 and 2. Pick N with $|x_n - a| < \delta/3$ and $|y_n - b| < \delta/3$ for all $n \geq N$. For such n, two applications of proposition 1.3.8(4) give

$$|x_n - y_n| \leq |x_n - a| + |a - b| + |b - y_n| < \frac{\delta}{3} + \frac{\delta}{3} + \frac{\delta}{3} = \delta$$

so $|f(x_n) - f(y_n)| < \epsilon/3$, which proposition 1.3.8(2) turns into $-\epsilon/3 \leq f(x_n) - f(y_n) \leq \epsilon/3$ for every $n \geq N$. The sequence $f(x_n) - f(y_n)$ converges to $g(a) - g(b)$ by theorem 2.1.6(2) with $c = -1$ and then theorem 2.1.6(1), so proposition 2.1.8(2) applied twice gives $-\epsilon/3 \leq g(a) - g(b) \leq \epsilon/3$, and proposition 1.3.8(2) converts this to $|g(a) - g(b)| \leq \epsilon/3 < \epsilon$. Thus $\delta/3$ answers ϵ for g, so g is uniformly continuous on $\overline{E}$, and by box 3.4.2 it is continuous there.
Step 5: uniqueness. Let $h\colon \overline{E} \to \mathbb{R}$ be continuous with $h(x) = f(x)$ for every $x \in E$, and let $a \in \overline{E}$. Choose a sequence (x_n) in E with $x_n \to a$, using proposition 3.1.9. Every x_n lies in $E \subseteq \overline{E}$, which is the domain of h, and h is continuous at a, so proposition 3.2.4 gives $h(x_n) \to h(a)$. But $h(x_n) = f(x_n) \to g(a)$ by Steps 1 and 2, so $h(a) = g(a)$ by proposition 2.1.4. Since a was an arbitrary point of $\overline{E}$, the functions h and g are equal, completing the proof.

The extension is unique for a reason that has nothing to do with uniform continuity. By proposition 3.1.9 every point of $\overline{E}$ is a limit of points of E, and proposition 3.2.4 then forces any two continuous functions agreeing on E to agree on $\overline{E}$. What uniform continuity provides is existence, and continuity does not provide it: the function $x \mapsto 1/x$ on $(0, 1)$ takes the value $n + 1$ at $1/(n + 1)$, so it is unbounded on $(0, 1)$ by theorem 1.4.1(2), whereas every continuous function on $[0, 1]$ is bounded by theorem 3.3.5. No continuous function on $[0, 1]$ restricts to it. The closure of $(0, 1)$ is $[0, 1]$ by example 3.1.7(1), so the extension that fails here is exactly the one the theorem asserts.

That computation is an instance of the third consequence, which explains both failures on a bounded domain in example 3.4.3 at once. Recall that a subset $S \subseteq \mathbb{R}$ is bounded when it is bounded above and below in the sense of definition 1.2.1. This is the same as the existence of an $M \geq 0$ with $|t| \leq M$ for every $t \in S$: given such an M, proposition 1.3.8(2) makes $-M$ a lower bound and M an upper bound, and conversely if $\ell \leq t \leq u$ for every $t \in S$ then $M = \max\{|\ell|, |u|\}$ works, since proposition 1.3.8(1) gives $t \leq u \leq |u| \leq M$ and $-t \leq -\ell \leq |\ell| \leq M$, whence $|t| \leq M$ by proposition 1.3.8(2).

Proposition 3.4.7: Uniform Continuity Carries Bounded Sets to Bounded Sets

Let $E \subseteq \mathbb{R}$ and let $f\colon E \to \mathbb{R}$ be uniformly continuous. If $B \subseteq E$ is bounded, then $f(B)$ is bounded. In particular, a uniformly continuous function on a bounded domain is bounded.

***Proof*:**
Suppose $f(B)$ is not bounded. Then for each $n \in \mathbb{N}$ there is a $b_n \in B$ with $|f(b_n)| > n$, since otherwise $|f(x)| \leq n$ would hold for every $x \in B$, making $-n$ a lower bound and n an upper bound for $f(B)$ by proposition 1.3.8(2). The sequence (b_n) lies in the bounded set B, so it is bounded, and theorem 2.2.9(1) gives a convergent subsequence $(b_{n_k})_{k \geq 1}$.
By proposition 2.3.3 the sequence (b_{n_k}) is Cauchy, and all of its terms lie in $B \subseteq E$, so

proposition 3.4.5 makes $(f(b_{n_k}))$ Cauchy and proposition 2.3.4 gives an $M \geq 0$ with $|f(b_{n_k})| \leq M$ for every k. Combining this with $|f(b_{n_k})| > n_k$ gives $n_k < M$ for every k.
This is impossible. The sequence $(1/n_k)_{k\geq 1}$ is a subsequence of $(1/n)_{n\geq 1}$, which converges to 0 by example 2.1.3(1), so $1/n_k \to 0$ by proposition 2.2.7(1). Applying definition 2.1.2 with $\epsilon = 1/(M+1)$, a positive number because $M \geq 0$, produces a k with $0 < 1/n_k < 1/(M+1)$, and proposition 1.1.2(7) inverts this to $M+1 < n_k$. Together with $n_k < M$ this gives $M+1 < M$, which is false. Hence $f(B)$ is bounded. Taking $B = E$ gives the final sentence, completing the proof.

Items (1) and (3) of example 3.4.3 are now settled in one line each, since both domains are bounded and both functions are unbounded on them. Item (2) is not settled this way, because $x \mapsto x^2$ is bounded on every bounded subset of $\mathbb{R}$. That item also shows the converse of proposition 3.4.7 to be false: a continuous function may carry bounded sets to bounded sets without being uniformly continuous.

A single quantitative condition implies uniform continuity outright, and it is the one that most concrete estimates produce.

Definition 3.4.8: [Real] Lipschitz

Let $E \subseteq \mathbb{R}$ and let $f\colon E \to \mathbb{R}$. Then f is *Lipschitz* on E if there is an $L \geq 0$ with

$$|f(x) - f(y)| \leq L|x-y| \qquad \text{for all } x, y \in E$$

Any such L is called a *Lipschitz constant* for f on E.

The absolute value $x \mapsto |x|$ on $\mathbb{R}$ is Lipschitz with $L = 1$, which is exactly proposition 1.3.8(5). An affine function $x \mapsto cx + d$ on $\mathbb{R}$ is Lipschitz with $L = |c|$, since $|(cx+d) - (cy+d)| = |c||x-y|$ by proposition 1.3.8(3). And for $R > 0$ the function $x \mapsto x^2$ on $[0, R]$ is Lipschitz with $L = 2R$, since for $x, y \in [0, R]$,

$$|x^2 - y^2| = |x+y||x-y| \leq 2R|x-y|$$

where the equality is proposition 1.3.8(3) and the inequality uses $0 \leq x+y \leq 2R$ together with $|x-y| \geq 0$. Compare the last instance with example 3.4.3(2): squaring is Lipschitz on each bounded interval, with a constant that grows with the interval, and no single constant serves all of $\mathbb{R}$.

Proposition 3.4.9: Lipschitz Implies Uniformly Continuous

Let $E \subseteq \mathbb{R}$ and let $f\colon E \to \mathbb{R}$ be Lipschitz on E. Then f is uniformly continuous on E.

***Proof*:**
Let $L \geq 0$ be a Lipschitz constant for f and put $L' = L + 1$, so that $L' > 0$. For $x, y \in E$ the number $|x-y|$ is nonnegative and $L \leq L'$, so $L|x-y| \leq L'|x-y|$ and therefore

$$|f(x) - f(y)| \leq L'|x-y|$$

Let $\epsilon > 0$ and put $\delta = \epsilon/L' > 0$. If $x, y \in E$ and $|x-y| < \delta$, then multiplying by $L' > 0$ and using proposition 1.1.2(5) gives $L'|x-y| < L'\delta = \epsilon$, so $|f(x) - f(y)| < \epsilon$. Hence δ answers ϵ in definition 3.4.1, as we sought to show.

The implication is strict, and the square root on $[0,1]$ separates the two conditions.

Example 3.4.10: The Square Root on $[0,1]$

Let $s\colon [0,1] \to \mathbb{R}$ be $s(x) = x^{1/2}$, which is defined because every $x \in [0,1]$ satisfies $x \geq 0$ (theorem 1.3.12 with $n = 2$).

First, s is uniformly continuous on $[0,1]$. By corollary 3.3.13 the function $x \mapsto x^{1/2}$ is continuous on $[0,\infty)$, so s is continuous by proposition 3.3.1. Now theorem 3.4.4 applies to s on $[0,1]$.

Second, s is not Lipschitz on $[0,1]$. Suppose $L \geq 0$ satisfied $|s(x) - s(y)| \leq L|x-y|$ for all $x, y \in [0,1]$. Fix $n \in \mathbb{N}$ and take $y = 0$ and $x = 1/n^2$, which lies in $[0,1]$ because $n \geq 1$ forces $n^2 \geq 1$. Since $(1/n)^2 = 1/n^2$ and $1/n \geq 0$, the uniqueness clause of theorem 1.3.12 gives $s(1/n^2) = 1/n$, and the same clause gives $s(0) = 0$. The assumed inequality reads

$$\frac{1}{n} \leq \frac{L}{n^2}$$

and multiplying by $n^2 > 0$ gives $n \leq L$. This holds for every $n \in \mathbb{N}$, contradicting theorem 1.4.1(2), which produces an $n \in \mathbb{N}$ with $n > L$. So no Lipschitz constant exists.

The failure is a statement about how large a δ a given ϵ permits. A Lipschitz function admits $\delta = \epsilon/L'$, which is proportional to ϵ, while the square root on $[0,1]$ needs a δ that shrinks faster than that near 0. How much faster is the subject of exercise 3.4.1 (1), where $\delta = \epsilon^2$ is shown to answer ϵ on all of $[0,\infty)$, a domain that is not bounded and to which theorem 3.4.4 does not apply.

Exercise 3.4.1

1. Let $x, y \geq 0$. Prove that $|x^{1/2} - y^{1/2}| \leq |x-y|^{1/2}$, and deduce that $x \mapsto x^{1/2}$ is uniformly continuous on $[0,\infty)$, with $\delta = \epsilon^2$ answering ϵ.
2. Let $E, F \subseteq \mathbb{R}$, let $f\colon E \to \mathbb{R}$ be uniformly continuous with $f(E) \subseteq F$, and let $g\colon F \to \mathbb{R}$ be uniformly continuous. Prove that $g \circ f$ is uniformly continuous on E.
3. Let $E \subseteq \mathbb{R}$ and let $f, g\colon E \to \mathbb{R}$ be uniformly continuous. Prove that $f + g$ is uniformly continuous on E. Then show that the product fg need not be, by taking $E = \mathbb{R}$ and $f = g$ the identity function.
4. Let $c > 0$. Prove that $x \mapsto 1/x$ is uniformly continuous on $[c,\infty)$ by exhibiting a Lipschitz constant, and prove that it is not uniformly continuous on $(0,\infty)$.
5. Let $f\colon (0,1] \to \mathbb{R}$ be continuous. Prove that f is uniformly continuous on $(0,1]$ if and only if there is a continuous $g\colon [0,1] \to \mathbb{R}$ with $g(x) = f(x)$ for every $x \in (0,1]$. Begin by showing that the closure of $(0,1]$ is $[0,1]$.
6. Let $E \subseteq \mathbb{R}$ and let $f\colon E \to \mathbb{R}$. Prove that f is uniformly continuous on E if and only if $f(x_n) - f(y_n) \to 0$ for every pair of sequences (x_n) and (y_n) in E with $x_n - y_n \to 0$.

4

Integration

A

The Construction by Dedekind Cuts

⋆ *Optional. This chapter is an aside; nothing later in the book depends on it.*

The main text builds $\mathbb{R}$ in section 1.1 as a set of equivalence classes of Cauchy sequences of rationals. This appendix carries the older construction, due to Dedekind, which builds $\mathbb{R}$ by partitioning $\mathbb{Q}$. It is self-contained and nothing in the book depends on it.

Two reasons to have it. The cut construction reaches the least upper bound property almost immediately, by taking a union of lower sets, where theorem 1.2.4 needed a bisection. Seeing that contrast is worth the detour. And theorem 1.5.2 identifies the field built here with the field built in the main text, so the appendix is an alternative route to the same object, which is a concrete use of a theorem that otherwise reads as an abstraction.

A.1 Cuts and the Field They Form

This appendix assumes the main text's definition 1.1.1 for the ordered-field axioms and the Archimedean property, its proposition 1.1.2 for their consequences, and its proposition 1.1.3 for the irrationality of a square root of two. Only the construction itself is redone here.

Definition A.1.1: [Dedekind] Cut

A *cut* is a pair $A|B$ of subsets of $\mathbb{Q}$ such that

1. $A \neq \emptyset$ and $B \neq \emptyset$;
2. $A \cup B = \mathbb{Q}$ and $A \cap B = \emptyset$;
3. $a < b$ for every $a \in A$ and every $b \in B$;
4. A has no largest element, that is, for every $a \in A$ there is $a' \in A$ with $a < a'$.

The set A is the *lower set* of the cut.

Condition (2) makes $B = \mathbb{Q} \setminus A$, so a cut is determined by its lower set alone, and condition (3) is a condition on that lower set: given (1) and (2), condition (3) holds if and only if A is closed downward, meaning that $p < q$ and $q \in A$ force $p \in A$. Indeed, if (3) holds and $p < q \in A$ with $p \notin A$, then $p \in B$, so $q < p$ by (3), contradicting trichotomy. Conversely, if A is closed downward and $a \in A$, $b \in B$ satisfy $b \leq a$, then $b \in A$, contradicting (2). Both formulations are used below.

Example A.1.2: Rational Cuts, the Cut at the Square Root of Two, and Two Non-Examples

1. For $q \in \mathbb{Q}$ put $q^* = \{p \in \mathbb{Q} \big| p < q\}$. Then $q^*|(\mathbb{Q} \setminus q^*)$ is a cut. It is nonempty since $q - 1 \in q^*$, and its complement is nonempty since $q \notin q^*$. It is closed downward, since $p < p' < q$ gives $p < q$. And it has no largest element: if $p < q$ then $p < (p+q)(1+1)^{-1} < q$ by proposition 1.1.2(8), so $(p+q)(1+1)^{-1}$ is an element of q^* exceeding p. This cut is called the *rational cut* at q. A cut is determined by its lower set, so q^* is written both for the set displayed above and for the cut, and therefore for the real number, it determines. Context fixes which is meant. The two cases used constantly below are 0^*, the set of negative rationals, and 1^*, the set of rationals less than 1.

2. Let $A = \{q \in \mathbb{Q} \big| q \leq 0 \text{ or } q^2 < 2\}$ and $B = \mathbb{Q} \setminus A$. Then $A|B$ is a cut, and it is not a rational cut. The set A contains 0, and $2 \notin A$ because $2^2 = 4 > 2$, so both A and B are nonempty. For downward closure, let $p < q$ with $q \in A$. If $p \leq 0$ then $p \in A$ by definition. If $p > 0$ then $q > p > 0$, so $q \in A$ forces $q^2 < 2$, and proposition 1.1.2(5) applied twice gives $p^2 < pq < q^2 < 2$, so $p \in A$. For the absence of a largest element, let $q \in A$. If $q \leq 0$ then $1 \in A$ (since $1^2 = 1 < 2$) and $1 > q$. If $q > 0$, so that $q^2 < 2$, put

$$q' = \frac{2q+2}{q+2}$$

which is a rational number since $q + 2 > 0$. Two computations settle the matter:

$$q' - q = \frac{2q+2-q(q+2)}{q+2} = \frac{2-q^2}{q+2}, \qquad (q')^2 - 2 = \frac{(2q+2)^2 - 2(q+2)^2}{(q+2)^2} = \frac{2(q^2-2)}{(q+2)^2}$$

Since $q^2 < 2$ and $q + 2 > 0$, the first is positive, so $q' > q$. Since $(q+2)^2 > 0$, the second is negative, so $(q')^2 < 2$ and $q' \in A$. Hence A has no largest element and $A|B$ is a cut. It is not the rational cut at any q: suppose $A = q^*$. Then $q \notin A$, so $q > 0$, and $q^2 \neq 2$ by proposition 1.1.3, so $q^2 > 2$. Form $q' = (2q+2)/(q+2)$ again. It is positive, since $2q + 2 > 0$

and $q+2>0$. The first display above gives $q'-q=(2-q^2)/(q+2)<0$, so $q'<q$. The second gives $(q')^2-2=2(q^2-2)/(q+2)^2>0$, so $q'\notin A$. But $q'<q$ means $q'\in q^*=A$, a contradiction.

3. The pair $\emptyset|\mathbb{Q}$ satisfies condition (2) of definition A.1.1, satisfies (3) and (4) vacuously, and fails condition (1). It is excluded deliberately: were it admitted, it would be a real number below every other, and the order constructed in definition A.1.4 would have a least element.

4. The pair $\{q\in\mathbb{Q}\big|q\le 0\}|\{q\in\mathbb{Q}\big|q>0\}$ satisfies (1), (2) and (3) and fails (4), since 0 is a largest element of the lower set. Without condition (4) each rational would give rise to two distinct cuts, q^* and $q^*\cup\{q\}$, and exercise A.1.1 (6) works out what then goes wrong with addition.

Part (2) is the cut that proposition 1.1.3 produces: A has no largest element, and B has no least element either (exercise A.1.1 (4)), because proposition 1.1.3 forces every element of B to satisfy $q^2>2$. Parts (1) and (2) together say that the cuts include copies of all the rational numbers and at least one object that is not among them.

Definition A.1.3: The Real Numbers

The set of all cuts of $\mathbb{Q}$ is denoted $\mathbb{R}$, and its elements are called *real numbers*.

Since a cut is determined by its lower set, the following abbreviation is used throughout: a real number is written α, its lower set is written A, and $p\in\alpha$ means $p\in A$. An operation on real numbers is then specified by saying which rationals belong to the lower set of the result, and part of verifying such a specification is checking that the set produced is the lower set of a cut.

Definition A.1.4: Order on the Real Numbers

For real numbers $\alpha=A|B$ and $\gamma=C|D$, write $\alpha<\gamma$ if $A\subseteq C$ and $A\ne C$, and $\alpha\le\gamma$ if $A\subseteq C$.

The order compares two real numbers by comparing how much of $\mathbb{Q}$ lies below each. Write $\sigma=A|B$ for the cut of example A.1.2(2), so that $A=\{q\in\mathbb{Q}\big|q\le 0 \text{ or } q^2<2\}$. Then $1^*<\sigma<2^*$. For the left inequality, a rational $p<1$ satisfies $p\le 0$ or $0<p<1$, and in the second case $p^2<1<2$. Hence $1^*\subseteq A$, while $1\in A$ and $1\notin 1^*$. For the right inequality, a rational $q\in A$ with $q>0$ has $q^2<2<4$, so $q<2$ by proposition 1.1.2(5), and a rational $q\le 0$ satisfies $q<2$ as well. Hence $A\subseteq 2^*$, while $3/2\in 2^*$ and $3/2\notin A$ because $(3/2)^2=9/4>2$.

Each arithmetic operation on $\mathbb{R}$ has to be specified by saying which rationals lie below the result, and the three operations differ sharply in how much care that takes. For addition the answer is the direct one: the rationals below $\alpha+\beta$ are the sums of a rational below α and a rational below β. For negation the direct answer fails. The set of negatives of the rationals not below α has a largest element whenever that complement has a least one, which condition (4) of definition A.1.1 forbids, so the extreme element has to be discarded, and the definition below does it by requiring a strict gap $r>0$. For multiplication the direct answer fails differently: a product of two rationals below α and β is large when both factors are large and negative, so the products of rationals below α and β are not bounded above at all. The definition therefore treats positive real numbers first and extends to the rest by the sign rules of proposition 1.1.2(4).

Theorem A.1.5: The Real Numbers Form an Ordered Field

For real numbers α and β define the following subsets of $\mathbb{Q}$.

1. $\alpha + \beta := \{r + s \big| r \in \alpha \text{ and } s \in \beta\}$.
2. $-\alpha := \{p \in \mathbb{Q} \big| \text{there is } r \in \mathbb{Q} \text{ with } r > 0 \text{ and } -p - r \notin \alpha\}$.
3. If $\alpha > 0^*$ and $\beta > 0^*$,
$$\alpha\beta := \{p \in \mathbb{Q} \big| p \leq rs \text{ for some } r \in \alpha \text{ and } s \in \beta \text{ with } r > 0 \text{ and } s > 0\}$$
Further, $\alpha \cdot 0^* := 0^*$ and $0^* \cdot \alpha := 0^*$; and for $\alpha, \beta \neq 0^*$ not both positive,
$$\alpha\beta := \begin{cases} (-\alpha)(-\beta) & \text{if } \alpha < 0^* \text{ and } \beta < 0^* \\ -\big((-\alpha)\beta\big) & \text{if } \alpha < 0^* < \beta \\ -\big(\alpha(-\beta)\big) & \text{if } \beta < 0^* < \alpha \end{cases}$$
4. If $\alpha > 0^*$,
$$\alpha^{-1} := \{p \in \mathbb{Q} \big| p \leq 0\} \cup \{p \in \mathbb{Q} \big| p > 0 \text{ and } bp < 1 \text{ for some } b \notin \alpha\}$$
and if $\alpha < 0^*$, $\alpha^{-1} := -\big((-\alpha)^{-1}\big)$.

Each of these subsets is the lower set of a cut. With these operations and with the order of definition A.1.4, the set $\mathbb{R}$ is an ordered field whose zero element is 0^* and whose unit element is 1^*.

***Proof*:**
Step 1: the order is transitive and satisfies trichotomy. Transitivity is transitivity of inclusion together with the observation that $A \subseteq C \subseteq E$ with $A \neq C$ and $C \neq E$ gives $A \neq E$: if $A = E$ then $C \subseteq E = A \subseteq C$, so $C = A$. For trichotomy, let $\alpha = A|B$ and $\gamma = C|D$ be cuts and suppose A is not contained in C. Take $p \in A$ with $p \notin C$. For any $q \in C$ we cannot have $p < q$, since C is closed downward and would then contain p. Nor is $q = p$ possible, since $p \notin C$. So $q < p$, and A closed downward gives $q \in A$. Hence $C \subseteq A$, and $C \neq A$ because $p \in A \setminus C$, so $\gamma < \alpha$. Thus at least one of $\alpha < \gamma$, $\alpha = \gamma$, $\gamma < \alpha$ holds, and at most one holds because $A \subseteq C$ and $C \subseteq A$ together give $A = C$.
Step 2: addition, and the additive axioms. First, $\alpha + \beta$ is the lower set of a cut. It is nonempty, being a set of sums of elements of two nonempty sets. It is not all of $\mathbb{Q}$: choose $r' \notin \alpha$ and $s' \notin \beta$. For $r \in \alpha$ and $s \in \beta$ we have $r < r'$ and $s < s'$ by definition A.1.1(3), hence $r+s < r'+s < r'+s'$, so $r' + s' \notin \alpha + \beta$. It is closed downward: if $q < r + s$ with $r \in \alpha$ and $s \in \beta$, then $q - s < r$, so $q - s \in \alpha$, and $q = (q - s) + s \in \alpha + \beta$. It has no largest element: given $r + s$, choose $t \in \alpha$ with $t > r$, and then $t + s > r + s$ lies in $\alpha + \beta$.
Commutativity and associativity are inherited from $\mathbb{Q}$, since $\alpha + \beta$ and $\beta + \alpha$ are the same set of sums, and $(\alpha + \beta) + \gamma$ and $\alpha + (\beta + \gamma)$ are both the set of all $r + s + t$ with $r \in \alpha$, $s \in \beta$, $t \in \gamma$.

For the identity, if $r \in \alpha$ and $s \in 0^*$ then $s < 0$, so $r + s < r$ and $r + s \in \alpha$. Hence $\alpha + 0^* \subseteq \alpha$. Conversely, given $p \in \alpha$ choose $r \in \alpha$ with $r > p$. Then $p - r < 0$, so $p - r \in 0^*$ and $p = r + (p - r) \in \alpha + 0^*$. Thus $\alpha + 0^* = \alpha$.

Next, $-\alpha$ is the lower set of a cut. It is nonempty: choose $s \notin \alpha$ and put $p = -s - 1$. Then $-p - 1 = s \notin \alpha$, so $p \in -\alpha$. It is not all of $\mathbb{Q}$: if $q \in \alpha$ and $-q \in -\alpha$, there would be $r > 0$ with $q - r \notin \alpha$, but $q - r < q$ and α is closed downward, so $q - r \in \alpha$. Hence $-q \notin -\alpha$. It is closed downward: if $q < p \in -\alpha$, take $r > 0$ with $-p - r \notin \alpha$. Then $-q - r > -p - r$, and since α is closed downward and does not contain $-p - r$, it does not contain $-q - r$ either, so $q \in -\alpha$. It has no largest element: with p and r as above, put $t = p + r/2$. Then $t > p$ and $-t - r/2 = -p - r \notin \alpha$, so $t \in -\alpha$.

Now $\alpha + (-\alpha) = 0^*$. For one inclusion, let $r \in \alpha$ and $p \in -\alpha$, and take $s > 0$ with $-p - s \notin \alpha$. Then $r < -p - s$ by definition A.1.1(3), so $r + p < -s < 0$ and $r + p \in 0^*$. For the other, let $v \in 0^*$ and put $w = -v/(1+1)$, which is a positive rational. The set S of integers n with $nw \in \alpha$ is nonempty and bounded above. Indeed, choose $a \in \alpha$ and $b \notin \alpha$. Since $\mathbb{Q}$ is Archimedean there is a positive integer m with $mw > -a$, so $-mw < a$ and hence $-mw \in \alpha$, giving $-m \in S$. There is likewise a positive integer k with $kw > b$, and every rational at least as large as b lies outside α because α is closed downward, so $nw \notin \alpha$ for every $n \geq k$ and S is bounded above by k. A nonempty set of integers bounded above has a largest element, so let n be the largest element of S: then $nw \in \alpha$ and $(n+1)w \notin \alpha$. Put $p = -(n+2)w$. Then $-p - w = (n+1)w \notin \alpha$, so $p \in -\alpha$, and $nw + p = nw - (n+2)w = -2w = v$. Hence $v \in \alpha + (-\alpha)$, and $\alpha + (-\alpha) = 0^*$.

Two consequences of what has just been proved are used repeatedly below. The additive inverse is unique: if $\alpha + \eta = 0^*$ then

$$\eta = 0^* + \eta = \big((-\alpha) + \alpha\big) + \eta = (-\alpha) + (\alpha + \eta) = (-\alpha) + 0^* = -\alpha$$

Consequently $-(-\alpha) = \alpha$, because $(-\alpha) + \alpha = 0^*$, and $-(\alpha + \beta) = (-\alpha) + (-\beta)$, because $(\alpha + \beta) + \big((-\alpha) + (-\beta)\big) = \big(\alpha + (-\alpha)\big) + \big(\beta + (-\beta)\big) = 0^*$.

Step 3: the order is compatible with addition. Let $\beta < \gamma$. Every element of $\alpha + \beta$ is $r + s$ with $r \in \alpha$ and $s \in \beta \subseteq \gamma$, so $\alpha + \beta \leq \alpha + \gamma$. Suppose the two were equal. Using only the properties of addition proved in Step 2,

$$\beta = 0^* + \beta = \big((-\alpha) + \alpha\big) + \beta = (-\alpha) + (\alpha + \beta) = (-\alpha) + (\alpha + \gamma) = \gamma$$

contradicting $\beta < \gamma$. So $\alpha + \beta < \alpha + \gamma$. In particular $\alpha > 0^*$ if and only if $-\alpha < 0^*$: adding $-\alpha$ to $0^* < \alpha$ gives $-\alpha < 0^*$, and adding α to $-\alpha < 0^*$ gives $0^* < \alpha$.

Step 4: multiplication of positive real numbers. Let $\alpha > 0^*$ and $\beta > 0^*$. Three observations about an arbitrary real number $\delta > 0^*$ are used throughout this step. Its lower set properly contains 0^*, so it contains some $q \geq 0$, and if $q = 0$ then, δ having no largest element, it contains a larger element, which is a positive rational. In either case δ contains a positive rational. Every rational outside δ is then positive, being larger than that one by definition A.1.1(3). And δ contains every nonpositive rational, since it is closed downward and contains a positive one.

The set $\alpha\beta$ is the lower set of a cut. It is nonempty, containing rs for positive $r \in \alpha$ and $s \in \beta$. It is not all of $\mathbb{Q}$: choose $r' \notin \alpha$ and $s' \notin \beta$, both positive by the previous paragraph. If $r's' \leq rs$ with $r \in \alpha$, $s \in \beta$, $r, s > 0$, then $r < r'$ and $s < s'$, so $rs < r's < r's'$ by proposition 1.1.2(5), a contradiction. Hence $r's' \notin \alpha\beta$. It is closed downward, since $q < p \leq rs$ gives $q \leq rs$. It has no largest element: given $p \leq rs$ as in the definition, choose $r' \in \alpha$ with $r' > r$. Then $r's > rs \geq p$ and $r's \in \alpha\beta$. Since $\alpha\beta$ contains the positive rational rs and contains every nonpositive rational, $\alpha\beta > 0^*$, which is the second order axiom of definition 1.1.1.

Commutativity holds because $rs = sr$ in $\mathbb{Q}$, so the condition defining $\alpha\beta$ and the condition defining $\beta\alpha$ are the same. For associativity, $p \in (\alpha\beta)\gamma$ means $p \leq ut$ for some positive $u \in \alpha\beta$ and positive $t \in \gamma$, and such a u satisfies $u \leq rs$ for positive $r \in \alpha$, $s \in \beta$. Then $ut \leq rst$, so $p \leq rst$. Conversely, if $p \leq rst$ with r, s, t positive elements of α, β, γ, then rs is a positive

element of $\alpha\beta$ and $p \leq (rs)t$, so $p \in (\alpha\beta)\gamma$. Hence $(\alpha\beta)\gamma$ is the set of p with $p \leq rst$ for some such r, s, t, and this description is symmetric in the three factors, so it also describes $\alpha(\beta\gamma)$.
For the unit, if $p \leq rs$ with $r \in \alpha$, $r > 0$ and $0 < s < 1$, then $rs < r$, so $p < r$ and $p \in \alpha$. Hence $\alpha 1^* \subseteq \alpha$. Conversely let $p \in \alpha$. If $p \leq 0$, choose positive $r \in \alpha$ and take $s = 1/2 \in 1^*$. Then $rs > 0 \geq p$. If $p > 0$, choose $r \in \alpha$ with $r > p$ and put $s = p/r$. Then $0 < s < 1$, so $s \in 1^*$, and $p = rs$. In both cases $p \in \alpha 1^*$, so $\alpha 1^* = \alpha$.
For distributivity among positive real numbers, let $\alpha, \beta, \gamma > 0^*$. Observe first that $rs \in \alpha\beta$ whenever $r \in \alpha$ is positive and $s \in \beta$ is arbitrary: for $s > 0$ this is the definition, and for $s \leq 0$ the product rs is nonpositive and $\alpha\beta$ contains every nonpositive rational. Now let $p \in \alpha(\beta + \gamma)$, so $p \leq rm$ with $r \in \alpha$ positive and $m \in \beta + \gamma$ positive. Write $m = s + t$ with $s \in \beta$ and $t \in \gamma$. Then $rm = rs + rt$ with $rs \in \alpha\beta$ and $rt \in \alpha\gamma$, so $rm \in \alpha\beta + \alpha\gamma$, and that set is closed downward, so p belongs to it. Conversely let $u \in \alpha\beta$ and $v \in \alpha\gamma$, say $u \leq r_1 s_1$ and $v \leq r_2 t_1$ with all of r_1, r_2, s_1, t_1 positive and lying in the respective real numbers. Let r be the larger of r_1 and r_2, an element of α. Then $u + v \leq r_1 s_1 + r_2 t_1 \leq r s_1 + r t_1 = r(s_1 + t_1)$, and $s_1 + t_1$ is a positive element of $\beta + \gamma$, so $u + v \in \alpha(\beta + \gamma)$. Hence $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$.
Step 5: multiplicative inverses of positive real numbers. Let $\alpha > 0^*$. Every rational outside α is then positive. The set α^{-1} is the lower set of a cut. It is nonempty, containing 0. It is not all of $\mathbb{Q}$: choose a positive $a \in \alpha$ and suppose $1/a \in \alpha^{-1}$. Since $1/a > 0$ there would be $b \notin \alpha$ with $b/a < 1$, that is $b < a$, contradicting $a < b$. It is closed downward: if $q < p \in \alpha^{-1}$ and $q \leq 0$ then $q \in \alpha^{-1}$, while if $q > 0$ then $p > 0$, so there is $b \notin \alpha$ with $bp < 1$, and $b > 0$ gives $bq < bp < 1$. It has no largest element: if $p \leq 0$, choose $b \notin \alpha$ and note that $q = 1/(2b)$ is positive with $bq = 1/2 < 1$, so $q \in \alpha^{-1}$ and $q > p$; if $p > 0$ with $bp < 1$ for some $b \notin \alpha$, then $p < 1/b$, and by proposition 1.1.2(8) any rational q with $p < q < 1/b$ lies in α^{-1} and exceeds p. Since α^{-1} contains a positive rational and every nonpositive one, $\alpha^{-1} > 0^*$.
The identity $\alpha\alpha^{-1} = 1^*$ rests on the following approximation of α from outside. *Claim: for every rational $\lambda > 1$ there are $r \in \alpha$ with $r > 0$ and $b \notin \alpha$ with $b \leq \lambda r$.* To see it, fix a positive $a_0 \in \alpha$ and some $b_0 \notin \alpha$, and put $w = (\lambda - 1)a_0 > 0$. Since $\mathbb{Q}$ is Archimedean there is a positive integer n_0 with $n_0 w > b_0 - a_0$, so $a_0 + n_0 w > b_0$ and therefore $a_0 + n_0 w \notin \alpha$, because α is closed downward and omits b_0. The set of positive integers n with $a_0 + nw \notin \alpha$ is thus nonempty. Let m be its least element. Put $r = a_0 + (m-1)w$ and $b = a_0 + mw$. Then $b \notin \alpha$ by the choice of m, and $r \in \alpha$: if $m = 1$ then $r = a_0 \in \alpha$, and if $m > 1$ then $m - 1$ is a positive integer smaller than m, so $a_0 + (m-1)w \in \alpha$ by the minimality of m. Also $r \geq a_0 > 0$, and

$$b = r + w = r + (\lambda - 1)a_0 \leq r + (\lambda - 1)r = \lambda r$$

which proves the claim.
Now let $p \in \alpha\alpha^{-1}$, say $p \leq rs$ with $r \in \alpha$, $s \in \alpha^{-1}$ both positive, and take $b \notin \alpha$ with $bs < 1$. Then $r < b$, so $rs < bs < 1$ and $p < 1$, giving $\alpha\alpha^{-1} \subseteq 1^*$. Conversely let $p < 1$. If $p \leq 0$ then $p \in \alpha\alpha^{-1}$, since that real number is positive and so contains every nonpositive rational. If $0 < p < 1$ then $1/p > 1$, so proposition 1.1.2(8) provides a rational λ with $1 < \lambda < 1/p$. Apply the claim to get $r \in \alpha$ positive and $b \notin \alpha$ with $b \leq \lambda r$. Then $pb \leq p\lambda r < r$ because $p\lambda < 1$ and $r > 0$, and dividing by the positive rational rb gives $p/r < 1/b$. Choose a rational s with $p/r < s < 1/b$, again by proposition 1.1.2(8). Then $s > 0$ and $bs < 1$, so $s \in \alpha^{-1}$, and $rs > p$, so $p \in \alpha\alpha^{-1}$. Hence $\alpha\alpha^{-1} = 1^*$.
Step 6: multiplication on all of $\mathbb{R}$. By the trichotomy of Step 1, a real number other than 0^* is either positive or negative, and in the second case it is $-(-\alpha)$ with $-\alpha$ positive by Step 3. So the four clauses of the statement define $\alpha\beta$ for every pair, and by Step 4 the result is again a real number.
Commutativity: for $\alpha, \beta > 0^*$ it is Step 4; if both are negative then $\alpha\beta = (-\alpha)(-\beta) =$

$(-\beta)(-\alpha) = \beta\alpha$; if $\alpha < 0^* < \beta$ then $\alpha\beta = -((-\alpha)\beta)$ and $\beta\alpha = -(\beta(-\alpha))$, which agree because $(-\alpha)\beta$ and $\beta(-\alpha)$ are products of positive real numbers and Step 4 gives commutativity there. The case $\beta < 0^* < \alpha$ is the same with the roles interchanged. And if either factor is 0^* both products are 0^*.

The sign rule $(-\alpha)\beta = -(\alpha\beta) = \alpha(-\beta)$ holds for all α, β. The second equality follows from the first by commutativity, so only the first needs checking, and $-0^* = 0^*$ handles the cases where either factor is 0^*. If $\alpha > 0^*$ and $\beta > 0^*$ then $-\alpha < 0^* < \beta$, so $(-\alpha)\beta = -((-(-\alpha))\beta) = -(\alpha\beta)$. If $\alpha > 0^* > \beta$ then $-\alpha < 0^*$ and $\beta < 0^*$, so $(-\alpha)\beta = (-(-\alpha))(-\beta) = \alpha(-\beta)$, while $\alpha\beta = -(\alpha(-\beta))$ by definition, so the two sides agree. If $\alpha < 0^* < \beta$ then $-\alpha > 0^*$ and $\alpha\beta = -((-\alpha)\beta)$ by definition, which is the assertion. If $\alpha < 0^*$ and $\beta < 0^*$ then $-\alpha > 0^* > \beta$, so $(-\alpha)\beta = -((-\alpha)(-\beta))$, while $\alpha\beta = (-\alpha)(-\beta)$ by definition, and again the two sides agree.

Associativity: if any of α, β, γ equals 0^*, both $(\alpha\beta)\gamma$ and $\alpha(\beta\gamma)$ equal 0^*. So let α, β, γ be nonzero and suppose the identity $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ holds for the triple (α, β, γ). It then holds for $(-\alpha, \beta, \gamma)$, since the sign rule turns the left side into $(-(\alpha\beta))\gamma = -((\alpha\beta)\gamma)$ and the right side into $-(\alpha(\beta\gamma))$. It holds for $(\alpha, -\beta, \gamma)$, since the two sides become $(-(\alpha\beta))\gamma = -((\alpha\beta)\gamma)$ and $\alpha(-(\beta\gamma)) = -(\alpha(\beta\gamma))$. And it holds for $(\alpha, \beta, -\gamma)$, since the two sides become $-((\alpha\beta)\gamma)$ and $\alpha(-(\beta\gamma)) = -(\alpha(\beta\gamma))$. Every nonzero real number is either positive or the negative of a positive one, so every triple of nonzero real numbers is obtained from a triple of positive ones by negating some of its entries, and the identity holds for positive triples by Step 4.

Unit and inverses: $0^* \subseteq 1^*$ and $0 \in 1^* \setminus 0^*$, so $1^* > 0^*$. In particular $1^* \neq 0^*$. Hence $\alpha 1^* = \alpha$ for $\alpha > 0^*$ by Step 4, while for $\alpha < 0^*$ the sign rule gives $\alpha 1^* = -((-\alpha)1^*) = -(-\alpha) = \alpha$, and $0^* \cdot 1^* = 0^*$. For $\alpha > 0^*$, Step 5 gives $\alpha\alpha^{-1} = 1^*$. For $\alpha < 0^*$ we have $-\alpha > 0^*$, so $(-\alpha)^{-1} > 0^*$ and hence $\alpha^{-1} = -((-\alpha)^{-1}) < 0^*$. Both factors being negative, $\alpha\alpha^{-1} = (-\alpha)(-(\alpha^{-1})) = (-\alpha)(-\alpha)^{-1} = 1^*$.

Distributivity: let $\alpha, \beta, \gamma \in \mathbb{R}$. If $\alpha = 0^*$ both sides of $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$ are 0^*. If $\alpha < 0^*$ and the identity is known for $-\alpha > 0^*$, then by the sign rule and the identity $-(\mu + \nu) = (-\mu) + (-\nu)$ of Step 2,

$$\alpha(\beta + \gamma) = -\big((-\alpha)(\beta + \gamma)\big) = -\big((-\alpha)\beta + (-\alpha)\gamma\big) = \alpha\beta + \alpha\gamma$$

So assume $\alpha > 0^*$. If β or γ equals 0^* the identity follows from $\alpha 0^* = 0^*$ and Step 2. If $\beta > 0^*$ and $\gamma > 0^*$ it is Step 4. If $\beta < 0^*$ and $\gamma < 0^*$ then $\beta + \gamma < 0^*$ by Step 3, while $-\beta$ and $-\gamma$ are positive, and using $-(\beta + \gamma) = (-\beta) + (-\gamma)$,

$$\alpha(\beta + \gamma) = -\big(\alpha((-\beta) + (-\gamma))\big) = -\big(\alpha(-\beta) + \alpha(-\gamma)\big) = -\big((-(\alpha\beta)) + (-(\alpha\gamma))\big) = \alpha\beta + \alpha\gamma$$

the last step being the Step 2 identity $-(\mu + \nu) = (-\mu) + (-\nu)$ together with $-(-\mu) = \mu$. There remains the case where β and γ have opposite signs, and by commutativity of addition it suffices to treat $\beta > 0^* > \gamma$. Put $\delta = \beta + \gamma$. If $\delta > 0^*$, then $\beta = \delta + (-\gamma)$ is a sum of two positive real numbers, so Step 4 gives $\alpha\beta = \alpha\delta + \alpha(-\gamma) = \alpha\delta - \alpha\gamma$, and adding $\alpha\gamma$ gives $\alpha\delta = \alpha\beta + \alpha\gamma$. If $\delta < 0^*$, then $-\gamma = \beta + (-\delta)$ is a sum of two positive real numbers, so $\alpha(-\gamma) = \alpha\beta + \alpha(-\delta)$, that is $-\alpha\gamma = \alpha\beta - \alpha\delta$, and rearranging again gives $\alpha\delta = \alpha\beta + \alpha\gamma$. If $\delta = 0^*$ then $\gamma = -\beta$, so $\alpha\beta + \alpha\gamma = \alpha\beta - \alpha\beta = 0^* = \alpha\delta$.

Steps 1 through 6 verify every clause of definition 1.1.1 for $\mathbb{R}$, completing the proof.

The theorem says that nothing was lost in passing from $\mathbb{Q}$ to $\mathbb{R}$: every identity and inequality that follows from the ordered field axioms, and therefore every consequence recorded in proposition 1.1.2, holds for real numbers. What was gained is not yet visible, since the least upper bound property plays no part above. It is proved in theorem 1.2.4, and it is the only thing about $\mathbb{R}$ assumed later in this book beyond the ordered field axioms.

One thing still has to be checked, namely that the copy of $\mathbb{Q}$ inside $\mathbb{R}$ given by the rational cuts is a sub-ordered-field carrying the original arithmetic of $\mathbb{Q}$. Without that, a rational number could not be treated as a real number without ambiguity.

Proposition A.1.6: The Rational Numbers Embed in the Real Numbers

The map $\mathbb{Q} \to \mathbb{R}$, $q \mapsto q^*$, is injective and satisfies, for all $q, r \in \mathbb{Q}$,

$$q^* + r^* = (q+r)^*, \qquad q^* r^* = (qr)^*, \qquad -(q^*) = (-q)^*$$

and $q < r$ if and only if $q^* < r^*$. Moreover $0 \mapsto 0^*$ and $1 \mapsto 1^*$.

***Proof*:**
That each q^* is a cut is example A.1.2(1), and 0^* and 1^* are the images of 0 and 1 by definition.
Order. If $q < r$ then every $p < q$ satisfies $p < r$, so $q^* \subseteq r^*$, and $q \in r^* \setminus q^*$, so $q^* < r^*$. Conversely suppose $q^* < r^*$. If $r \leq q$ then every $p < r$ satisfies $p < q$, so $r^* \subseteq q^*$, that is $r^* \leq q^*$, and this contradicts $q^* < r^*$ by the trichotomy established in Step 1 of theorem A.1.5. Hence $q < r$ by trichotomy in $\mathbb{Q}$. In particular the map is injective, since $q \neq r$ forces $q < r$ or $r < q$ and hence $q^* \neq r^*$.
Addition. If $p < q$ and $p' < r$ then $p + p' < q + r$, so $q^* + r^* \subseteq (q+r)^*$. Conversely let $t < q + r$ and put $h = (q + r - t)(1+1)^{-1}$, a positive rational. Then $q - h < q$ and $r - h < r$, and $(q-h) + (r-h) = q + r - 2h = t$, so $t \in q^* + r^*$.
Negation. A rational p lies in $-(q^*)$ exactly when there is $s > 0$ with $-p - s \notin q^*$, that is with $-p - s \geq q$, that is with $p \leq -q - s$. Such an s exists if and only if $p < -q$: if $p < -q$ take $s = (-q-p)(1+1)^{-1} > 0$, for which $-q - s > p$. Conversely, such an s gives $p \leq -q - s < -q$. Hence $-(q^*) = \{p \big| p < -q\} = (-q)^*$.
Multiplication. Suppose first $q > 0$ and $r > 0$, so that $q^* > 0^*$ and $r^* > 0^*$ by the order statement already proved. If $p \leq st$ with $0 < s < q$ and $0 < t < r$, then $st < qt < qr$ by proposition 1.1.2(5), so $p < qr$ and $p \in (qr)^*$. Conversely let $p < qr$. If $p \leq 0$, take $s = q(1+1)^{-1}$ and $t = r(1+1)^{-1}$, which are positive elements of q^* and r^* with $st > 0 \geq p$. If $p > 0$, put $c = p(qr)^{-1}$, so $0 < c < 1$. By proposition 1.1.2(8) choose a rational μ with $c < \mu < 1$, and then, since $c\mu^{-1} < 1$, a rational ν with $c\mu^{-1} < \nu < 1$. Put $s = \mu q$ and $t = \nu r$. Then $0 < s < q$ and $0 < t < r$, and $\mu\nu > c$ gives $st = \mu\nu qr > c\,qr = p$, so $p \in q^* r^*$. Hence $q^* r^* = (qr)^*$ for positive q and r.
For the remaining sign cases, if $q = 0$ or $r = 0$ then both sides are 0^*. If $q < 0 < r$ then $q^* < 0^*$, so by the definition of multiplication and the negation rule just proved,

$$q^* r^* = -\big((-(q^*))r^*\big) = -\big((-q)^* r^*\big) = -\big(((-q)r)^*\big) = \big(-((-q)r)\big)^* = (qr)^*$$

where the last step is $-((-q)r) = qr$ from proposition 1.1.2(4). The case $r < 0 < q$ is identical with the roles interchanged. If $q < 0$ and $r < 0$ then $q^* r^* = (-(q^*))(-(r^*)) = (-q)^*(-r)^* = ((-q)(-r))^* = (qr)^*$, again by proposition 1.1.2(4) in the last step, as we sought to show.

From now on a rational number q and the real number q^* are identified, and the star is retained only where the distinction between the two is at issue. Under this identification $\mathbb{Q}$ is a subfield of $\mathbb{R}$ on which the order of definition A.1.4 restricts to the order of $\mathbb{Q}$, and example A.1.2(2) exhibits an element of $\mathbb{R}$ that is not in $\mathbb{Q}$.

The construction is not finished until the cut field is shown to have the property the main text's does, and this is where the cut route pays off: the supremum is a union, and no sequence has to be built.

Proposition A.1.7: The Cut Field Is Complete

Every nonempty set of cuts that is bounded above has a least upper bound.

***Proof*:**
Let $\mathcal{C}$ be a nonempty set of cuts, bounded above by the cut β with lower set B. Let A be the union of the lower sets of the members of $\mathcal{C}$, and let γ be the pair $A|(\mathbb{Q} \setminus A)$.

γ is a cut. A is nonempty because $\mathcal{C}$ is and each lower set is. It is not all of $\mathbb{Q}$: every member of $\mathcal{C}$ is at most β, so every lower set is contained in B, hence $A \subseteq B \neq \mathbb{Q}$. It is closed downward, being a union of downward-closed sets. And it has no largest element: given $p \in A$, choose a member of $\mathcal{C}$ whose lower set contains p. That lower set has no largest element, so it contains some $r > p$, and $r \in A$.

γ is an upper bound. Each member of $\mathcal{C}$ has its lower set contained in A, which is exactly the statement that it is at most γ in the order of definition A.1.4.

γ is least. Let δ be an upper bound of $\mathcal{C}$, with lower set D. Every lower set of a member of $\mathcal{C}$ is contained in D, so their union A is contained in D, which is $\gamma \leq \delta$, completing the proof.

So the cut field is a complete ordered field, and theorem 1.5.2 applies to it and to the field of section 1.1 together: the two are isomorphic by a unique order-preserving isomorphism, and the appendix has constructed the same $\mathbb{R}$ by a different route.

A.1.8 Remark: How the Two Constructions Compare Cuts are not the only route from $\mathbb{Q}$ to $\mathbb{R}$. A second construction, the one Terence Tao follows in *Analysis I*, calls a sequence $q_1, q_2, q_3, \ldots$ of rationals *Cauchy* when for every rational $\epsilon > 0$ there is an index N with $-\epsilon < q_n - q_m < \epsilon$ for all $n, m \geq N$. It calls two Cauchy sequences $q_1, q_2, \ldots$ and $q'_1, q'_2, \ldots$ equivalent when for every rational $\epsilon > 0$ there is an index N with $-\epsilon < q_n - q'_n < \epsilon$ for all $n \geq N$. And it takes a real number to be an equivalence class. That construction uses the arithmetic of $\mathbb{Q}$ and a notion of small difference, and nothing about the order, which is why it is the one that generalizes to the completion of a metric space. The cut construction uses the order of $\mathbb{Q}$ and nothing else. The two produce the same theory for the purposes of this book: every property of $\mathbb{R}$ used below is derived either from the ordered field axioms verified in theorem A.1.5 or from the least upper bound property proved in theorem 1.2.4.

Exercise A.1.1

1. Decide which of the following subsets $A \subseteq \mathbb{Q}$ are lower sets of cuts, and justify each answer against definition A.1.1.

    1. $A = \{q \in \mathbb{Q} | q \leq 3\}$.
    2. $A = \{q \in \mathbb{Q} | q \leq 0 \text{ or } q^2 < 3\}$.
    3. $A = \{q \in \mathbb{Q} | q^2 < 2\}$.

2. Let α be a real number with $\alpha > 0^*$. Show that there is a rational $r > 0$ with $r^* \leq \alpha$. Deduce that there is no real number α satisfying $0^* < \alpha < r^*$ for every rational $r > 0$.

3. Let $\alpha = A|B$ be a cut. Show that $\alpha = q^*$ for some $q \in \mathbb{Q}$ if and only if B has a least element.

4. Let $\alpha = A|B$ be the cut of example A.1.2(2), so that B is the set of rationals $q > 0$ with $q^2 > 2$. Show that B has no least element, by running the computation of example A.1.2(2) with a $q \in B$. Combined with exercise A.1.1 (3), this gives a second proof that α is not a rational cut.

5. Let $\alpha = A|B$ be a cut and let $r \in \mathbb{Q}$. Show that $\{a + r \big| a \in A\}$ is the lower set of a cut, and that this cut is $\alpha + r^*$.

6. Suppose condition (4) of definition A.1.1 were dropped. Show that for each $q \in \mathbb{Q}$ the set $\{p \in \mathbb{Q} \big| p \leq q\}$ would then satisfy conditions (1), (2) and (3), so that each rational would give rise to two objects. Writing α for that set and defining addition as in theorem A.1.5(1), show that $\alpha + 0^* \neq \alpha$, and identify which of the two objects attached to q the sum $\alpha + 0^*$ is.

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Bibliography

[1] Nathanael Chwojko-Srawley. *Preliminary Chapter*. 2023. URL: /Users/nathanaelchwojko-srawkey/website-nate/nate-website/src/assets/latex/notes/preliminary_chapter.